

UMC 0.11 um Logic/Mixed_Mode AI Advanced Enhancement RVT 5 V MOSFET SPICE Model Document

**Version 0.2 Phase1
2012/06/06**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2012/06/06	Laney Fan Wei Lun Hung	Original release.

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

5 V n-ch MOSFET
5 V p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.2.4

For detailed descriptions of these compact model equations and parameters, please refer to this material.

BSIM3V3 MOSFET Model User Manual

In BSIM3V3.2.4, there are two types of models that can be used. One is a binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is a global model, which means one scalable parameter set is provided and it can cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for their many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the silicon data, which can be found in the following sections.

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

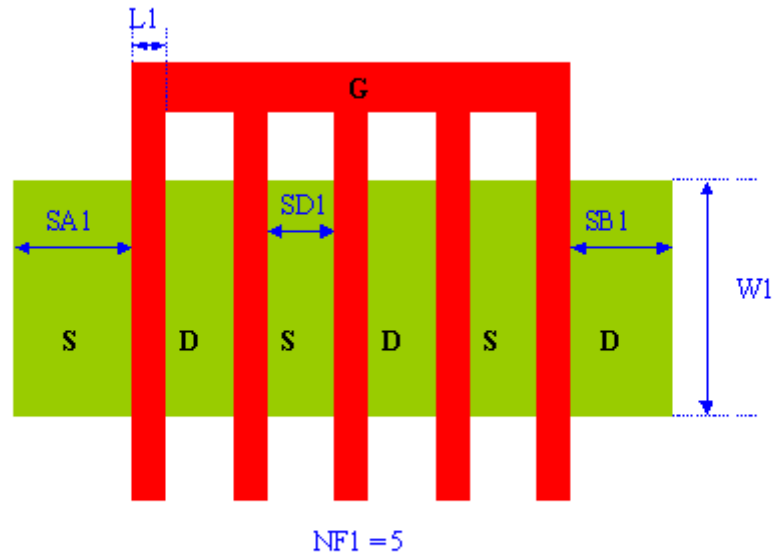
Synopsys HSPICE release 2007.09
Cadence SPECTRE version 10.1.1.111.isr8 32bit
Mentor Graphics ELDO version 6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET
FNFP: Fast n-ch MOSFET and Slow p-ch MOSFET
SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

Follow these steps to invoke the model file during simulation.

(a) Example for Odd Finger Number



(b) Example for Even Finger Number

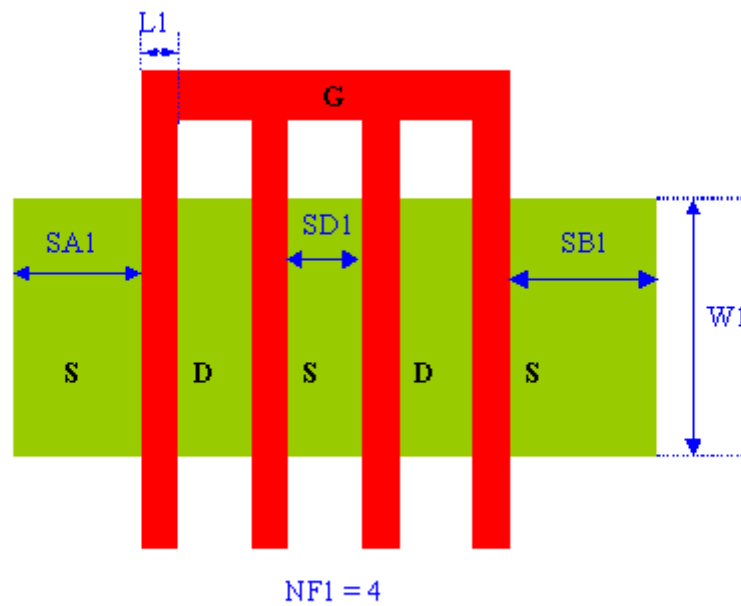


Figure 1-1 Layout example for LOD effect

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
.lib "/ModelFileName.lib" CaseName
, where CaseName can be TT, FF, SS, FNFP, or SNFP.
- Define device condition by 'Mxx' statement;

Example 1: For odd finger Number device (Refer to Figure 1-1 (a))

```
M1 Vdd Vgg Vss Vbb DeviceName W='W1*NF1' L=L1 NF=NF1 SA=SA1  
+ SB=SB1 SD=SD1  
+ PS='2*(SA1+W1)+(SD1+W1)*(NF1-1)'  
+ PD='2*(SB1+W1)+(SD1+W1)*(NF1-1)'  
+ AS='SA1*W1+SD1*W1*(NF1-1)/2'  
+ AD='SB1*W1+SD1*W1*(NF1-1)/2'
```

Example 2: For even finger number device (Refer to Figure 1-1(b))

```
M1 Vdd Vgg Vss Vbb DeviceName W='W1*NF1' L=L1 NF=NF1 SA=SA1  
+ SB=SB1 SD=SD1  
+ PS='2*(SA1+W1)+2*(SD1+W1)*((NF1)/2-1)+2*(SB1+W1)'  
+ PD='(SD1+W1)*NF1'  
+ AS='SA1*W1+SD1*W1*((NF1)/2-1)+SB1*W1'  
+ AD='SD1*W1*(NF1)/2'
```

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
SetOption1 options scale=0.9
include "/ModelFileName.lib.scs" section=CaseName
, where CaseName can be TT, FF, SS, FNFP, or SNFP.
- Define device condition by 'Mxx' statement;

Example 1: For odd finger number device (Refer to Figure 1-1 (a))

```
M1 ( Vdd Vgg Vss Vbb) DeviceName w=W1*NF1 l=L1 nf=NF1 sa=SA1
+ sb=SB1 sd=SD1 sca=SCA1 scb=SCB1 scc=SCC1
+ ps=2*(SA1+W1)+(SD1+W1)*(NF1-1)
+ pd=2*(SB1+W1)+(SD1+W1)*(NF1-1)
+ as=SA1*W1+SD1*W1*(NF1-1)/2
+ ad=SB1*W1+SD1*W1*(NF1-1)/2
```

Example 2: For even finger number device (Refer to Figure 1-1(b))

```
M1 (Vdd Vgg Vss Vbb) DeviceName w=W1*NF1 l=L1 nf=NF1 sa=SA1
+ sb=SB1 sd=SD1
+ ps=2*(SA1+W1)+2*(SD1+W1)*((NF1)/2-1)+2*(SB1+W1)
+ pd=(SD1+W1)*NF1
+ as=SA1*W1+SD1*W1*((NF1)/2-1)+SB1*W1
+ ad=SD1*W1*(NF1)/2
```

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
.lib "/ModelFileName.lib.eldo" CaseName
, where CaseName can be TT, FF, SS, FNFP, or SNFP.
- Define device condition by 'Mxx' statement;

Example 1: For odd finger number device (Refer to Figure 1-1 (a))

```
M1 Vdd Vgg Vss Vbb DeviceName W='W1*NF1' L=L1 NF=NF1 SA=SA1
+ SB=SB1 SD=SD1 + PS='2*(SA1+W1)+(SD1+W1)*(NF1-1)'
+ PD='2*(SB1+W1)+(SD1+W1)*(NF1-1)'
+ AS='SA1*W1+SD1*W1*(NF1-1)/2'
+ AD='SB1*W1+SD1*W1*(NF1-1)/2'
```

Example 2: For even finger number device (Refer to Figure 1-1(b))

```
M1 Vdd Vgg Vss Vbb DeviceName W='W1*NF1' L=L1 NF=NF1 SA=SA1
+ SB=SB1 SD=SD1
+ PS='2*(SA1+W1)+2*(SD1+W1)*((NF1)/2-1)+2*(SB1+W1)'
```

+ PD='(SD1+W1)*NF1'
+ AS='SA1*W1+SD1*W1*((NF1)/2-1)+SB1*W1'
+ AD='SD1*W1*(NF1)/2'

ModelFileName for 5 V MOSFET: l110ae_rvt50_v021

DeviceName for 5 V n-ch MOSFET: n_50_rvtl110ae

DeviceName for 5 V p-ch MOSFET: p_50_rvtl110ae

As stated above, one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Note: Three choices of thick oxide IO devices are available, 2.5V, 3.3V and HG 3.3V, 5V, which cannot coexist on the same chip. If any design requirement is necessary, please discuss in advance with UMC through CE department.

Geometry range (After 90% shrinkage) :

For 5 V n-ch MOSFET:

$$\begin{aligned}0.488 \text{ um} &\leq L_{\text{DES}} \leq 10 \text{ um} \\0.350 \text{ um} &\leq W_{\text{DES}} \leq 10 \text{ um} \\0.324 \text{ um} &\leq SA, SB\end{aligned}$$

For 5 V P-ch MOSFET:

$$\begin{aligned}0.488 \text{ um} &\leq L_{\text{DES}} \leq 10 \text{ um} \\0.350 \text{ um} &\leq W_{\text{DES}} \leq 10 \text{ um} \\0.324 \text{ um} &\leq SA, SB\end{aligned}$$

Note: It is recommended to adopt multi-finger structures if $W_{\text{DES}} > 10 \text{ um}$.

Adopt multi-finger structures if $W_{\text{DES}} > 10 \text{ um}$.
 $0.324 \text{ um} \leq SA, SB \leq 2 \text{ um}$

Note: $SAREF = SBREF = 3.6 \text{ um}$ is the reference dimension of the STI-stress (LOD) effect.

Voltage range:

For 5 V n-ch MOSFET:

$$\begin{aligned}0 \text{ V} &\leq V_{\text{GS}} \leq 5 \text{ V}(*1.1) \\0 \text{ V} &\leq V_{\text{DS}} \leq 5 \text{ V}(*1.1) \\-3 \text{ V} &\leq V_{\text{BS}} \leq 0 \text{ V}\end{aligned}$$

For 5 V p-ch MOSFET:

$$\begin{aligned}0 \text{ V} &\geq V_{\text{GS}} \geq -5 \text{ V}(*1.1) \\0 \text{ V} &\geq V_{\text{DS}} \geq -5 \text{ V}(*1.1) \\3 \text{ V} &\geq V_{\text{BS}} \geq 0 \text{ V}\end{aligned}$$

Temperature range:

$$-40 \text{ C} \sim +125 \text{ C}$$

Feature\ GDSII	Drawing in shrinkable L130AE GDS database (unit : um)	L110AE GDSII =0.9*(L130AE GDS) Dimension (unit : um)	L110AE SPICE Dimension (unit : um)
Core (W/L)	10/0.12	9/0.108	HS(L + 8 nm): 9/0.116 LL/SP(L +12 nm): 9/0.12
2.5V I/O (W/L)	10/0.24	9/0.216	(L +12 nm): 9/0.228
3.3V I/O (W/L)	10/0.34	9/0.306	(L +12 nm): 9/0.318
5V I/O (W/L)	10/0.542	9/0.488	(L +12 nm): 9/0.5
HG 3.3V I/O (W/L)	HG IO-PMOS : 10/0.32 HG IO-NMOS : 10/0.34	HG IO-PMOS : 9/0.228 HG IO-NMOS : 9/0.306	HG IO-PMOS (L +12 nm): 9/0.3 HG IO-NMOS (L +12 nm): 9/0.318
Salicide and Non-salicide resistor	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
Mixed-Mode Device	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
RF Device	(Width/Length)	0.9*(Width/Length)	0.9*(Width/Length)
Contact size	0.16*0.16	0.144*0.144	0.144*0.144
Via size(1X;2X)	0.2*0.2; 0.4*0.4	0.18*0.18; 0.36*0.36	MVA1(W -0.02um) : 0.16*0.2 Other 1X MVA : 0.18*0.18 2X MVA (W -0.04um): 0.28*0.44
Metal size	Width/Space	0.9*(Width/Space)	0.9*(Width/Space)
Slotting inserting	Metal line width>5.556um	Metal line width>5um	

Note:

1. Because process bias make drawn and physical dimension are different, please refer to interconnect capacitance model (G-04 document in individual DSM) for precise interconnection-capacitance extraction.
2. Please refer to SPICE model (G-05 document in individual DSM) for precise device characteristics.

2 5 V N-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW1PA	BSW1PA	BSW1PA	BSW1PA
Lot ID	D6FW8	D6FW8	D6FW8	D6FW8
Wafer Number	#14, #15	#14, #15	#14, #15	#14, #15

2.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at $I_D = 300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for NMOS and $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for PMOS, respectively.

Table 2-2 Electrical parameters for NMOS

All parameters are given for T=25 °C , V_{BS}=0 V unless specified otherwise.

Target & Centered Value are given for SA=3.6u SB=3.6u

Measured & Extracted Value are given for SA=3.6u SB=3.6u

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
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Long channel device (W/L= 10 μ m / 10 μ m)

V _{TLIN}	V _{DS} = 0.1 V	V	0.818	0.817	0.818	0.818
I _{ON}	V _{DS} = 5 V	μA/ μm	57.90	57.70	57.87	57.90

Wide and short channel device (W/L= 10 μ m / 0.5 μ m)

V _{TLIN}	V _{DS} = 0.1 V	V	0.869	0.863	0.864	0.869
V _{TSAT}	V _{DS} = 5 V	V	0.782	0.777	0.777	0.782
Body Effect V _T shift	V _{DS} = 5 V V _{BS} = 0 ~ -5 V	V	--	0.504	0.559	0.575
DIBL	V _{DS} = 0.1~5 V	V	0.087	0.085	0.087	0.087
I _{ON}	V _{DS} =V _{GS} = 5 V	μA/μm	535.00	534.80	532.85	535.00
I _{OFF}	V _{DS} = 5 V, V _{GS} =0 V	A/μm	6.5e-14	6.2E-14	6.5E-14	6.5E-14

Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)

V _{TLIN}	V _{DS} = 0.1 V	V	0.815	0.810	0.809	0.815
V _{TSAT}	V _{DS} = 5 V	V	0.799	0.796	0.789	0.796
I _{ON}	V _{DS} =V _{GS} = 5 V	μA/μm	51.43	51.31	51.73	51.43

Narrow and short channel device (W/L= 0.35 μ m / 0.5 μ m)

V _{TLIN}	V _{DS} = 0.1 V	V	0.833	0.833	0.823	0.833
V _{TSAT}	V _{DS} = 5 V	V	0.749	0.755	0.747	0.749
I _{ON}	V _{DS} =V _{GS} = 5 V	μA/μm	611.43	612.63	616.09	611.34
I _{OFF}	V _{DS} = 5 V, V _{GS} =0 V	A/μm	--	2.1E-12	1.1E-13	1.1E-13

Please set gminde=**1.18e-13** in spectre simulator for achieving the consistent I_{OFF} result with hspice's.

2.3 MOSFET DC Model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

WL	10	5	1	0.8	0.6	0.5
10	X	X	X	X	X	X
5	X		X			X
1	X	X	X	X	X	X
0.8	X		X			X
0.6	X	X	X	X	X	X
0.43	X		X		X	X
0.35	X	X	X	X	X	X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}				I_D vs. V_{DS}			
	Start	Stop	Step		Start	Stop	Step
V_{DS} (V)	0.1	5	4.9	V_{DS} (V)	0	5.5	0.05
V_{GS} (V)	-0.5	5.5	0.05	V_{GS} (V)	1.1	5	0.39
V_{BS} (V)	0	-3	-0.6	V_{BS} (V)	0	-3	-3

2.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

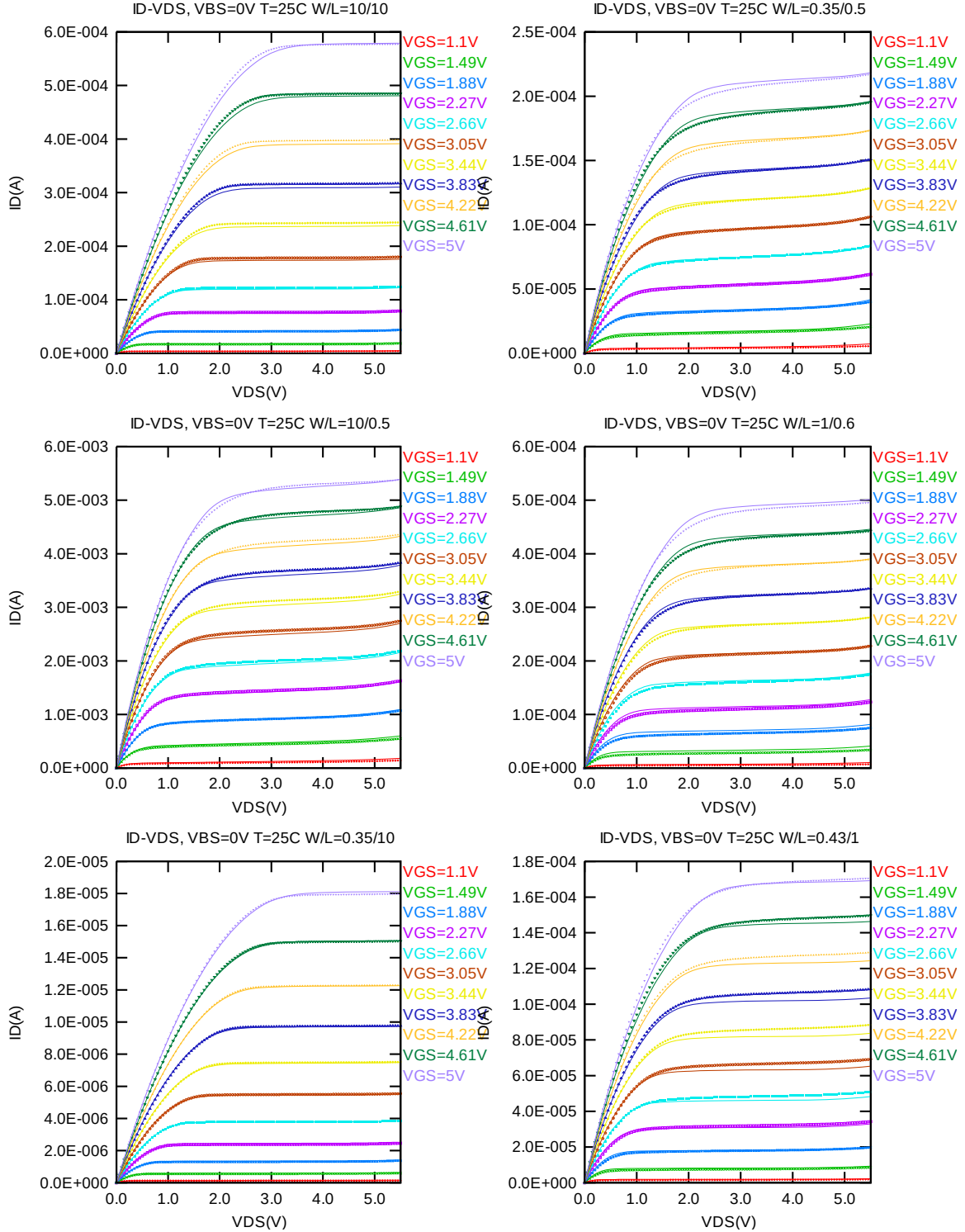


Figure 2-1 I_D vs. V_{DS} at $V_{BS}=0V$ for NMOS

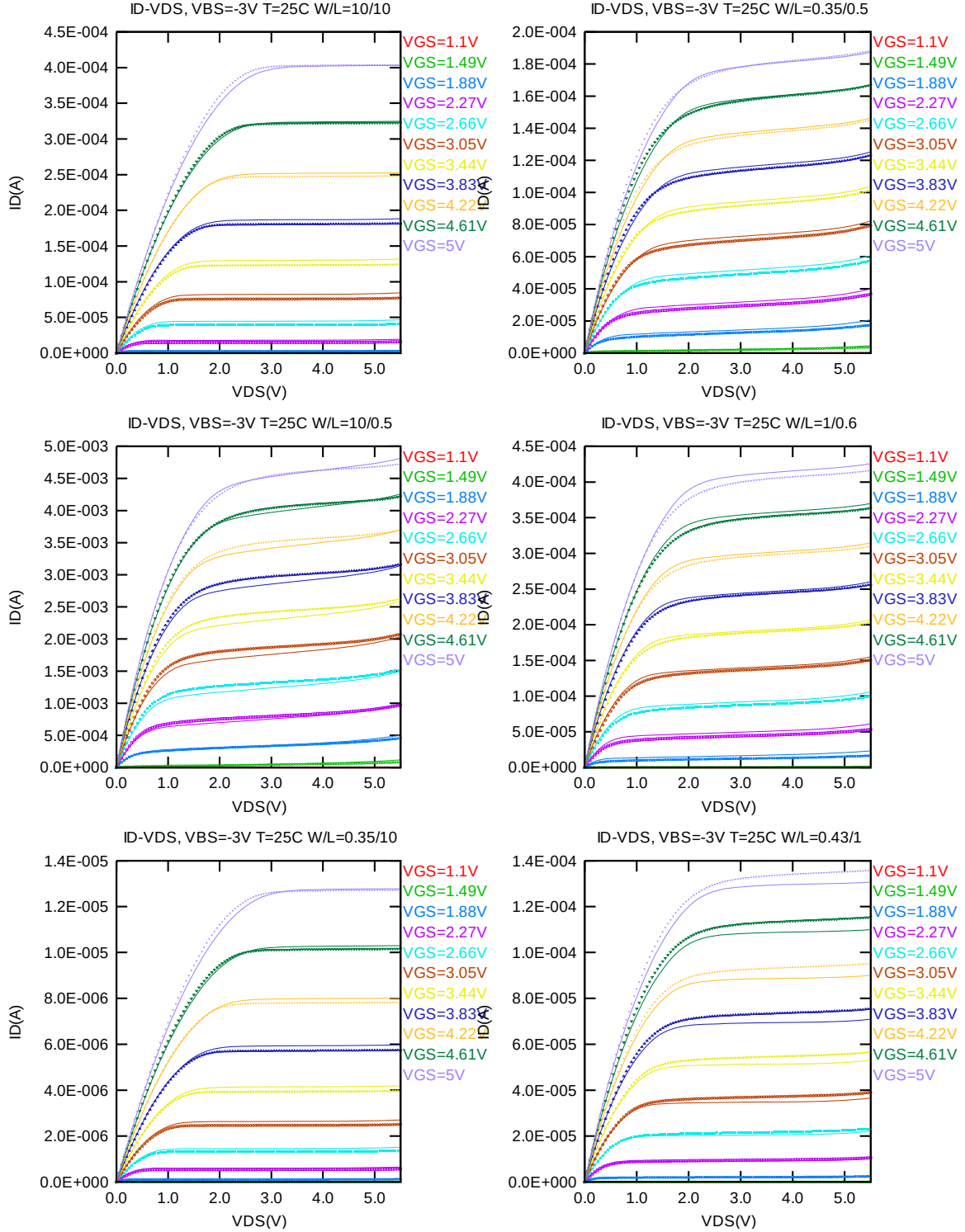


Figure 2-2 I_D vs. V_{DS} at $V_{BS} = -3V$ for NMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

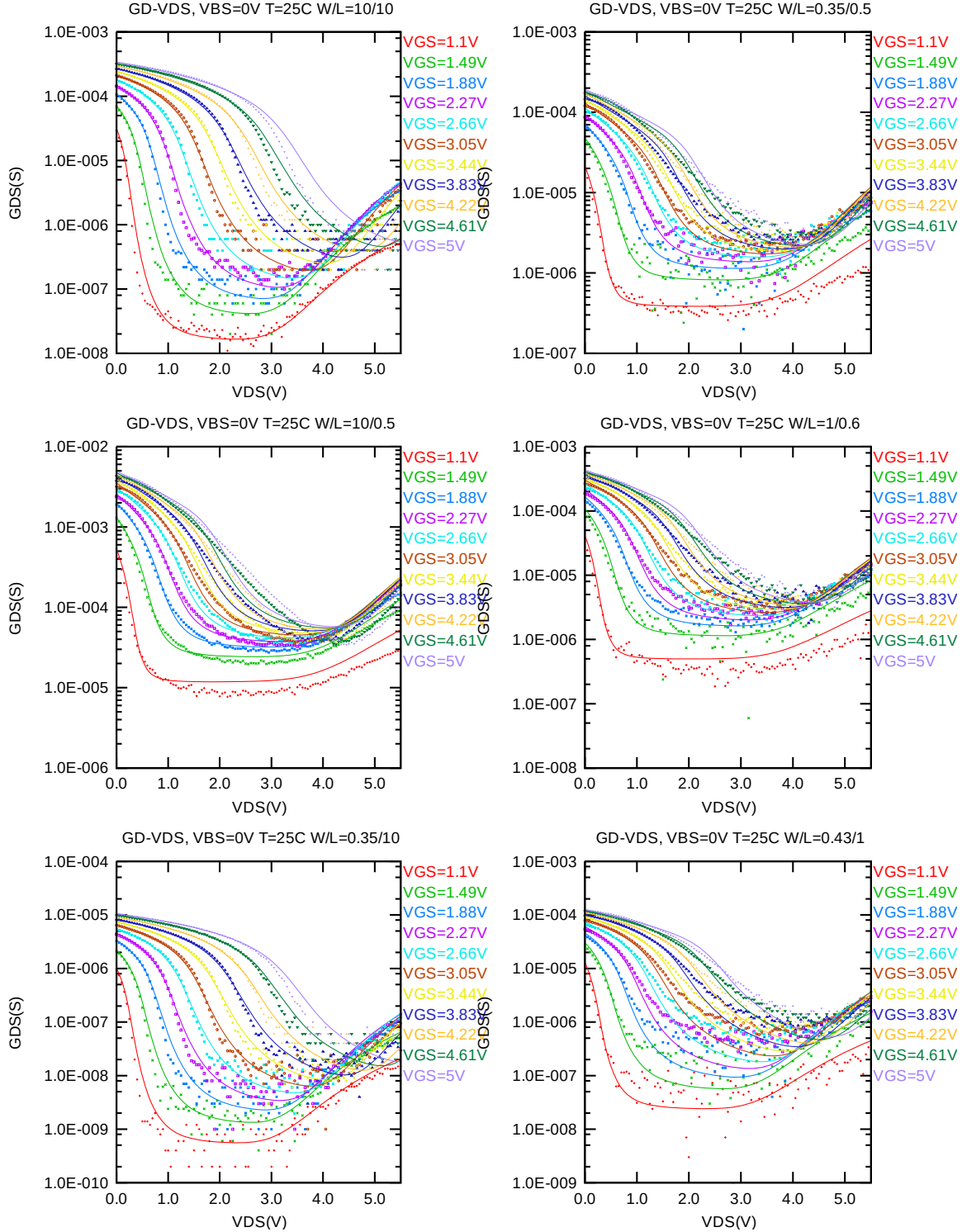


Figure 2-3 G_{DS} vs. V_{DS} at $V_{BS}=0V$ for NMOS

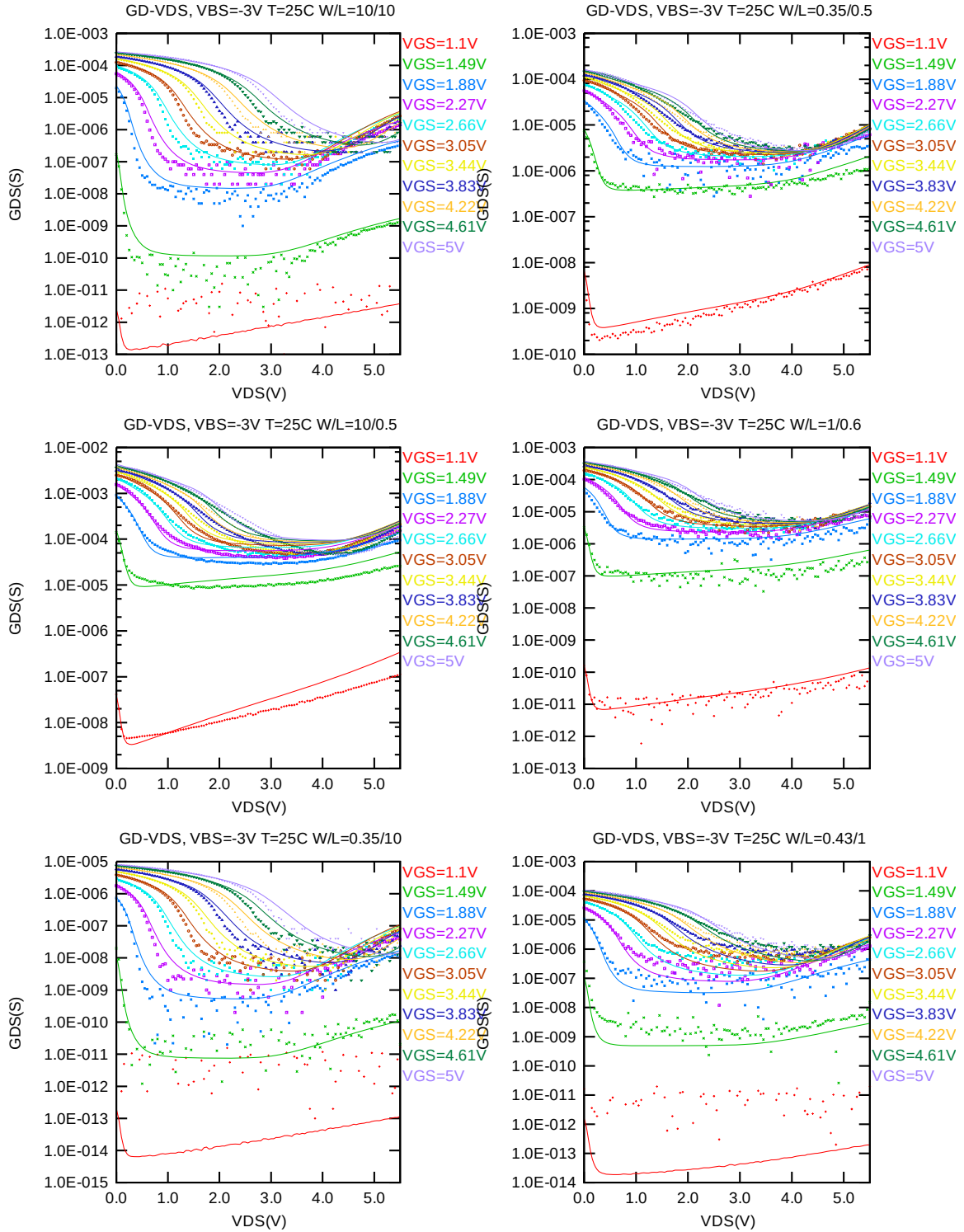


Figure 2-4 G_{DS} vs. V_{DS} at $V_{BS}=-3$ V for NMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

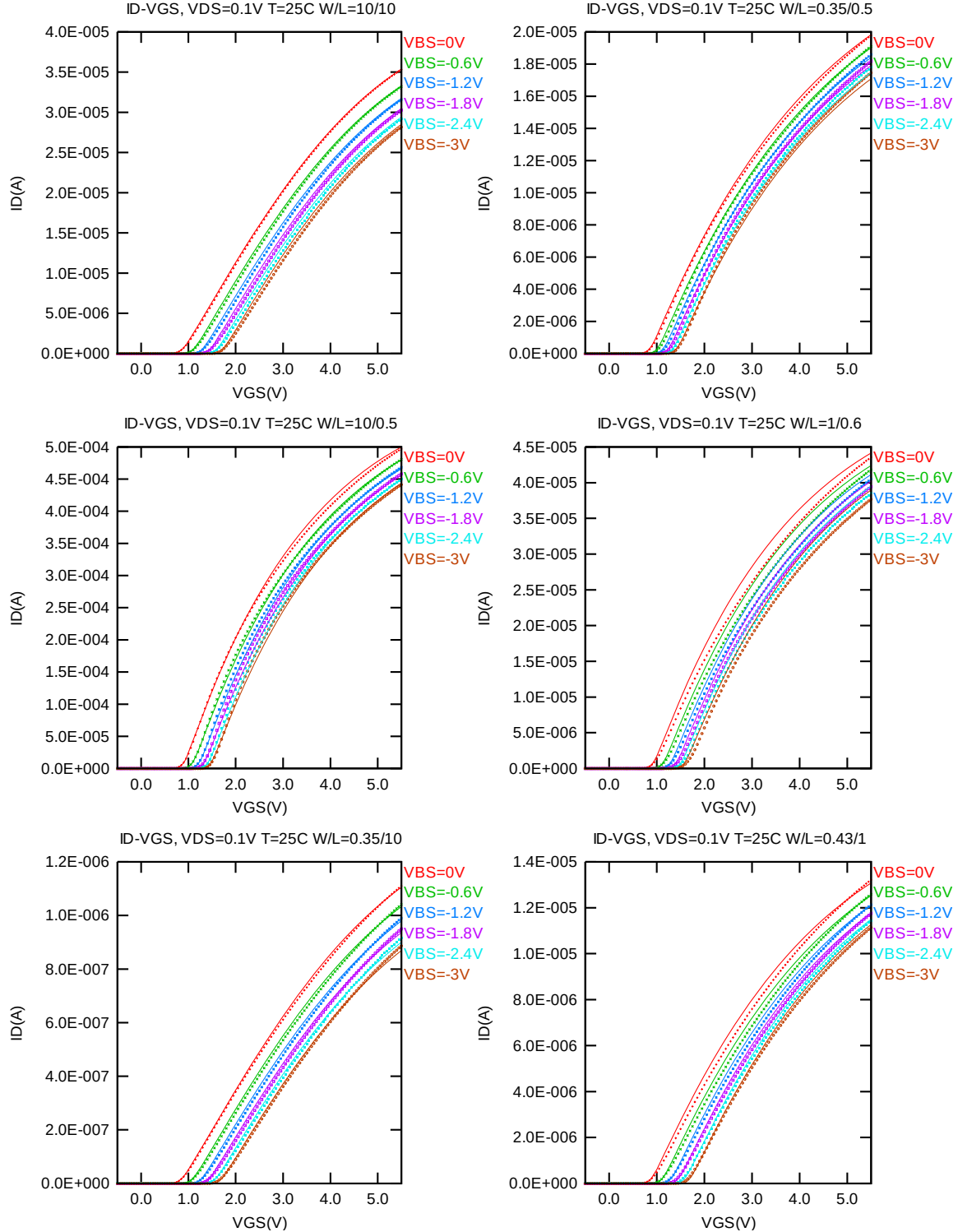


Figure 2-5 I_D vs. V_{GS} at $V_{DS}=0.1V$ for NMOS

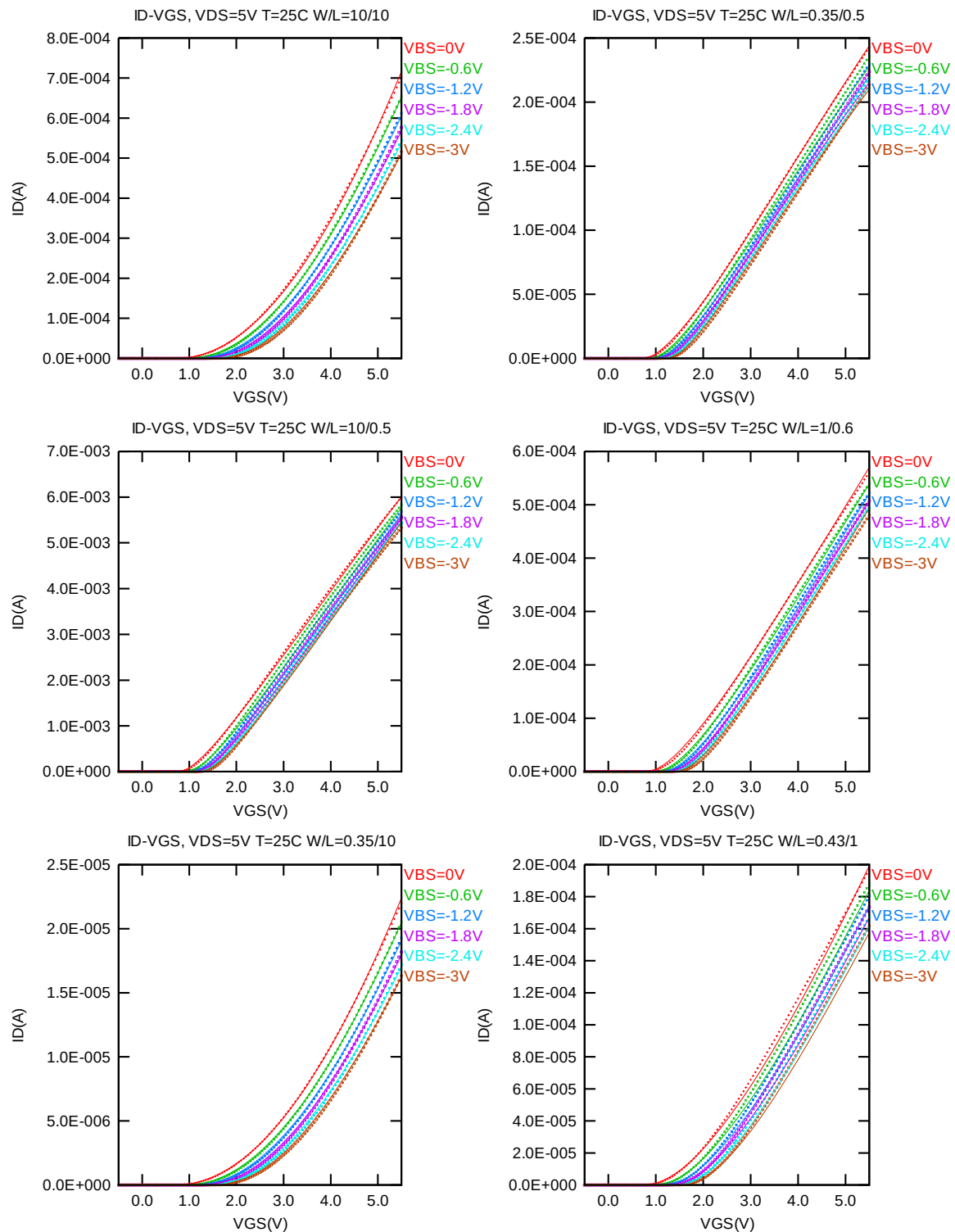


Figure 2-6 I_D vs. V_{GS} at $V_{DS}=5V$ for NMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

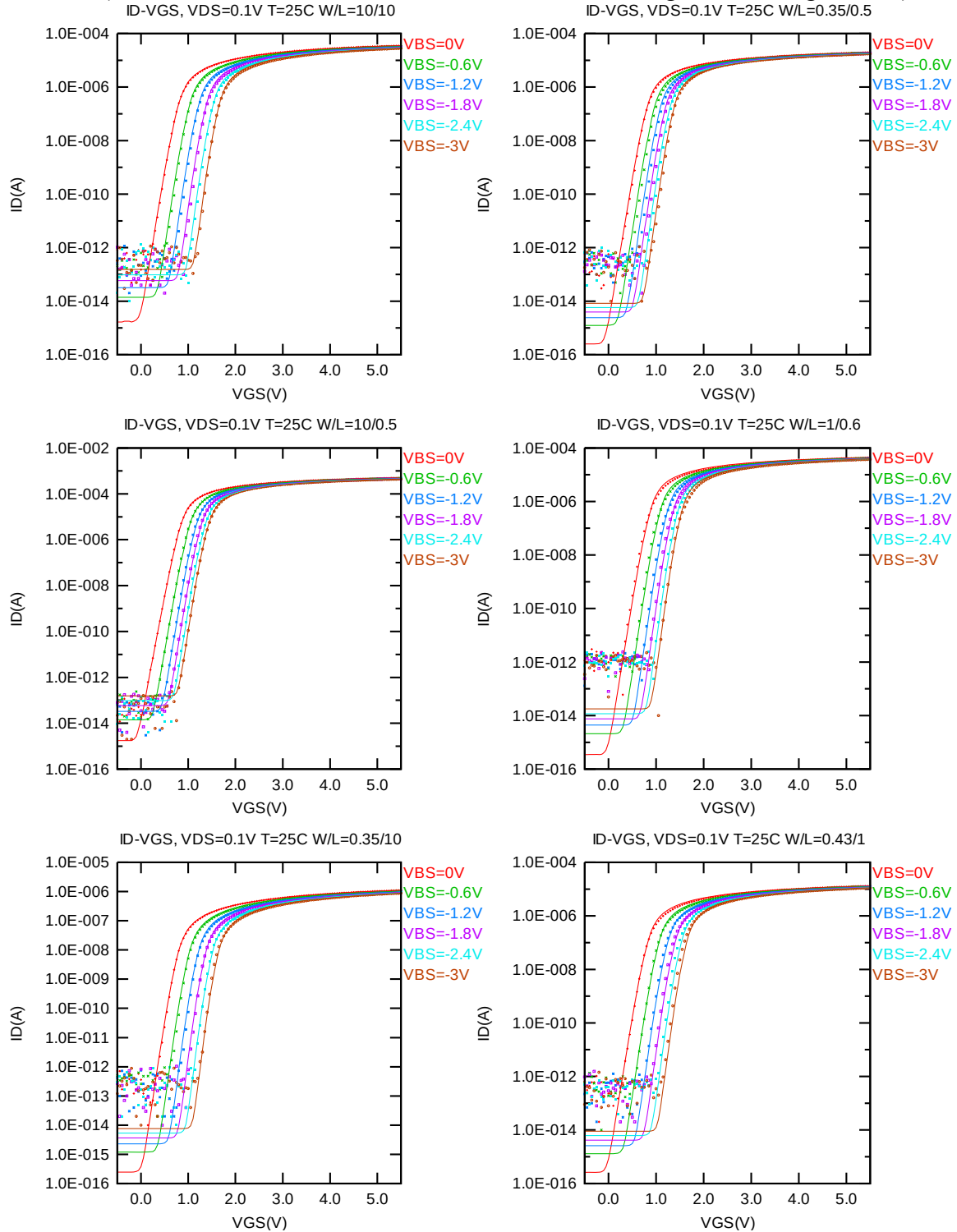


Figure 2-7 Log scaled I_D vs. V_{GS} at $V_{DS}=0.1$ V for NMOS

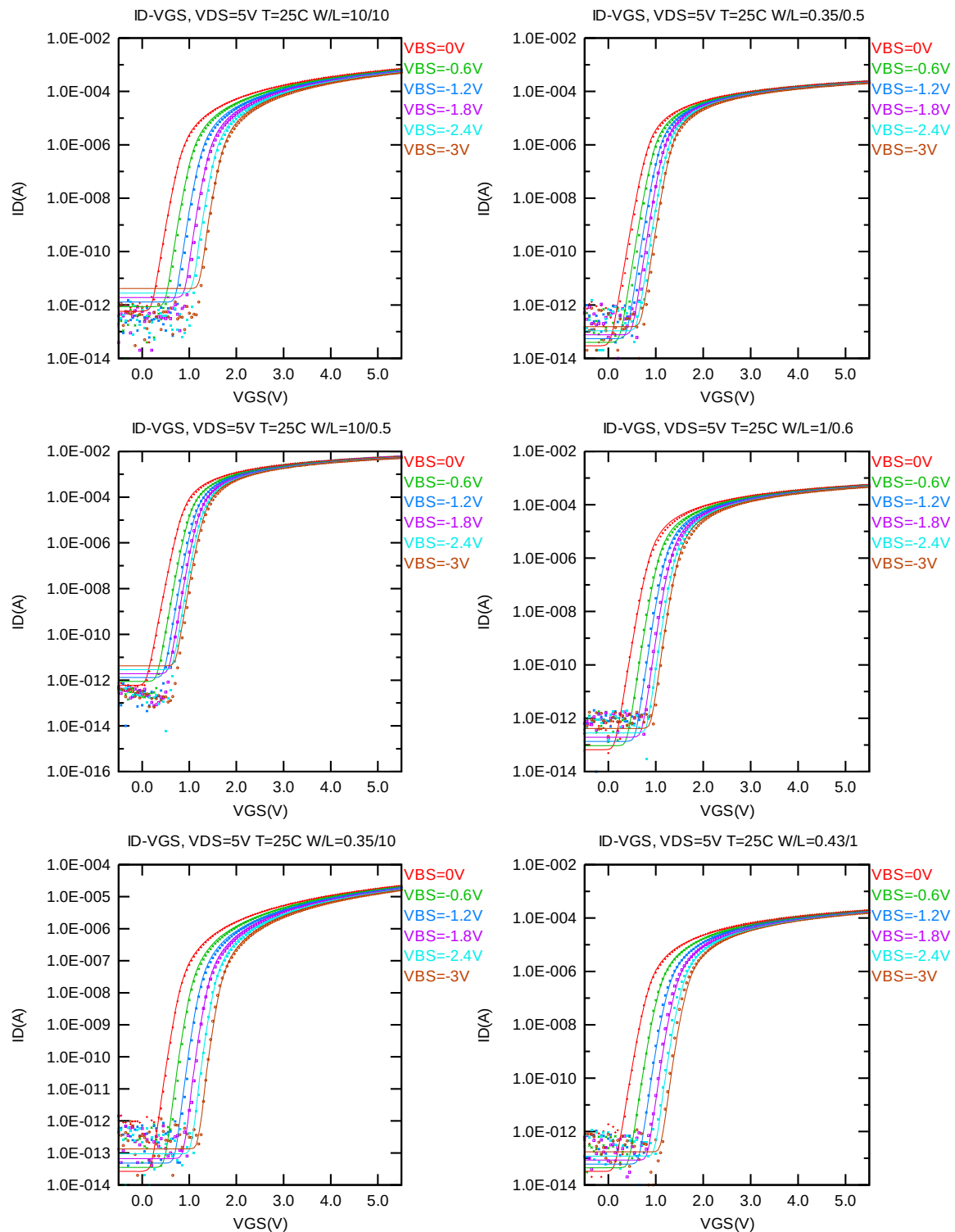


Figure 2-8 Log scaled I_D vs. V_{GS} at $V_{DS}=5$ V for NMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

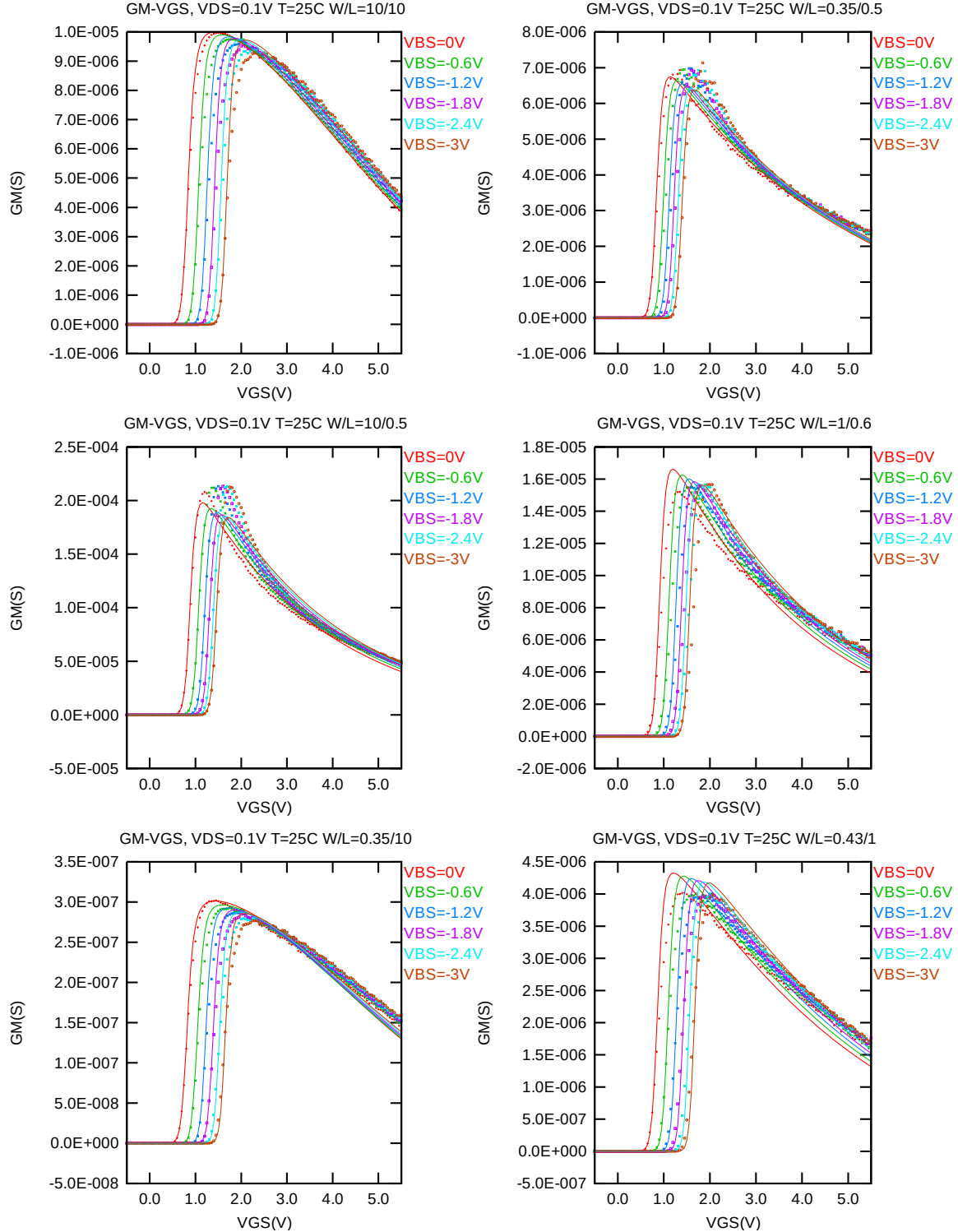


Figure 2-9 G_M vs. V_{GS} at $V_{DS}=0.1V$ for NMOS

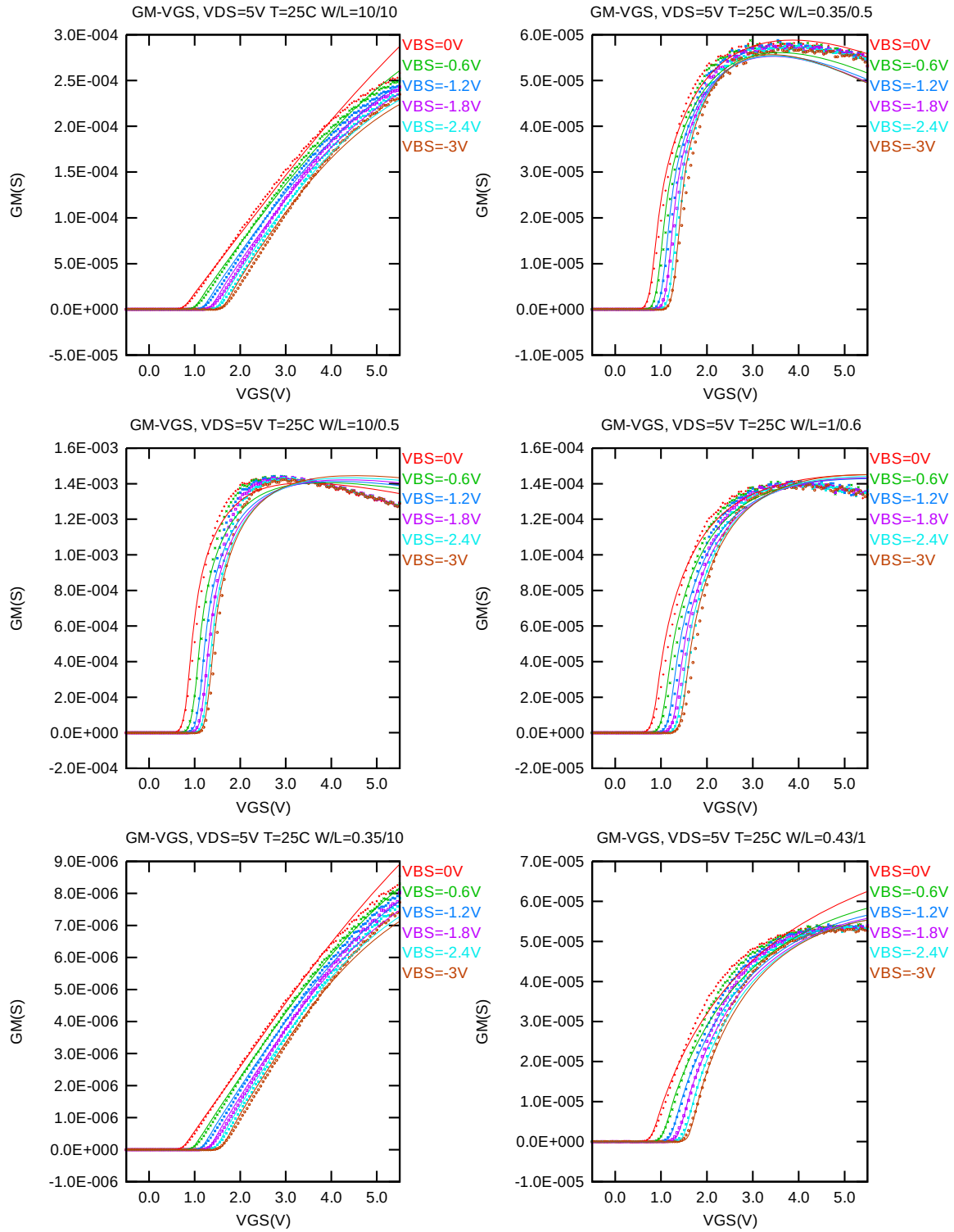


Figure 2-10 G_M vs. V_{GS} at $V_{DS}=5V$ for NMOS

The L and W in the plots refer to $LDES$ and $WDES$ respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300\text{nA} \cdot WDES \cdot 0.9 / (LDES \cdot 0.9 + 12\text{nm})$ at 25°C . The plots show the physical dimension.

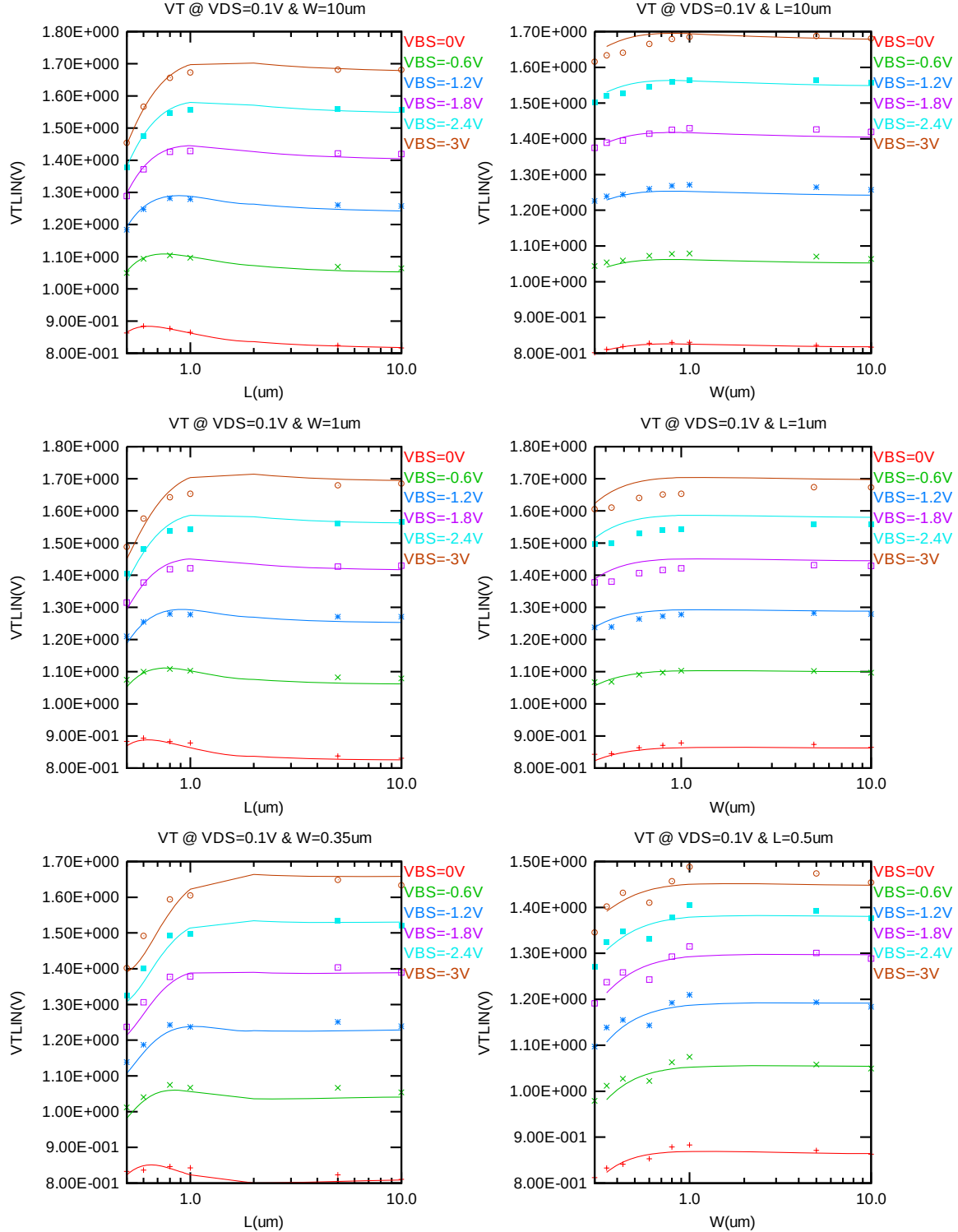


Figure 2-11 V_{TLIN} vs. channel length & channel width for low drain bias ($V_{DS}=0.1$ V) for NMOS

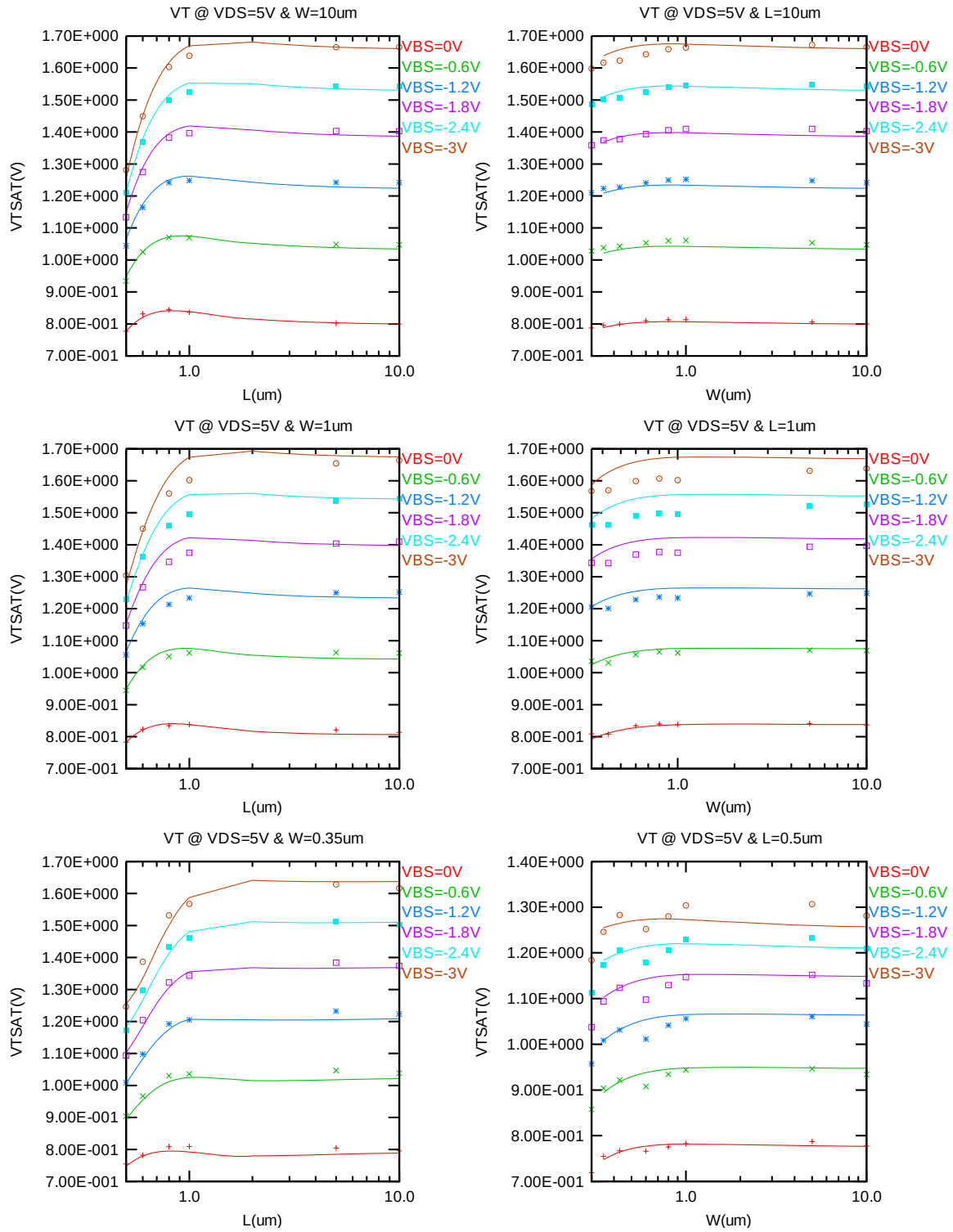


Figure 2-12 V_{TSAT} vs. channel length & channel width for high drain bias ($V_{DS}=5$ V) for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS}=V_{GS}=5$ V for $V_{BS}=0$ V and $V_{BS}=-3$ V at 25 C.(The dimensions are after 90% shrink and 12nm sizing channel length back.)

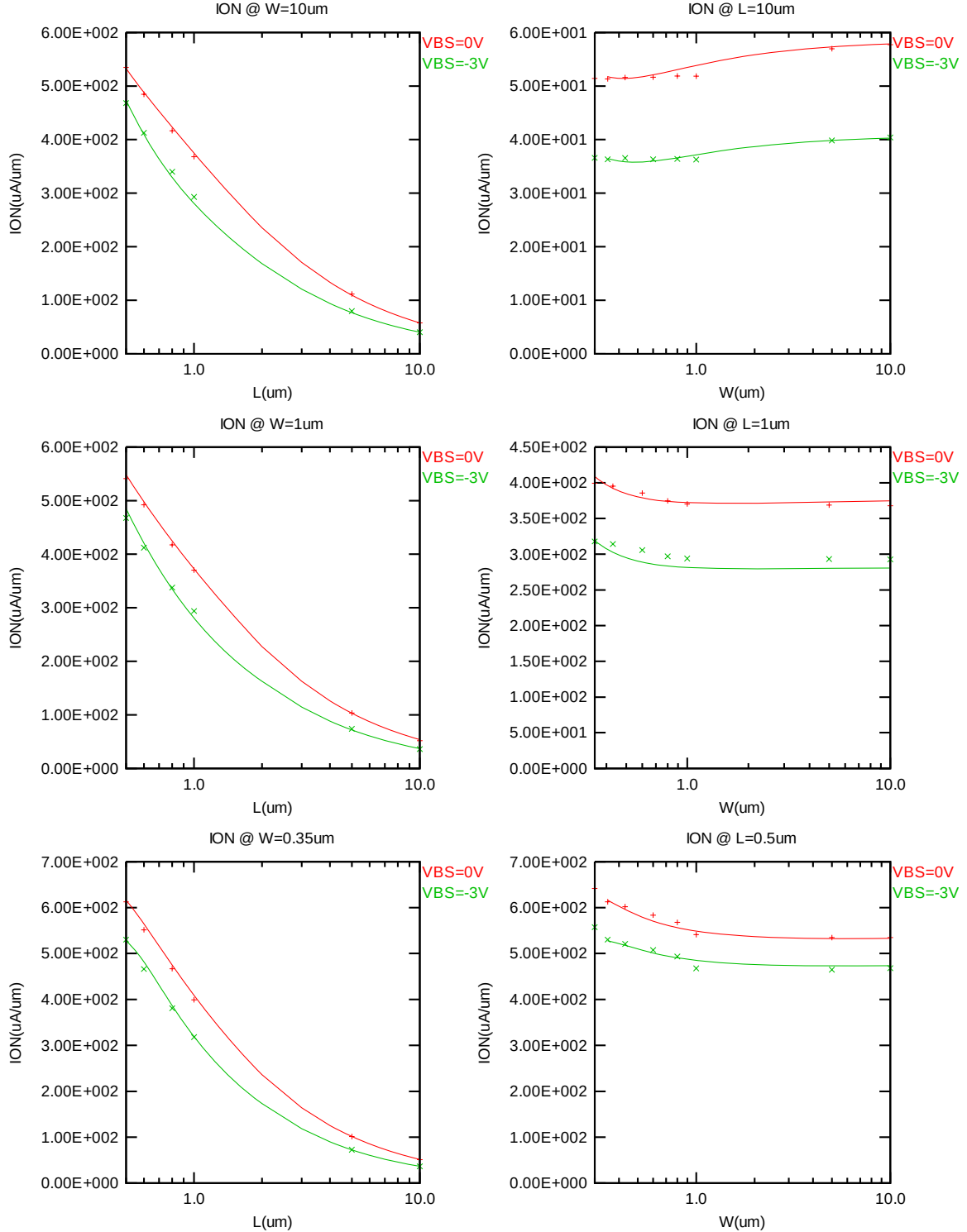


Figure 2-13 Saturation current I_{ON} vs. channel length and channel width for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS}=5\text{ V}$ and $V_{GS}=0\text{ V}$ for $V_{BS}=0\text{ V}$ to -3 V at 25 C . A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.

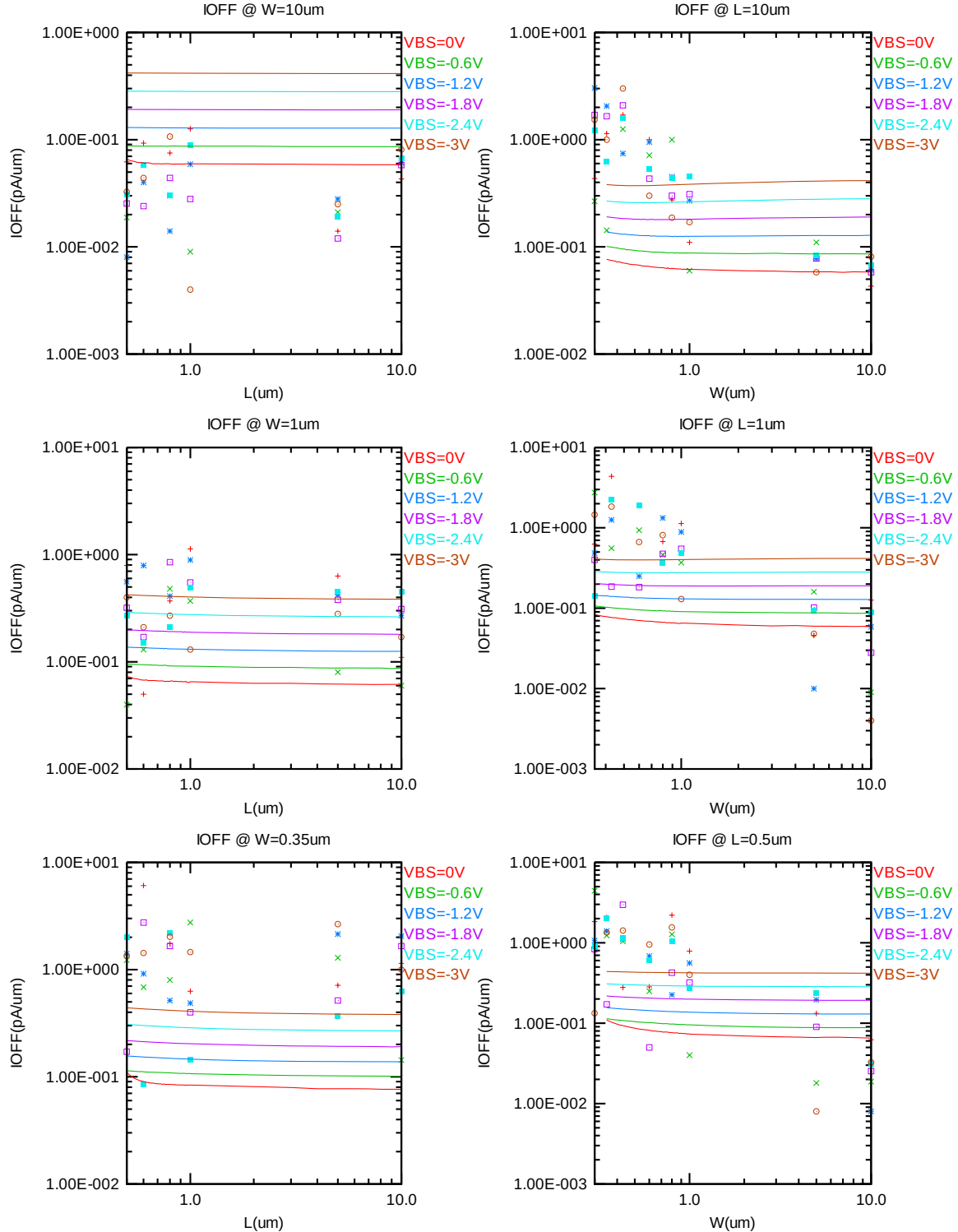


Figure 2-14 I_{OFF} vs. channel length and channel width for NMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at $I_D=300nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 12nm)$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

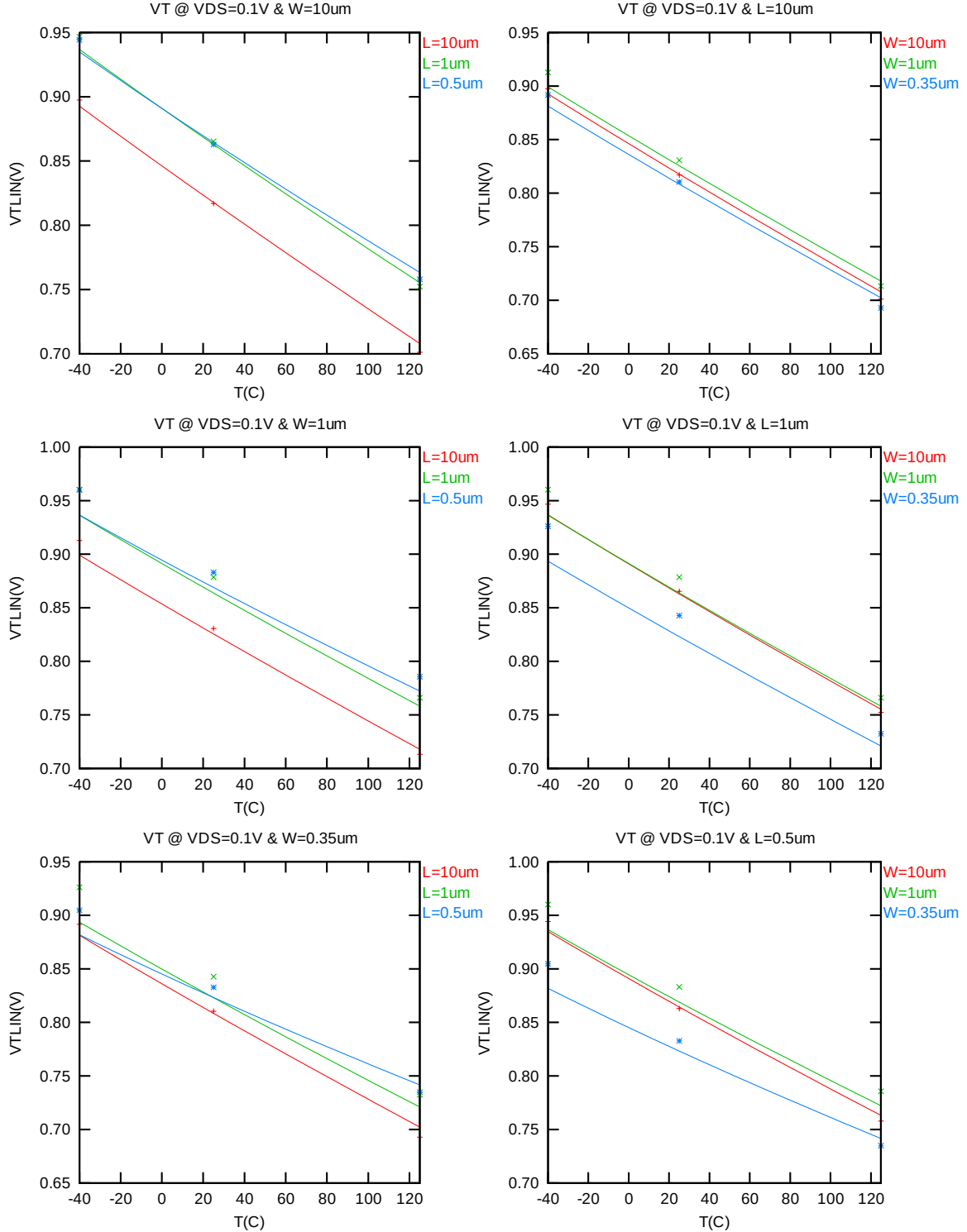


Figure 2-15 V_{TLIN} vs. temperature for NMOS

V_{TSAT} vs. temperature

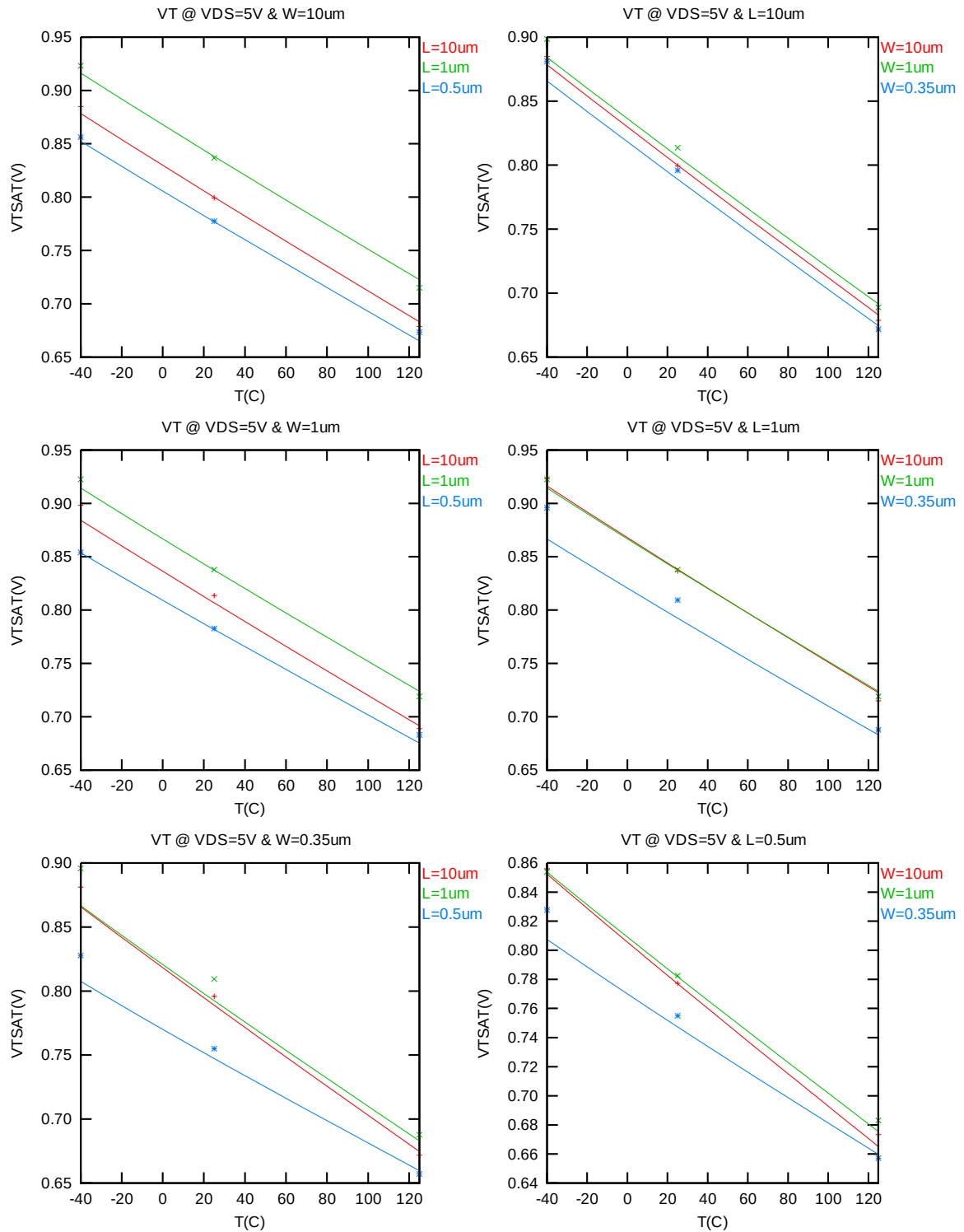


Figure 2-16 V_{TSAT} vs. temperature for NMOS

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS}=V_{DS}=5$ V, $V_{BS}=0$ V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

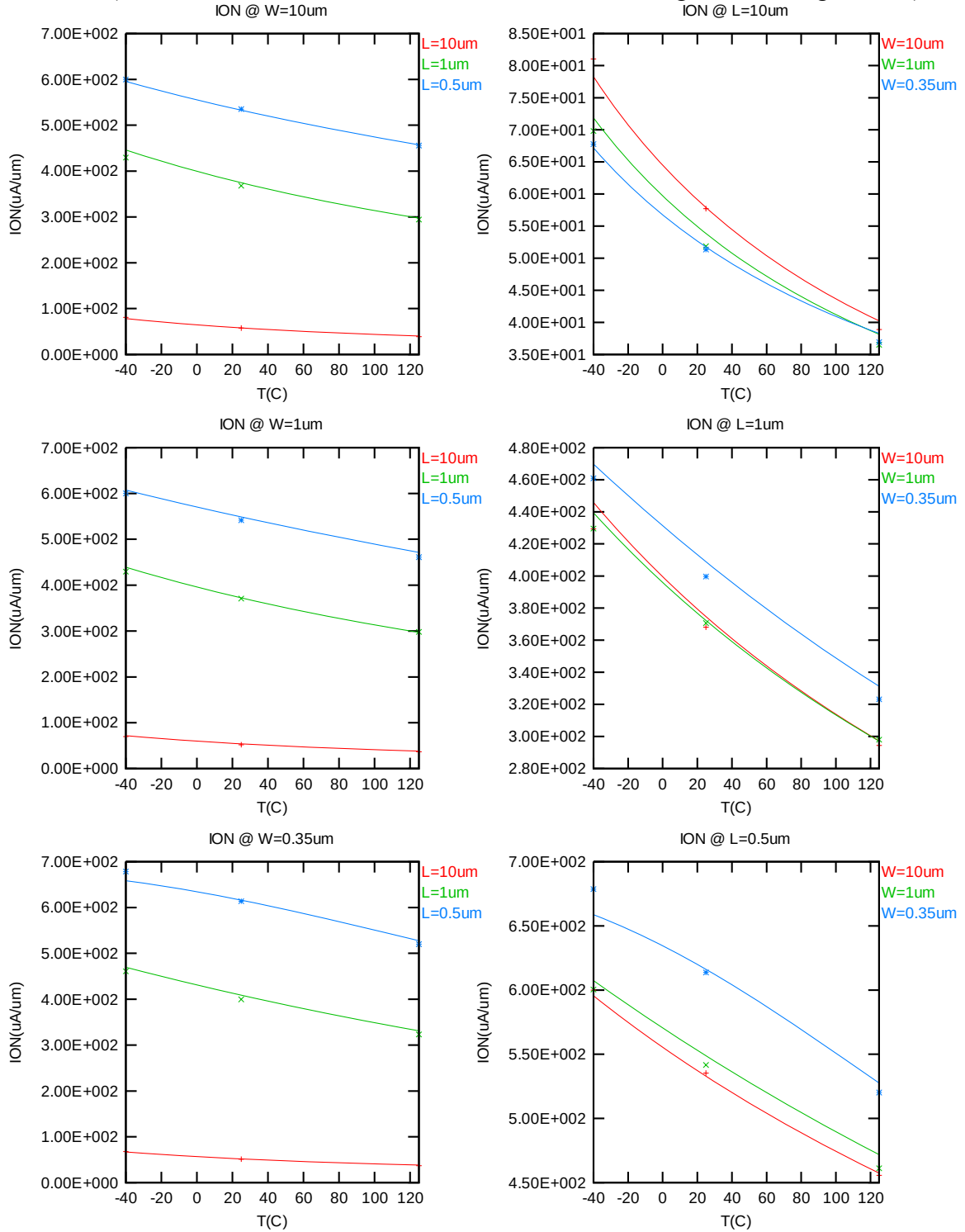


Figure 2-17 Saturation current vs. temperature for NMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at $V_{\text{DS}}=5\text{ V}$, $V_{\text{GS}}=V_{\text{BS}}=0\text{ V}$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

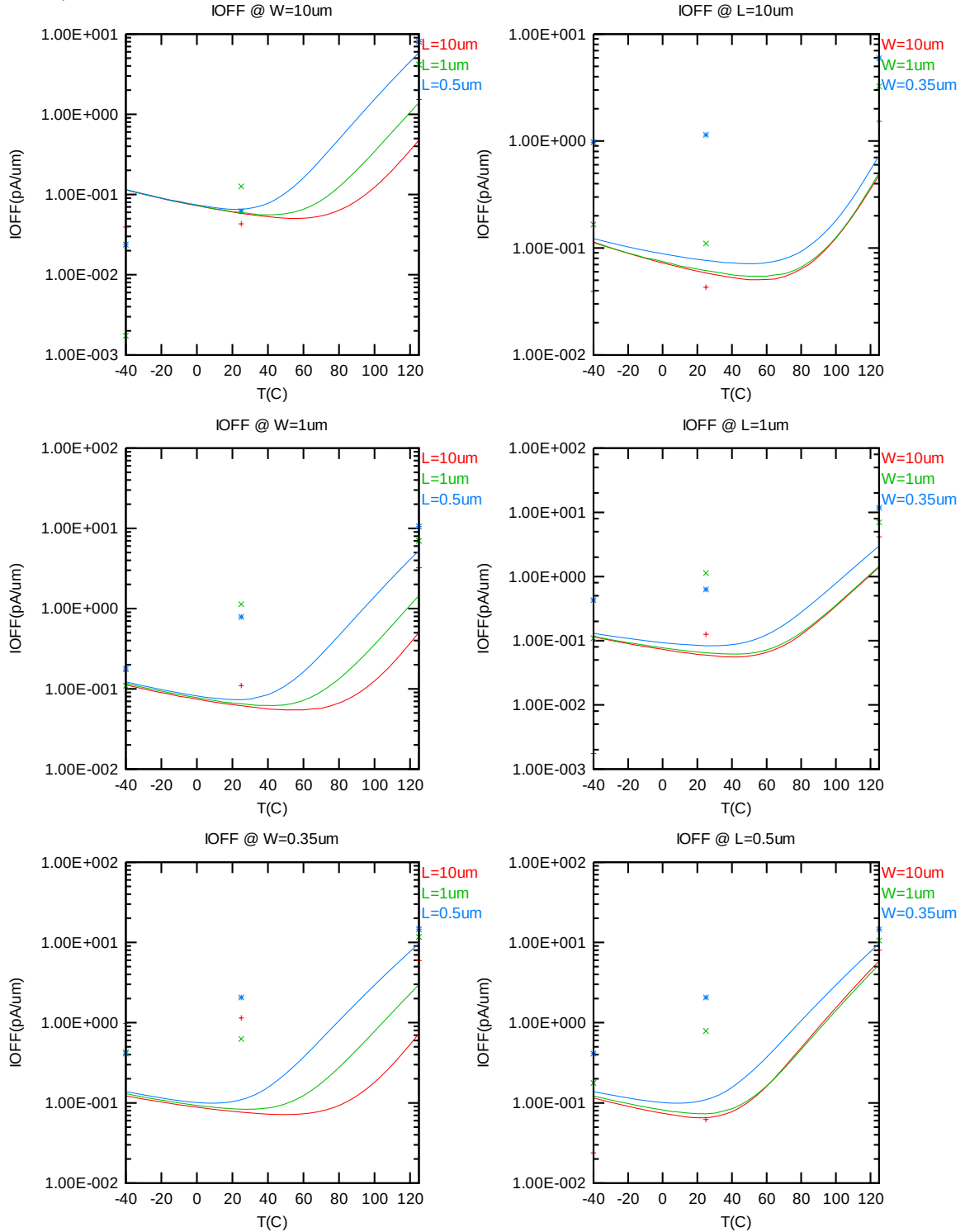


Figure 2-18 I_{OFF} vs. temperature for NMOS

2.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

- a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- b. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- c. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- d. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- e. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 2-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D accuracy

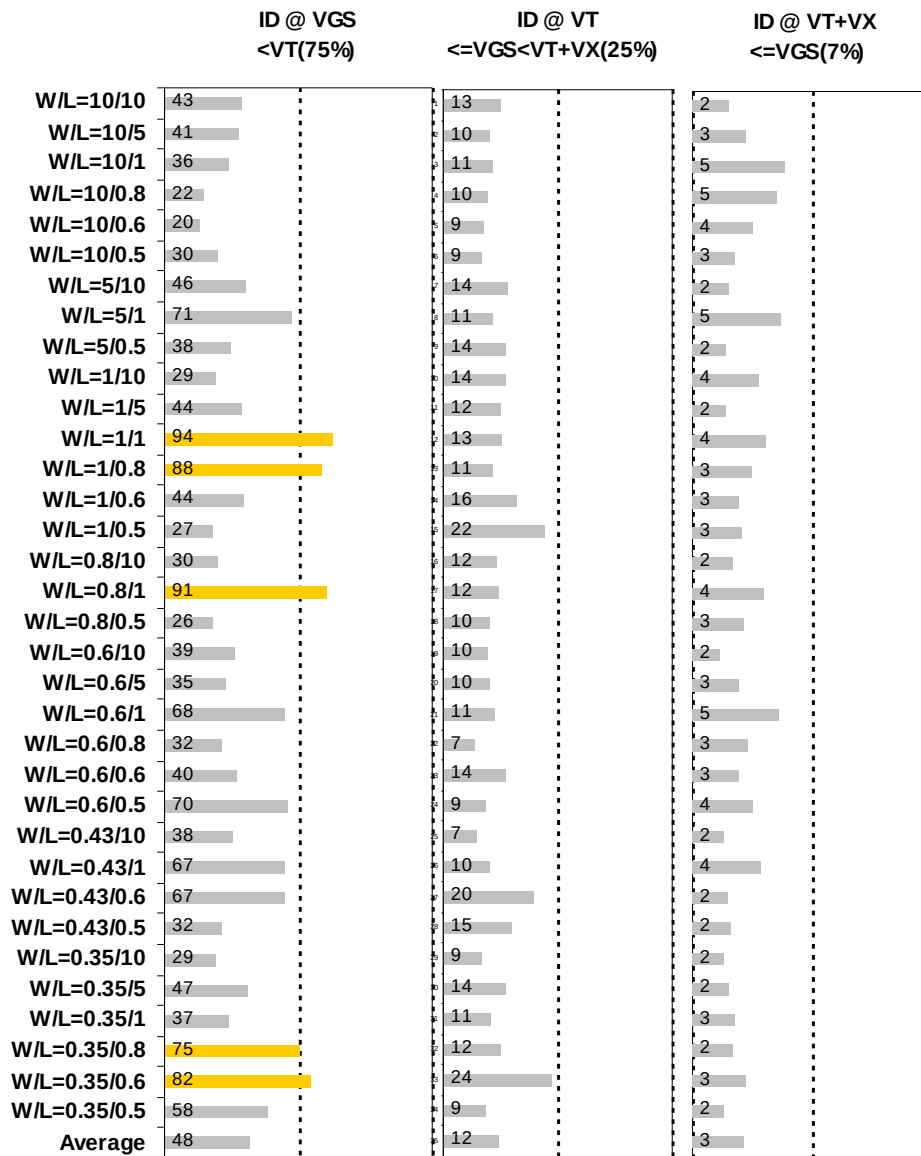


Figure 2-19 I_D accuracy at 25 C for NMOS

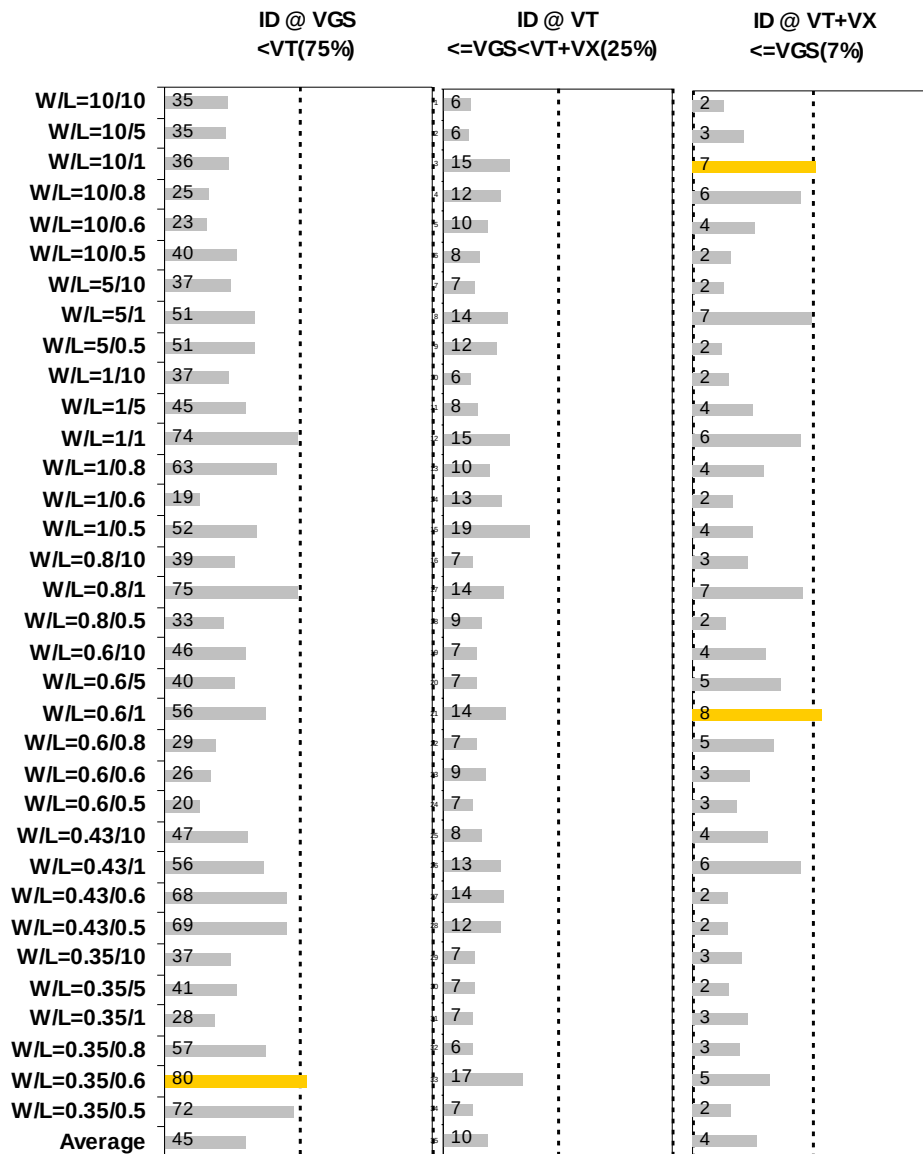


Figure 2-20 I_D accuracy at 125 C for NMOS

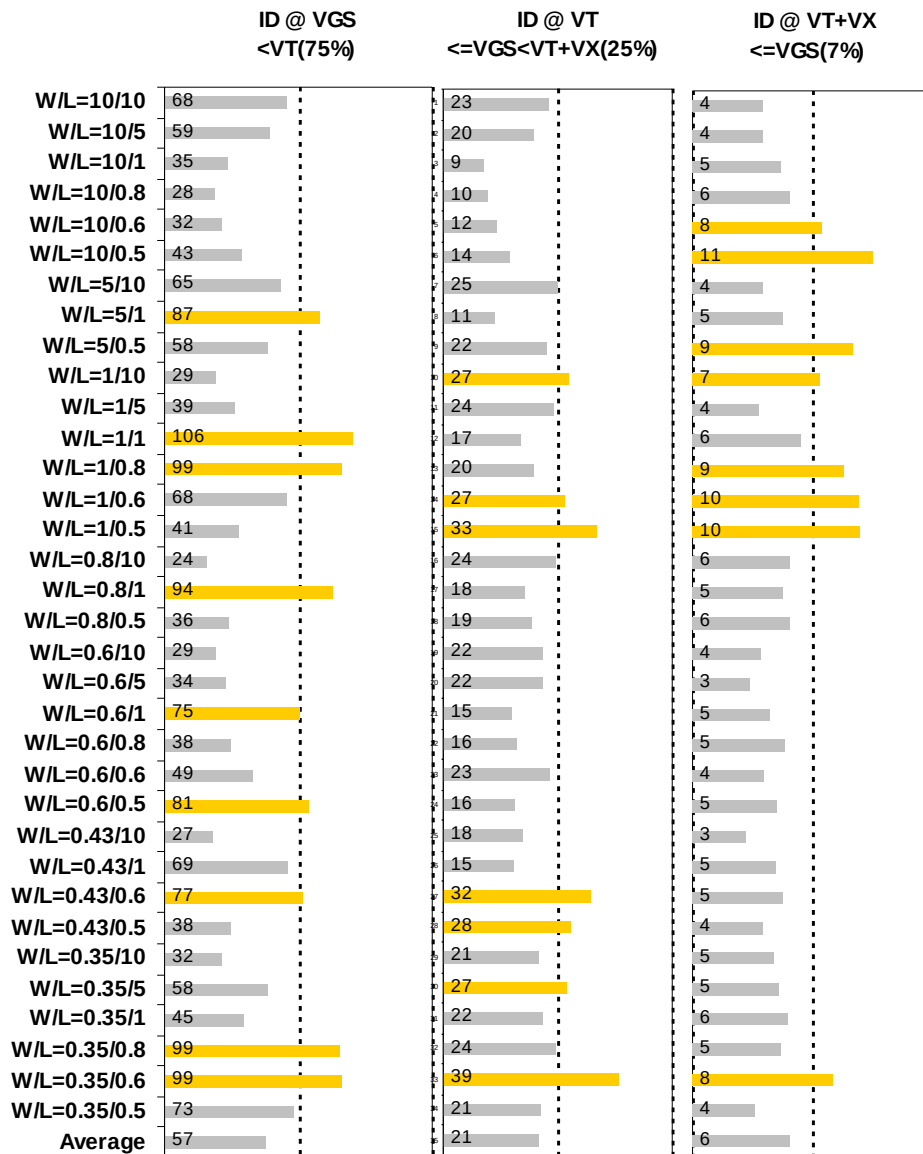


Figure 2-21 I_D accuracy at -40 C for NMOS

G_M and G_{DS} accuracy

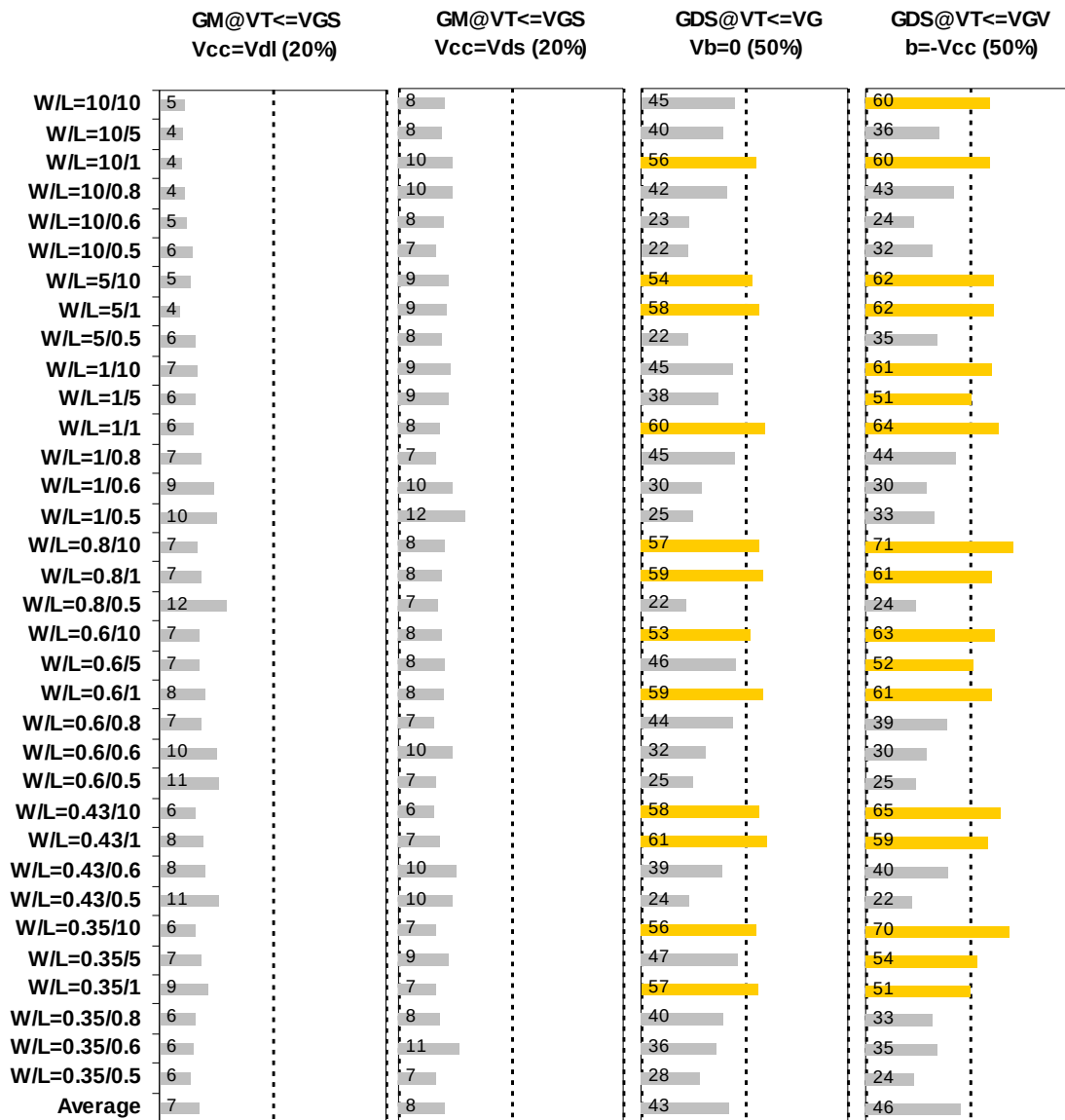


Figure 2-22 G_M and G_{DS} accuracy at 25 C for NMOS

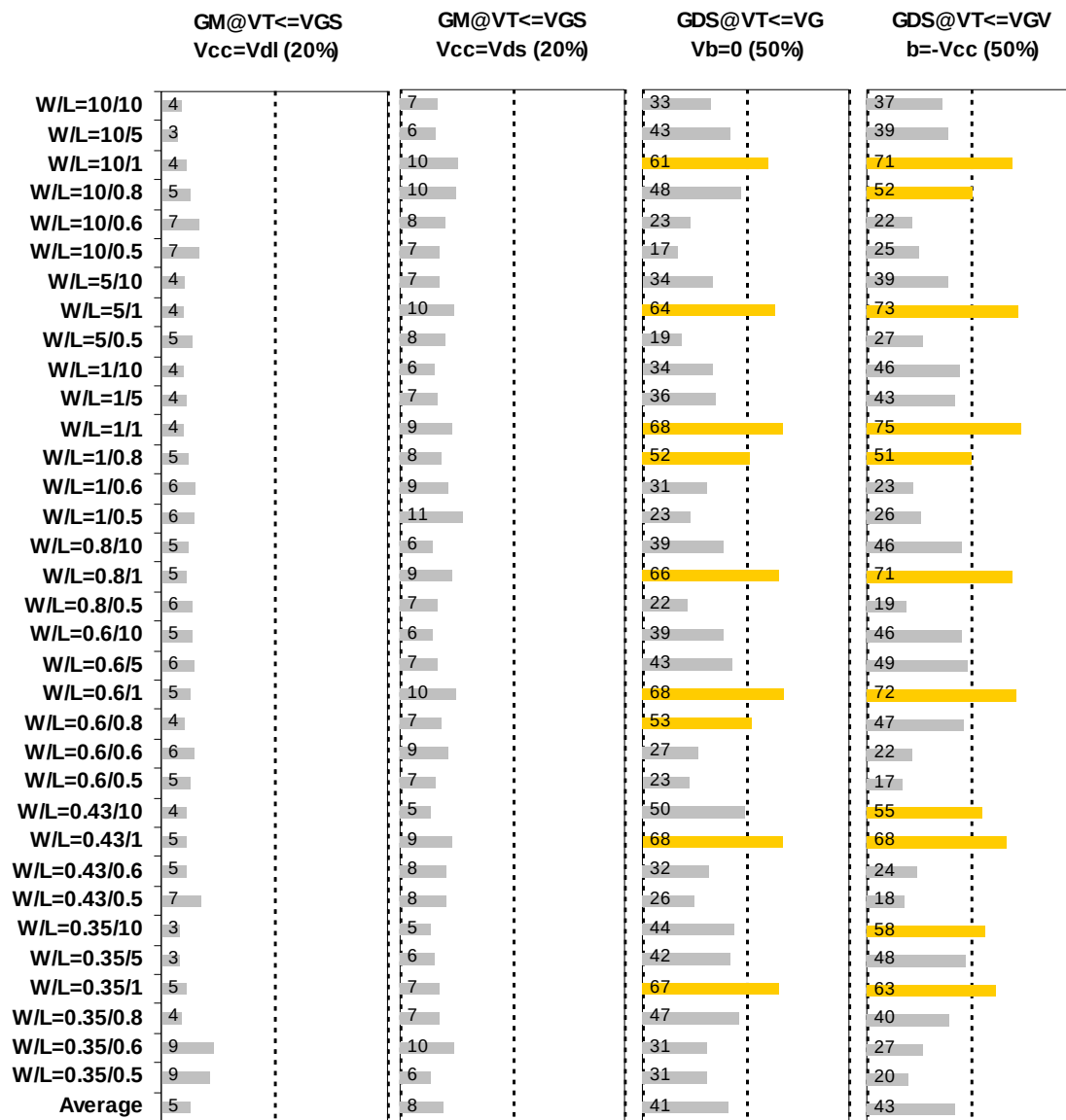


Figure 2-23 G_M and G_{DS} accuracy at 125 C for NMOS

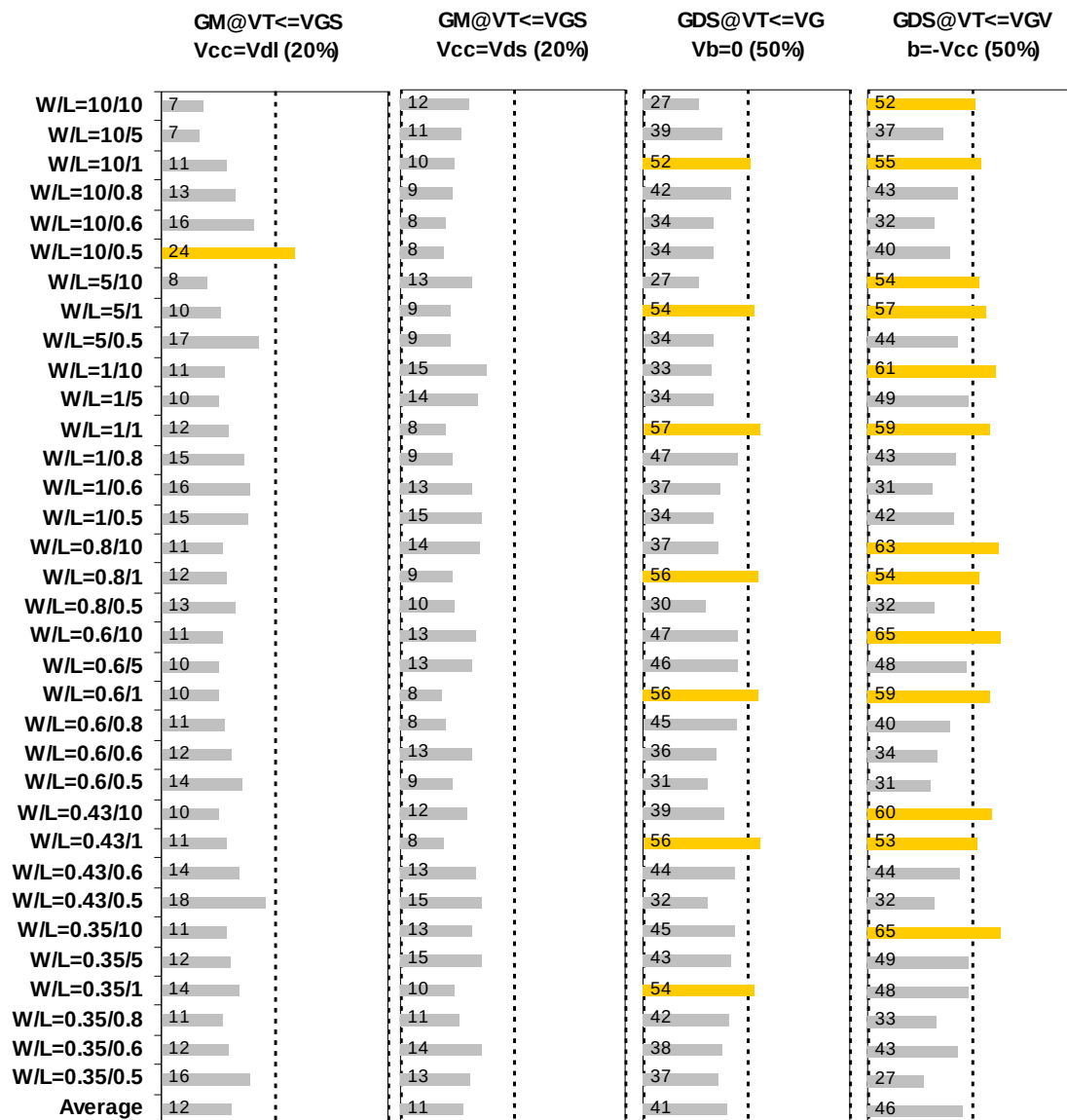


Figure 2-24 G_M and G_{DS} accuracy at -40 C for NMOS

2.4 MOSFET Capacitance Model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Measurement structure for NMOS C_{gg} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components, C _{GDS}	80	80

Table 2-7 Large area measurement structure for NMOS C_{gc} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	.5	100
	2	100
	1	100
	2	2.08
	.5	2.08
	2	1.72
	.5	1.72
	2	1.36
	.5	1.36
	2	1
	.5	1
	2	.64
	.5	.64
	2	.35
	.5	.35

2.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized in Table 2-8 below.

Table 2-8 Measurement conditions for NMOS capacitance model

V_{BIAS} [V]	From -5.5 to 5.5 V in 0.1 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

2.4.3 T_{OX} extraction

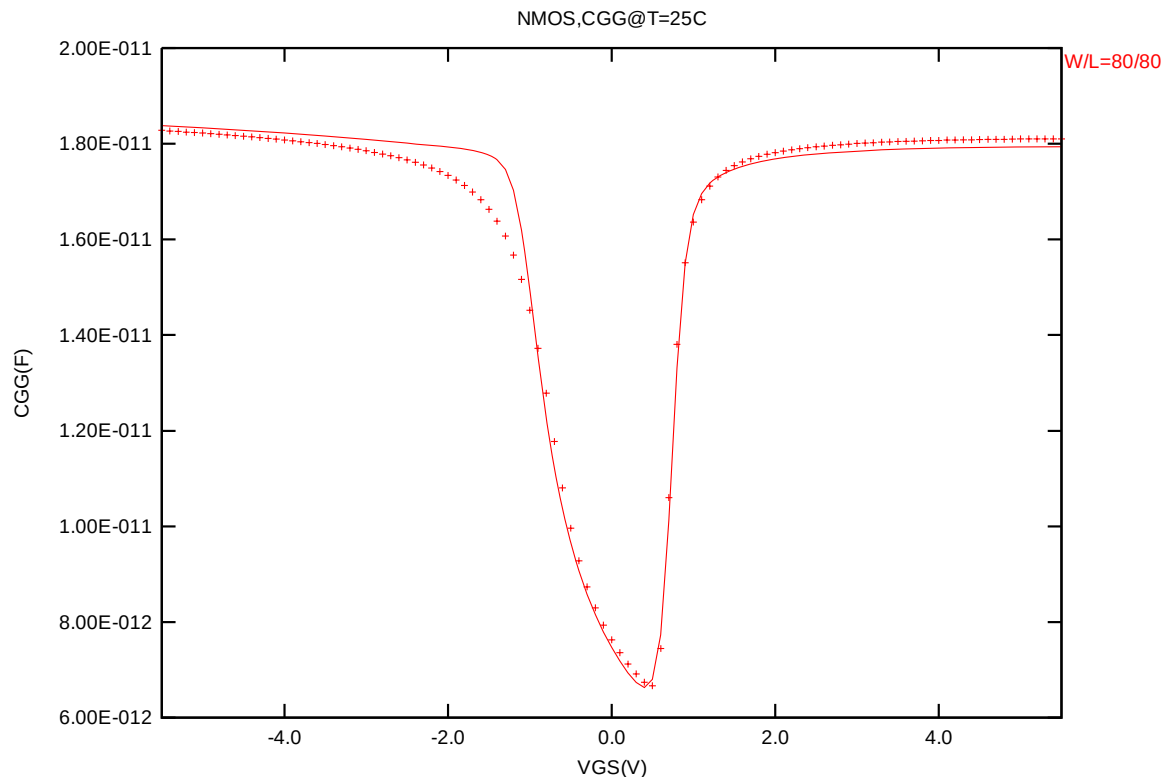
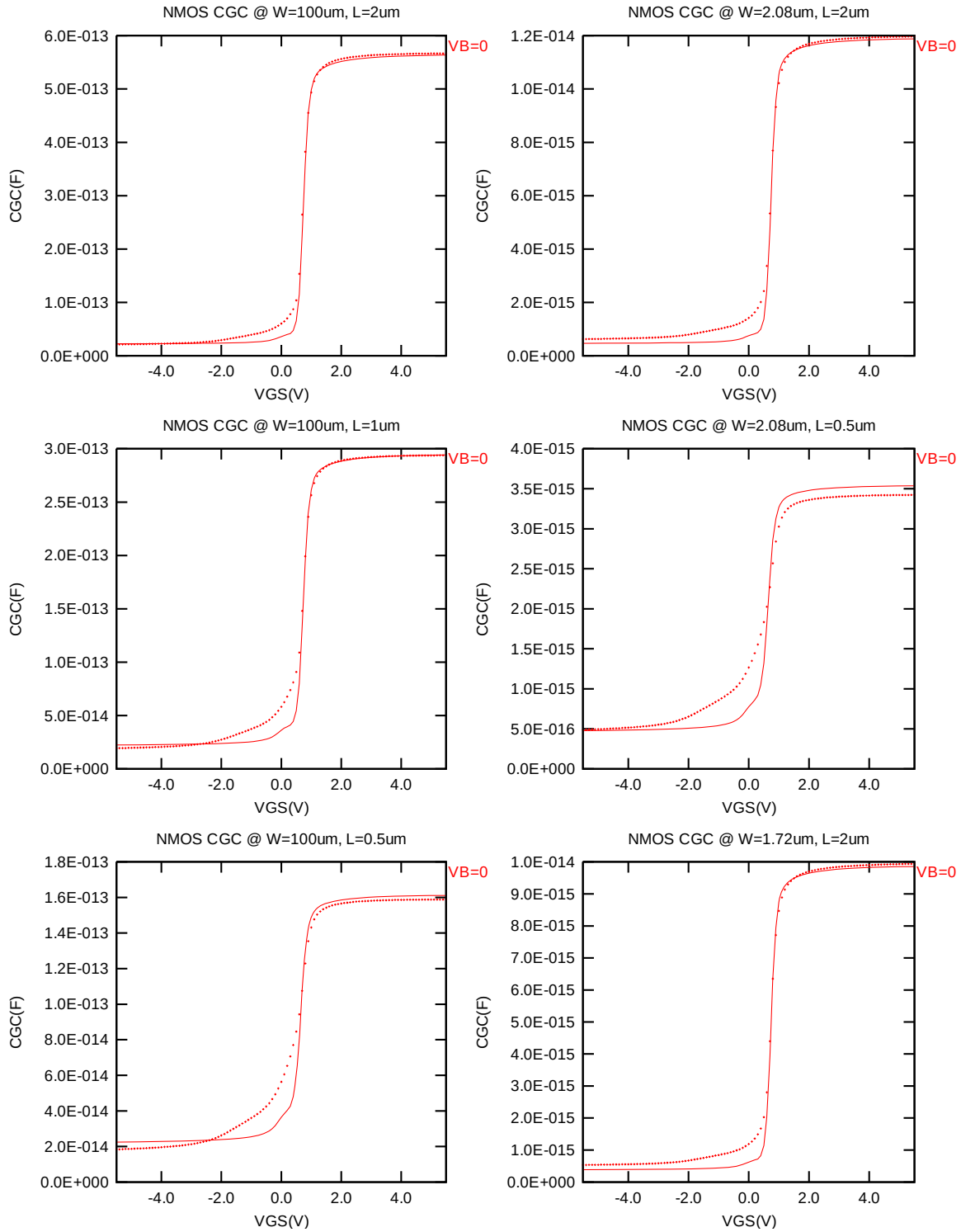
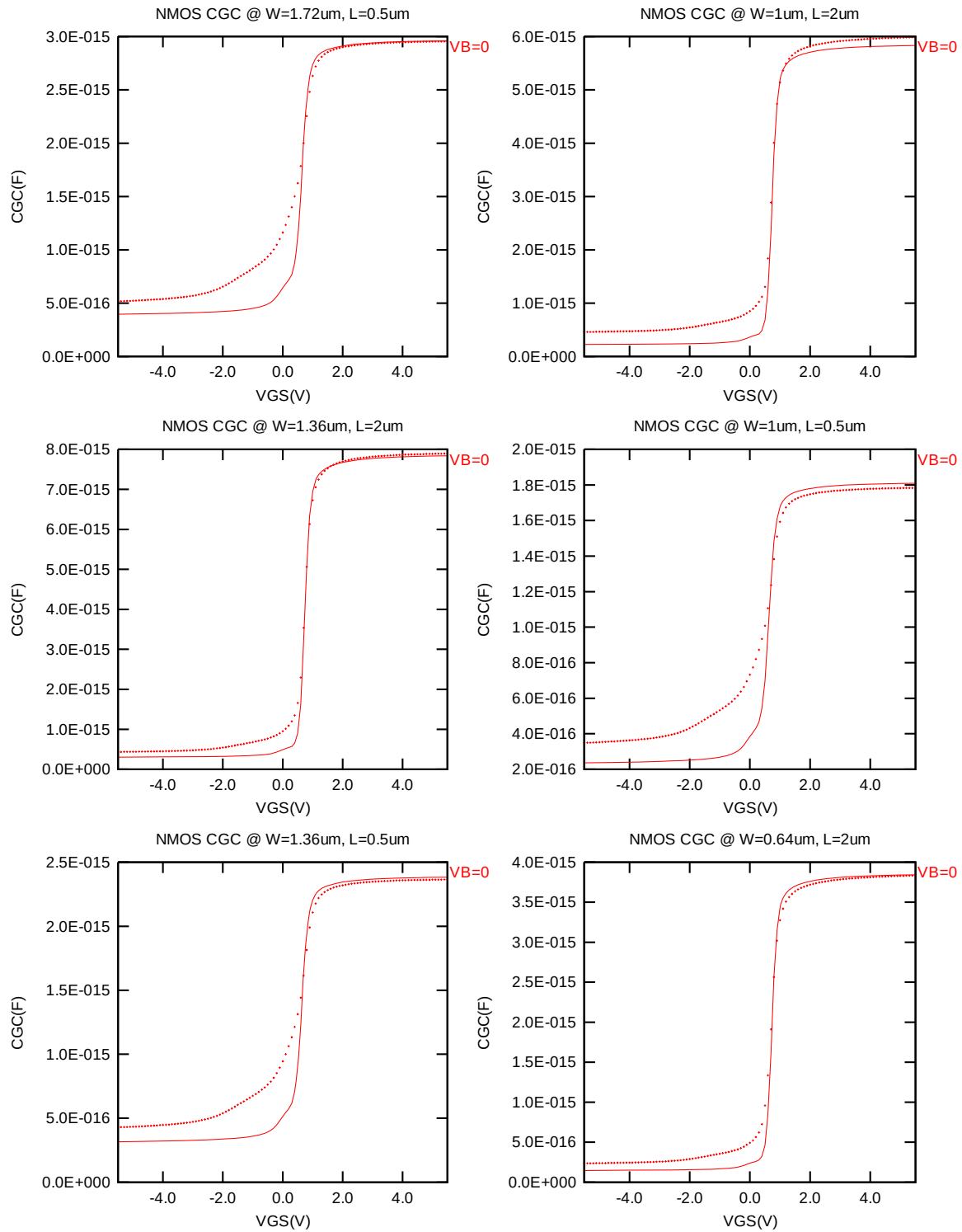


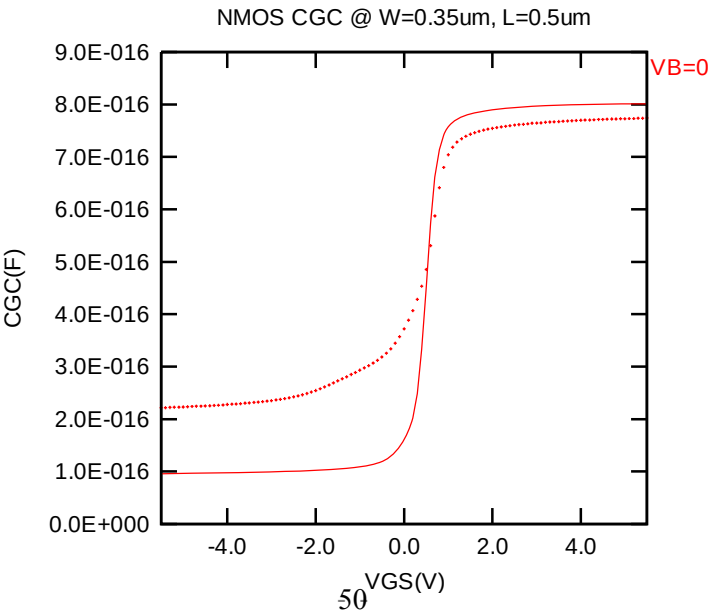
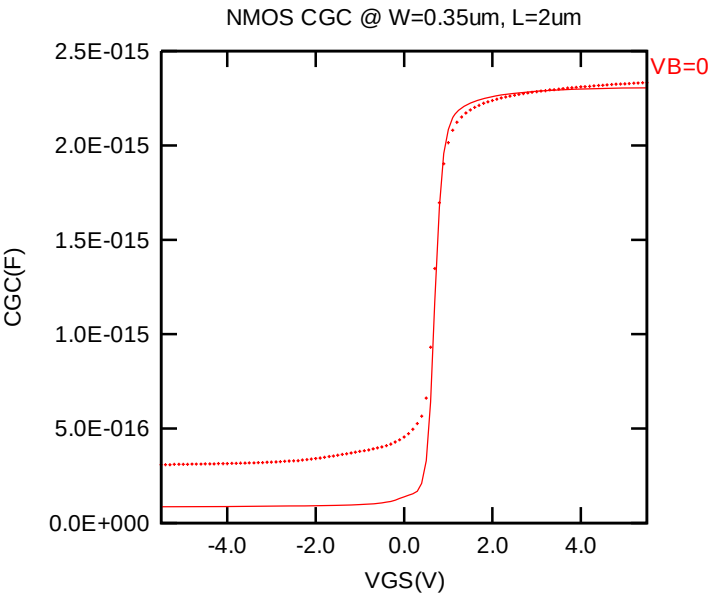
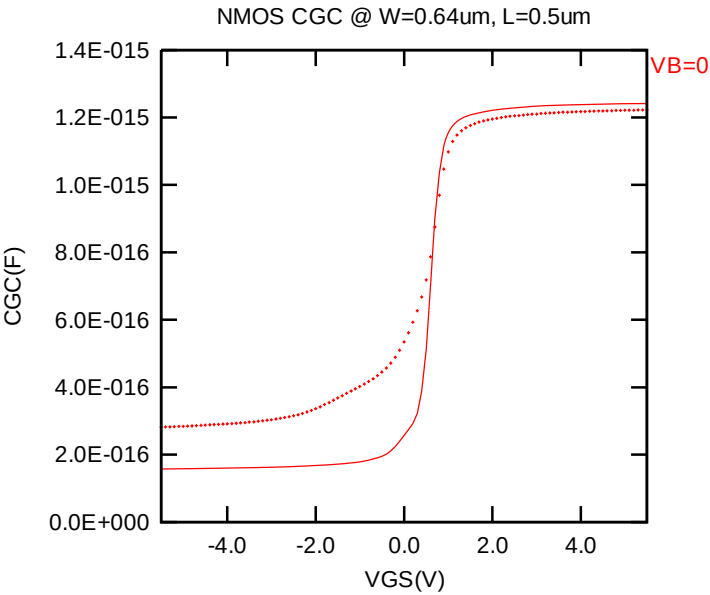
Figure 2-25 Measured and simulated C_{GG} curves

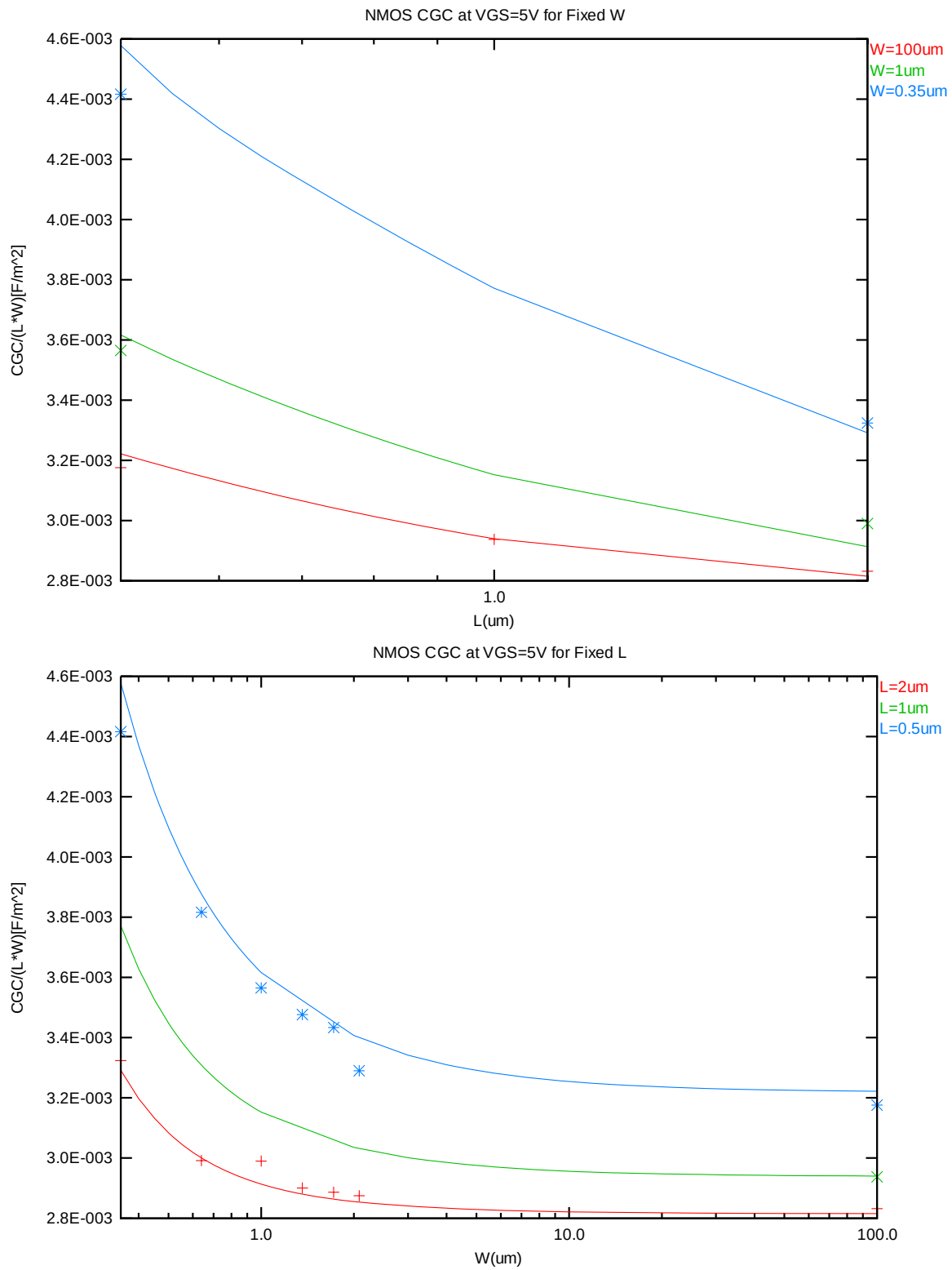
2.4.4 Width and length AC model accuracy

Width and length AC model accuracy









2.5 Parasitic Source / Drain Junction Capacitance Model

2.5.1 Test devices

The following test structures were used for parameter extraction.

Table 2-9 Junction capacitor structures for NMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	5.00E-08	2.00E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.76E-08	1.82E-02	Poly finger structure

2.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-10 Junction capacitor measurement conditions for NMOS

V _{BIAS} [V]	VH: pwell & VL: n+ From 0 to 5.5 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-40,25,125

2.5.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from silicon data.

Table 2-11 Extracted parameters from junction capacitor measurement for NMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	Unit	Value	Parameter	Unit	Value	Parameter	Unit	Value
CJ	F/m ²	7.64E-04	CJSW	F/m	9.64E-11	CJSWG	F/m	1.76E-10
PB	V	6.72E-01	PBSW	V	5.71E-01	PBSWG	V	6.12E-01
MJ	--	2.87E-01	MJSW	--	7.84E-02	MJSWG	--	5.15E-01
TCJ	1/K	1.12E-03	TCJSW	1/K	5.17E-04	TCJSWG	1/K	1.44E-03
TPB	V/K	1.70E-03	TPBSW	V/K	2.75E-03	TPBSWG	V/K	1.25E-03

CJ, CJSW vs. Vbias plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

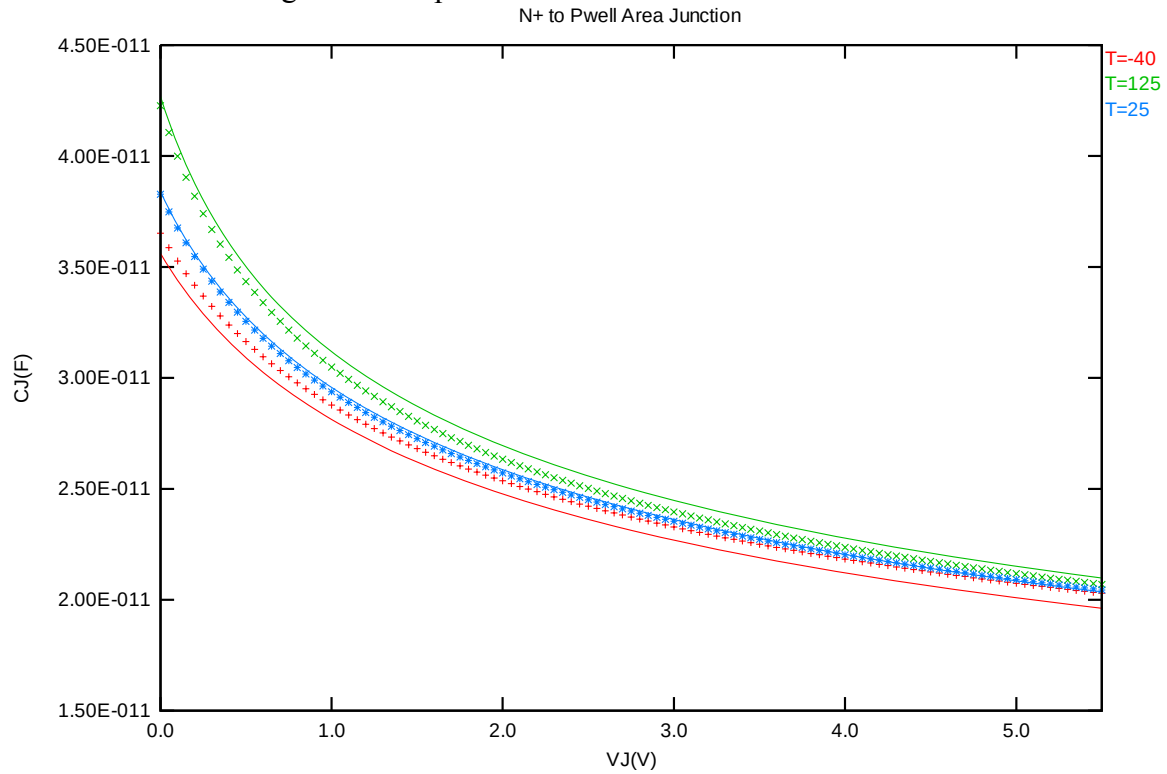


Figure 2-27 Bottom area junction capacitance at -40 C, 25 C and 125 C for NMOS

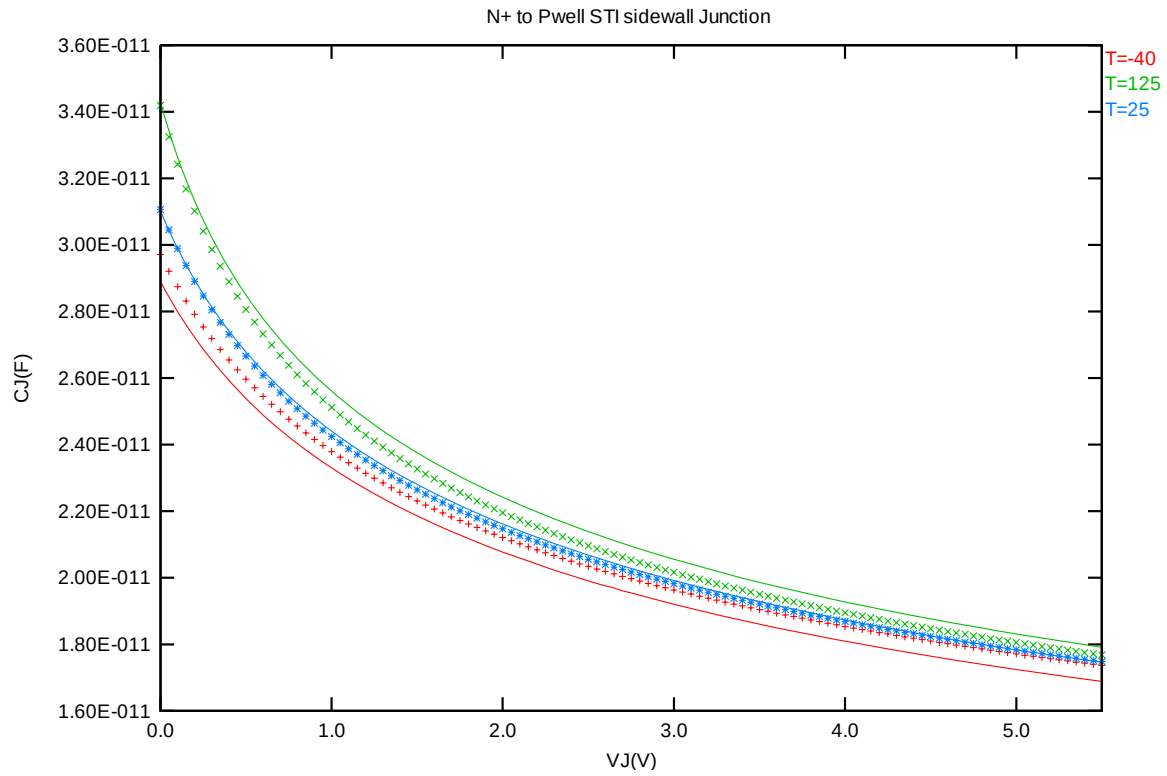


Figure 2-28 STI bounded sidewall junction capacitance at -40 C, 25 C and 125 C for NMOS

3 5 V P-ch MOSFET

3.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for PMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	BSW1PA	BSW1PA	BSW1PA	BSW1PA
Lot ID	D6FW8	D6FW8	D6FW8	D6FW8
Wafer Number	#14, #15	#14, #15	#14, #15	#14, #15

3.2 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at $I_D = 300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for NMOS and $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ for PMOS, respectively.

All parameters are given for $T=25\text{ C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Table 3-2 Electrical parameters for PMOS

All parameters are given for $T=25\text{ }^{\circ}\text{C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Target & Centered Value are given for $SA=3.6\mu$ $SB=3.6\mu$

Measured & Extracted Value are given for $SA=3.6\mu$ $SB=3.6\mu$

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
-----------	-----------	------	--------	----------------	-----------------	----------------

Long channel device ($W/L=10\text{ }\mu\text{ m} / 10\text{ }\mu\text{ m}$)

V_{TLIN}	$V_{DS} = -0.1\text{ V}$	V	-0.865	-0.864	-0.863	-0.865
I_{ON}	$V_{DS} = -5\text{ V}$	$\mu\text{A} / \mu\text{m}$	-12.08	-12.08	-12.09	-12.08

Wide and short channel device ($W/L=10\text{ }\mu\text{ m} / 0.5\text{ }\mu\text{ m}$)

V_{TLIN}	$V_{DS} = -0.1\text{ V}$	V	-0.835	-0.834	-0.834	-0.835
V_{TSAT}	$V_{DS} = -5\text{ V}$	V	-0.801	-0.798	-0.796	-0.801
Body Effect V_T shift	$V_{DS} = -5\text{ V}$ $V_{BS} = 0 \sim -5\text{ V}$	V	--	-0.928	-1.198	-1.189
DIBL	$V_{DS} = -0.1 \sim -5\text{ V}$	V	-0.034	-0.037	-0.038	-0.034
I_{ON}	$V_{DS}=V_{GS} = -5\text{ V}$	$\mu\text{A}/\mu\text{m}$	-210.00	-210.40	-210.76	-210.00
I_{OFF}	$V_{DS} = -5\text{ V}$, $V_{GS}=0\text{ V}$	$\text{A}/\mu\text{m}$	-4.000e-12	-1.4E-12	-1.5E-12	-4.0E-12

Narrow and long channel device ($W/L=0.35\text{ }\mu\text{ m} / 10\text{ }\mu\text{ m}$)

V_{TLIN}	$V_{DS} = -0.1\text{ V}$	V	-0.883	-0.878	-0.877	-0.883
V_{TSAT}	$V_{DS} = -5\text{ V}$	V	-0.873	-0.868	-0.871	-0.876
I_{ON}	$V_{DS}=V_{GS} = -5\text{ V}$	$\mu\text{A}/\mu\text{m}$	-12.20	-12.18	-12.20	-12.20

Narrow and short channel device ($W/L=0.35\text{ }\mu\text{ m} / 0.5\text{ }\mu\text{ m}$)

V_{TLIN}	$V_{DS} = -0.1\text{ V}$	V	-0.841	-0.838	-0.837	-0.841
V_{TSAT}	$V_{DS} = -5\text{ V}$	V	-0.802	-0.803	-0.797	-0.801
I_{ON}	$V_{DS}=V_{GS} = -5\text{ V}$	$\mu\text{A}/\mu\text{m}$	-203.54	-205.84	-206.08	-203.54
I_{OFF}	$V_{DS} = -5\text{ V}$, $V_{GS}=0\text{ V}$	$\text{A}/\mu\text{m}$	-4.477e-12	-1.7E-12	-1.5E-12	-3.9E-12

Please set $gmindc=8e-12$ in spectre simulator for achieving the consistent I_{OFF} result with hspice's.

3.3 MOSFET DC Model

3.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 3-3 Geometry of PMOS devices measured for DC parameter extraction

WL	10	5	1	0.8	0.6	0.5
10	X	X	X	X	X	X
5	X		X			X
1	X	X	X	X	X	X
0.8	X		X			X
0.6	X	X	X	X	X	X
0.43	X		X		X	X
0.35	X	X	X	X	X	X

3.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 3-4 Test conditions for IV curves of PMOS

I_D vs. V_{GS}				I_D vs. V_{DS}			
	Start	Stop	Step		Start	Stop	Step
V_{DS} (V)	-0.1	-5	-4.9	V_{DS} (V)	0	-5.5	-0.05
V_{GS} (V)	0.5	-5.5	-0.05	V_{GS} (V)	-1.1	-5	-0.39
V_{BS} (V)	0	3	0.6	V_{BS} (V)	0	3	3

3.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension (LDES *0.9+12nm and WDES *0.9) respectively.

I_D vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

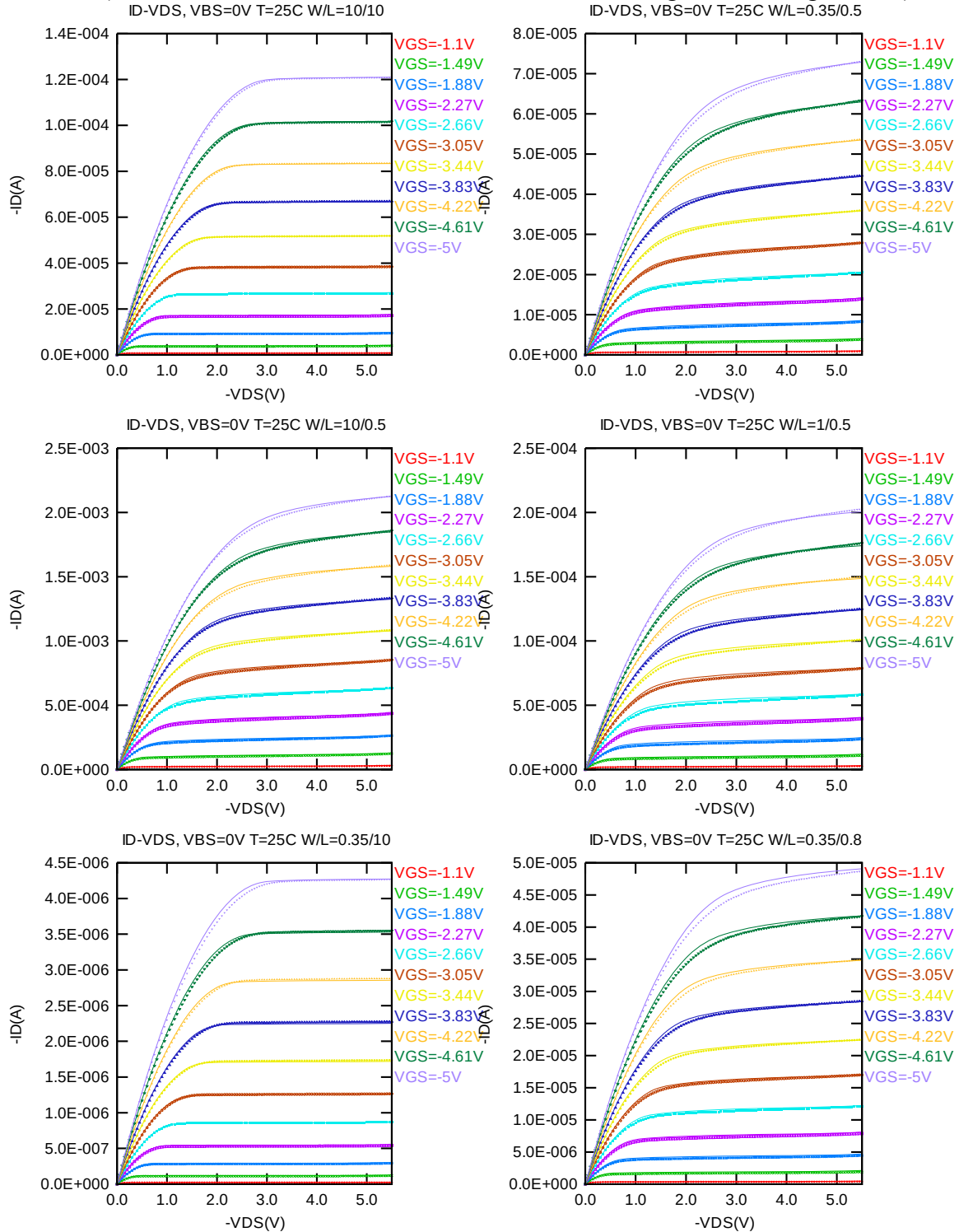


Figure 3-1 I_D vs. V_{DS} at $V_{BS}=0V$ for PMOS

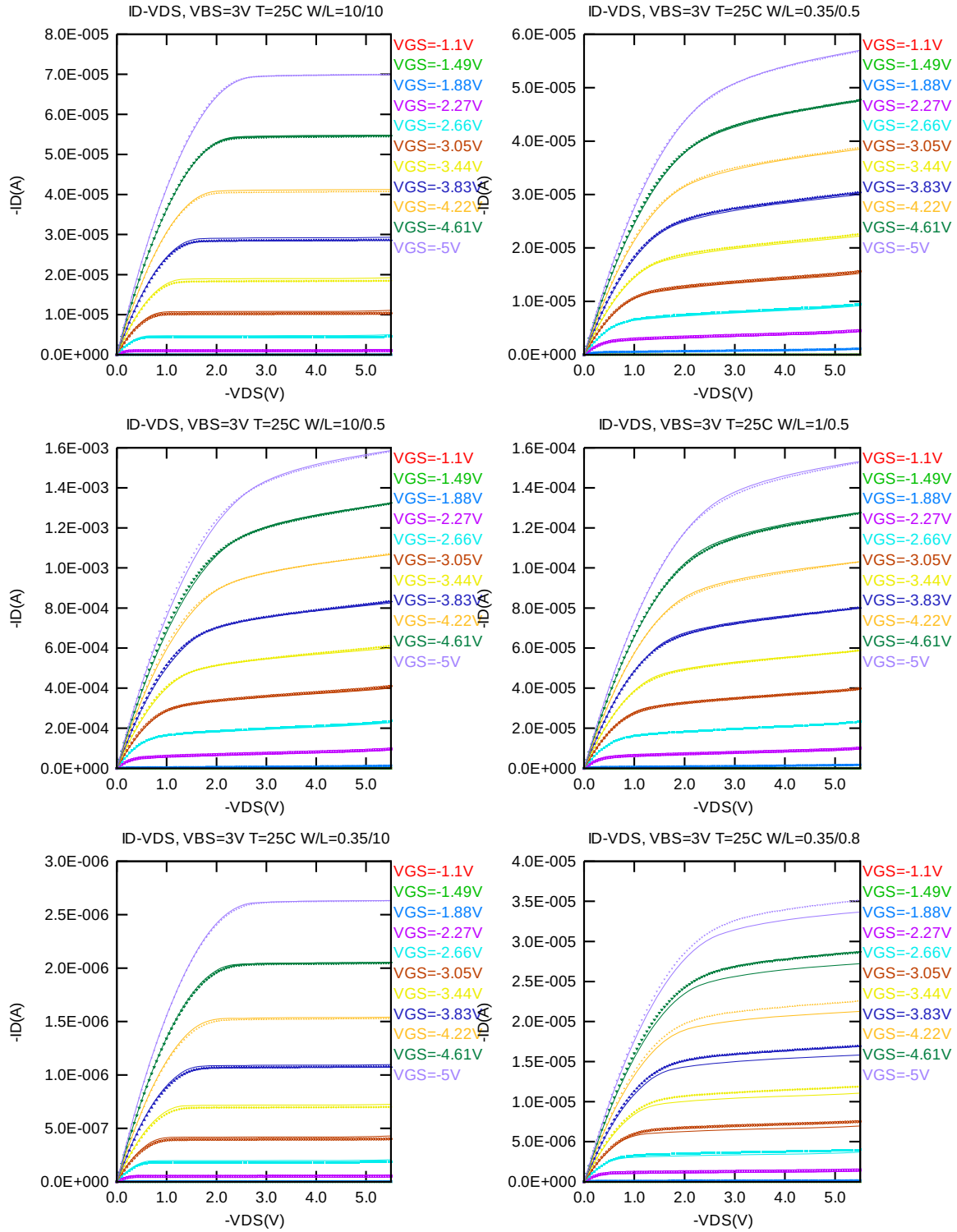


Figure 3-2 I_D vs. V_{DS} at $V_{BS}=3$ V for PMOS

G_{DS} vs. V_{DS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

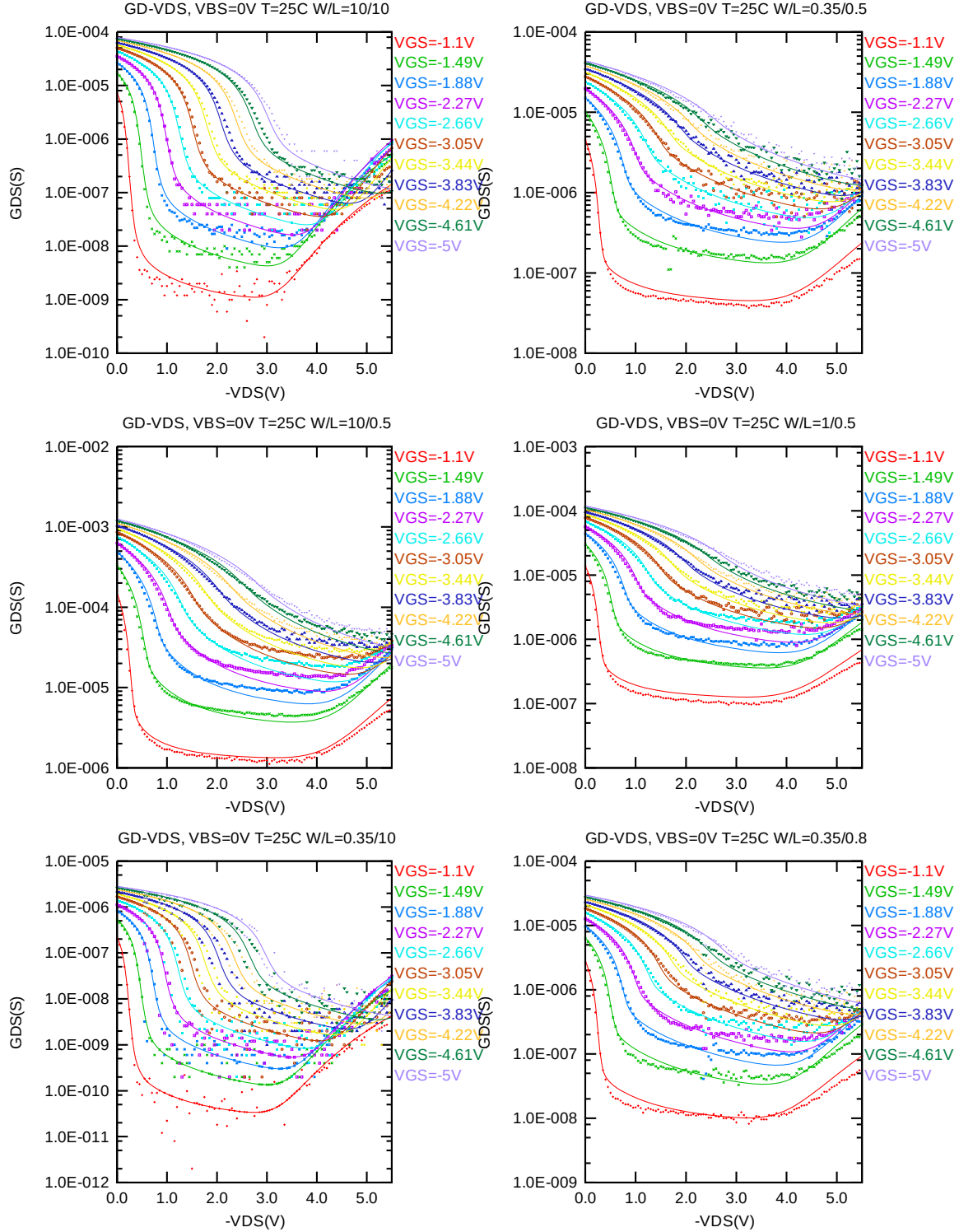


Figure 3-3 G_{DS} vs. V_{DS} at $V_{BS}=0V$ for PMOS

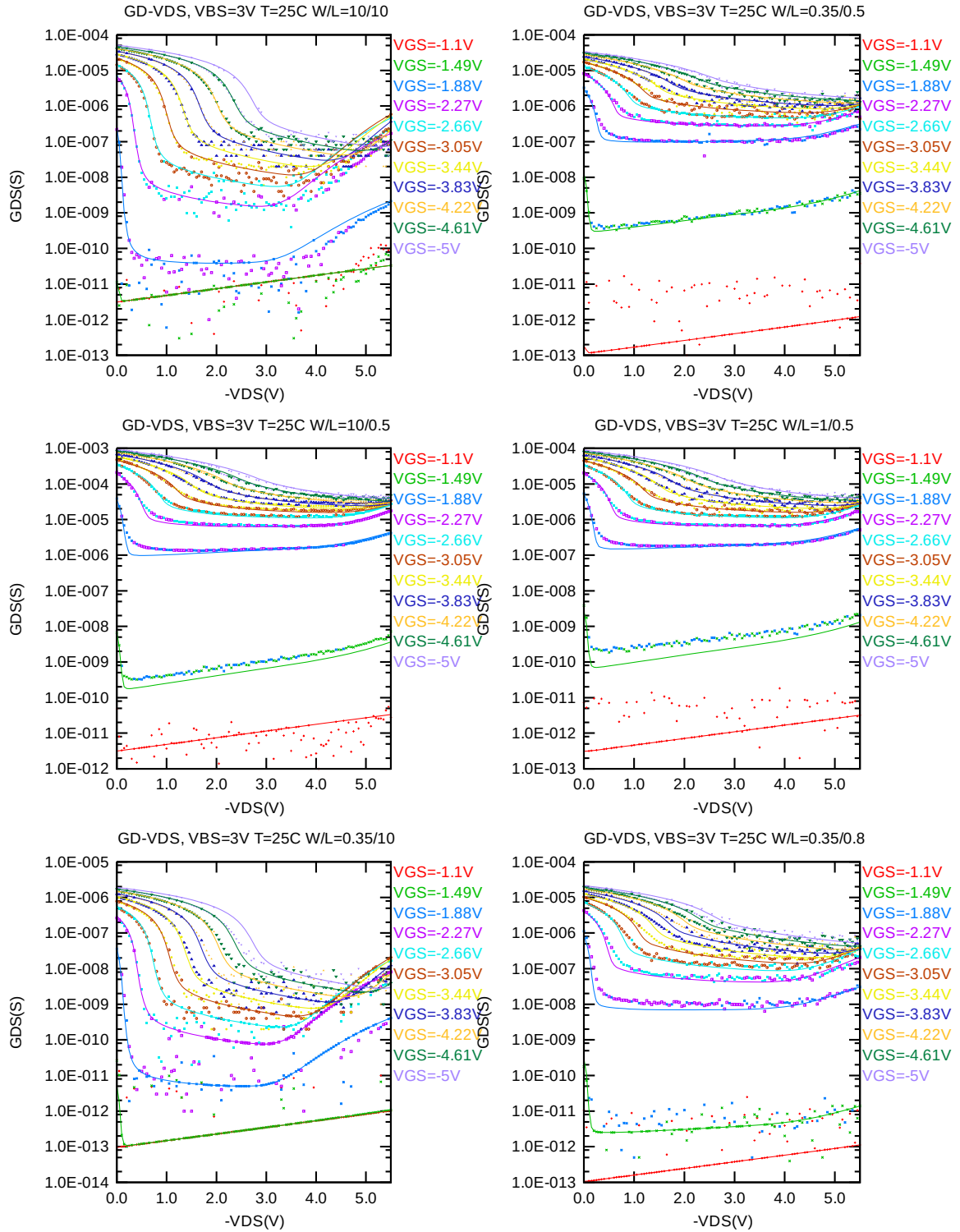


Figure 3-4 G_{DS} vs. V_{DS} at $V_{BS}=3V$ for PMOS

I_D vs. V_{GS} plots (linear scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

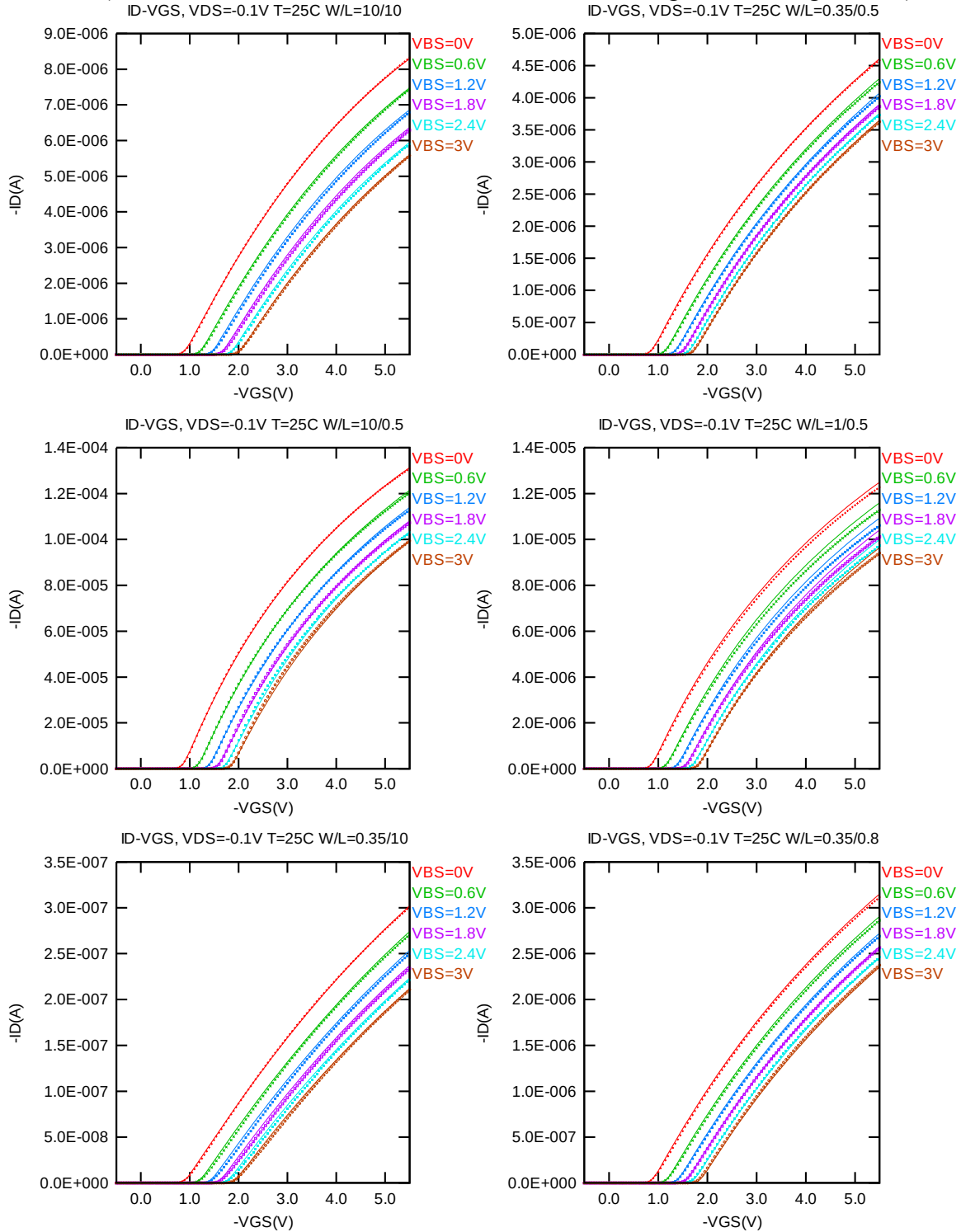


Figure 3-5 I_D vs. V_{GS} at $V_{DS} = -0.1V$ for PMOS

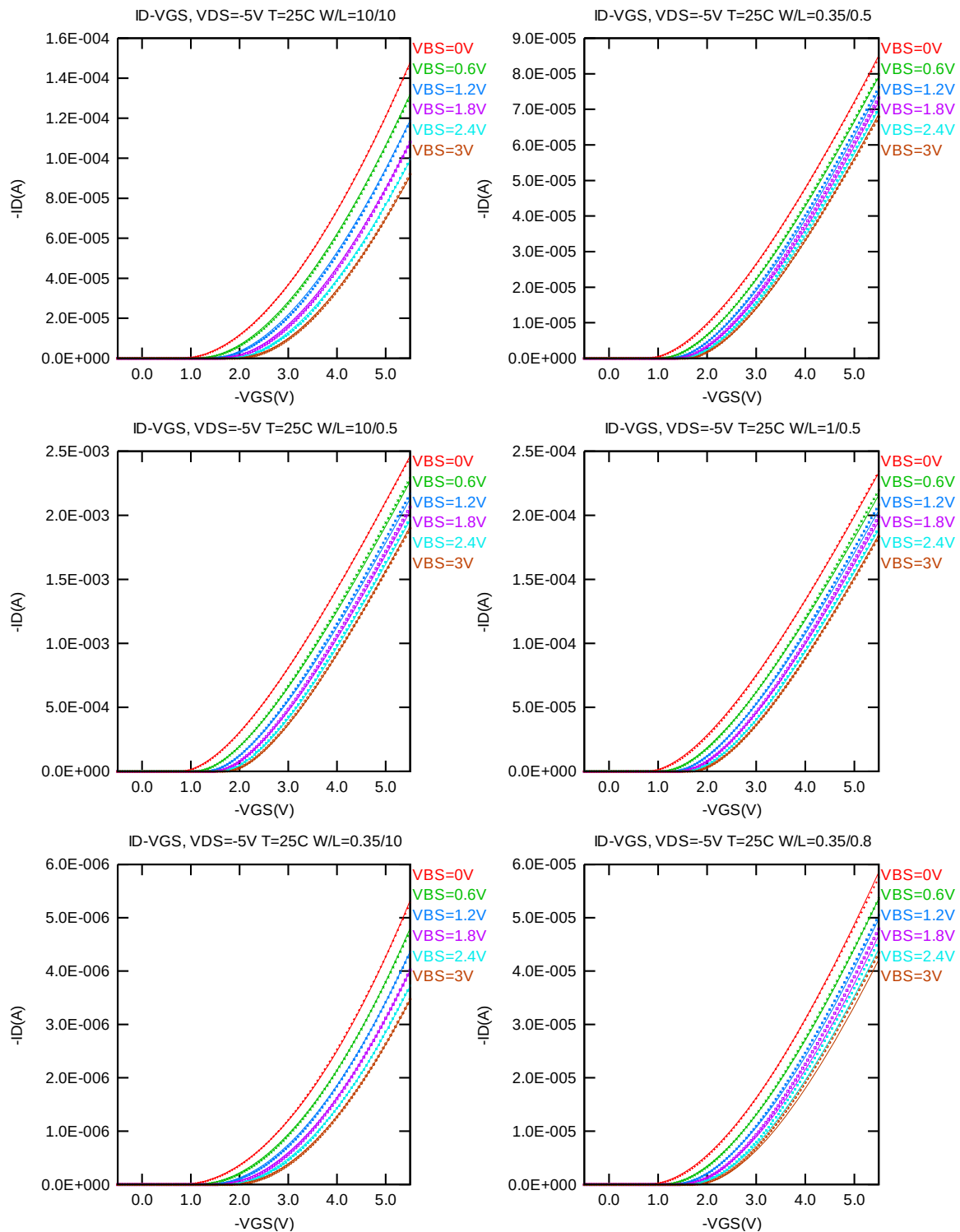


Figure 3-6 I_D vs. V_{GS} at $V_{DS} = -5V$ for PMOS

I_D vs. V_{GS} plots (logarithmic scale)

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

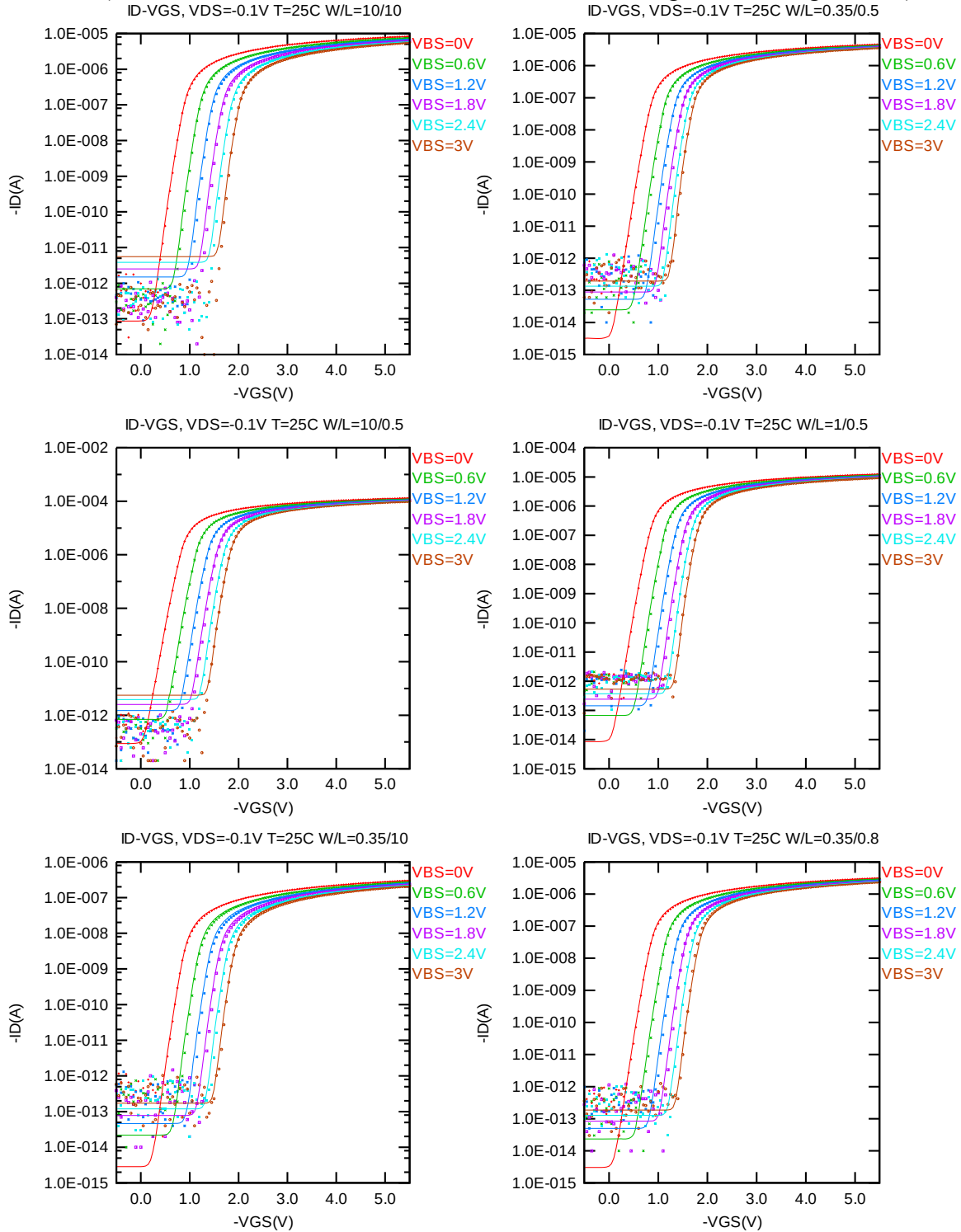


Figure 3-7 Log scaled I_D vs. V_{GS} at $V_{DS} = -0.1$ V for PMOS

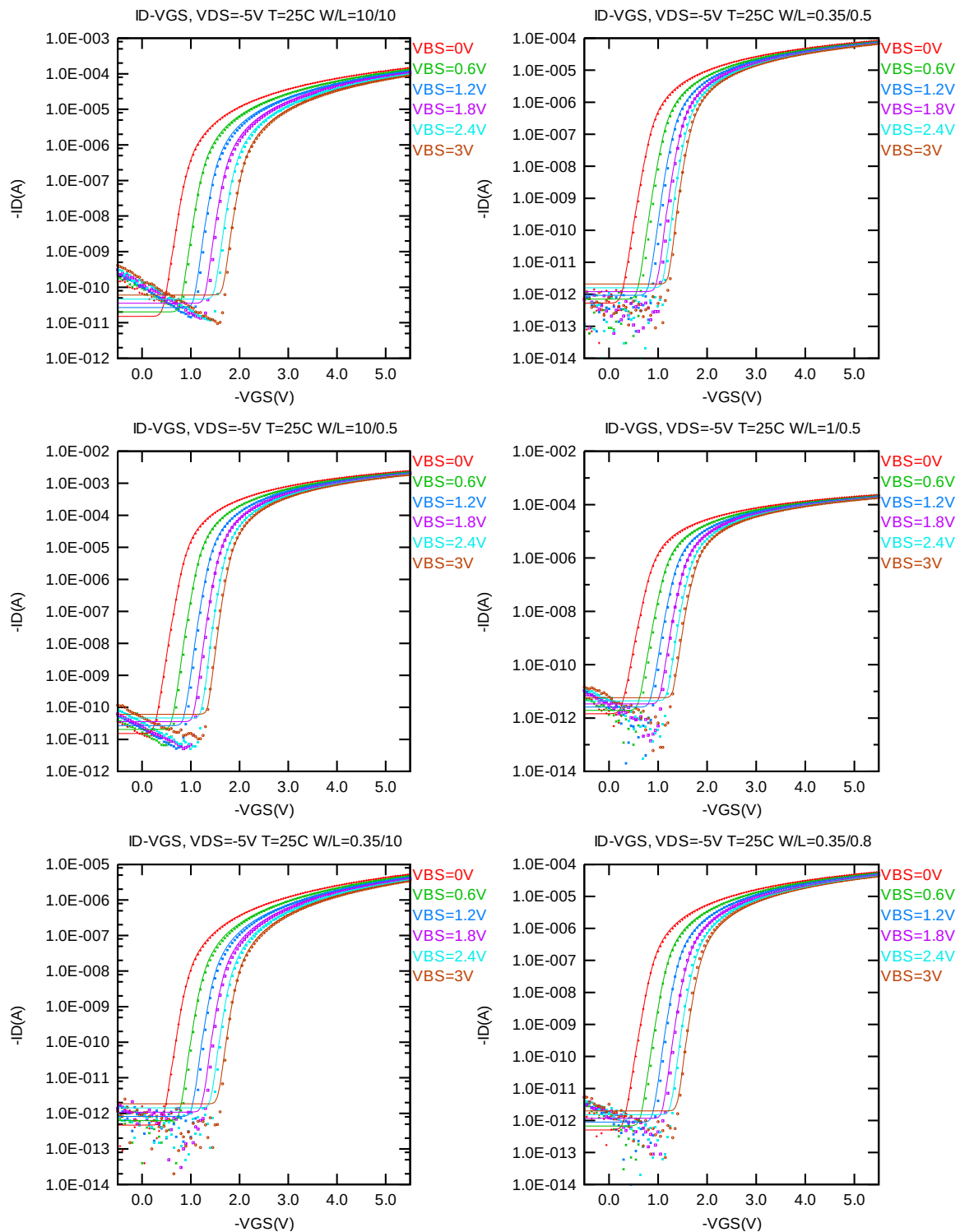


Figure 3-8 Log scaled I_D vs. V_{GS} at $V_{DS} = -5$ V for PMOS

G_M vs. V_{GS} plots

The following plots show measured and simulated output I-V curves for selected devices at 25 C. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

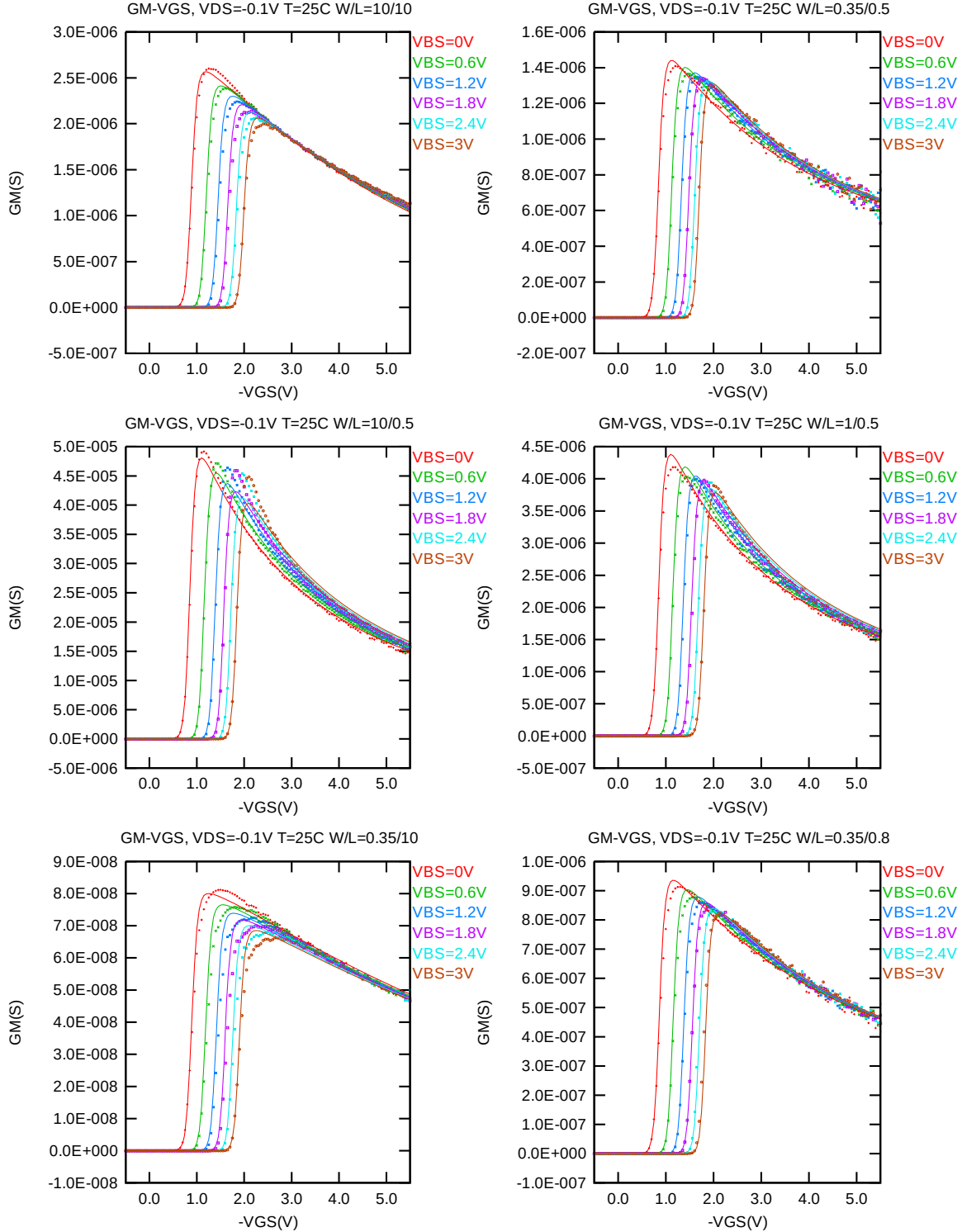


Figure 3-9 G_M vs. V_{GS} at $V_{DS} = -0.1V$ for PMOS

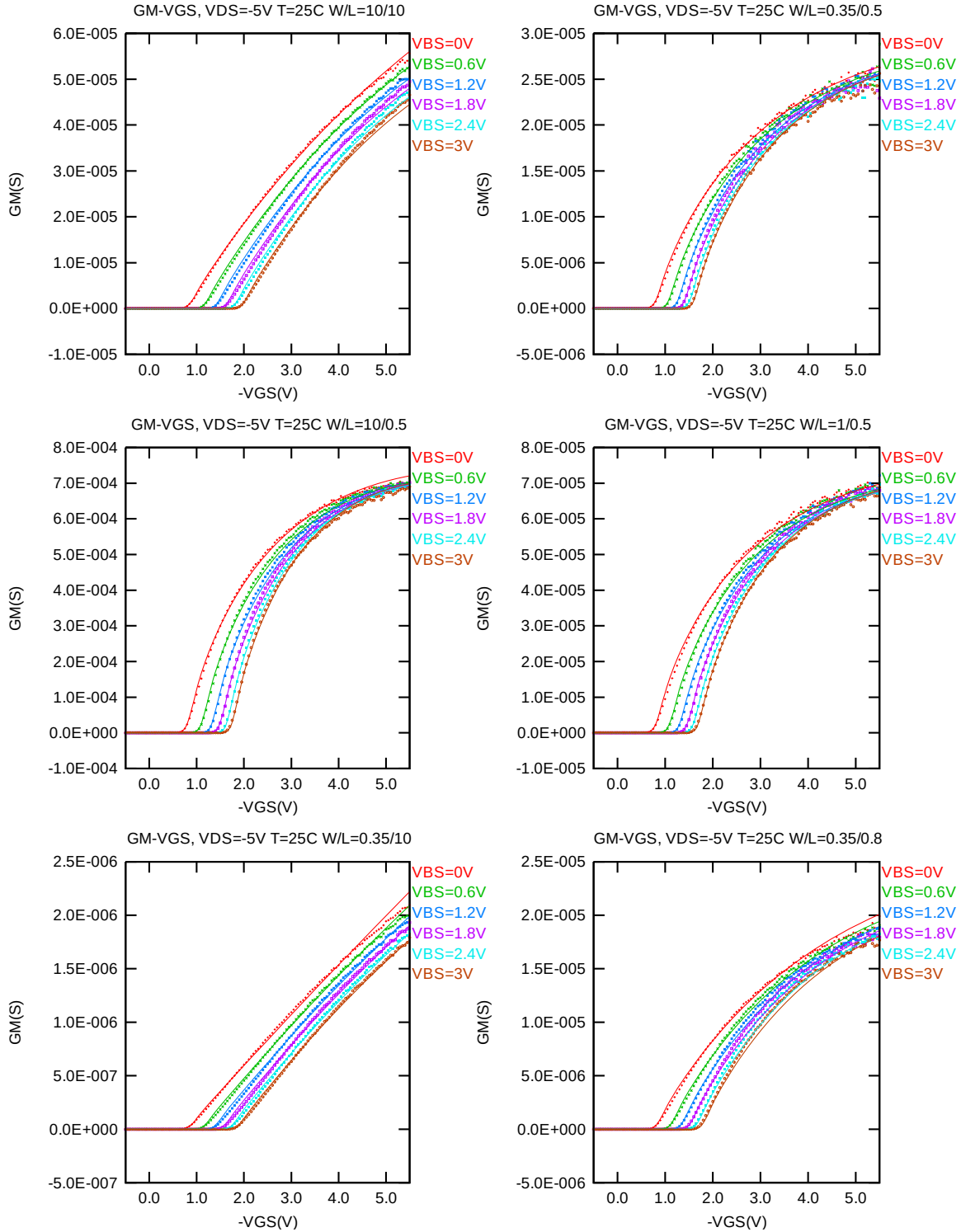


Figure 3-10 G_M vs. V_{GS} at $V_{DS} = -5V$ for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ at 25°C . The plots show the physical dimension.

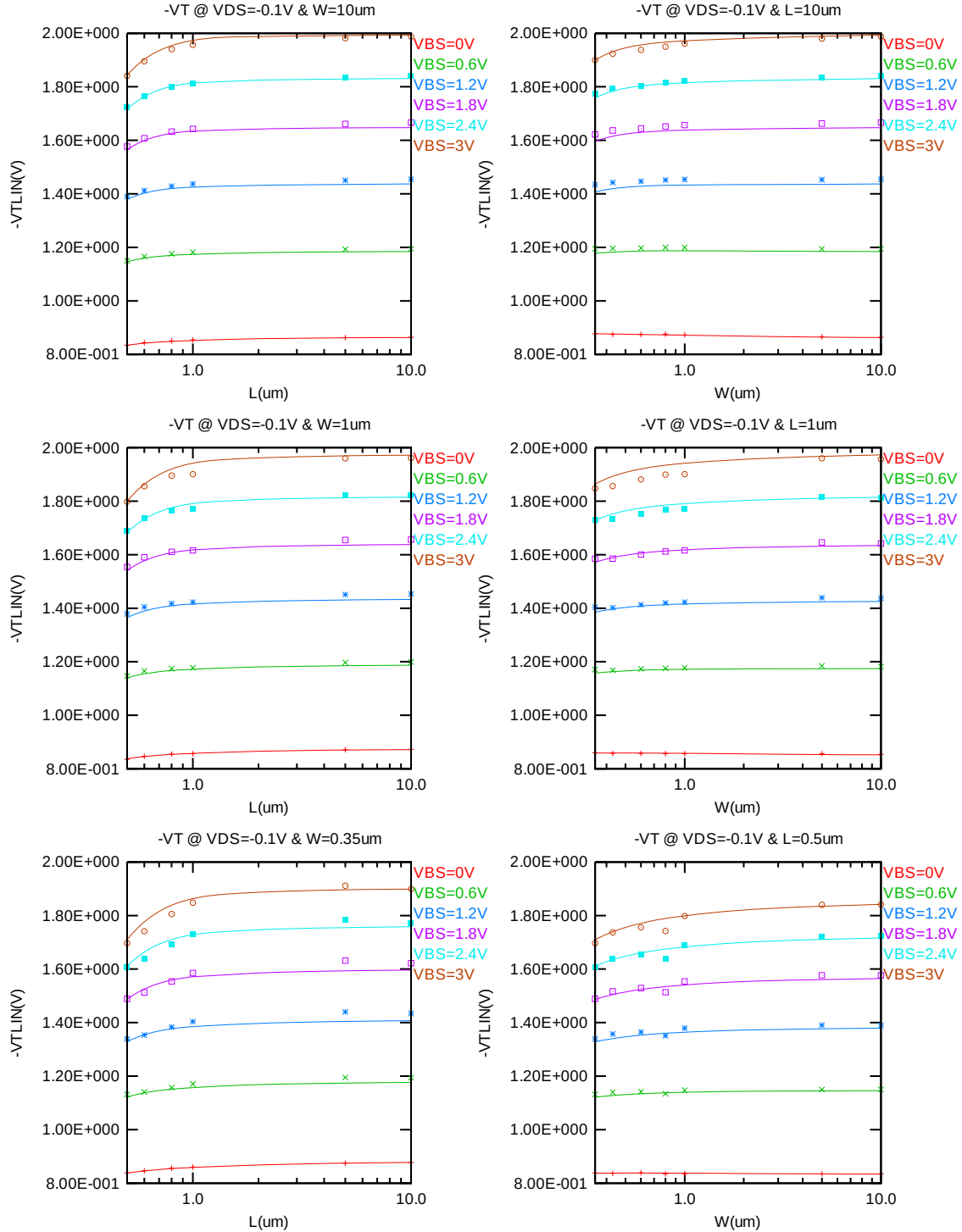


Figure 3-11 $V_{T LIN}$ vs. channel length & channel width for low drain bias ($V_{DS} = -0.1\text{V}$) for PMOS

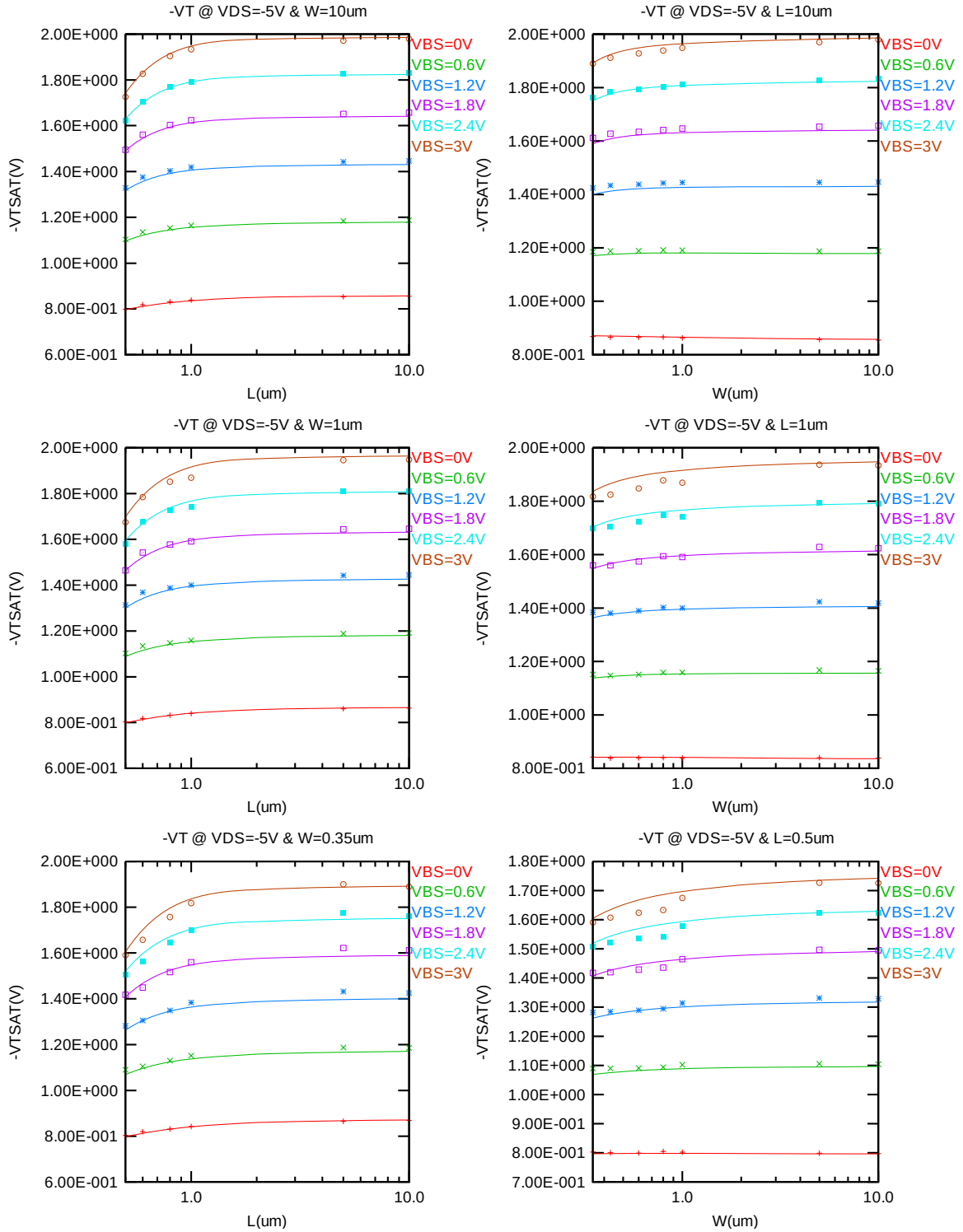


Figure 3-12 V_{TSAT} vs. channel length & channel width for high drain bias ($V_{DS} = -5V$) for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS}=V_{GS}=-5$ V for $V_{BS}=0$ V and $V_{BS}=3$ V at 25 C.(The dimensions are after 90% shrink and 12nm sizing channel length back.)

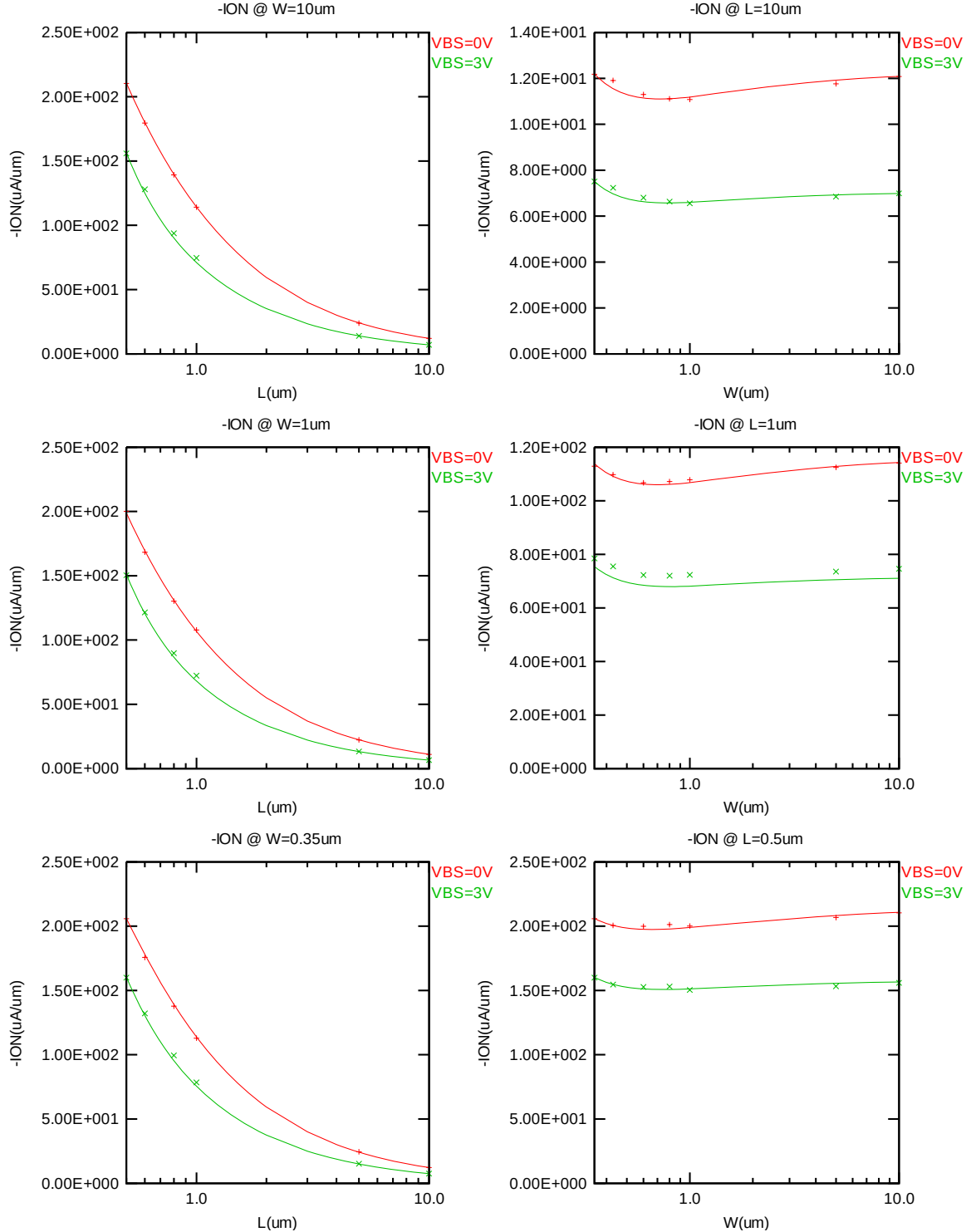


Figure 3-13 Saturation current I_{ON} vs. channel length and channel width for PMOS

The L and W in the plots refer to LDES and WDES respectively.

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{\text{DS}}=5\text{ V}$ and $V_{\text{GS}}=0\text{ V}$ for $V_{\text{BS}}=0\text{ V}$ to 3 V at 25 C . A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.

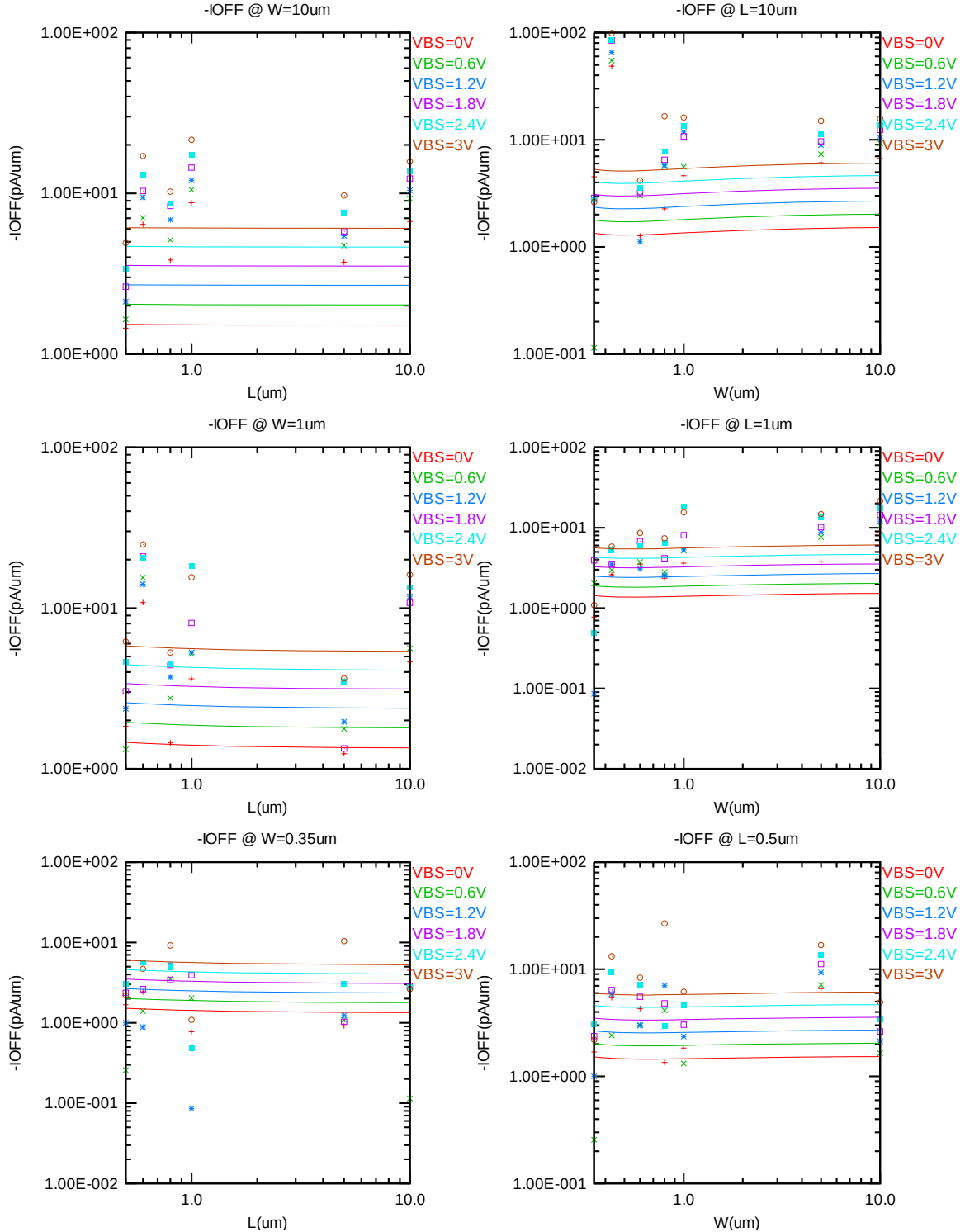


Figure 3-14 I_{OFF} vs. channel length and channel width for PMOS

V_{TLIN} vs. temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at $ID = -70nA * WDES * 0.9 / (LDES * 0.9 + 12nm)$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

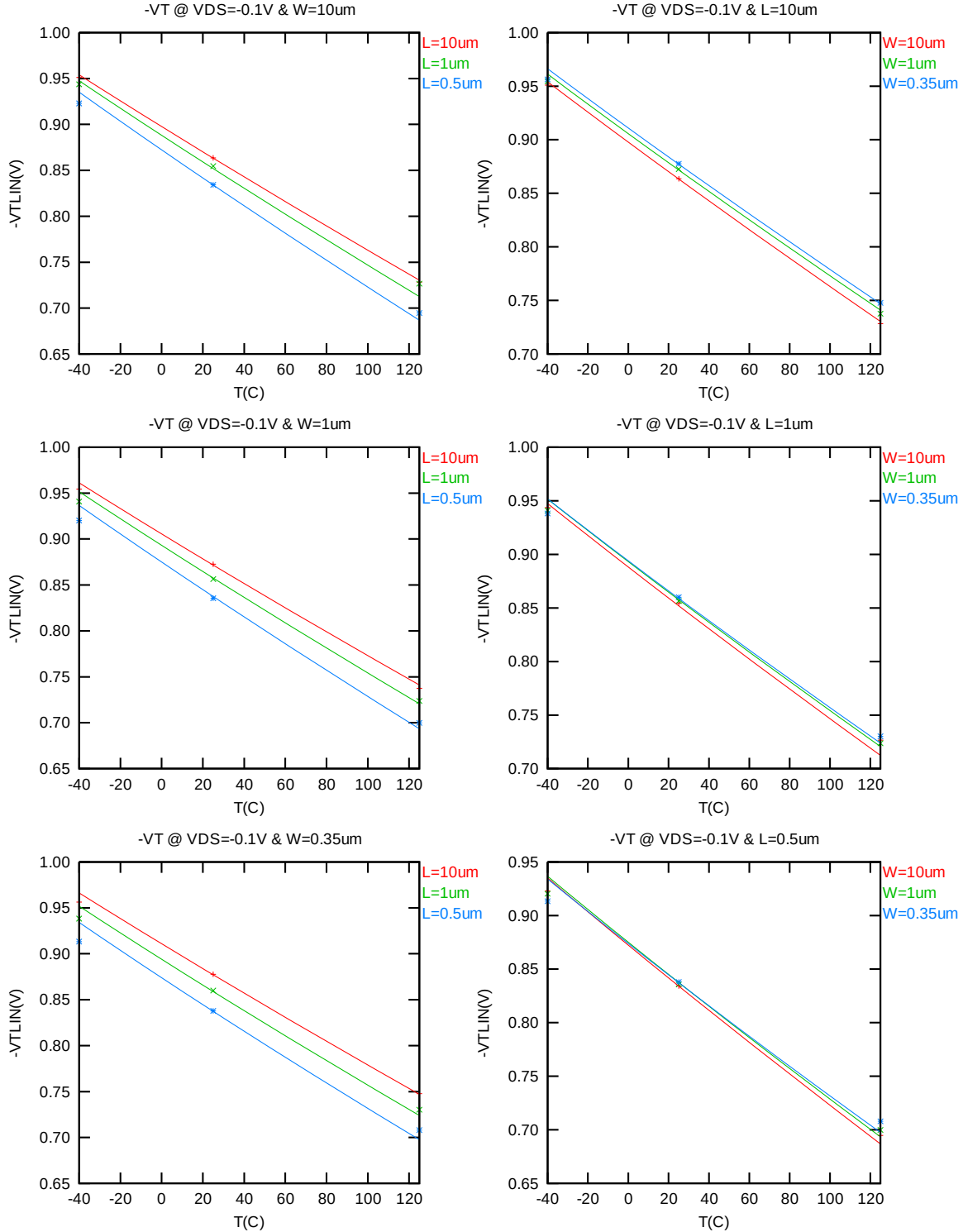


Figure 3-15 V_{TLIN} vs. temperature for PMOS

V_{TSAT} vs. temperature

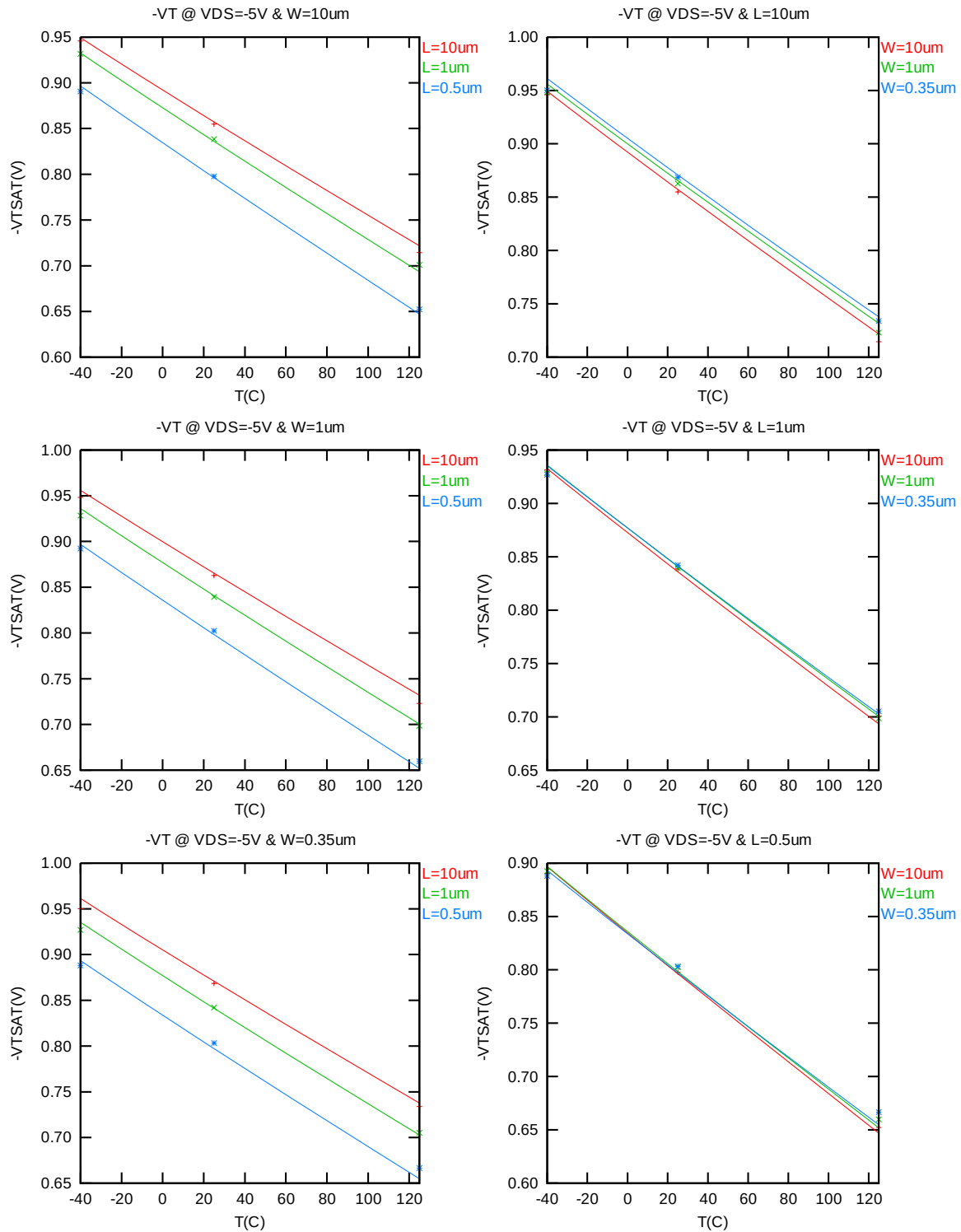


Figure 3-16 V_{TSAT} vs. temperature for PMOS

I_{ON} vs. temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS}=V_{DS}=5$ V, $V_{BS}=0$ V. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

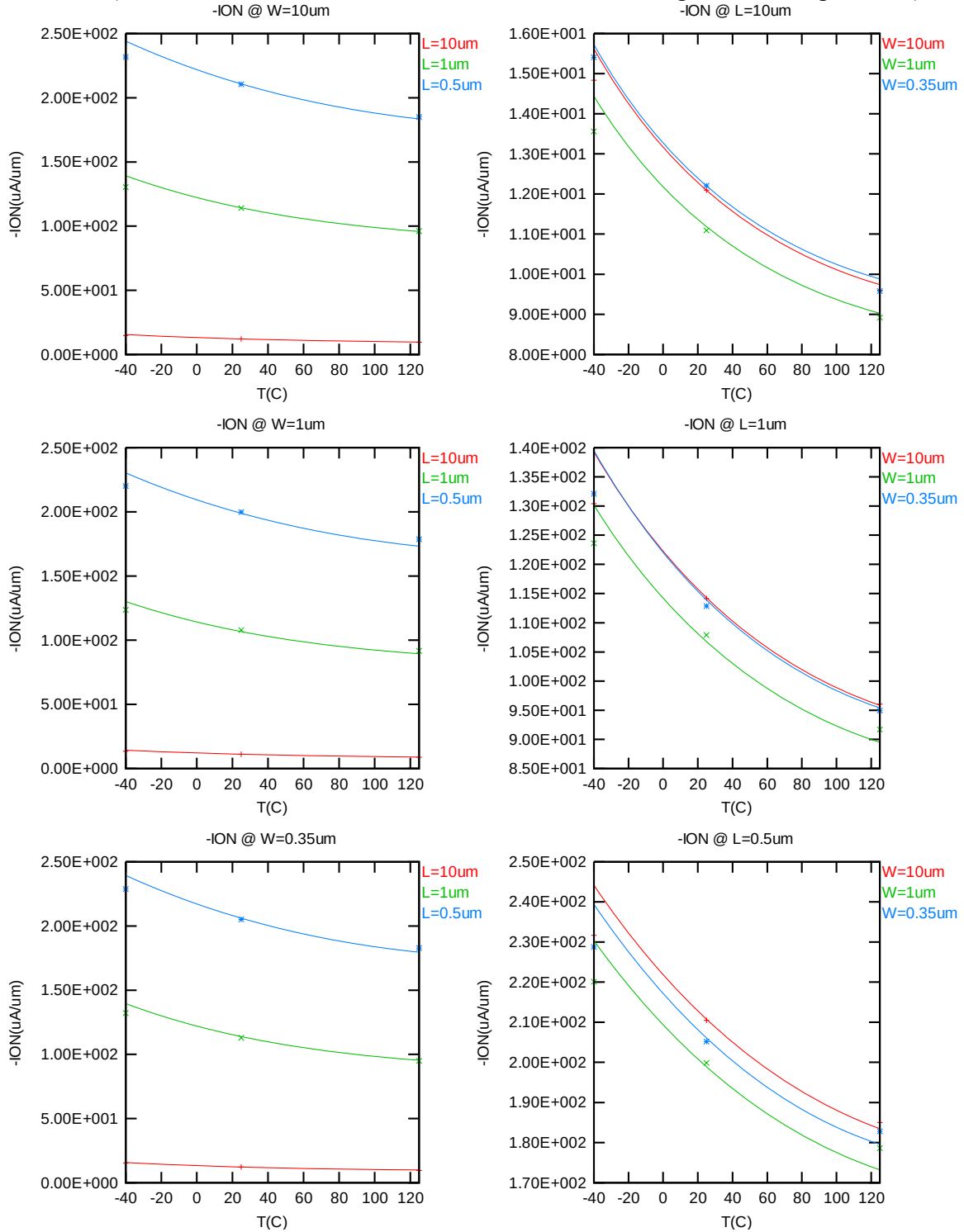


Figure 3-17 Saturation current vs. temperature for PMOS

I_{OFF} vs. temperature

The following plots show the temperature dependence of I_{OFF} extracted at $V_{\text{DS}}=5\text{ V}$, $V_{\text{GS}}=V_{\text{BS}}=0\text{ V}$. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

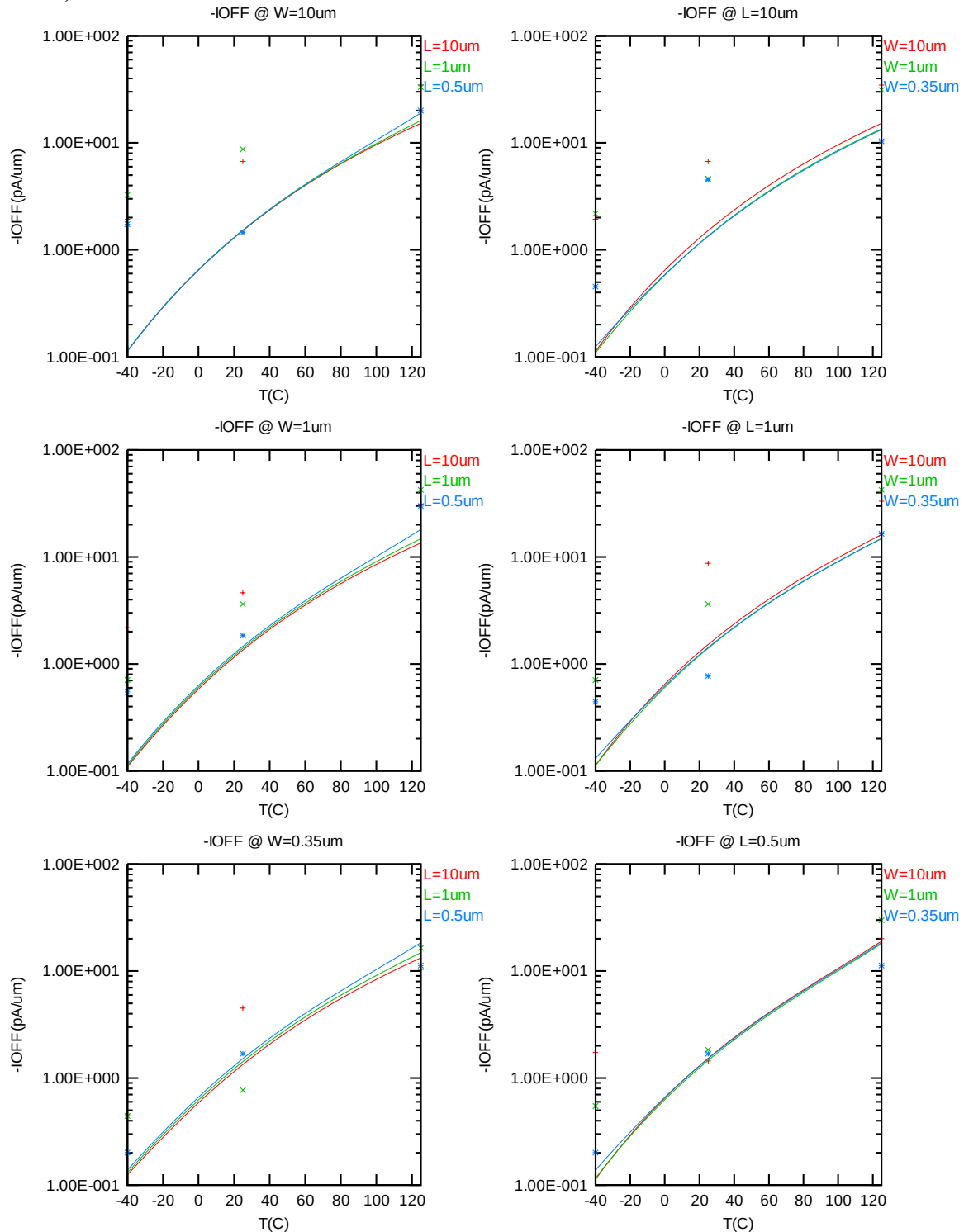


Figure 3-18 I_{OFF} vs. temperature for PMOS

3.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation regions. These criteria are explained below.

1) I_D accuracy criteria

- f. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- g. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- h. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- i. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- j. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 3-5 Requirements for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D accuracy

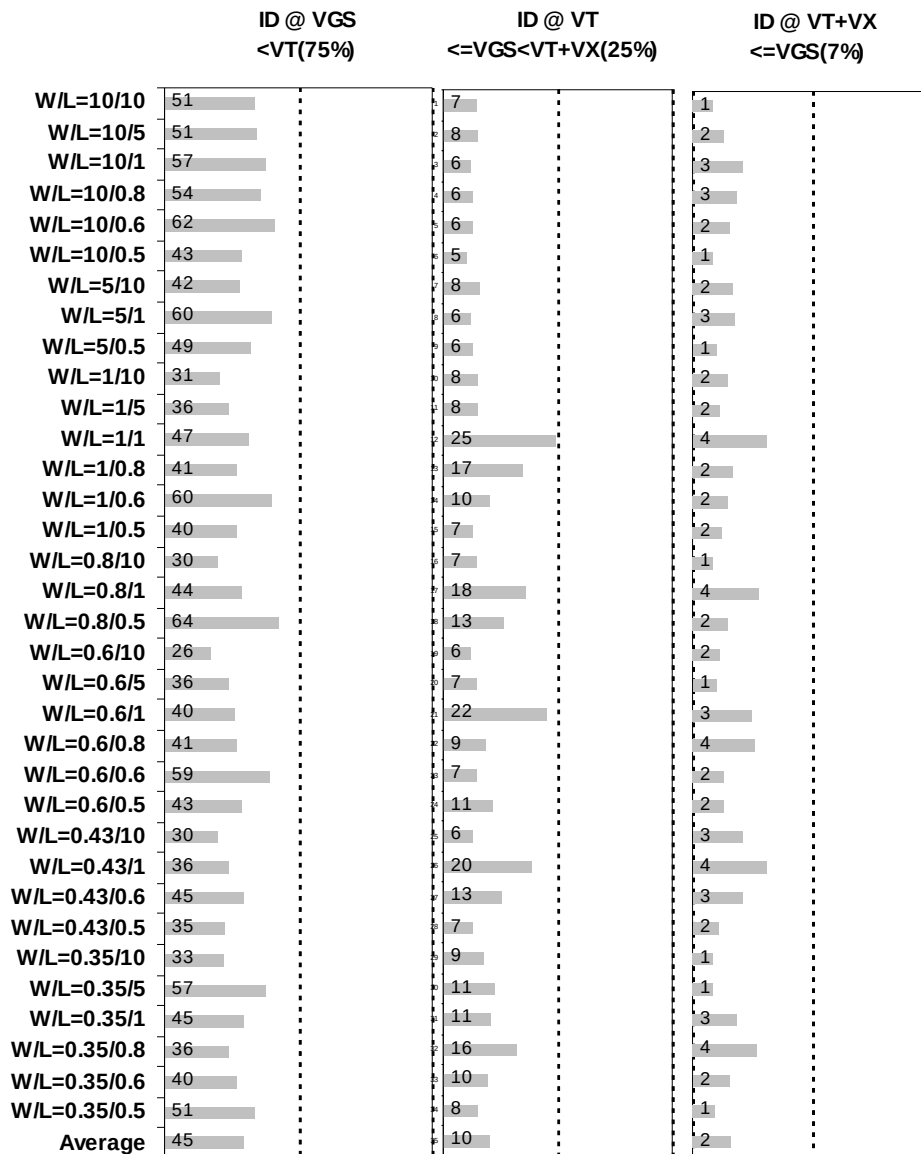


Figure 3-19 I_D accuracy at 25 C for PMOS

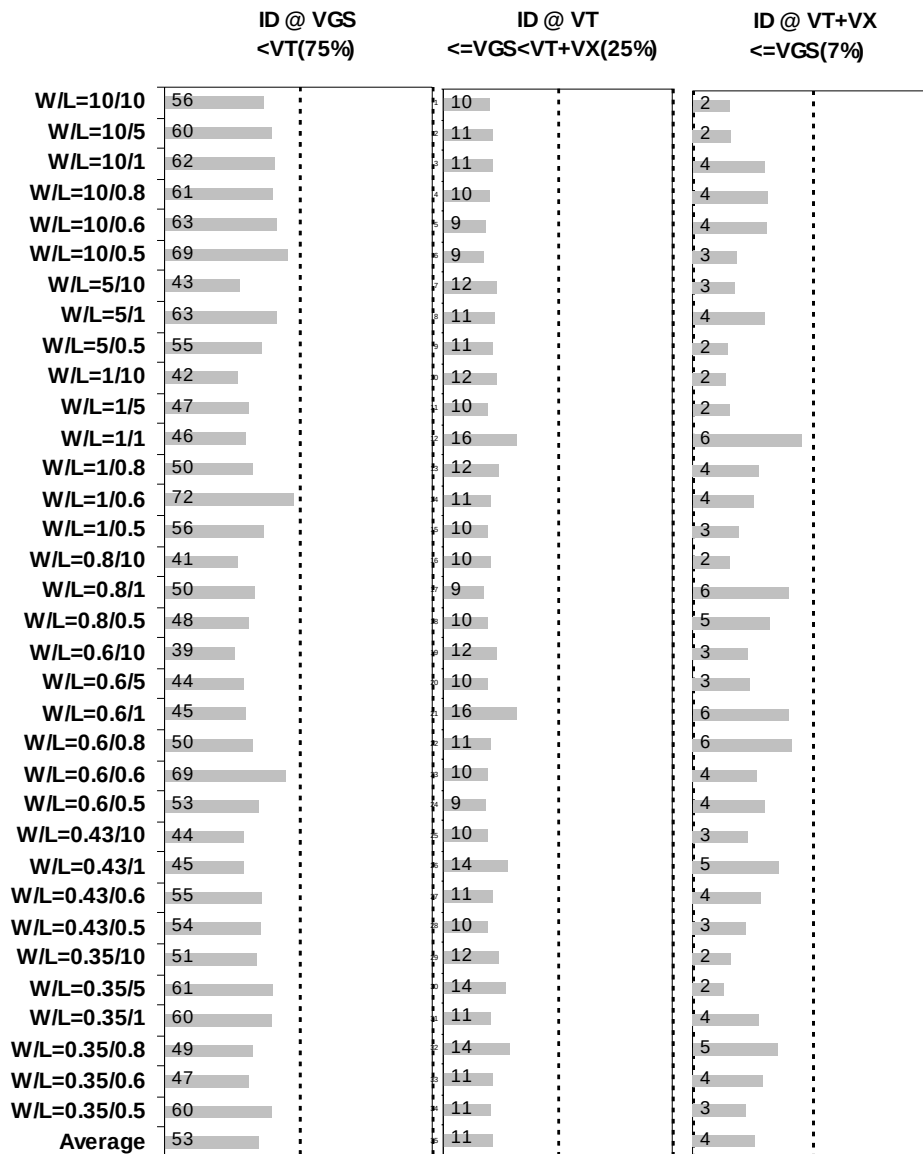


Figure 3-20 I_D accuracy at 125 C for PMOS

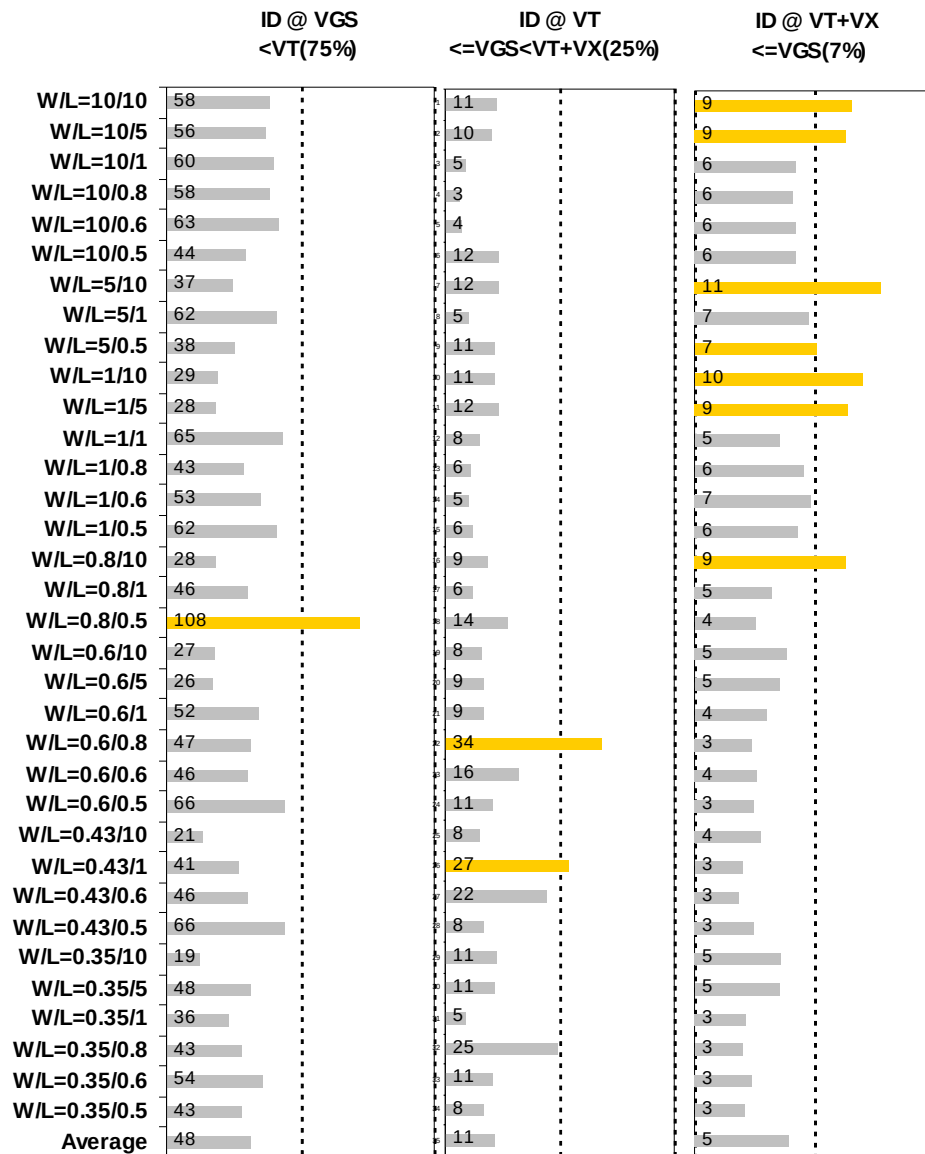


Figure 3-21 I_D accuracy at -40 C for PMOS

G_M and G_{DS} accuracy

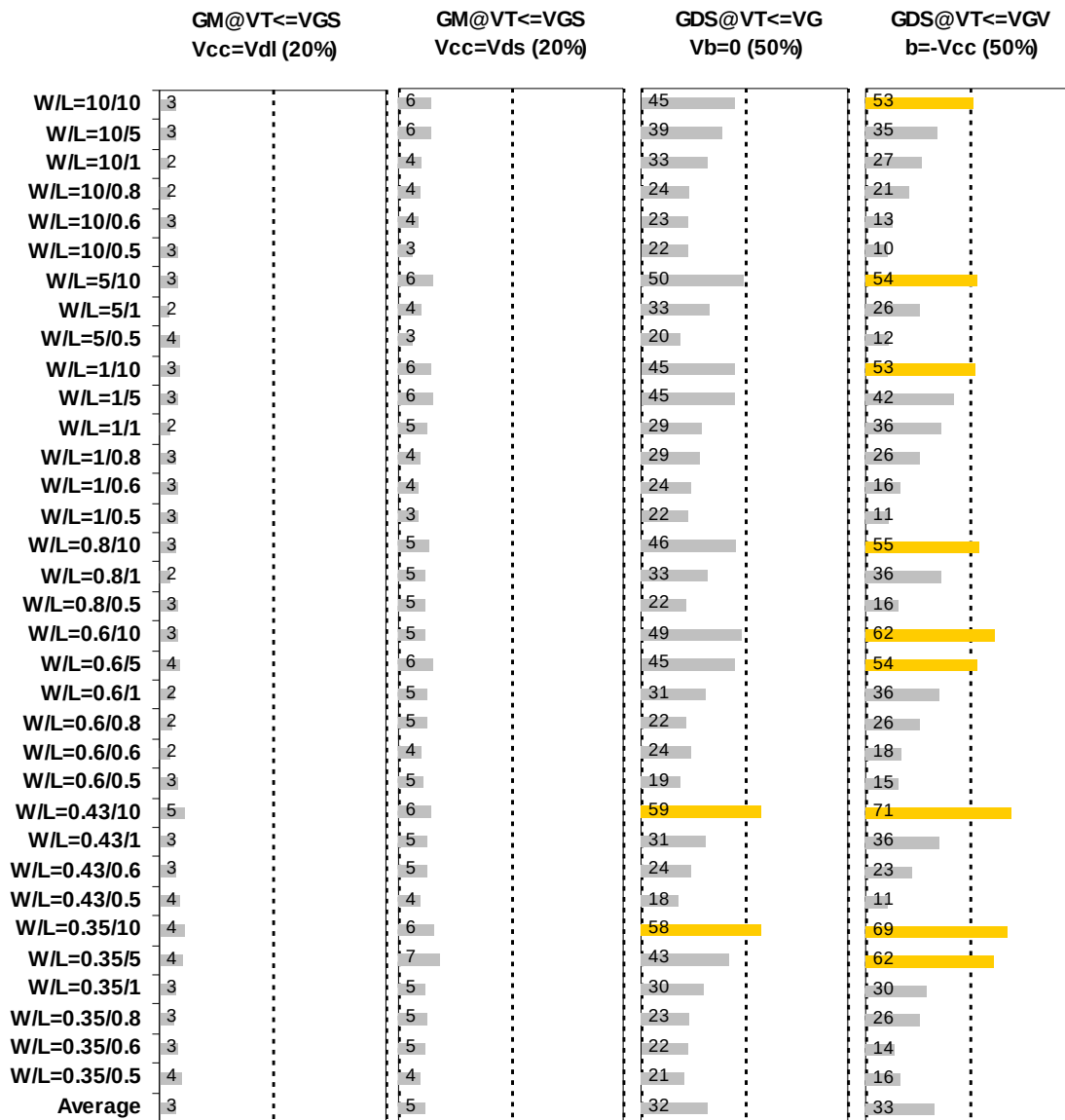


Figure 3-22 G_M and G_{DS} accuracy at 25 C for PMOS

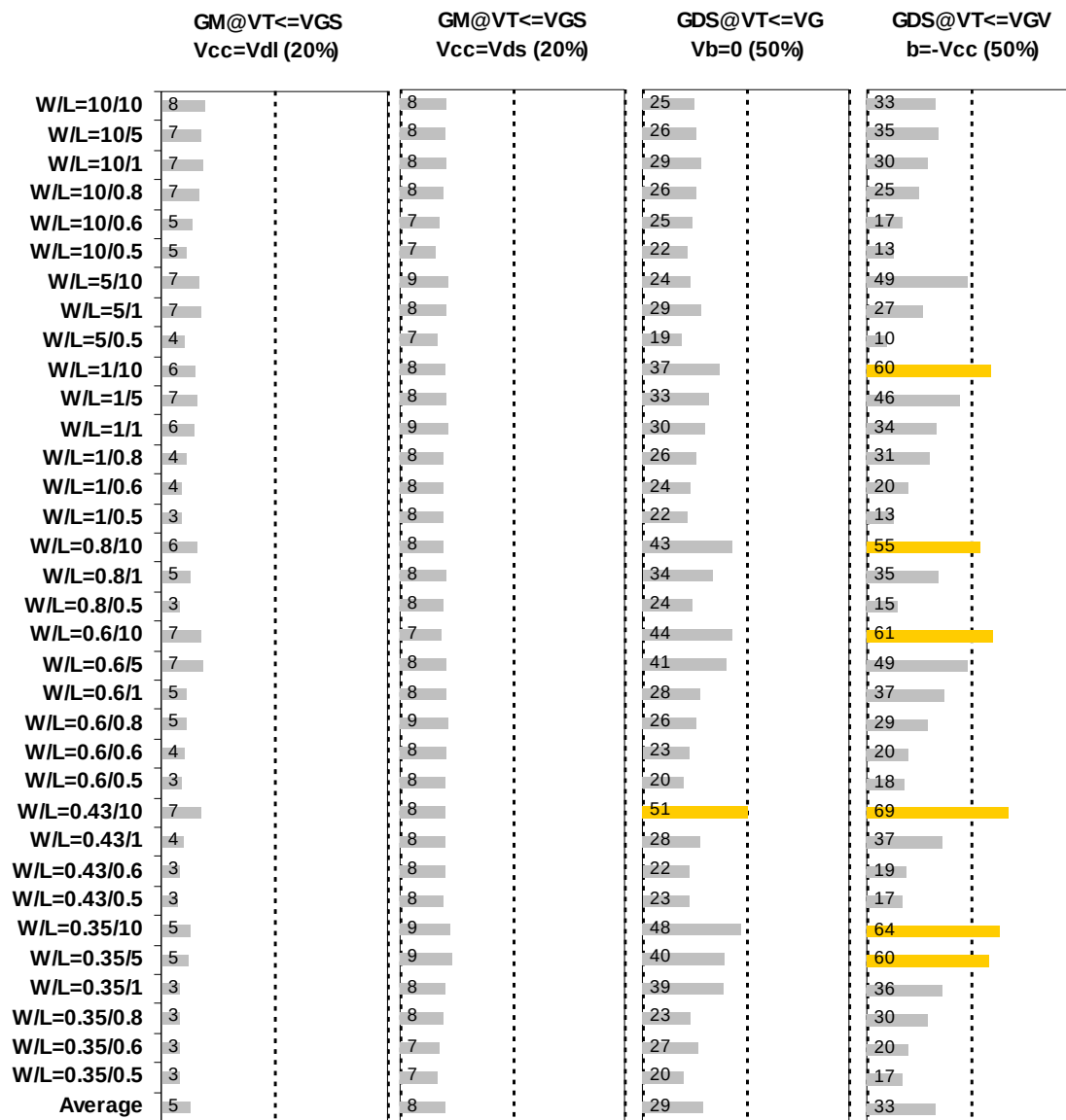


Figure 3-23 G_M and G_{DS} accuracy at 125 C for PMOS

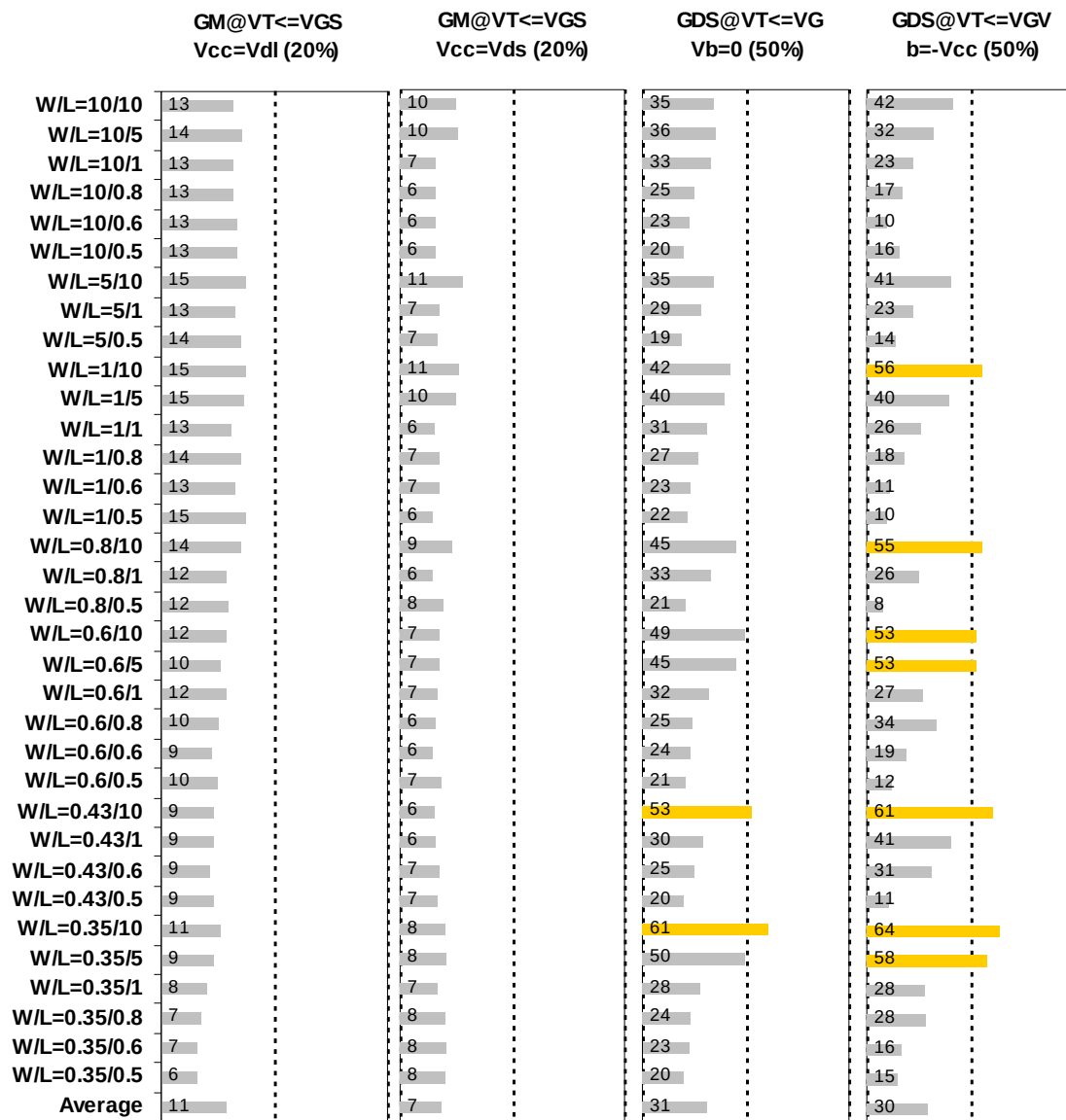


Figure 3-24 G_M and G_{DS} accuracy at -40 C for PMOS

3.4 MOSFET Capacitance Model

3.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 3-6 Measurement structure for PMOS C_{gg} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components, C _{GDS}	80	80

Table 3-7 Large area measurement structure for PMOS C_{gc} capacitance model

Structure	L[μm]	W[μm]
Gate capacitance components	.5	100
	2	100
	1	100
	2	2.08
	.5	2.08
	2	1.72
	.5	1.72
	2	1.36
	.5	1.36
	2	1
	.5	1
	2	.64
	.5	.64
	2	.35
	.5	.35

3.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized in Table 2-8 below.

Table 3-8 Measurement conditions for PMOS capacitance model

V_{BIAS} [V]	From -5.5 to 5.5 V in 0.1 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

3.4.3 T_{OX} extraction

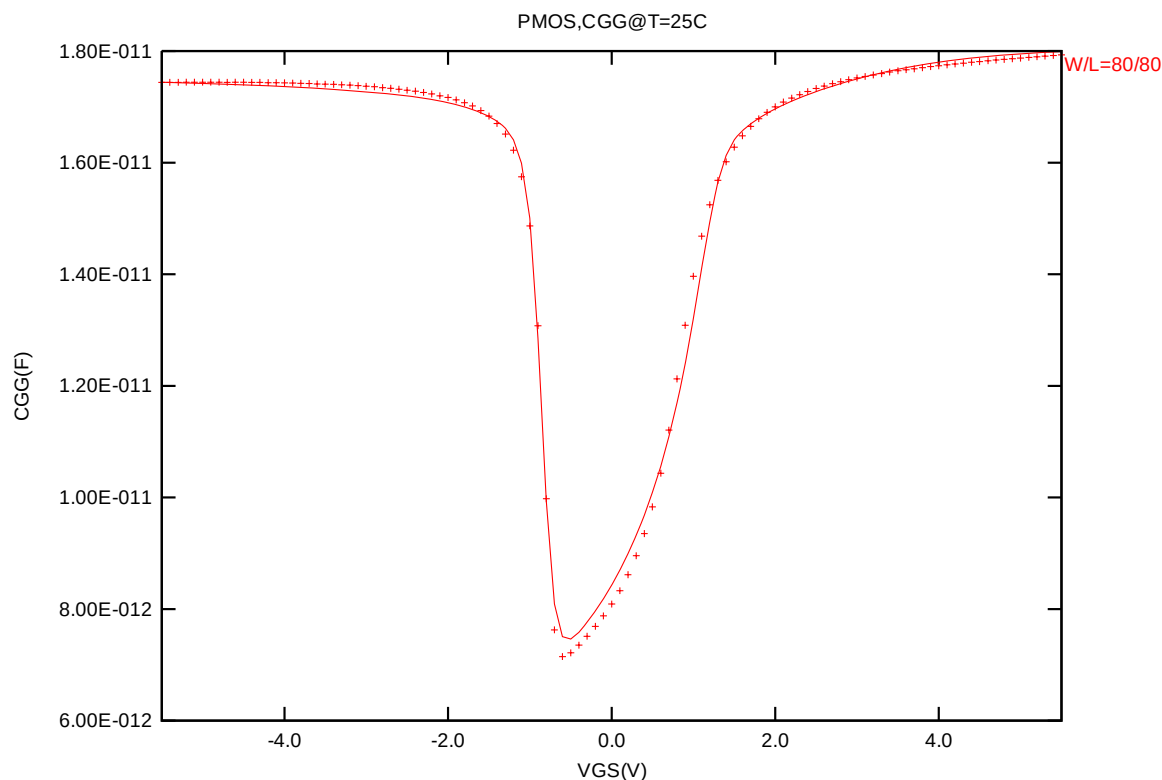
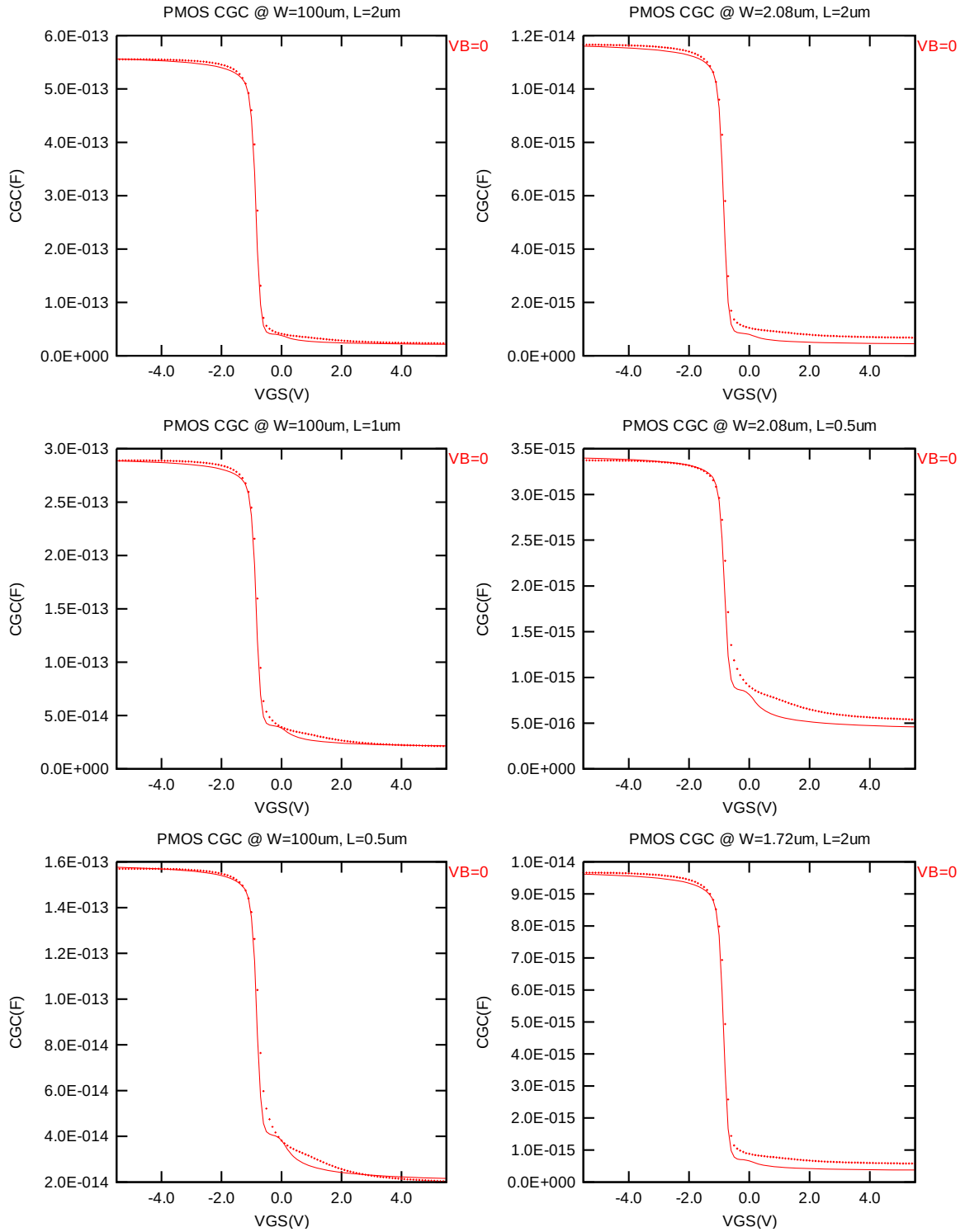
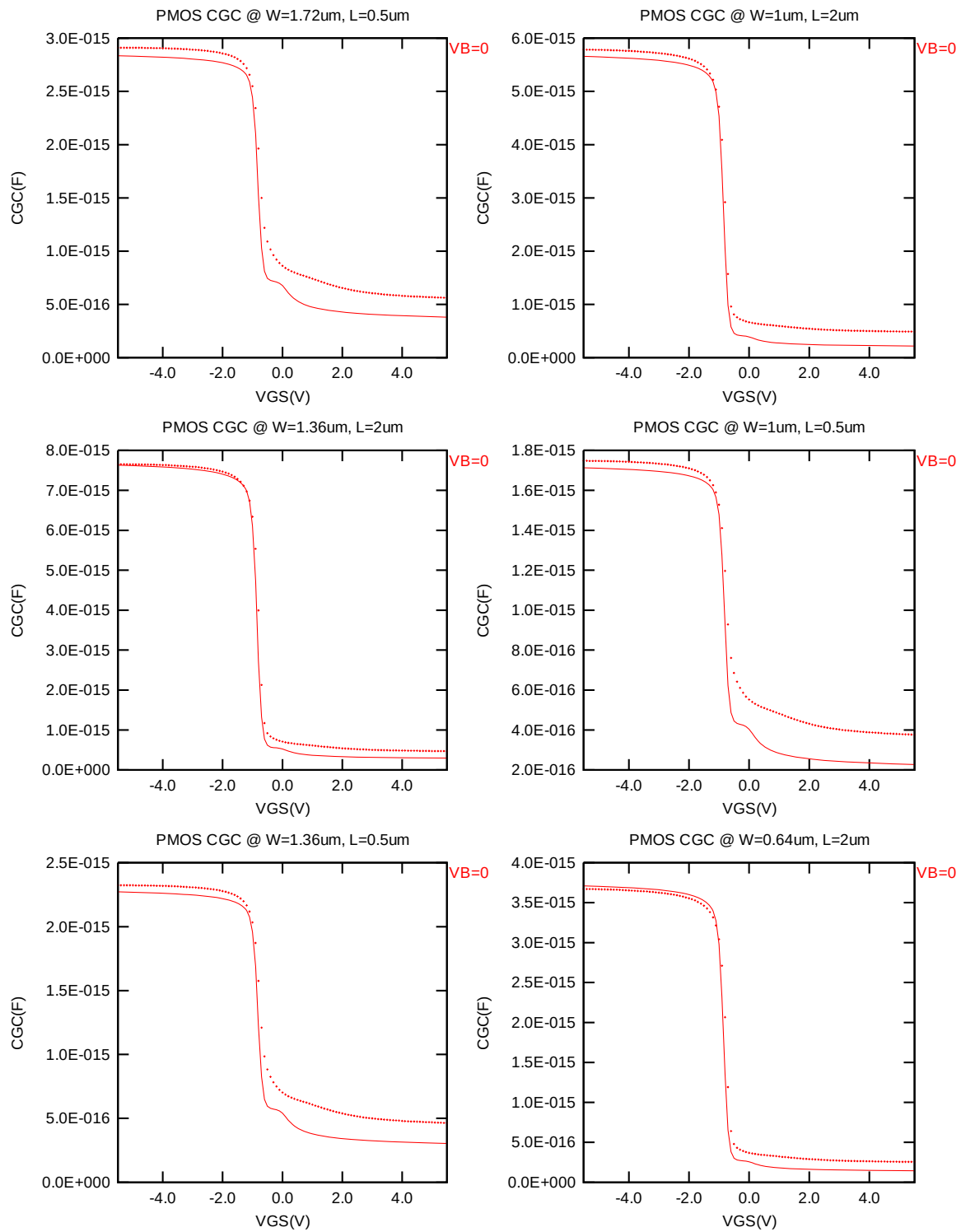


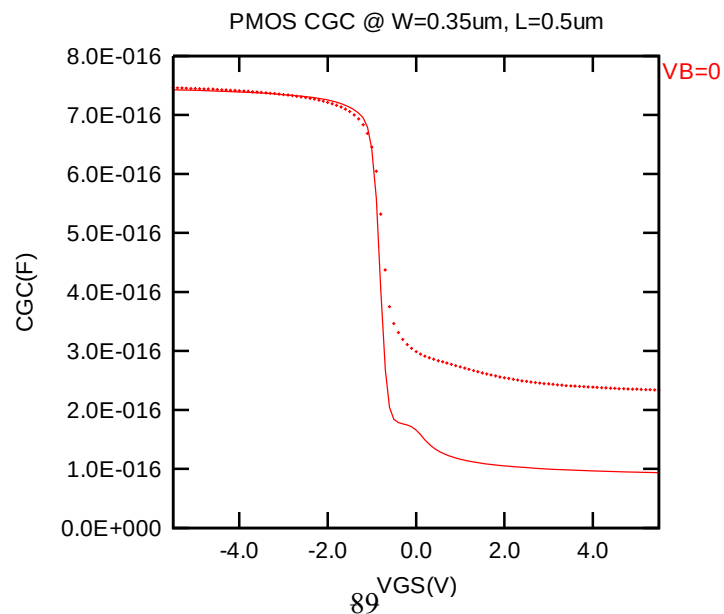
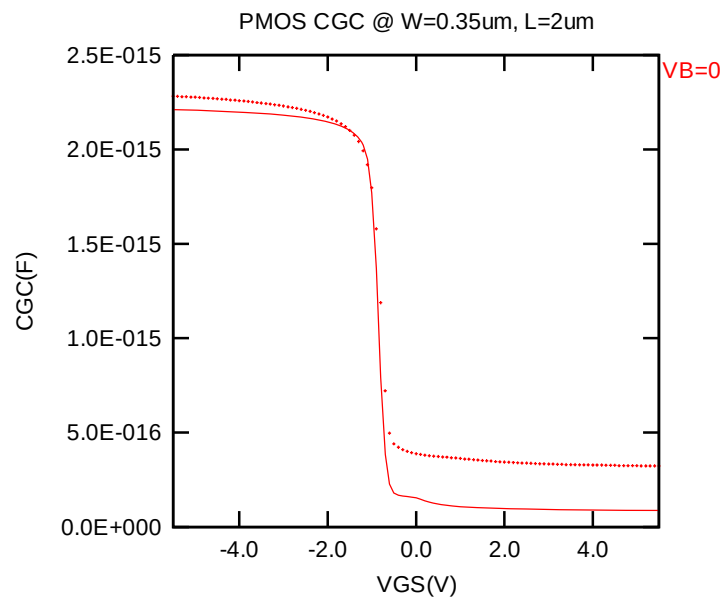
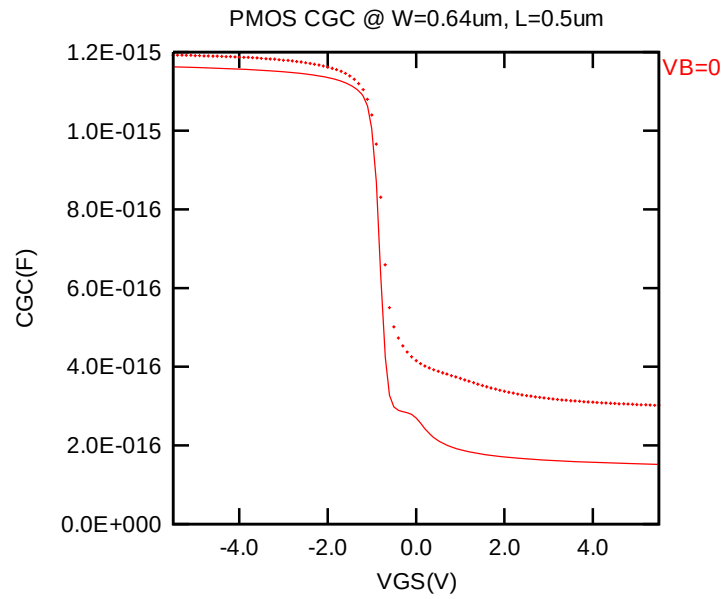
Figure 3-25 Measured and simulated C_{GG} curves

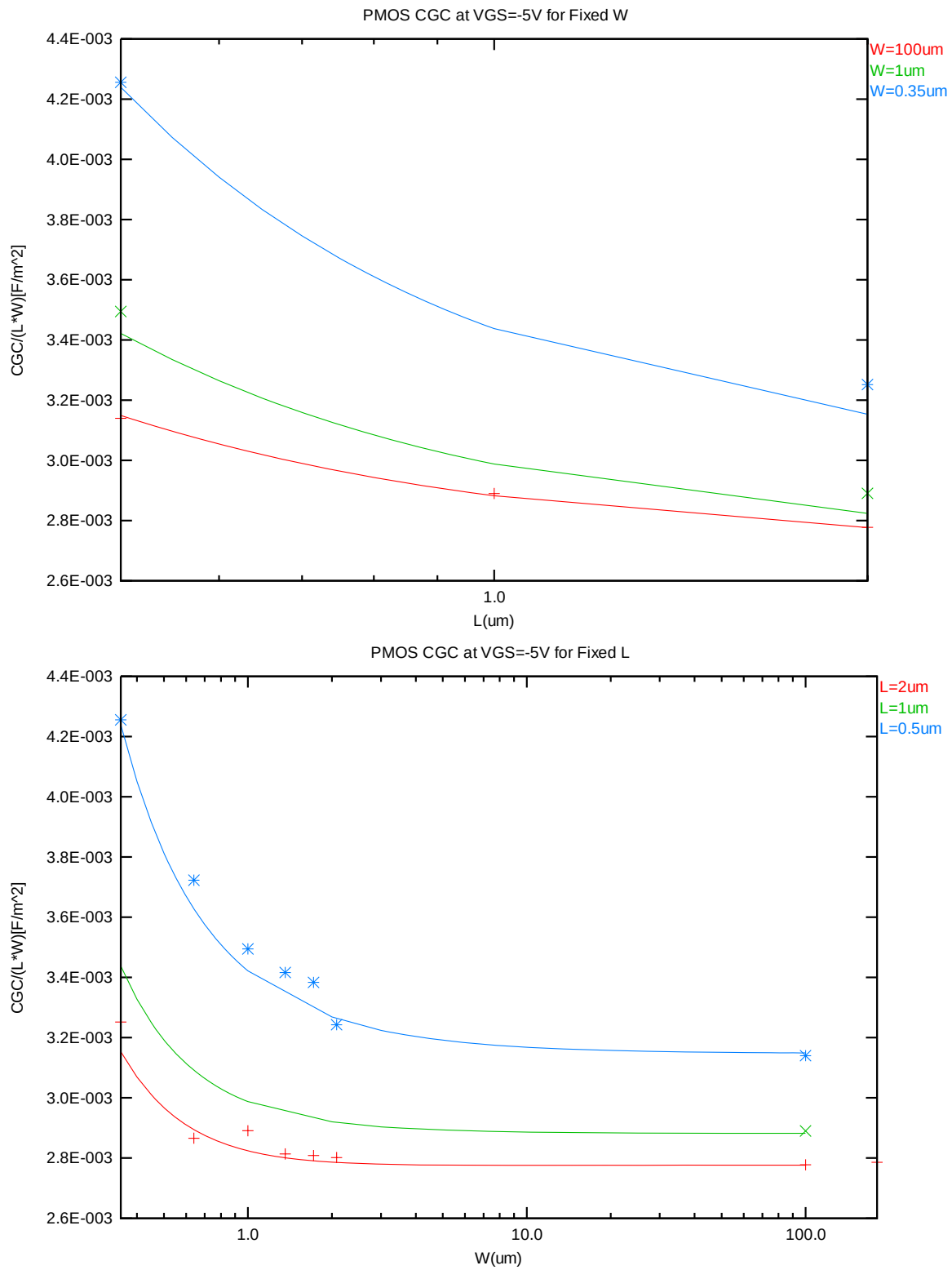
3.4.4 Width and length AC model accuracy

Width and length AC model accuracy









3.5 Parasitic Source / Drain Junction Capacitance Model

3.5.1 Test devices

The following test structures were used for parameter extraction.

Table 3-9 Junction capacitor structures for PMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	5.00E-08	2.00E-03	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.76E-8	1.82E-2	Poly finger structure

3.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 3-10 Junction capacitor measurement conditions for PMOS

V _{BIAS} [V]	VH: nwell & VL: p+ From 0 to 5.5 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-40,25,125

3.5.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from silicon data.

Table 3-11 Extracted parameters from junction capacitor measurement for PMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	Unit	Value	Parameter	Unit	Value	Parameter	Unit	Value
CJ	F/m ²	9.60E-04	CJSW	F/m	7.35E-11	CJSWG	F/m	1.96E-10
PB	V	6.69E-01	PBSW	V	8.11E-01	PBSWG	V	9.14E-01
MJ	--	3.00E-01	MJSW	--	5.78E-02	MJSWG	--	5.67E-01
TCJ	1/K	1.05E-03	TCJSW	1/K	1.10E-04	TCJSWG	1/K	1.54E-03
TPB	V/K	1.55E-03	TPBSW	V/K	1.03E-03	TPBSWG	V/K	1.74E-03

CJ, CJSW vs. Vbias plots

The following plots show the variation of bottom area junction and STI bounded sidewall at different bias voltages and temperatures.

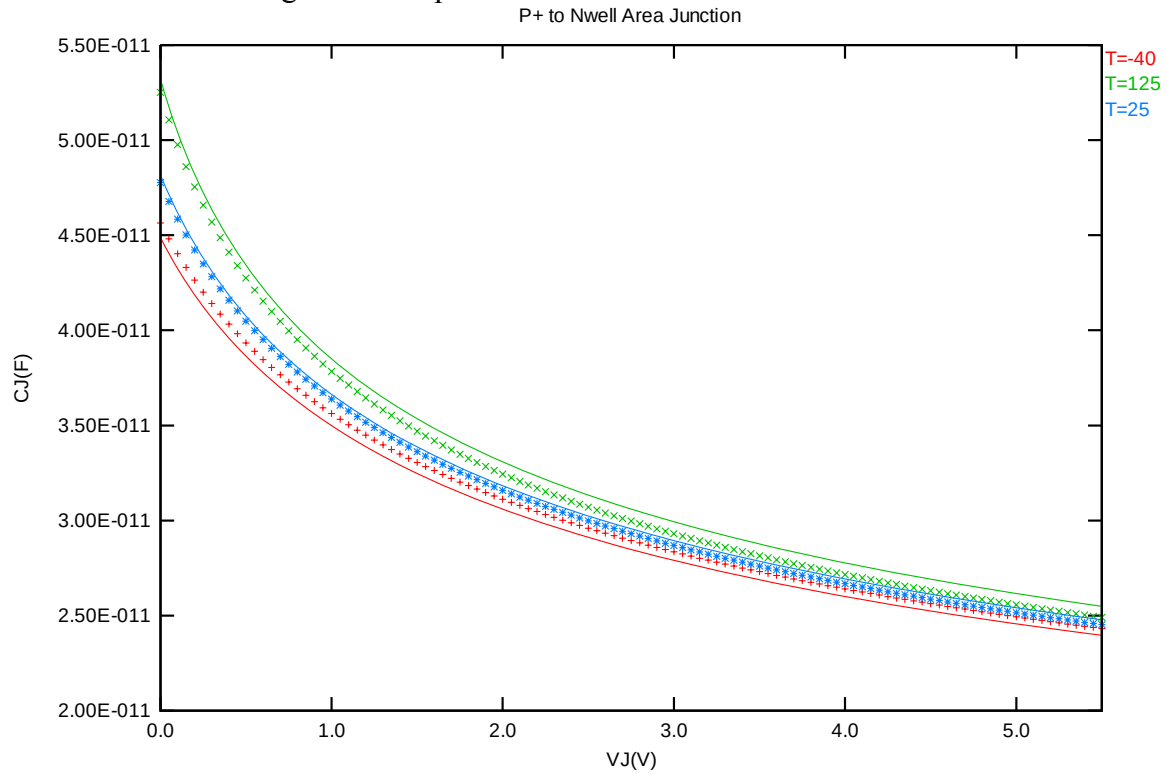


Figure 3-27 Bottom area junction capacitance at -40 C, 25 C and 125 C for PMOS

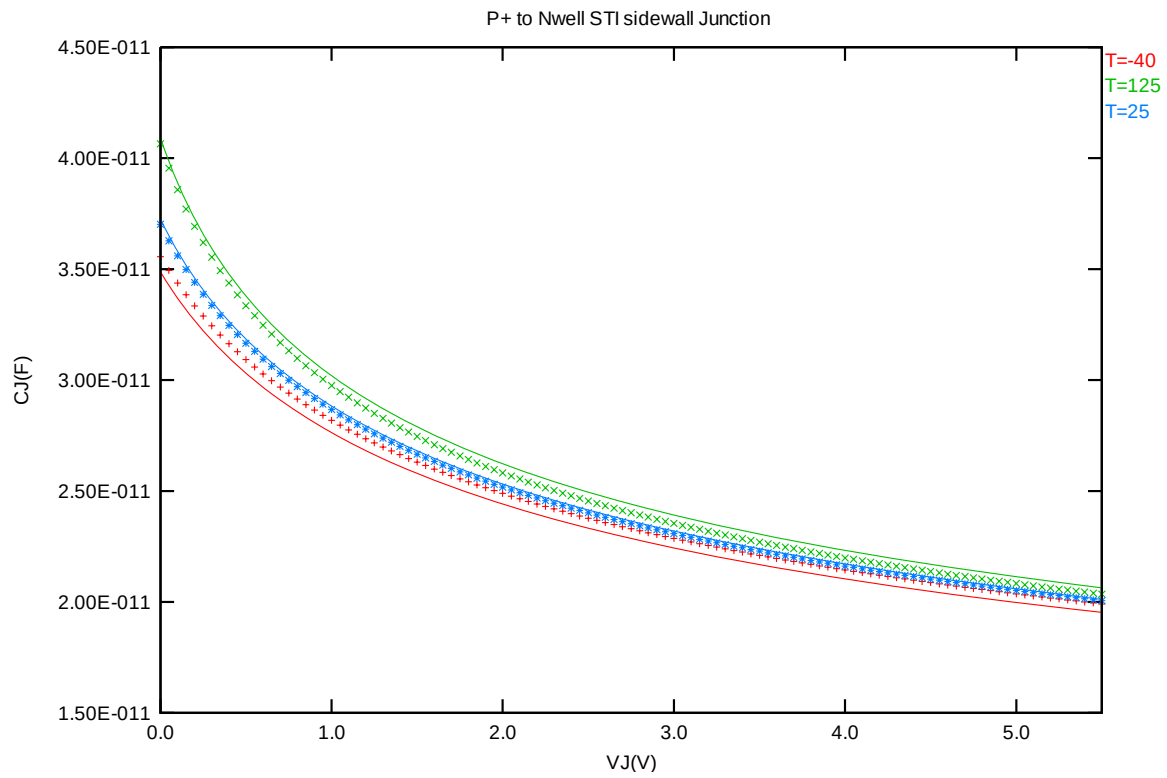


Figure 3-28 STI bounded sidewall junction capacitance at -40 C, 25 C and 125 C for PMOS

4 Worst Case Model

4.1 Parameter Variations

Table 4-1 Corner parameters

	n_50_rvtl110ae	TT	SS	FF	SNFP	FNSP
1	tox	1.19E-08	+3.57E-10	-3.57E-10	+0.00E+00	+0.00E+00
2	xl	-7.00E-10	+1.50E-08	-1.50E-08	+8.00E-09	-8.00E-09
3	xw	-1.00E-08	-1.98E-08	+1.21E-08	-8.36E-09	+6.61E-09
4	wwl	6.07E-21	-2.78E-22	-4.81E-22	+4.23E-22	-5.43E-22
5	vth0	7.46E-01	+2.46E-02	-2.68E-02	+1.31E-02	-1.36E-02
6	pvth0	-2.55E-15	-6.23E-16	+3.36E-16	-6.62E-16	+5.88E-16
7	k3	7.24E-01	+1.05E+00	-1.21E+00	+5.57E-01	-5.79E-01
8	nlx	1.43E-07	+5.44E-08	-4.77E-08	+2.44E-08	-2.26E-08
9	leta0	-7.45E-09	+1.18E-09	-1.25E-09	+5.95E-13	-1.19E-09
10	peta0	-5.23E-16	-5.73E-16	+3.78E-16	+1.16E-16	+2.38E-16
11	u0	3.46E-02	-4.51E-03	+4.30E-03	-2.42E-03	+2.39E-03
12	lrds	5.72E-05	+5.87E-05	-2.80E-05	+2.14E-05	-1.47E-05
13	prds	-8.48E-12	+0.00E+00	+0.00E+00	+0.00E+00	+0.00E+00
14	ags	1.73E-01	+2.59E-02	-2.19E-02	+1.22E-02	-1.12E-02
15	wb0	9.57E-15	-5.42E-15	+4.52E-15	-2.59E-15	+2.35E-15
16	lvsat	-3.59E-03	-8.28E-04	+7.76E-04	-7.49E-04	+7.68E-04
17	pvsat	1.41E-09	-5.51E-10	+2.87E-10	-1.33E-10	+6.53E-11
18	cgso	4.30E-11	-10.0%	+10.0%	+0.0%	+0.0%
19	cgdo	4.30E-11	-10.0%	+10.0%	+0.0%	+0.0%
20	cgdl	1.00E-10	-10.0%	+10.0%	+0.0%	+0.0%

21	cgs1	1.00E-10	-10.0%	+10.0%	+0.0%	+0.0%
22	cj	7.64E-04	+10.0%	-10.0%	+10.0%	-10.0%
23	cjsw	9.64E-11	+10.0%	-10.0%	+10.0%	-10.0%
24	cjswg	1.76E-10	+10.0%	-10.0%	+10.0%	-10.0%
25	jtsswgs	2.35E-09	-6.05E-08	+3.12E-07	-5.31E-08	+3.09E-07
26	jtsswgd	2.35E-09	-6.05E-08	+3.12E-07	-5.31E-08	+3.09E-07

	p_50_rvtl110ae	TT	SS	FF	SNFP	FNSP
1	tox	1.24E-08	+3.72E-10	-3.72E-10	+0.00E+00	+0.00E+00
2	xl	2.40E-08	+2.00E-08	-2.00E-08	-1.00E-08	+1.00E-08
3	xw	-1.00E-08	-7.85E-10	+5.33E-10	+2.25E-10	-4.27E-10
4	vth0	-8.19E-01	-4.11E-02	+4.19E-02	+2.12E-02	-2.08E-02
5	lvth0	7.28E-09	-1.40E-08	+1.27E-08	+6.32E-09	-6.81E-09
6	pvth0	1.83E-15	-1.19E-15	+2.36E-15	+1.10E-15	-8.52E-16
7	k3	1.05E+00	+2.79E-01	-3.98E-01	-1.99E-01	+1.93E-01
8	u0	1.17E-02	-8.20E-04	+7.68E-04	+4.72E-04	-4.80E-04
9	pu0	-4.54E-16	-2.04E-16	+1.75E-16	+8.86E-17	-9.43E-17
10	wu0	0.00E+00	-1.85E-10	+1.72E-10	+8.87E-11	-8.96E-11
11	vsat	6.49E+04	-3.53E+03	+1.44E+03	+1.82E+03	-2.32E+03
12	pvsat	-4.21E-09	-8.16E-10	+6.15E-10	+3.07E-10	-3.42E-10
13	ags	1.00E-01	+6.03E-03	-6.12E-03	-3.15E-03	+3.08E-03
14	wags	4.35E-09	+2.24E-10	-3.28E-10	-1.44E-10	+1.34E-10
15	rdsw	2.49E+03	+4.79E+02	-2.43E+02	-1.44E+02	+1.99E+02
16	eta0	3.86E-11	+1.46E-05	+0.00E+00	+0.00E+00	+3.04E-04
17	cgso	2.84E-11	-10.0%	+10.0%	+0.0%	+0.0%
18	cgdo	2.84E-11	-10.0%	+10.0%	+0.0%	+0.0%
19	cgs1	1.18E-10	-10.0%	+10.0%	+0.0%	+0.0%

20	cgdl	1.18E-10	-10.0%	+10.0%	+0.0%	+0.0%
21	cj	9.60E-04	+10.0%	-10.0%	-10.0%	+10.0%
22	cjsw	7.35E-11	+10.0%	-10.0%	-10.0%	+10.0%
23	cjswg	1.96E-10	+10.0%	-10.0%	-10.0%	+10.0%
24	jtsswgs	5.24E-07	-4.90E-07	+7.35E-06	+7.35E-06	-4.90E-07
25	jtsswgd	5.24E-07	-4.90E-07	+7.35E-06	+7.35E-06	-4.90E-07

4.2 Wafer Data vs. Corner Plots

Threshold voltage in linear region V_{TLIN}

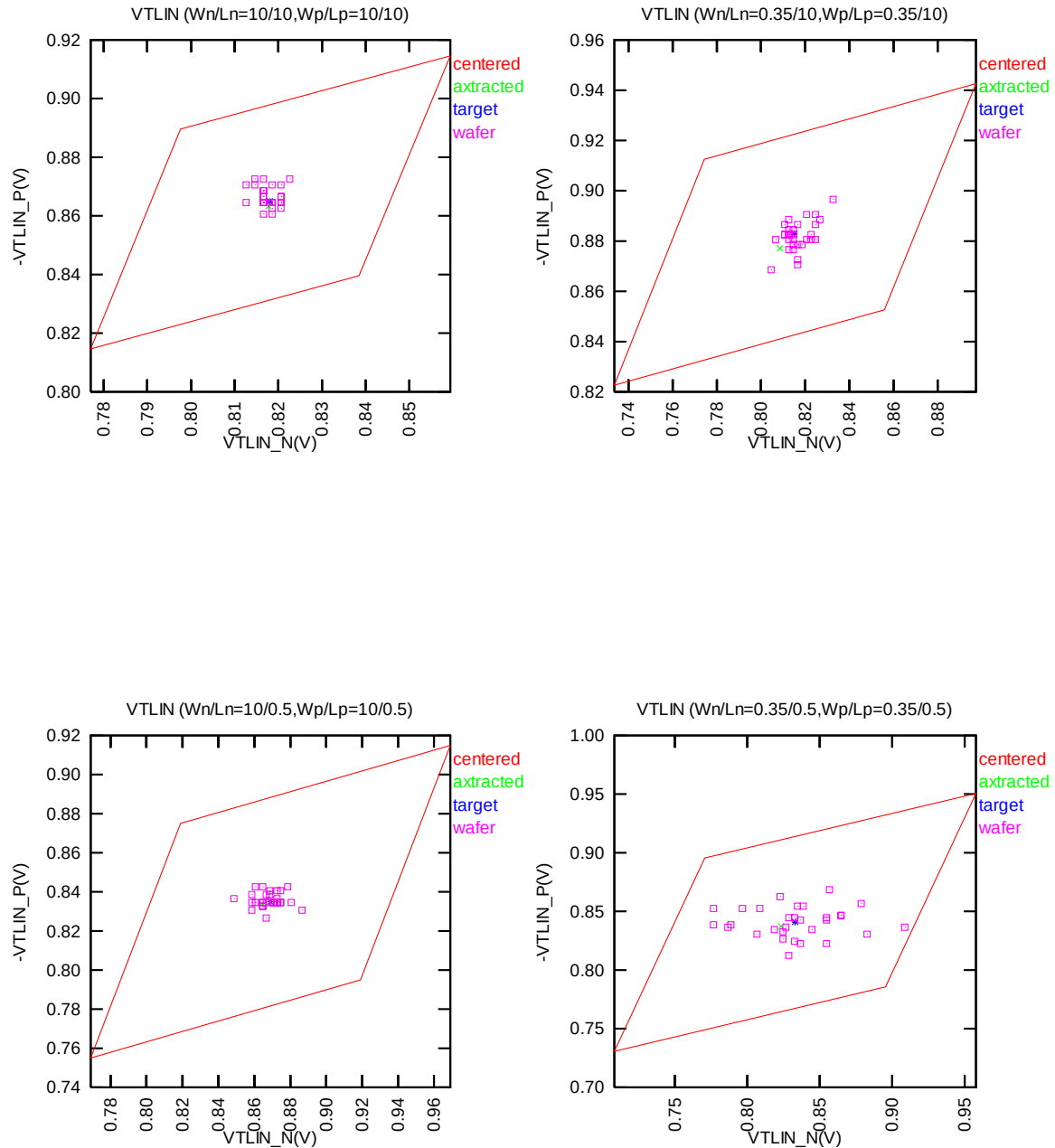


Figure 4-1 Corner plots of threshold voltage at low drain bias

Threshold voltage in saturation region V_{TSAT}

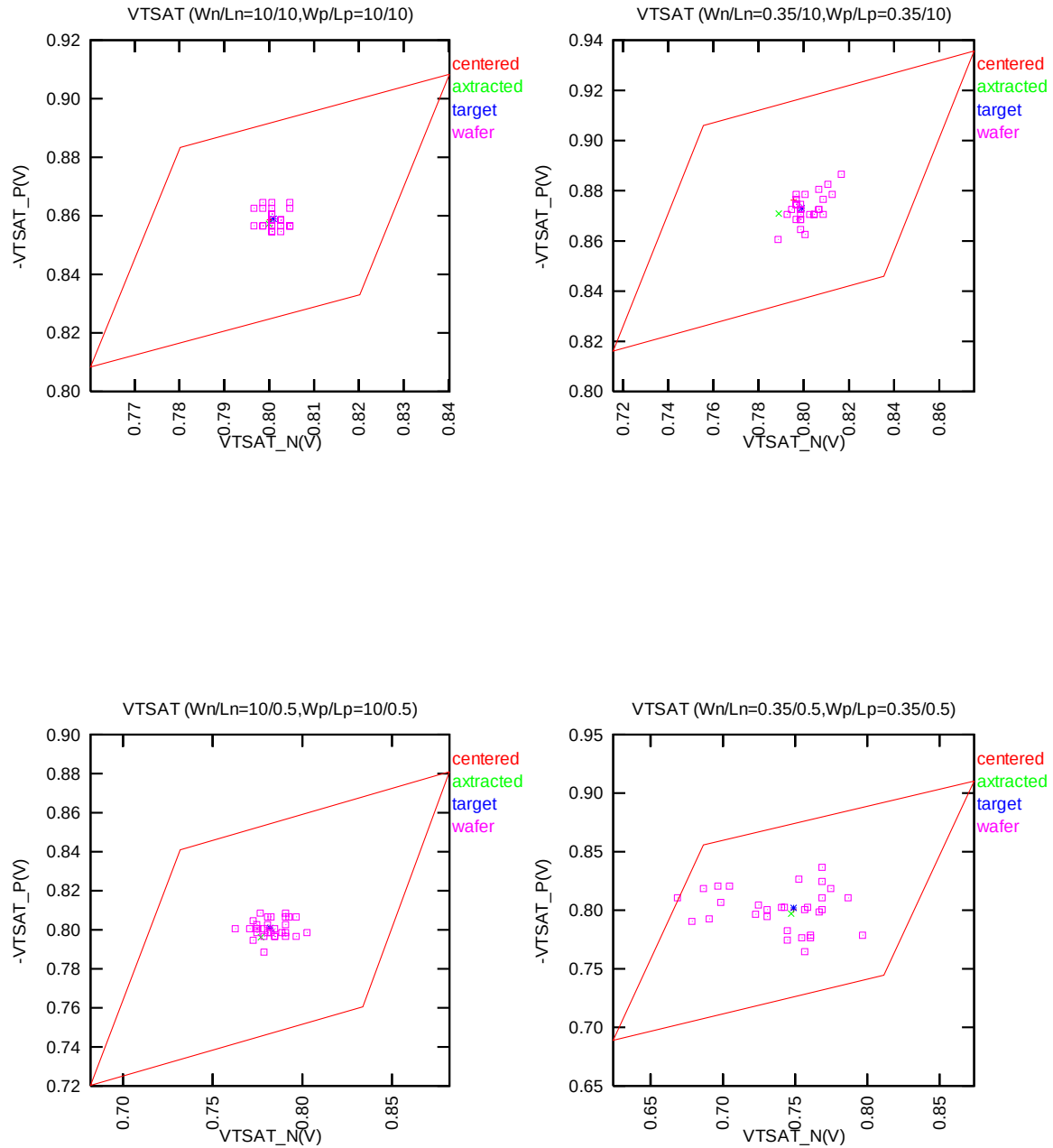


Figure 4-2 Corner plots of threshold voltage at high drain bias

Saturation current I_{ON}

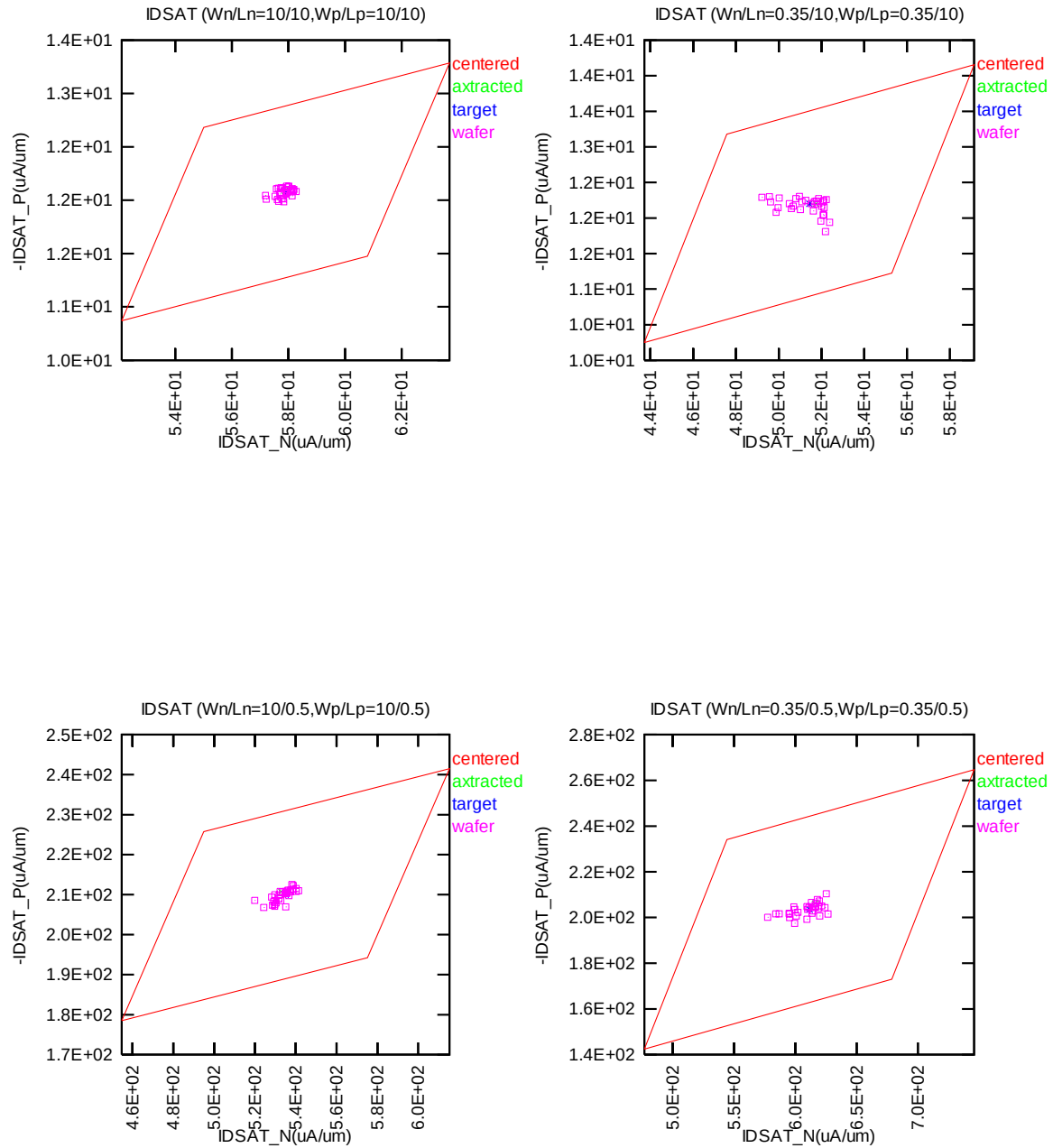


Figure 4-3 Corner plots of on-current

Off state current I_{OFF}

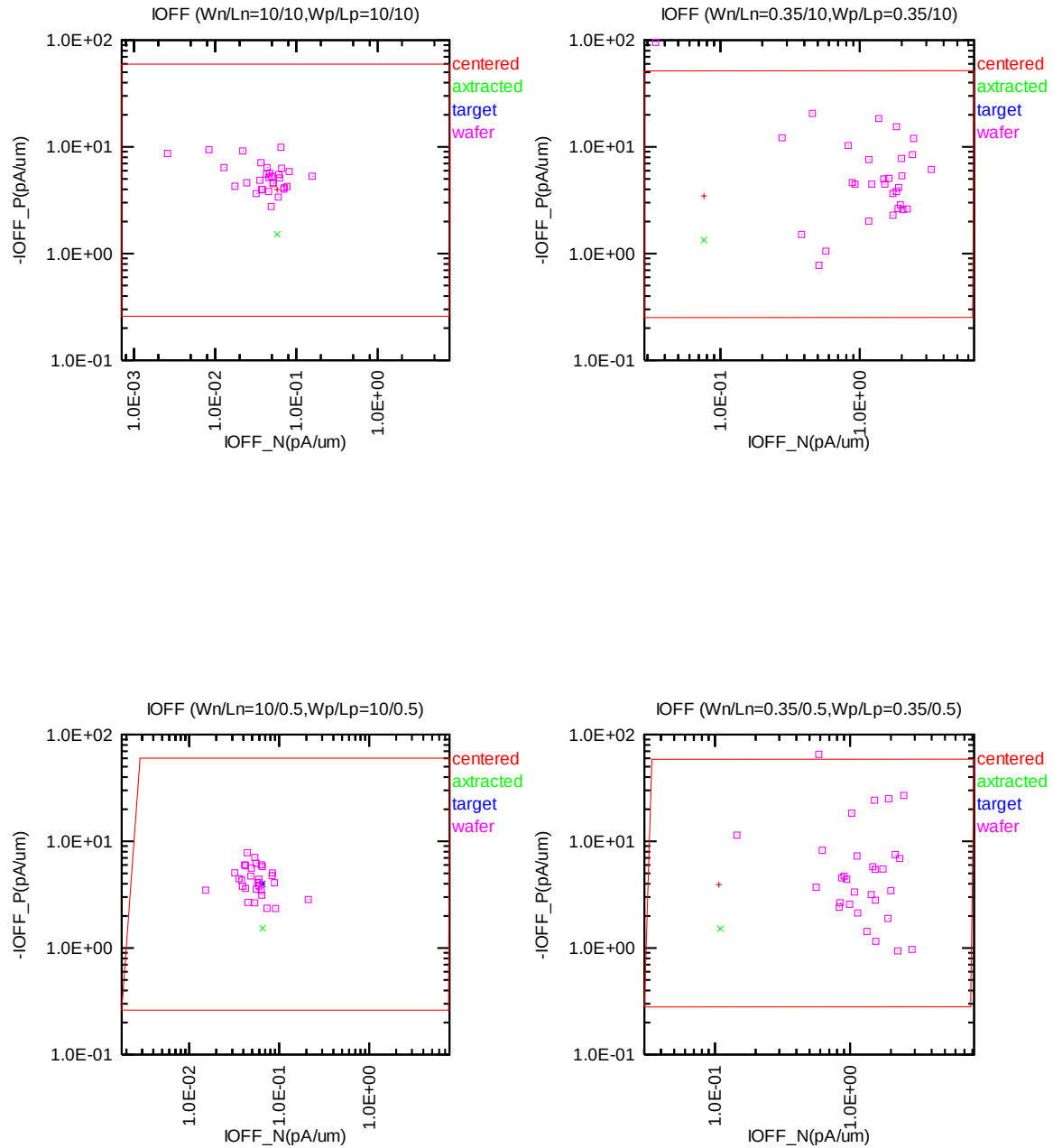


Figure 4-4 Corner plots of off-current

Note: Due to the GIDL effect, the corner graph of off-current is slightly strange.

4.3 Device Performance Reference Tables

Note: The model simulation results in the following tables have already reflected silicon data that is sizing channel length back for 12nm. It means: The real silicon size of W/L=9µm/0.5µm would be the same as the model simulation results in size of W/L=9µm/0.488µm as below table (The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 4-2 NMOS ID current at $V_{GS}=5\text{ V}$, $V_{DS}=2.5\text{ V}$ and $V_{GS}=5\text{ V}$, $V_{DS}=5\text{ V}$.

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
IDSAT (µA)	VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	9/9	131.04	774.67	100.31	574.22	74.26	400.42
SS	9/9	109.16	697.85	83.73	516.40	62.22	359.82
FF	9/9	154.45	850.75	118.01	631.97	87.07	441.39
SNFP	9/9	120.64	735.64	92.42	545.35	68.51	380.62
FNSP	9/9	141.65	813.60	108.36	603.06	80.11	420.23
TT	9/0.488	1728.40	5377.80	1541.10	4813.40	1324.50	4134.30
SS	9/0.488	1373.10	4597.50	1213.40	4089.30	1033.40	3478.40
FF	9/0.488	2112.50	6150.60	1900.10	5537.20	1648.40	4798.90
SNFP	9/0.488	1548.90	4989.50	1374.80	4451.50	1176.20	3806.10
FNSP	9/0.488	1914.20	5764.60	1714.30	5175.20	1480.20	4464.20
TT	0.35/9	4.47	25.87	3.49	19.95	2.66	14.79
SS	0.35/9	3.44	21.98	2.70	16.97	2.07	12.61
FF	0.35/9	5.62	29.72	4.38	22.94	3.31	16.99
SNFP	0.35/9	3.96	23.90	3.10	18.46	2.37	13.72
FNSP	0.35/9	5.00	27.83	3.90	21.45	2.96	15.85
TT	0.35/0.488	71.74	228.96	63.59	213.97	51.54	183.23
SS	0.35/0.488	51.63	181.34	44.57	166.88	34.93	139.05
FF	0.35/0.488	94.13	275.89	85.35	261.06	71.29	228.73
SNFP	0.35/0.488	61.19	205.29	53.45	190.43	42.42	160.49
FNSP	0.35/0.488	82.82	252.53	74.36	237.52	61.38	206.19

Table 4-3 NMOS ID current density at $V_{GS}=5\text{ V}$, $V_{DS}=2.5\text{ V}$ and $V_{GS}=5\text{ V}$, $V_{DS}=5\text{ V}$.

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
IDSAT ($\mu\text{A}/\mu\text{m}$)	VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	9/9	14.56	86.07	11.15	63.80	8.25	44.49
SS	9/9	12.13	77.54	9.30	57.38	6.91	39.98
FF	9/9	17.16	94.53	13.11	70.22	9.67	49.04
SNFP	9/9	13.40	81.74	10.27	60.59	7.61	42.29
FNSP	9/9	15.74	90.40	12.04	67.01	8.90	46.69
TT	9/0.488	192.04	597.53	171.23	534.82	147.17	459.37
SS	9/0.488	152.57	510.83	134.82	454.37	114.82	386.49
FF	9/0.488	234.72	683.40	211.12	615.24	183.16	533.21
SNFP	9/0.488	172.10	554.39	152.76	494.61	130.69	422.90
FNSP	9/0.488	212.69	640.51	190.48	575.02	164.47	496.02
TT	0.35/9	12.76	73.90	9.98	57.01	7.60	42.24
SS	0.35/9	9.83	62.81	7.72	48.49	5.92	36.03
FF	0.35/9	16.05	84.91	12.50	65.55	9.46	48.54
SNFP	0.35/9	11.31	68.27	8.86	52.75	6.78	39.21
FNSP	0.35/9	14.28	79.53	11.14	61.28	8.45	45.28
TT	0.35/0.488	204.96	654.17	181.70	611.34	147.25	523.51
SS	0.35/0.488	147.51	518.11	127.35	476.80	99.81	397.29
FF	0.35/0.488	268.93	788.26	243.84	745.89	203.69	653.51
SNFP	0.35/0.488	174.82	586.54	152.71	544.09	121.19	458.54
FNSP	0.35/0.488	236.63	721.51	212.45	678.63	175.36	589.11

Table 4-4 NMOS V_{TLIN} and V_{TSAT} .

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
VTH (V)	VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	9/9	0.893	0.879	0.819	0.801	0.709	0.684
SS	9/9	0.933	0.918	0.860	0.841	0.754	0.727
FF	9/9	0.854	0.840	0.777	0.760	0.664	0.641
SNFP	9/9	0.913	0.898	0.839	0.821	0.731	0.705
FNSP	9/9	0.874	0.860	0.798	0.780	0.687	0.663
TT	9/0.488	0.940	0.858	0.869	0.782	0.768	0.671
SS	9/0.488	1.037	0.956	0.969	0.882	0.873	0.774
FF	9/0.488	0.842	0.759	0.769	0.682	0.665	0.568
SNFP	9/0.488	0.989	0.909	0.919	0.834	0.820	0.724
FNSP	9/0.488	0.891	0.808	0.819	0.732	0.716	0.619
TT	0.35/9	0.887	0.872	0.815	0.795	0.708	0.681
SS	0.35/9	0.966	0.950	0.897	0.875	0.796	0.764
FF	0.35/9	0.808	0.793	0.733	0.715	0.623	0.598
SNFP	0.35/9	0.927	0.911	0.855	0.835	0.752	0.722
FNSP	0.35/9	0.848	0.833	0.774	0.755	0.666	0.640
TT	0.35/0.488	0.891	0.809	0.833	0.749	0.752	0.662
SS	0.35/0.488	1.010	0.930	0.958	0.874	0.887	0.794
FF	0.35/0.488	0.770	0.688	0.708	0.624	0.621	0.532
SNFP	0.35/0.488	0.951	0.869	0.895	0.811	0.819	0.728
FNSP	0.35/0.488	0.831	0.749	0.771	0.687	0.685	0.596

Table 4-5 NMOS ID current at $V_{GS}=0$ and $V_{DS}=0.1$ V or $V_{DS}=5$ V (off current).

NMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
IOFF (A)	VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	9/9	1.61E-15	1.03E-12	4.27E-15	5.26E-13	3.65E-12	4.55E-12
SS	9/9	2.22E-16	1.07E-14	1.50E-15	1.07E-14	2.62E-12	3.00E-12
FF	9/9	1.90E-13	1.37E-10	1.93E-13	6.88E-11	5.75E-12	4.26E-11
SNFP	9/9	1.67E-16	7.11E-15	1.89E-15	7.11E-15	2.95E-12	3.41E-12
FNSP	9/9	1.87E-13	1.35E-10	1.89E-13	6.79E-11	4.83E-12	4.10E-11
TT	9/0.488	1.61E-15	1.04E-12	1.13E-14	5.83E-13	1.22E-11	4.72E-11
SS	9/0.488	2.22E-16	1.07E-14	1.44E-15	1.42E-14	3.44E-12	9.82E-12
FF	9/0.488	1.92E-13	1.38E-10	2.76E-13	7.02E-11	6.07E-11	2.95E-10
SNFP	9/0.488	1.67E-16	7.11E-15	3.22E-15	2.84E-14	5.74E-12	1.89E-11
FNSP	9/0.488	1.90E-13	1.37E-10	2.19E-13	6.90E-11	2.85E-11	1.52E-10
TT	0.35/9	2.46E-16	4.32E-14	3.78E-16	2.68E-14	2.05E-13	2.55E-13
SS	0.35/9	2.01E-16	9.99E-15	2.22E-16	9.99E-15	1.27E-13	1.50E-13
FF	0.35/9	6.55E-15	4.58E-12	7.21E-15	2.31E-12	5.07E-13	1.82E-12
SNFP	0.35/9	2.01E-16	9.99E-15	2.53E-16	9.99E-15	1.51E-13	1.79E-13
FNSP	0.35/9	6.38E-15	4.47E-12	6.65E-15	2.25E-12	3.22E-13	1.56E-12
TT	0.35/0.488	2.57E-16	4.86E-14	1.36E-15	3.74E-14	7.72E-13	3.17E-12
SS	0.35/0.488	2.01E-16	9.99E-15	2.84E-16	1.05E-14	1.85E-13	4.95E-13
FF	0.35/0.488	7.82E-15	5.48E-12	2.50E-14	2.89E-12	5.84E-12	2.75E-11
SNFP	0.35/0.488	2.01E-16	9.99E-15	4.72E-16	1.20E-14	3.20E-13	1.10E-12
FNSP	0.35/0.488	7.59E-15	5.33E-12	1.22E-14	2.72E-12	2.18E-12	1.09E-11

gmindc=1e-15

Table 4-6 PMOS ID current at $V_{GS}=-5$ V, $V_{DS}=-2.5$ V and $V_{GS}=-5$ V, $V_{DS}=-5$ V.

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-IDSAT (μ A)	-VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	9/9	27.50	155.46	22.45	120.42	19.15	97.11
SS	9/9	23.60	139.78	19.32	108.30	16.53	87.32
FF	9/9	31.71	171.09	25.84	132.54	21.98	106.94
SNFP	9/9	29.49	163.35	24.03	126.48	20.44	101.92
FNSP	9/9	25.57	147.56	20.93	114.36	17.91	92.31
TT	9/0.488	526.85	2180.90	457.27	1887.00	406.41	1645.80
SS	9/0.488	421.74	1858.00	366.21	1603.00	326.44	1394.70
FF	9/0.488	643.66	2492.80	560.26	2171.00	497.85	1903.90
SNFP	9/0.488	582.83	2338.70	505.90	2029.00	448.66	1772.50
FNSP	9/0.488	473.61	2020.20	411.37	1745.00	366.65	1520.60
TT	0.35/9	0.97	6.09	0.79	4.73	0.68	3.84
SS	0.35/9	0.77	5.11	0.63	3.97	0.54	3.22
FF	0.35/9	1.19	7.07	0.97	5.50	0.83	4.46
SNFP	0.35/9	1.07	6.59	0.88	5.11	0.75	4.14
FNSP	0.35/9	0.87	5.60	0.71	4.35	0.61	3.53
TT	0.35/0.488	18.99	82.60	16.15	71.24	14.02	62.17
SS	0.35/0.488	12.21	59.07	10.26	49.87	8.87	42.79
FF	0.35/0.488	26.80	105.22	23.09	92.61	20.16	82.18
SNFP	0.35/0.488	22.73	94.06	19.42	81.93	16.87	72.01
FNSP	0.35/0.488	15.49	70.91	13.11	60.55	11.38	52.47

Table 4-7 PMOS ID current density at $V_{GS}=-5$ V, $V_{DS}=-2.5$ V and $V_{GS}=-5$ V, $V_{DS}=-5$ V.

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-IDSAT (μ A/ μ m)	-VDS	2.5V	5V	2.5V	5V	2.5V	5V
TT	9/9	3.06	17.27	2.49	13.38	2.13	10.79
SS	9/9	2.62	15.53	2.15	12.03	1.84	9.70
FF	9/9	3.52	19.01	2.87	14.73	2.44	11.88
SNFP	9/9	3.28	18.15	2.67	14.05	2.27	11.32
FNSP	9/9	2.84	16.40	2.33	12.71	1.99	10.26
TT	9/0.488	58.54	242.32	50.81	209.67	45.16	182.87
SS	9/0.488	46.86	206.44	40.69	178.11	36.27	154.97
FF	9/0.488	71.52	276.98	62.25	241.22	55.32	211.54
SNFP	9/0.488	64.76	259.86	56.21	225.44	49.85	196.94
FNSP	9/0.488	52.62	224.47	45.71	193.89	40.74	168.96
TT	0.35/9	2.76	17.41	2.26	13.52	1.94	10.97
SS	0.35/9	2.20	14.61	1.81	11.35	1.56	9.20
FF	0.35/9	3.39	20.21	2.77	15.70	2.37	12.74
SNFP	0.35/9	3.06	18.82	2.50	14.61	2.14	11.84
FNSP	0.35/9	2.48	16.00	2.04	12.43	1.75	10.09
TT	0.35/0.488	54.25	235.99	46.13	203.54	40.05	177.63
SS	0.35/0.488	34.87	168.76	29.32	142.48	25.34	122.26
FF	0.35/0.488	76.58	300.63	65.97	264.61	57.61	234.80
SNFP	0.35/0.488	64.93	268.73	55.49	234.07	48.20	205.75
FNSP	0.35/0.488	44.27	202.61	37.47	173.01	32.53	149.90

Table 4-8 PMOS V_{TLIN} and V_{TSAT} .

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-VTH (V)	-VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	9/9	0.955	0.950	0.865	0.859	0.732	0.723
SS	9/9	1.004	0.999	0.915	0.909	0.784	0.775
FF	9/9	0.906	0.901	0.814	0.808	0.680	0.666
SNFP	9/9	0.931	0.926	0.840	0.833	0.706	0.692
FNSP	9/9	0.980	0.975	0.890	0.883	0.757	0.749
TT	9/0.488	0.933	0.899	0.835	0.801	0.691	0.654
SS	9/0.488	1.012	0.978	0.915	0.881	0.774	0.736
FF	9/0.488	0.855	0.820	0.755	0.720	0.608	0.570
SNFP	9/0.488	0.894	0.859	0.795	0.760	0.649	0.612
FNSP	9/0.488	0.973	0.939	0.875	0.841	0.732	0.695
TT	0.35/9	0.971	0.966	0.882	0.876	0.752	0.743
SS	0.35/9	1.030	1.024	0.943	0.936	0.815	0.805
FF	0.35/9	0.913	0.908	0.822	0.816	0.690	0.676
SNFP	0.35/9	0.942	0.937	0.852	0.846	0.721	0.707
FNSP	0.35/9	1.001	0.995	0.912	0.906	0.783	0.774
TT	0.35/0.488	0.935	0.894	0.841	0.801	0.704	0.660
SS	0.35/0.488	1.039	1.000	0.951	0.910	0.823	0.776
FF	0.35/0.488	0.828	0.786	0.731	0.689	0.588	0.544
SNFP	0.35/0.488	0.881	0.840	0.786	0.744	0.646	0.602
FNSP	0.35/0.488	0.988	0.948	0.896	0.856	0.762	0.718

Table 4-9 PMOS ID current at $V_{GS}=0$ and $V_{DS}=-0.1$ V or $V_{DS}=-5$ V (off current).

PMOS	Temp. (°C)	-40 °C		25 °C		125 °C	
-IOFF (A)	-VDS	0.1V	5V	0.1V	5V	0.1V	5V
TT	9/9	1.02E-14	2.64E-12	2.05E-13	3.56E-11	3.95E-12	3.57E-10
SS	9/9	8.88E-16	1.81E-13	1.35E-14	2.32E-12	9.81E-13	2.39E-11
FF	9/9	1.51E-13	3.95E-11	3.08E-12	5.35E-10	4.51E-11	5.35E-09
SNFP	9/9	1.51E-13	3.95E-11	3.08E-12	5.35E-10	4.48E-11	5.35E-09
FNSP	9/9	8.88E-16	1.81E-13	1.38E-14	2.32E-12	1.07E-12	2.40E-11
TT	9/0.488	1.03E-14	2.66E-12	2.20E-13	3.60E-11	2.06E-11	3.88E-10
SS	9/0.488	8.88E-16	1.81E-13	1.63E-14	2.35E-12	5.74E-12	3.20E-11
FF	9/0.488	1.53E-13	3.99E-11	3.17E-12	5.41E-10	1.06E-10	5.50E-09
SNFP	9/0.488	1.52E-13	3.99E-11	3.14E-12	5.40E-10	7.87E-11	5.46E-09
FNSP	9/0.488	8.88E-16	1.81E-13	1.90E-14	2.35E-12	9.36E-12	3.80E-11
TT	0.35/9	5.38E-16	9.86E-14	7.13E-15	1.21E-12	1.50E-13	1.21E-11
SS	0.35/9	2.26E-16	1.55E-14	6.52E-16	8.82E-14	4.97E-14	8.34E-13
FF	0.35/9	5.30E-15	1.35E-12	1.04E-13	1.81E-11	1.54E-12	1.81E-10
SNFP	0.35/9	5.29E-15	1.34E-12	1.04E-13	1.81E-11	1.53E-12	1.81E-10
FNSP	0.35/9	2.22E-16	1.58E-14	6.52E-16	8.82E-14	5.25E-14	8.38E-13
TT	0.35/0.488	5.86E-16	1.10E-13	8.53E-15	1.37E-12	7.45E-13	1.47E-11
SS	0.35/0.488	2.26E-16	1.64E-14	7.67E-16	9.79E-14	1.57E-13	1.11E-12
FF	0.35/0.488	6.01E-15	1.53E-12	1.23E-13	2.06E-11	5.19E-12	2.12E-10
SNFP	0.35/0.488	5.99E-15	1.53E-12	1.20E-13	2.06E-11	3.22E-12	2.08E-10
FNSP	0.35/0.488	2.29E-16	1.67E-14	8.60E-16	9.84E-14	2.82E-13	1.34E-12

gmindc=1e-15

5 Shallow Trench Isolation (STI) or LOD Stress Effect

5.1 Simulation Results for NMOS at SA=SB=0.324 μm (SAREF=3.6 μm)

The simulation results for the NMOS characterization are summarized in the table below.

Table 5-1 Simulation results for NMOS

All parameters are given for $T=25\text{ }^{\circ}\text{C}$, $V_{\text{BS}}=0\text{ V}$ unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
-----------	-----------	------	------------------

Long channel device ($W/L=10\text{ }\mu\text{m} / 10\text{ }\mu\text{m}$)

V_{TLIN}	$V_{\text{DS}}=0.1\text{ V}$	V	0.819
I_{ON}	$V_{\text{DS}}=5\text{ V}$	$\mu\text{A}/\mu\text{m}$	57.57

Wide and short channel device ($W/L=10\text{ }\mu\text{m} / 0.5\text{ }\mu\text{m}$)

V_{TLIN}	$V_{\text{DS}}=0.1\text{ V}$	V	0.885
V_{TSAT}	$V_{\text{DS}}=5\text{ V}$	V	0.798
Body Effect V_{T} shift	$V_{\text{DS}}=5\text{ V}$ $V_{\text{BS}}=0 \sim -5\text{ V}$	V	0.589
DIBL	$V_{\text{DS}}=0.1\sim 5\text{ V}$	V	0.087

I_{ON}	$V_{\text{DS}}=V_{\text{GS}}=5\text{ V}$	$\mu\text{A}/\mu\text{m}$	497.33
I_{OFF}	$V_{\text{DS}}=5\text{ V}$, $V_{\text{GS}}=0\text{ V}$	$\text{A}/\mu\text{m}$	6.3E-14

Narrow and long channel device ($W/L=0.35\text{ }\mu\text{m} / 10\text{ }\mu\text{m}$)

V_{TLIN}	$V_{\text{DS}}=0.1\text{ V}$	V	0.816
V_{TSAT}	$V_{\text{DS}}=5\text{ V}$	V	0.797
I_{ON}	$V_{\text{DS}}=V_{\text{GS}}=5\text{ V}$	$\mu\text{A}/\mu\text{m}$	51.14

Narrow and short channel device ($W/L=0.35\text{ }\mu\text{m} / 0.5\text{ }\mu\text{m}$)

V_{TLIN}	$V_{\text{DS}}=0.1\text{ V}$	V	0.849
V_{TSAT}	$V_{\text{DS}}=5\text{ V}$	V	0.765
I_{ON}	$V_{\text{DS}}=V_{\text{GS}}=5\text{ V}$	$\mu\text{A}/\mu\text{m}$	579.89
I_{OFF}	$V_{\text{DS}}=5\text{ V}$, $V_{\text{GS}}=0\text{ V}$	$\text{A}/\mu\text{m}$	1.0E-13

5.1.1 Model verification plots for NMOS

The L and W in the plots refer to LDES and WDES respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at $I_D=300\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ at 25°C .

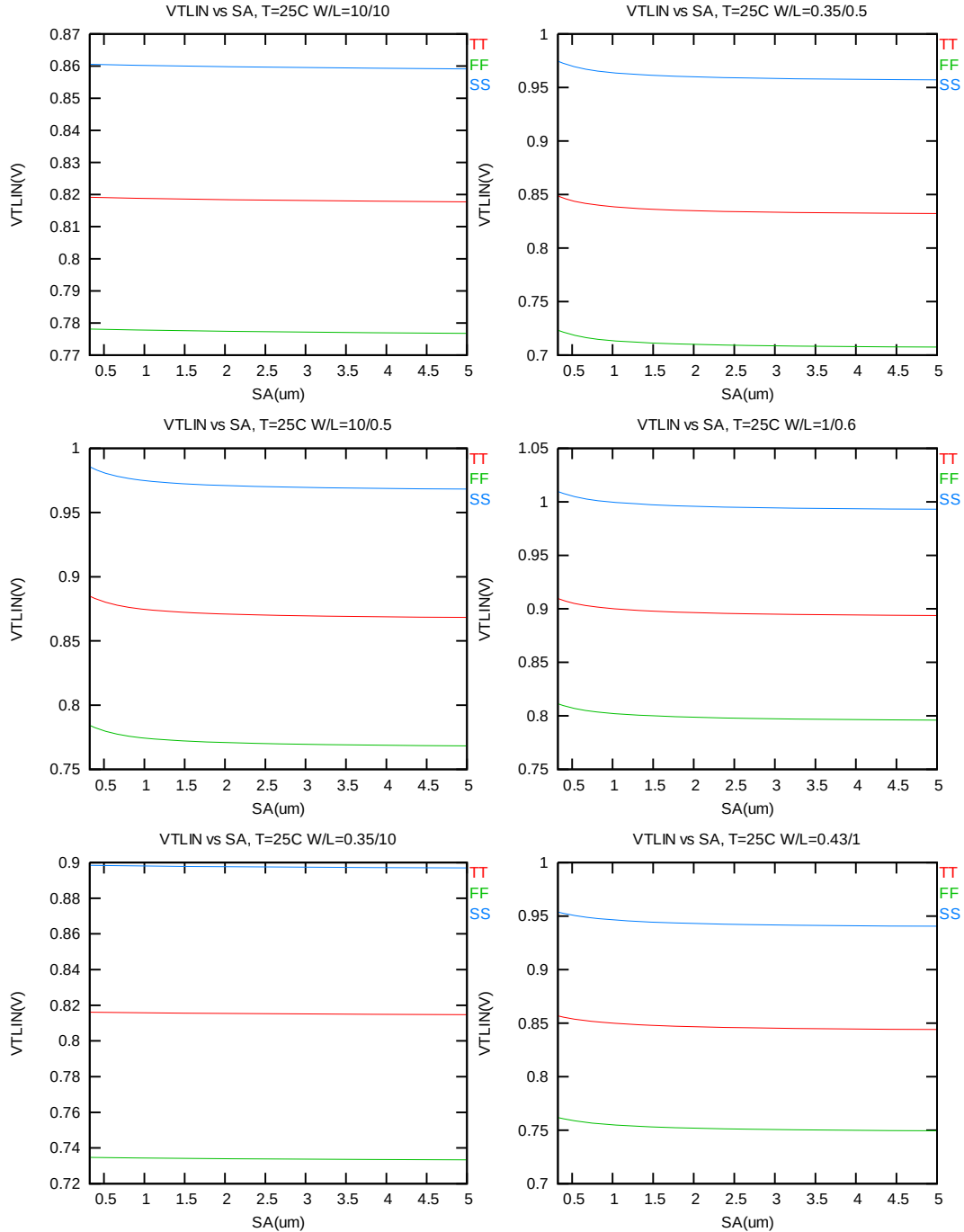


Figure 5-1 V_T vs. SA for low drain bias ($V_{DS}=0.1\text{ V}$) for NMOS

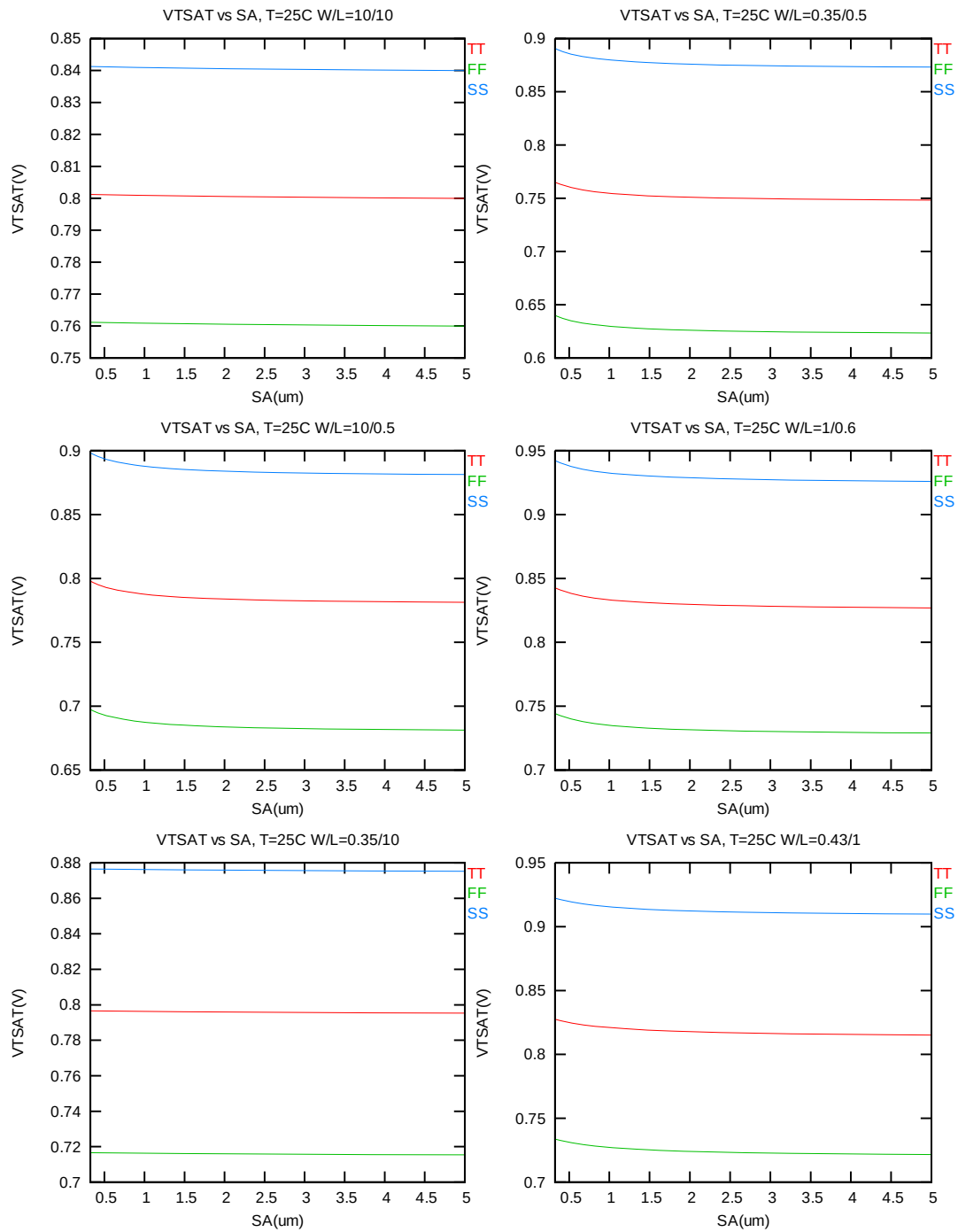


Figure 5-2 V_T vs. SA for high drain bias ($V_{DS}=5$ V) for NMOS

I_{ON} vs. SA

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}=5$ V for $V_{BS}=0$ at 25C.

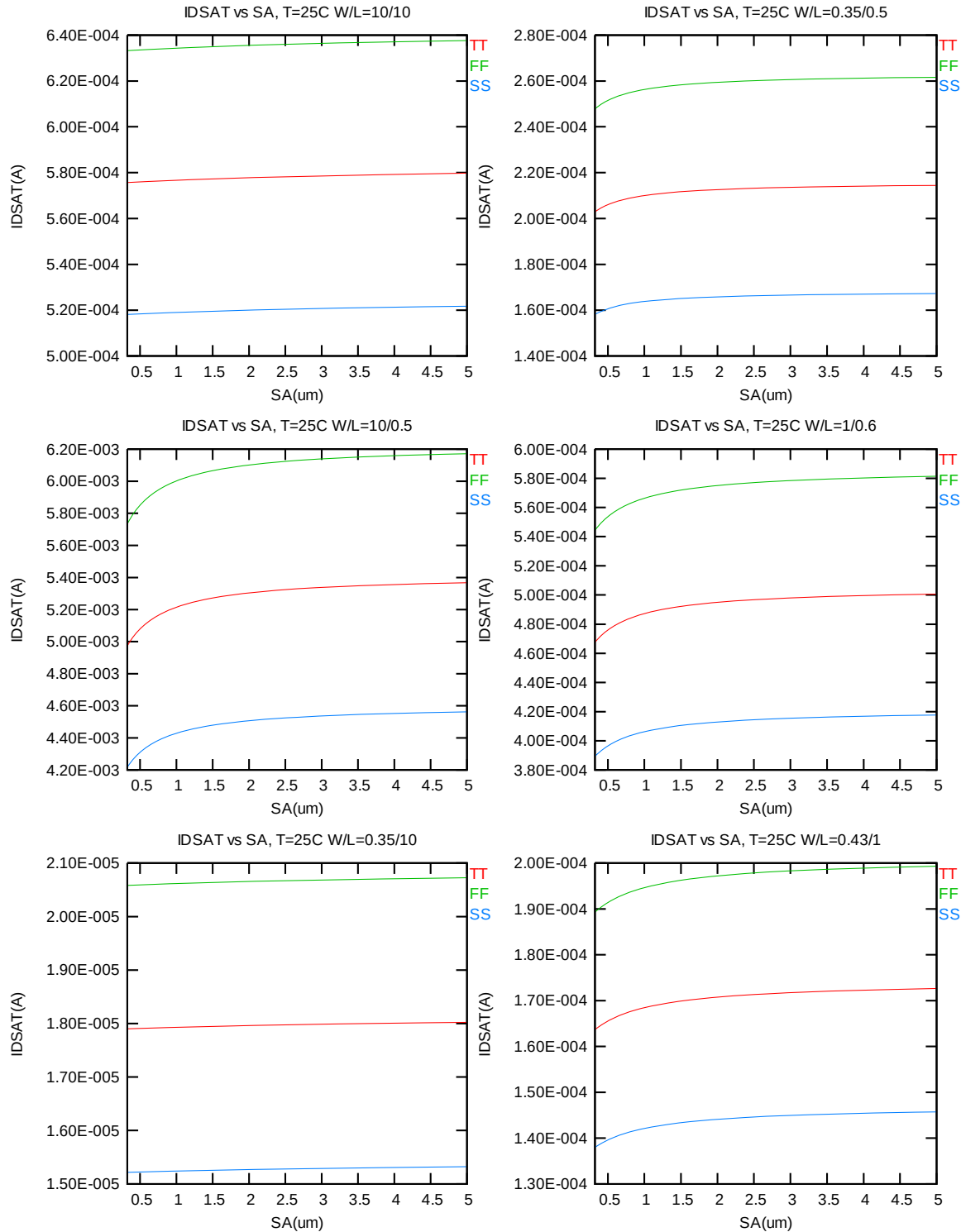


Figure 5-3 Saturation current I_{ON} vs. SA for NMOS

5.2 Simulation Results for PMOS at SA=0.324 um (SAREF=3.6um)

The simulation results for the PMOS characterization are summarized in the table below.

Table 5-2 Simulation results for PMOS

All parameters are given for T=25 °C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
-----------	-----------	------	------------------

Long channel device (W/L= 10 μ m / 10 μ m)

V _{TLIN}	V _{DS} = -0.1 V	V	-0.864
I _{ON}	V _{DS} = -5 V	μA/ μm	-12.24

Wide and short channel device (W/L= 10 μ m / 0.5 μ m)

V _{TLIN}	V _{DS} = -0.1 V	V	-0.832
V _{TSAT}	V _{DS} = -5 V	V	-0.797
Body Effect V _T shift	V _{DS} = -5 V V _{BS} = 0 ~ 5 V	V	-1.208
DIBL	V _{DS} = -0.1~-5 V	V	-0.035
I _{ON}	V _{DS} =V _{GS} = -5 V	μA/μm	-224.99
I _{OFF}	V _{DS} = -5 V, V _{GS} =0 V	A/μm	-4.0E-12

Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)

V _{TLIN}	V _{DS} = -0.1 V	V	-0.882
V _{TSAT}	V _{DS} = -5 V	V	-0.876
I _{ON}	V _{DS} =V _{GS} = -5 V	μA/μm	-12.36

Narrow and short channel device (W/L= 0.35 μ m / 0.5 μ m)

V _{TLIN}	V _{DS} = -0.1 V	V	-0.837
V _{TSAT}	V _{DS} = -5 V	V	-0.796
I _{ON}	V _{DS} =V _{GS} = -5 V	μA/μm	-217.64
I _{OFF}	V _{DS} = -5 V, V _{GS} =0 V	A/μm	-5.0E-12

5.2.1 Model verification plots for PMOS

The L and W in the plots refer to LDES and WDES respectively

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at $I_D = -70\text{nA} \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12\text{nm})$ at 25°C .

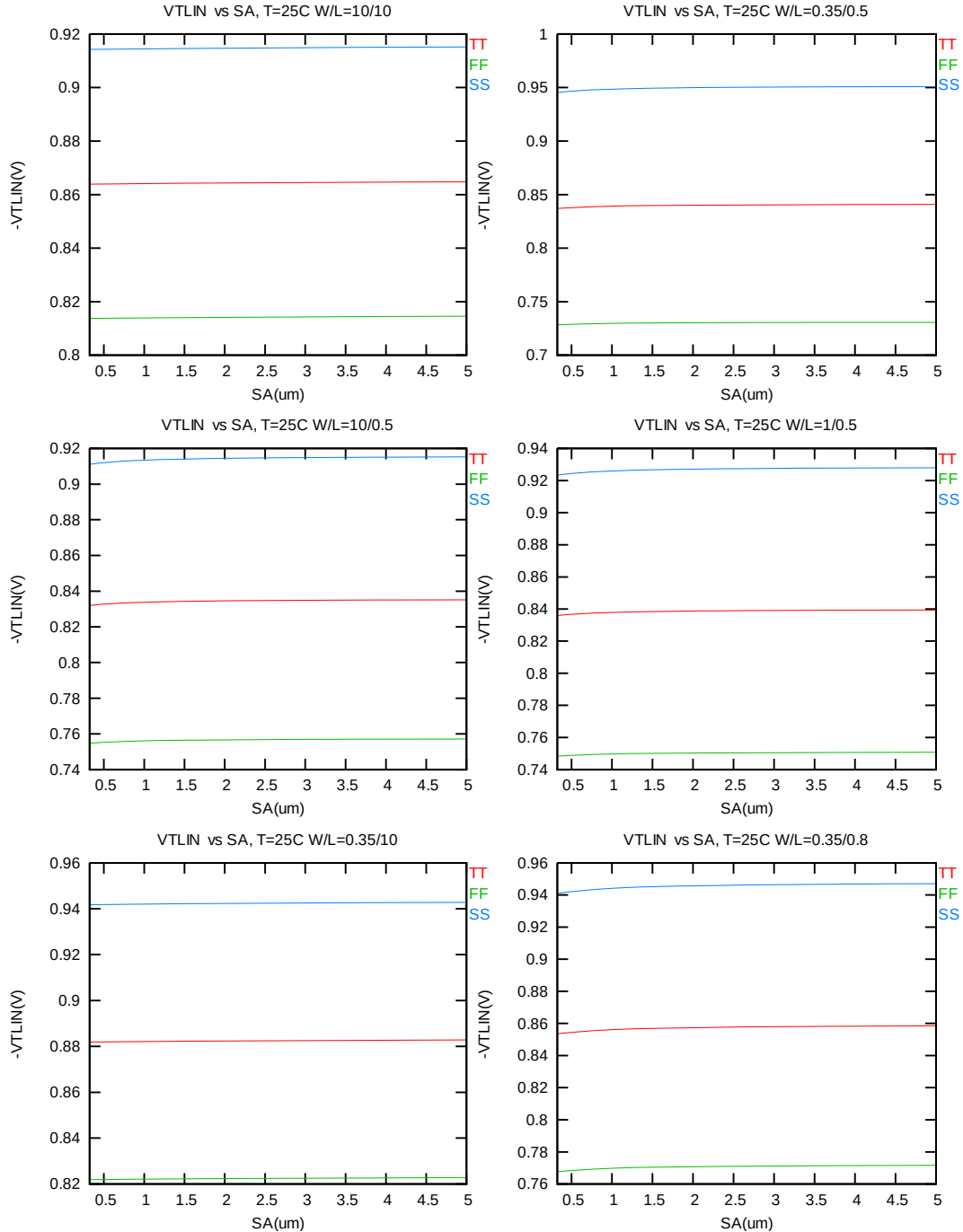


Figure 5-4 V_T vs. SA for low drain bias ($V_{DS} = -0.1\text{ V}$) for PMOS

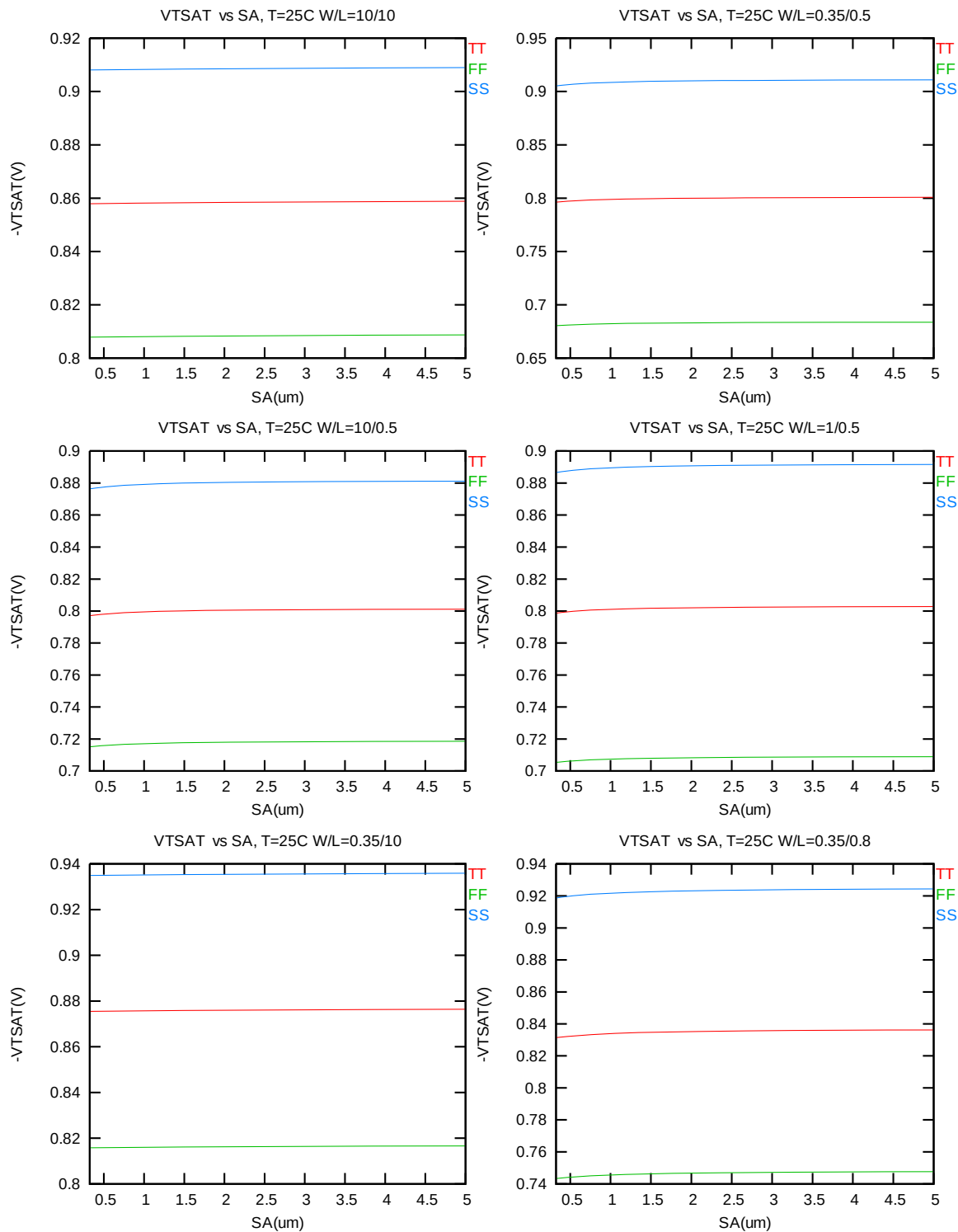


Figure 5-5 V_T vs. SA for high drain bias ($V_{DS} = -5$ V) for PMOS

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}=-5$ V for $V_{BS}=0$ at 25C.

I_{ON} vs. SA

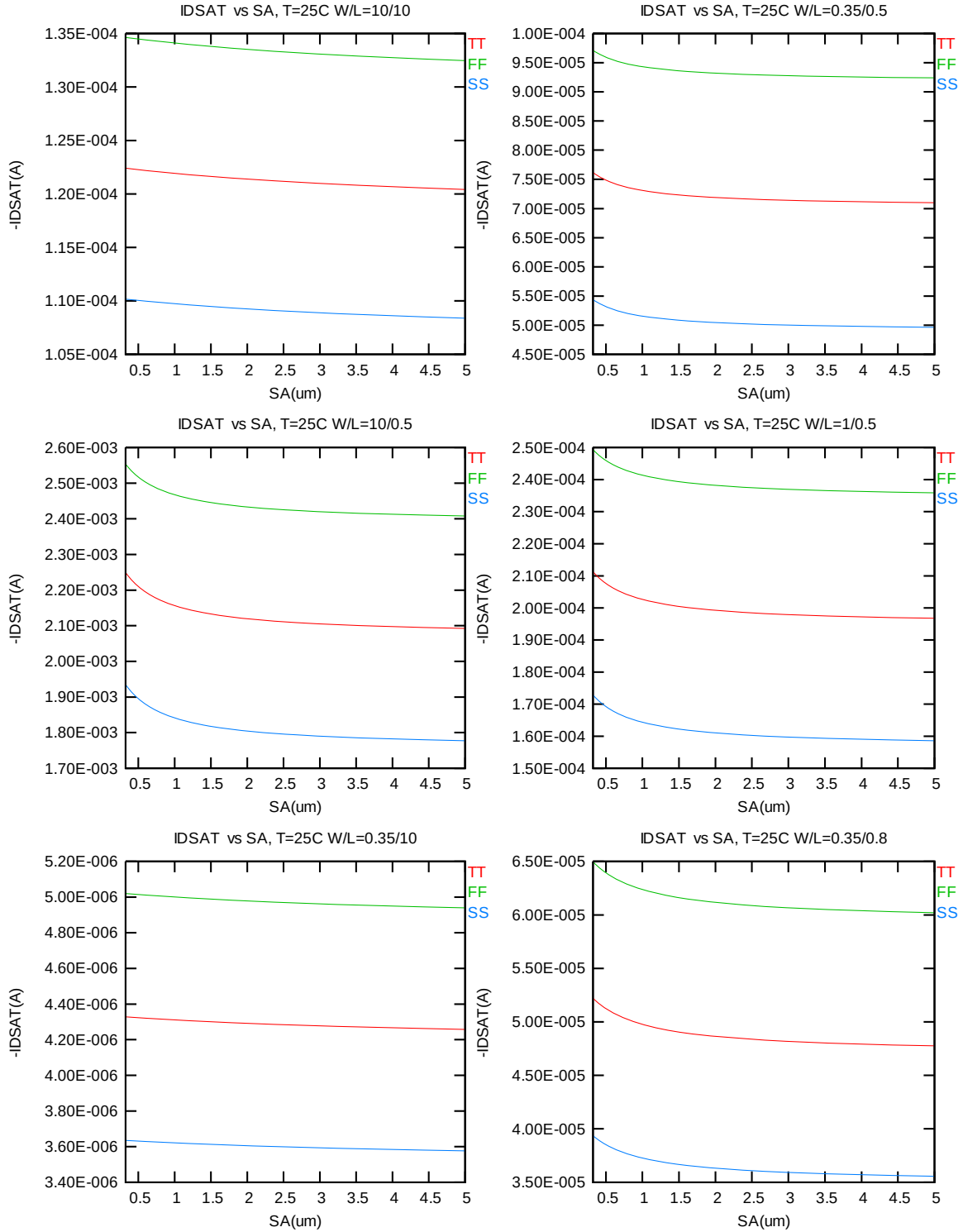


Figure 5-6 Saturation current I_{ON} vs. SA for PMOS

5.3 V_T and I_{ON} Variation v. SA for both N/PMOS

Delta V_T is defined as $V_T(SA) - V_T(3.6\mu m)$ for NMOS and $-[V_T(SA) - V_T(3.6\mu m)]$ for PMOS. Delta I_{ON} is defined as $[I_{ON}(SA) - I_{ON}(3.6\mu m)] / I_{ON}(3.6\mu m) * 100\%$.

Delta V_T and I_{ON} vs. SA

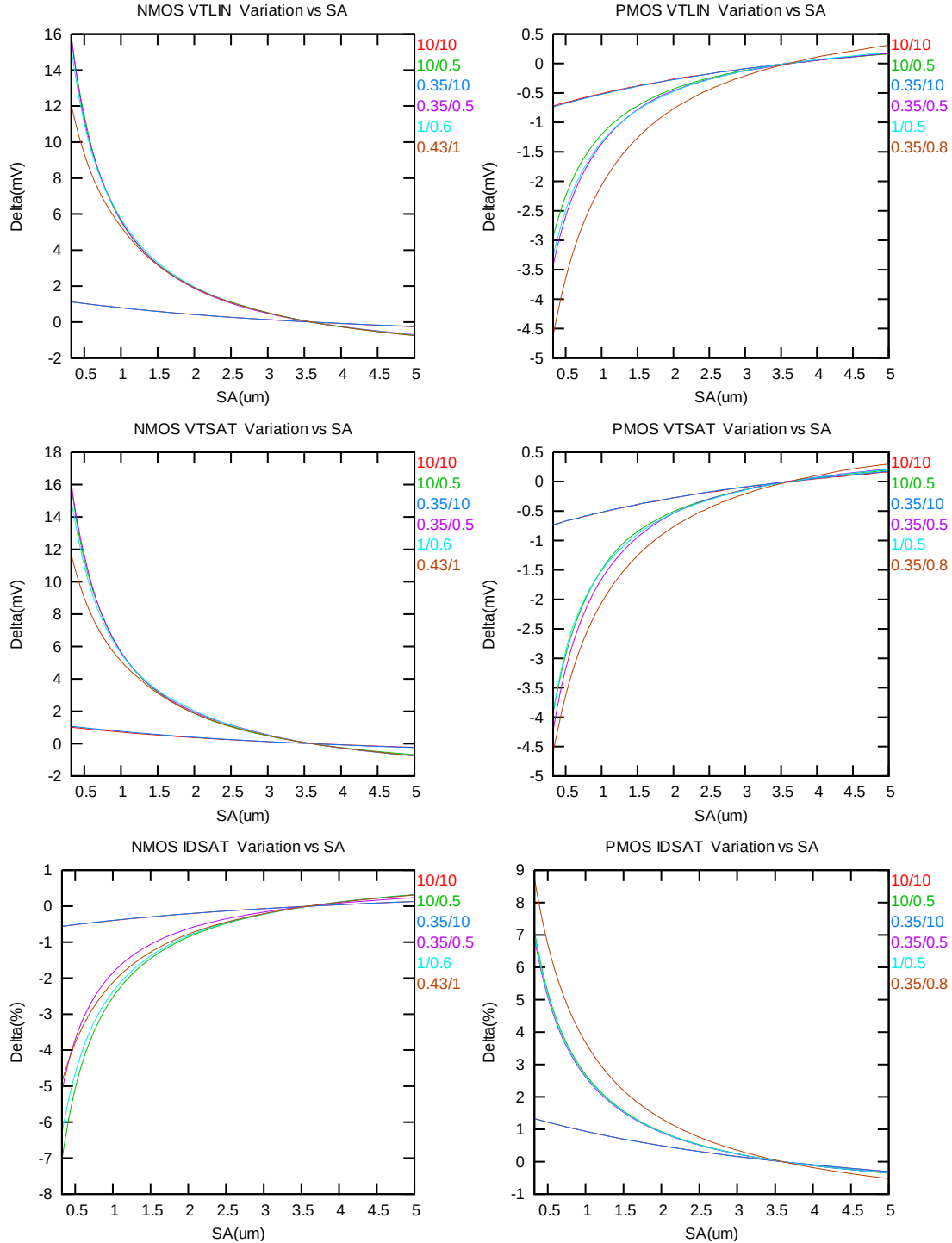


Figure 5-7 Delta V_T and I_{ON} vs. SA for both N/PMOS

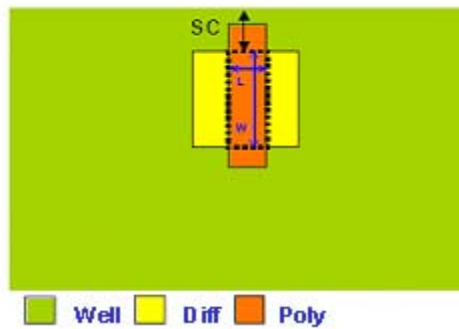
6 Well Proximity Effect (WPE)

6.1 Simulation Results for NMOS at SC=0.4um/0.75um for 1V/1P structure (SCREF=2um)

Note:

1. SA and SB are set to 0.324um in all the following plots and tables.
2. 1V structure: device is close to only one well edge, which is vertical relative to the poly of device (Fig. 6-1a); 1P structure: device is close to only one well edge, which is parallel to the poly of device (Fig. 6-1b)

(a) 1V structure



(b) 1P structure

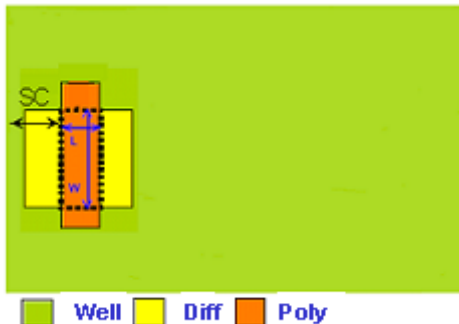


Figure 6-1 Layout example for WPE effect

The simulation results for the NMOS characterization are summarized in the table below.

Table 6-1 Simulation results for NMOS

All parameters are given for $T=25^{\circ}\text{C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Parameter	Condition	Unit	Simulation Value	
			1V structure	1P structure

Long channel device ($W/L=10\text{ }\mu\text{m} / 10\text{ }\mu\text{m}$)

V_{TLIN}	$V_{DS}=0.1\text{ V}$	V	0.825	0.820
I_{ON}	$V_{DS}=5\text{ V}$	$\mu\text{A}/\mu\text{m}$	57.36	57.55

Wide and short channel device ($W/L=10\text{ }\mu\text{m} / 0.5\text{ }\mu\text{m}$)

V_{TLIN}	$V_{DS}=0.1\text{ V}$	V	0.891	0.914
V_{TSAT}	$V_{DS}=5\text{ V}$	V	0.804	0.827
Body Effect V_T shift	$V_{DS}=5\text{ V}$ $V_{BS}=0 \sim -5\text{ V}$	V	0.589	0.589
DIBL	$V_{DS}=0.1\sim 5\text{ V}$	V	0.087	0.087

I_{ON}	$V_{DS}=V_{GS}=5\text{ V}$	$\mu\text{A}/\mu\text{m}$	496.33	492.27
I_{OFF}	$V_{DS}=5\text{ V}, V_{GS}=0\text{ V}$	$\text{A}/\mu\text{m}$	6.3E-14	6.1E-14

Narrow and long channel device ($W/L=0.35\text{ }\mu\text{m} / 10\text{ }\mu\text{m}$)

V_{TLIN}	$V_{DS}=0.1\text{ V}$	V	0.963	0.817
V_{TSAT}	$V_{DS}=5\text{ V}$	V	0.944	0.797
I_{ON}	$V_{DS}=V_{GS}=5\text{ V}$	$\mu\text{A}/\mu\text{m}$	46.49	51.12

Narrow and short channel device ($W/L=0.35\text{ }\mu\text{m} / 0.5\text{ }\mu\text{m}$)

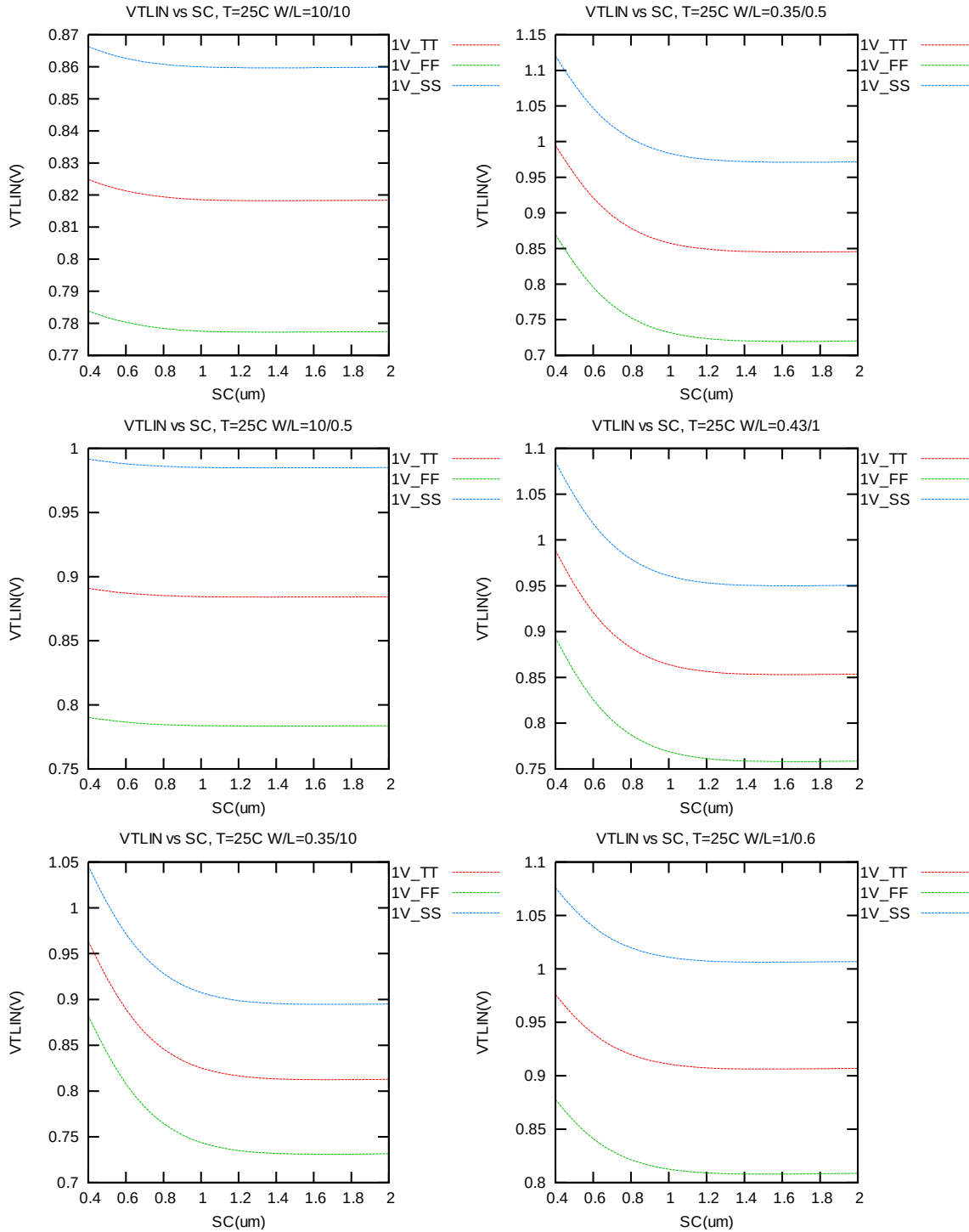
V_{TLIN}	$V_{DS}=0.1\text{ V}$	V	0.995	0.878
V_{TSAT}	$V_{DS}=5\text{ V}$	V	0.911	0.794
I_{ON}	$V_{DS}=V_{GS}=5\text{ V}$	$\mu\text{A}/\mu\text{m}$	558.51	575.69
I_{OFF}	$V_{DS}=5\text{ V}, V_{GS}=0\text{ V}$	$\text{A}/\mu\text{m}$	8.5E-14	9.2E-14

6.1.1 Model simulation plots for NMOS

The L and W in the plots refer to LDES and WDES respectively.

V_T vs. SC

The following figures show the SC dependence of V_{tgm} was extracted by gm maximum and saturation threshold voltage extracted at $I_D=300nA \cdot W_{DES} \cdot 0.9 / (L_{DES} \cdot 0.9 + 12nm)$ at $25^\circ C$.



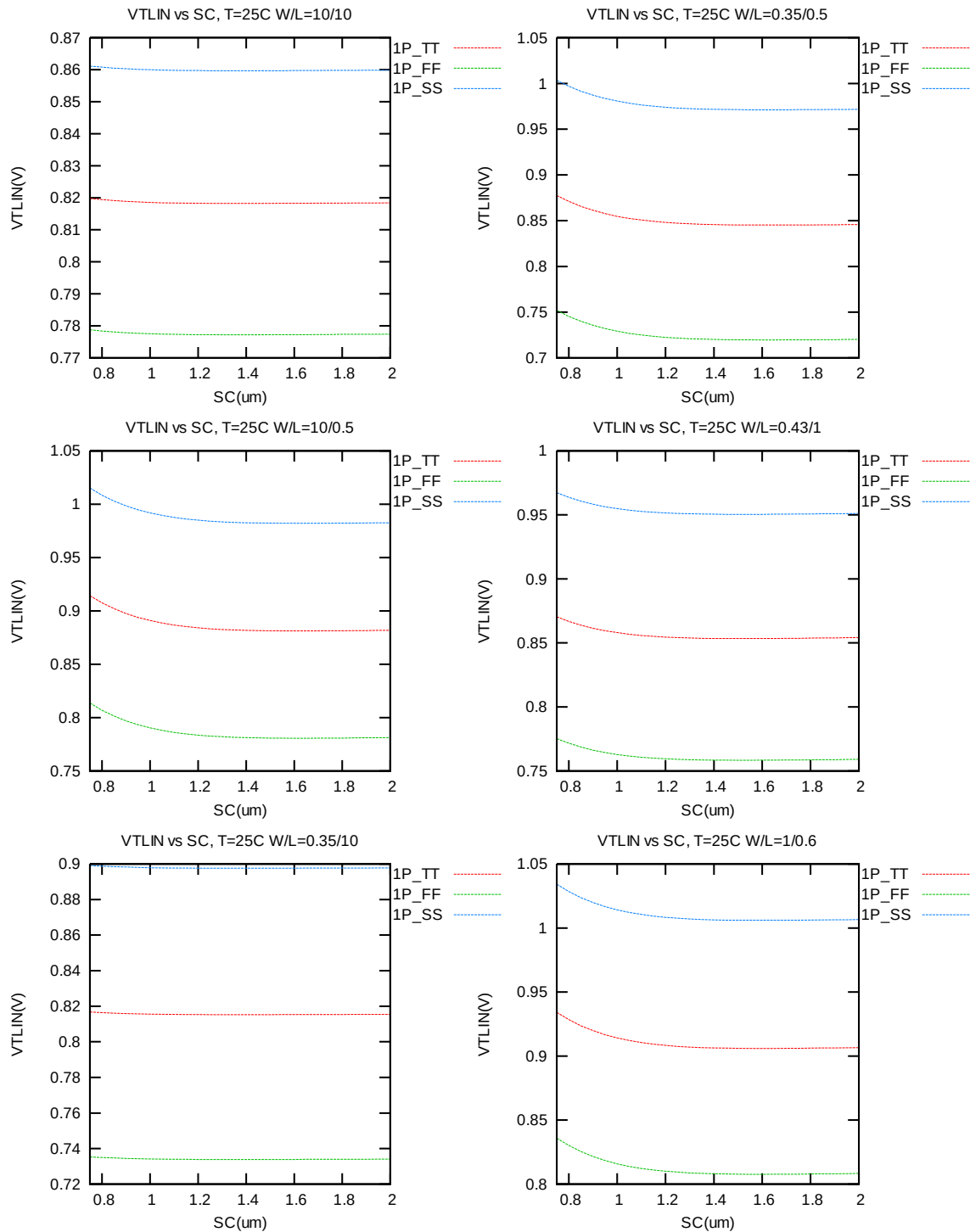
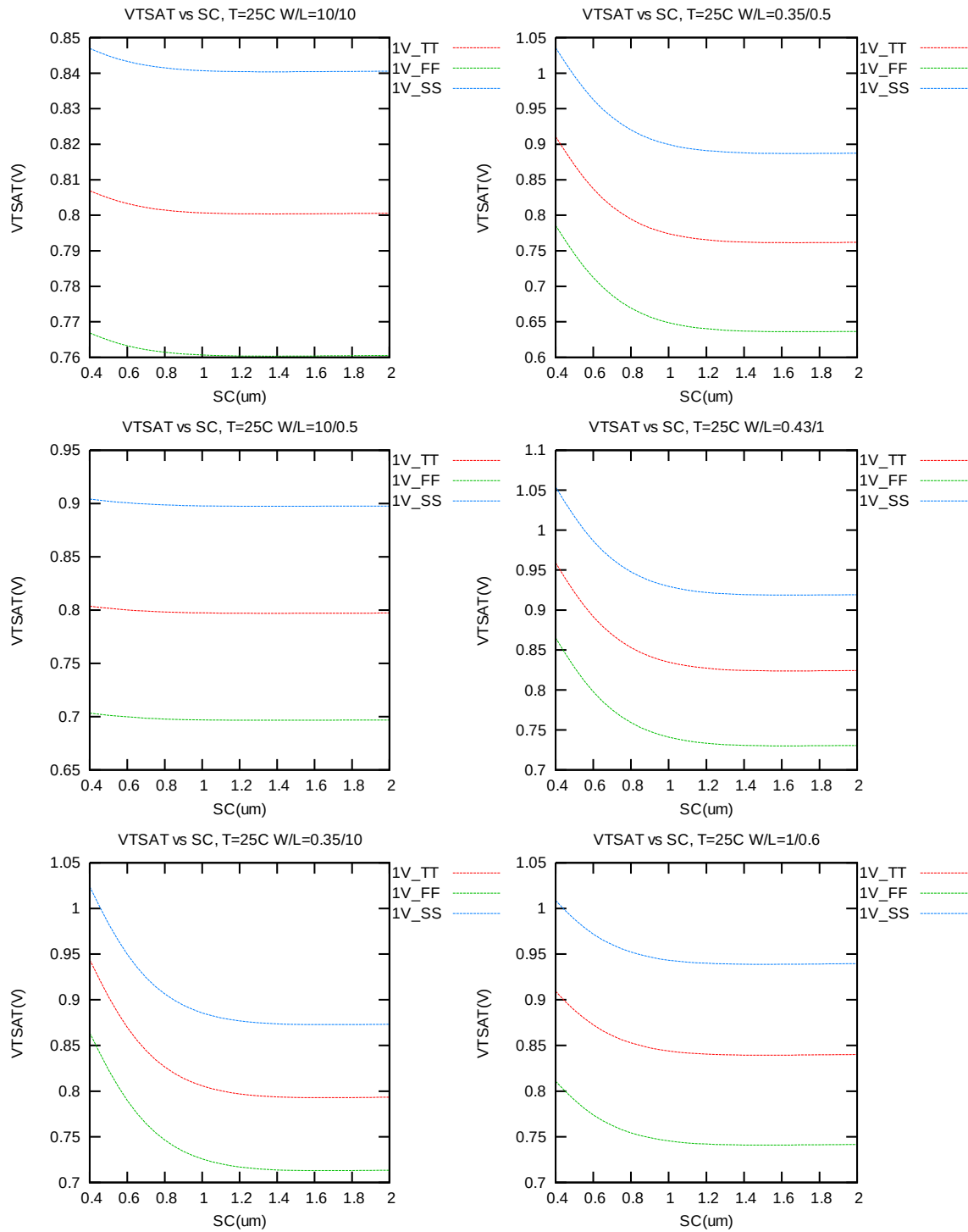


Figure 6-2 V_T vs. SC for low drain bias ($V_{DS}=0.1$ V) for NMOS



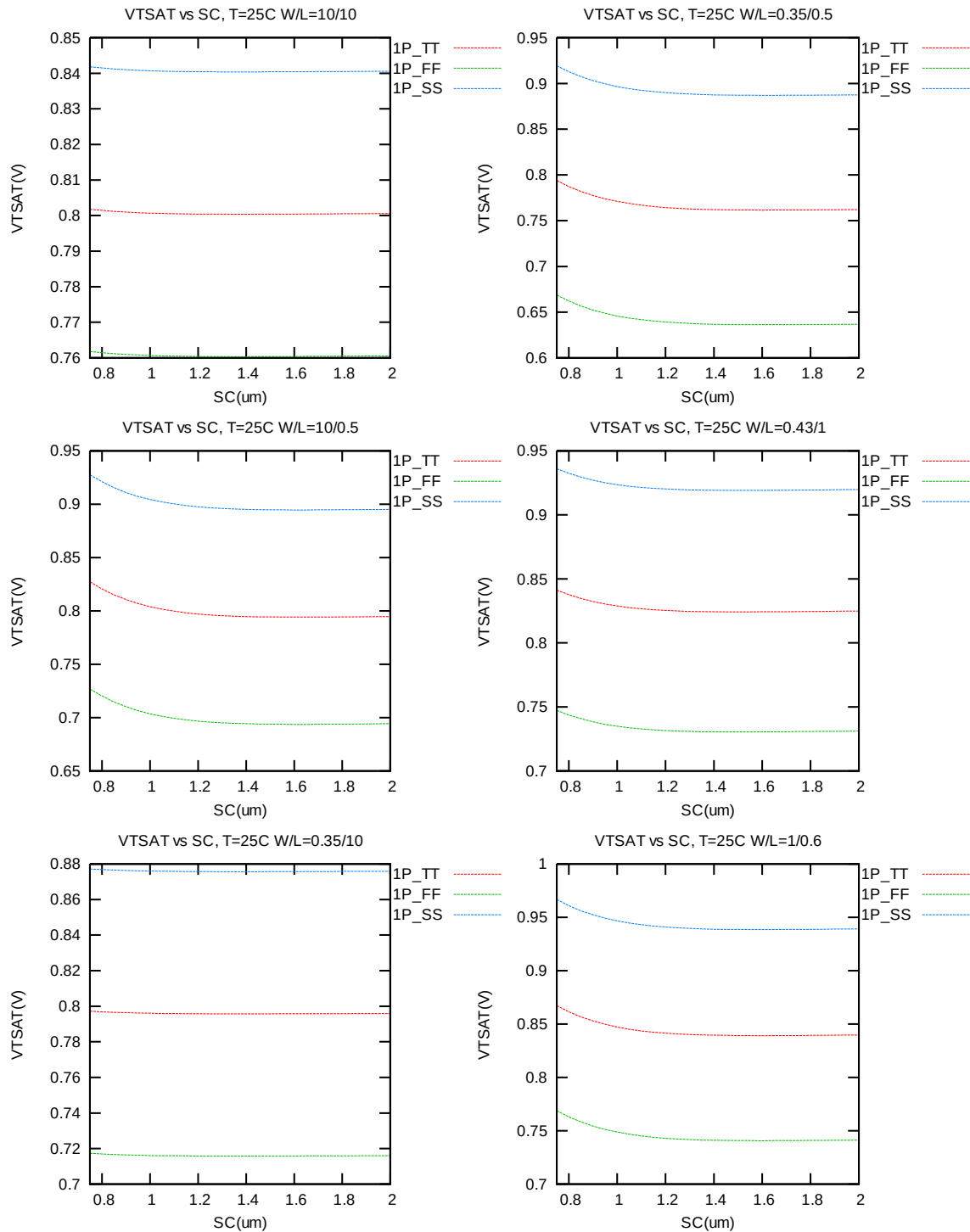
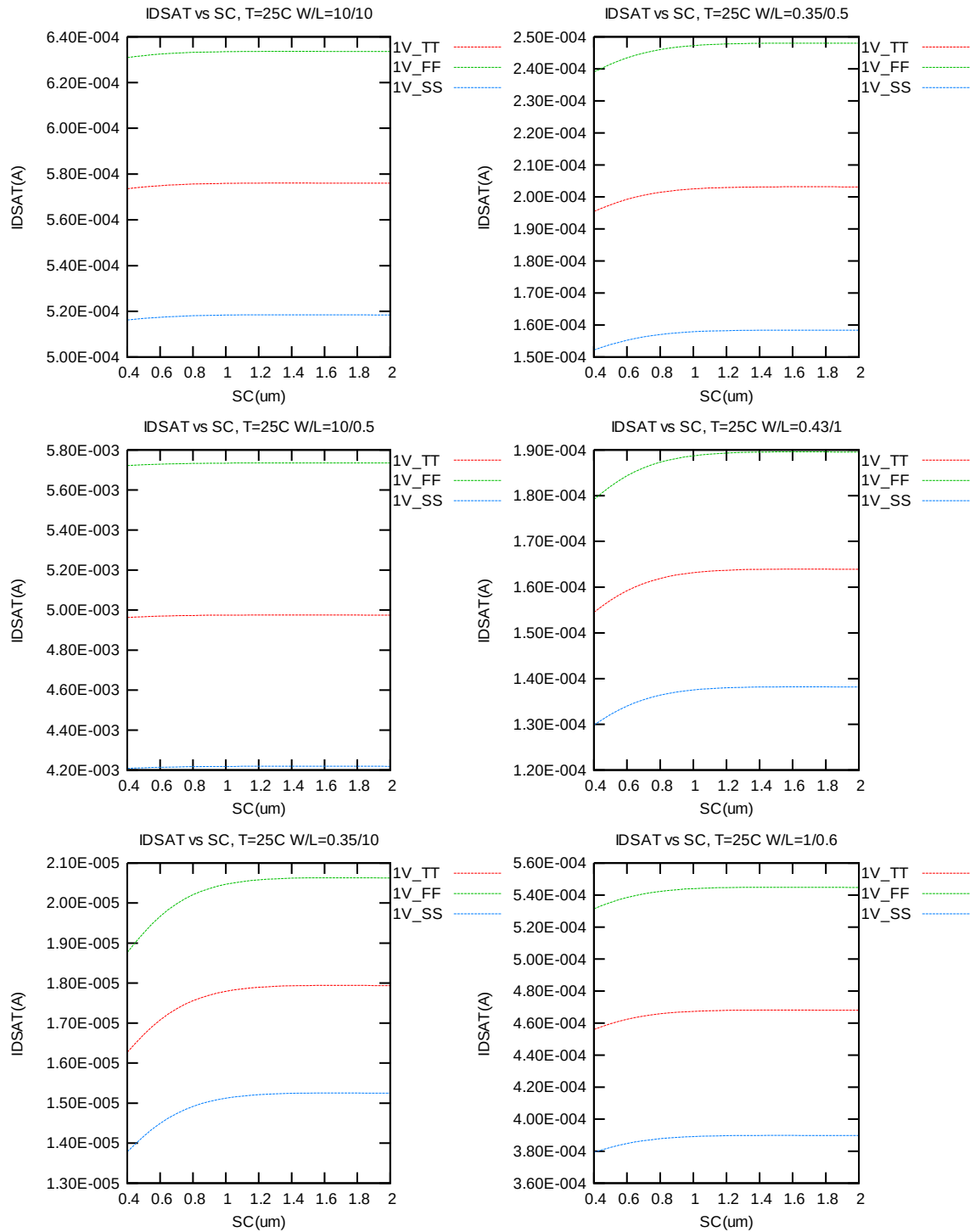


Figure 6-3 V_T vs. SC for high drain bias ($V_{DS}=5$ V) for NMOS

I_{ON} vs. SC

The following plots show the SC dependence of I_{ON} extracted at $V_{DS}=V_{GS}=5$ V for $V_{BS}=0$ at 25C.



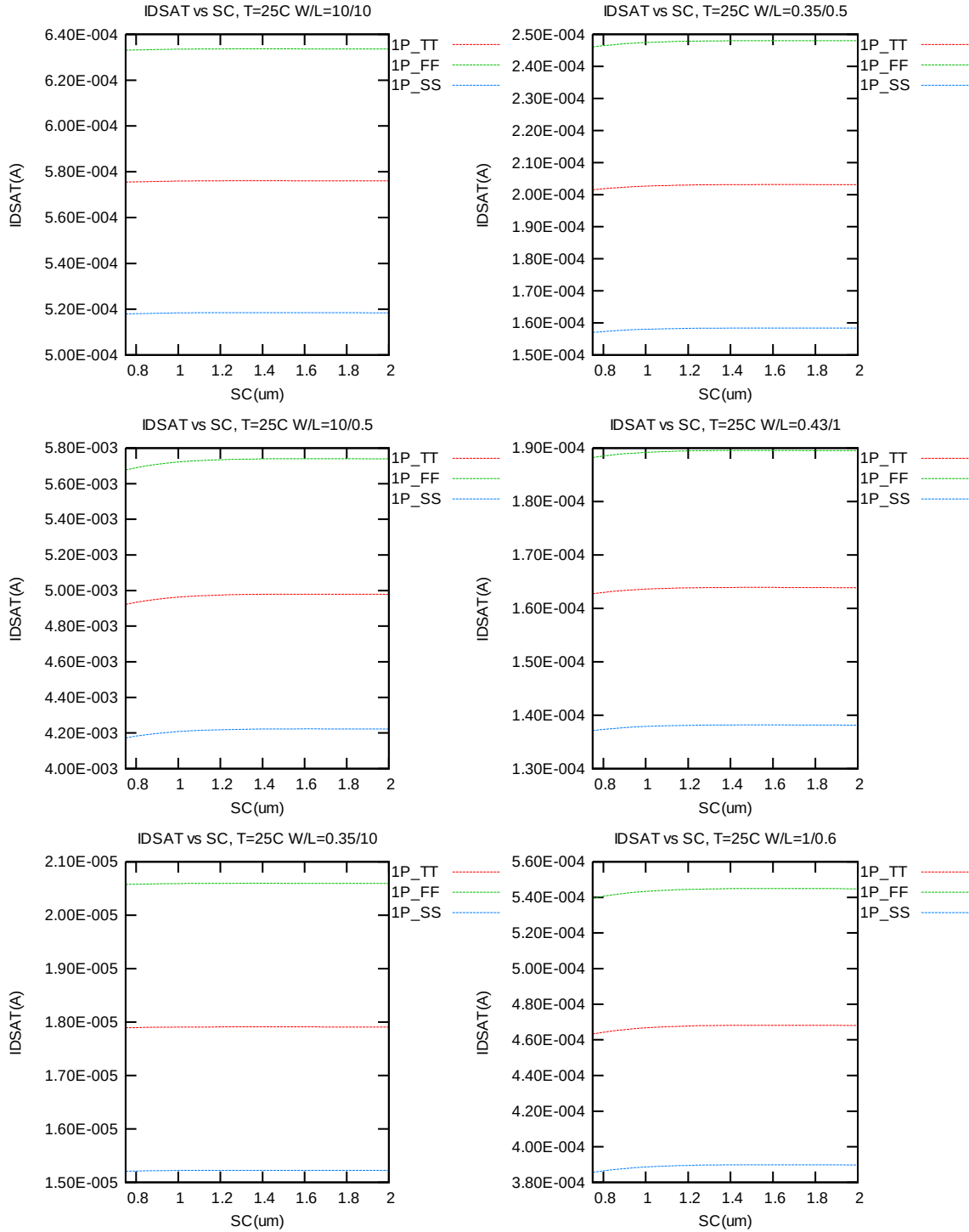


Figure 6-4 Saturation current I_{ON} vs. SC for NMOS

6.2 Simulation Results for PMOS at SC=0.4um/0.75um for 1V/1P structure (SCREF=2um)

The simulation results for the PMOS characterization are summarized in the table below.

Table 6-2 Simulation results for PMOS

All parameters are given for T=25 °C , V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value	
			1V structure	1H structure

Long channel device (W/L= 10 μ m / 10 μ m)

V _{TLIN}	V _{DS} = -0.1 V	V	-0.866	-0.863
I _{ON}	V _{DS} = -5 V	μA/ μm	-12.22	-12.25

Wide and short channel device (W/L= 10 μ m / 0.5 μ m)

V _{TLIN}	V _{DS} = -0.1 V	V	-0.834	-0.847
V _{TSAT}	V _{DS} = -5 V	V	-0.800	-0.812
Body Effect V _T shift	V _{DS} = -5 V V _{BS} = 0 ~ 5 V	V	-1.208	-1.208
DIBL	V _{DS} = -0.1~-5 V	V	-0.035	-0.035
I _{ON}	V _{DS} =V _{GS} = -5 V	μA/μm	-224.75	-223.50
I _{OFF}	V _{DS} = -5 V, V _{GS} =0 V	A/μm	-4.0E-12	-4.0E-12

Narrow and long channel device (W/L= 0.35 μ m / 10 μ m)

V _{TLIN}	V _{DS} = -0.1 V	V	-0.972	-0.881
V _{TSAT}	V _{DS} = -5 V	V	-0.966	-0.875
I _{ON}	V _{DS} =V _{GS} = -5 V	μA/μm	-11.66	-12.37

Narrow and short channel device (W/L= 0.35 μ m / 0.5 μ m)

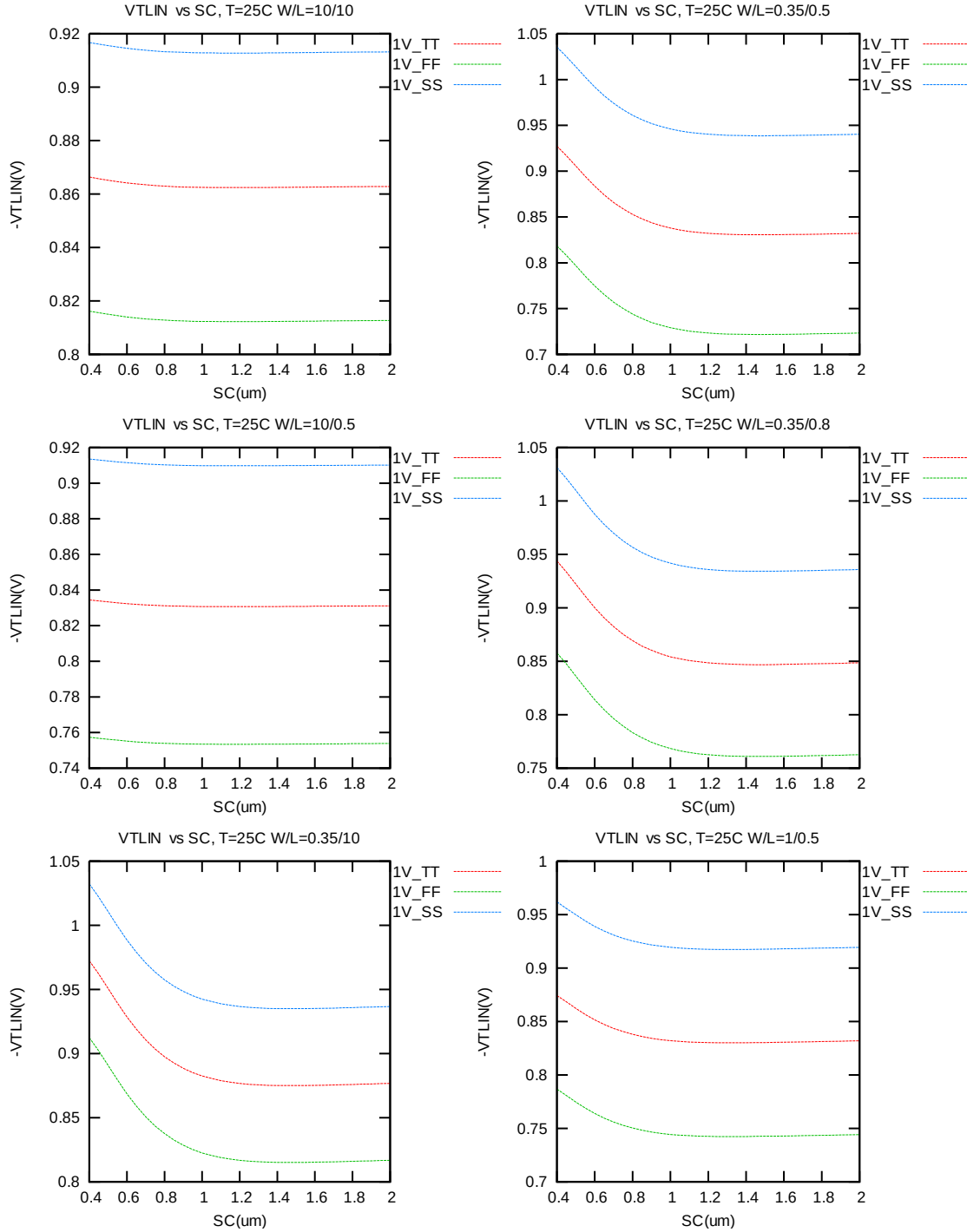
V _{TLIN}	V _{DS} = -0.1 V	V	-0.927	-0.852
V _{TSAT}	V _{DS} = -5 V	V	-0.886	-0.811
I _{ON}	V _{DS} =V _{GS} = -5 V	μA/μm	-209.07	-216.19
I _{OFF}	V _{DS} = -5 V, V _{GS} =0 V	A/μm	-5.0E-12	-5.0E-12

6.2.1 Model simulation plots for PMOS

The L and W in the plots refer to LDES and WDES respectively.

V_T vs. SC

The following figures show the SC dependence of V_{tgm} was extracted by gm maximum and saturation threshold voltage extracted at $I_D = -70nA * W_{DES} * 0.9 / (L_{DES} * 0.9 + 12nm)$ at 25°C.



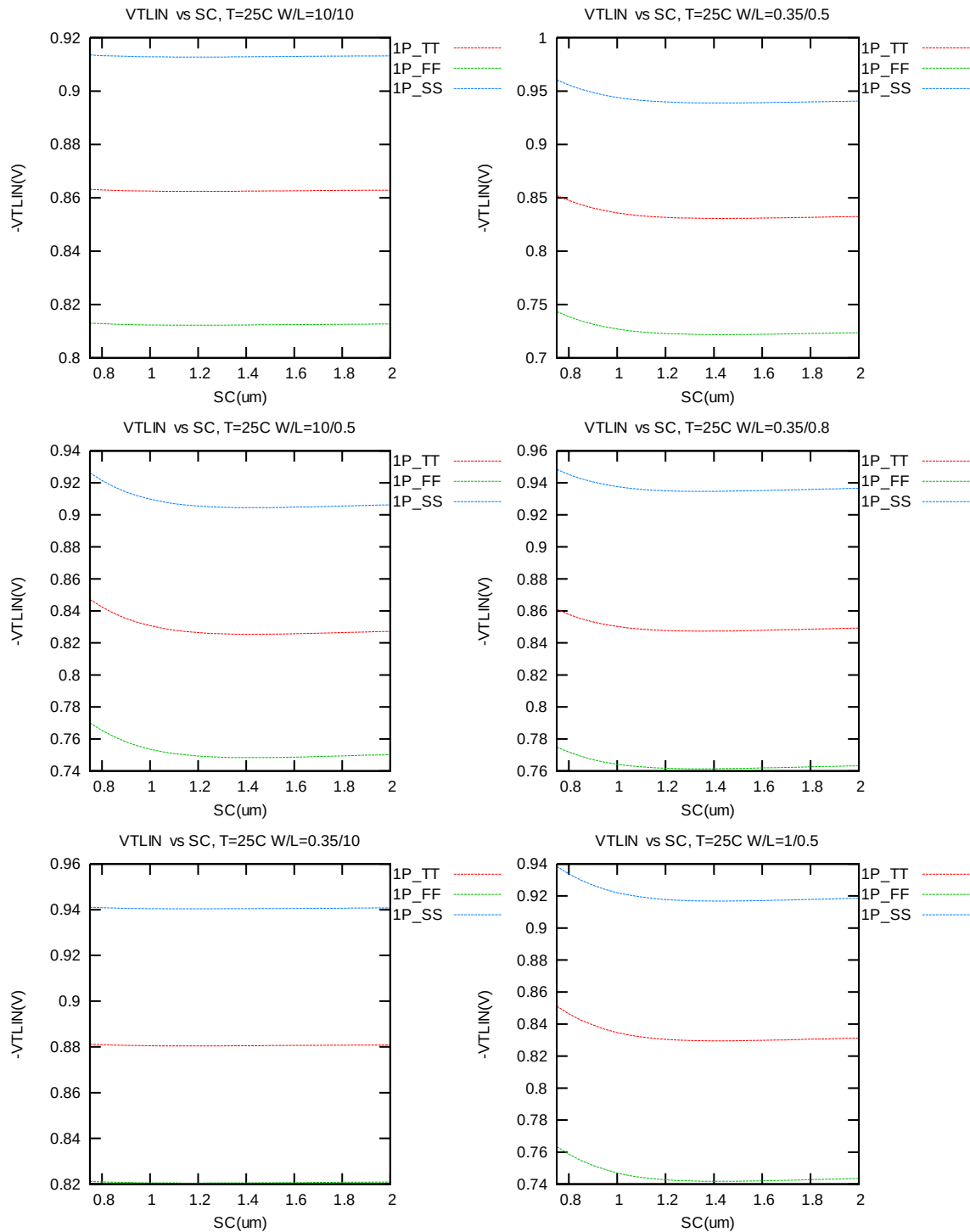
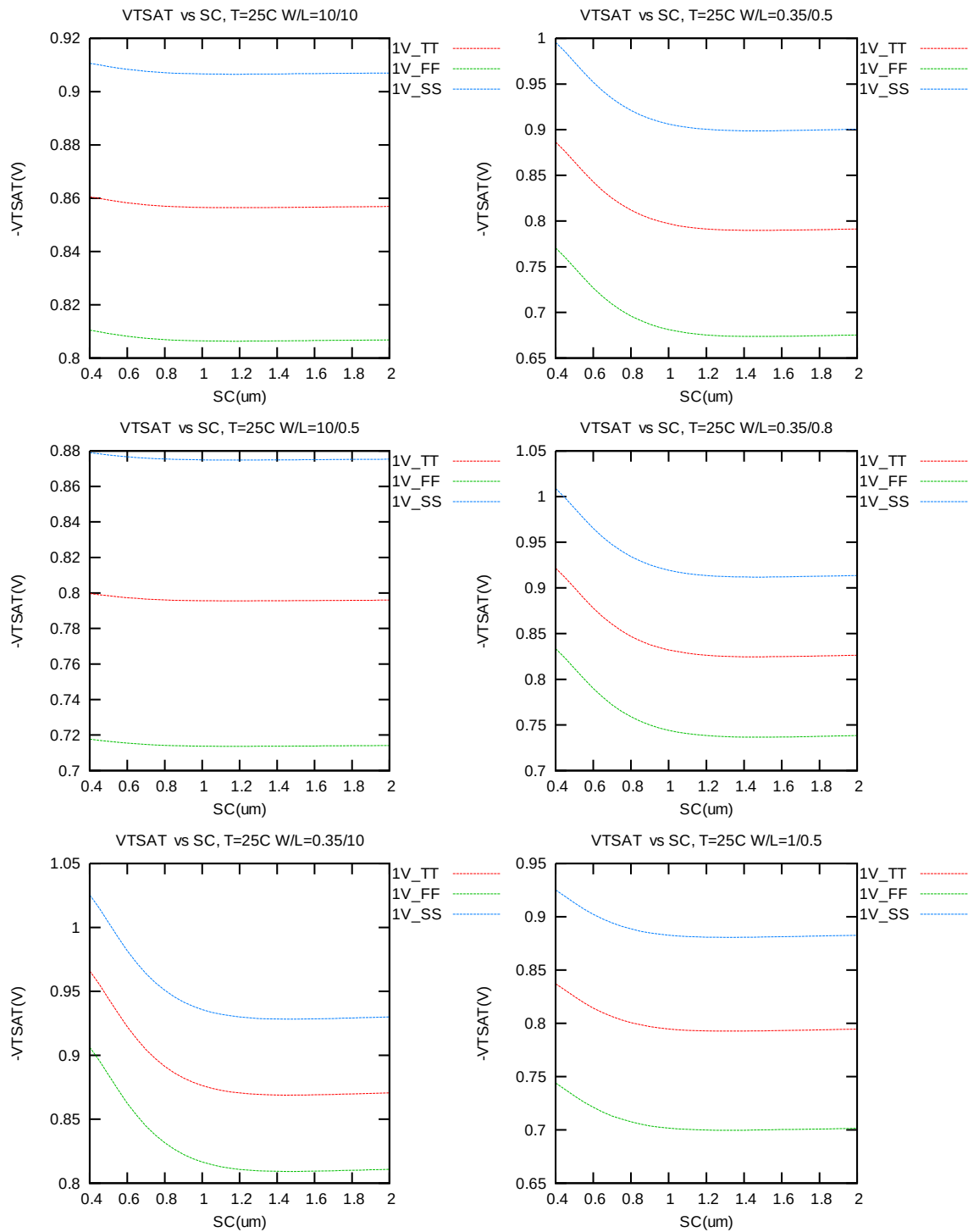


Figure 6-5 V_T vs. SC for low drain bias ($V_{DS} = -0.1$ V) for PMOS



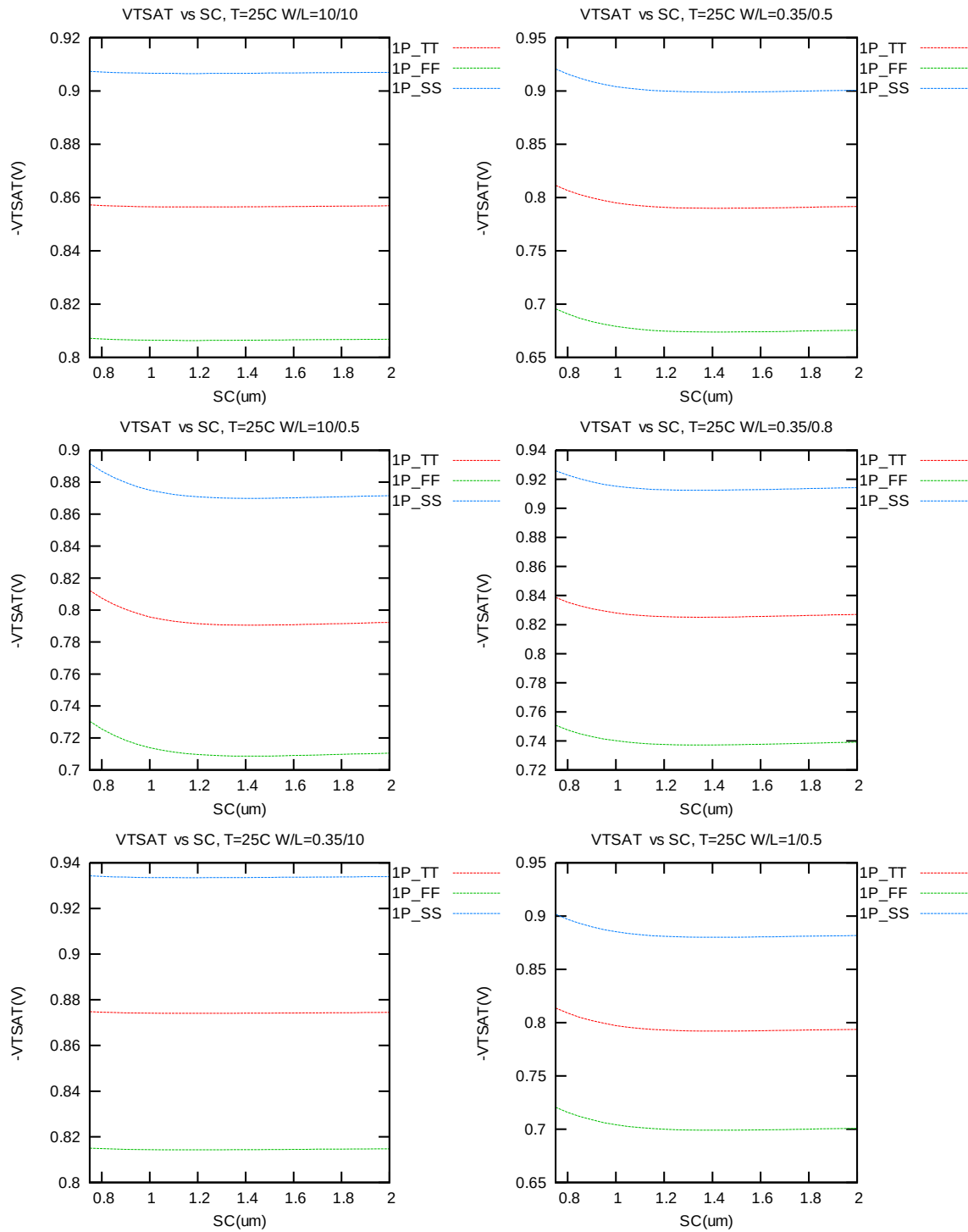
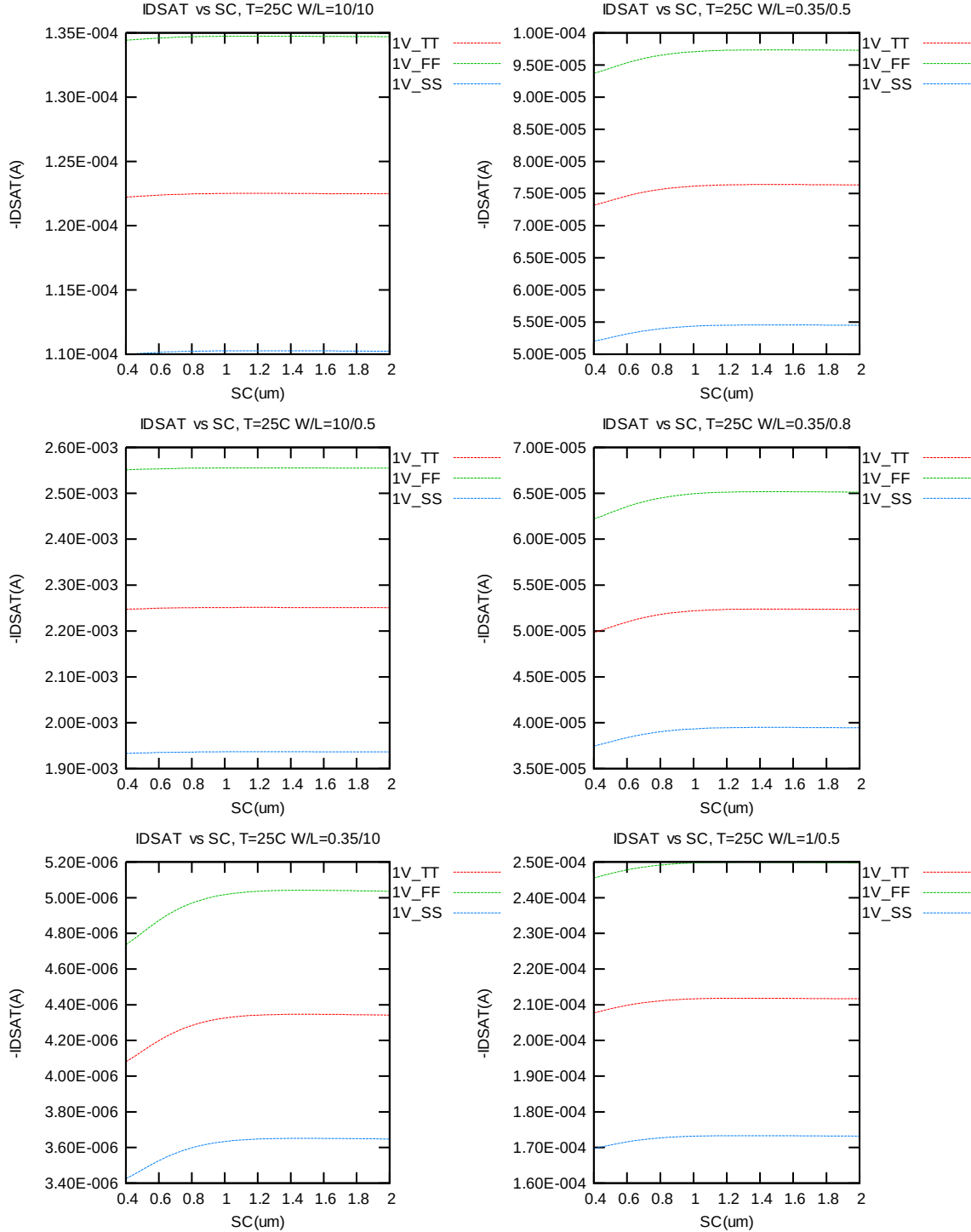


Figure 6-6 V_T vs. SC for high drain bias ($V_{DS} = -5$ V) for PMOS

The following plots show the SC dependence of I_{ON} extracted at $V_{DS}=V_{GS}=-5$ V for $V_{BS}=0$ at 25C.



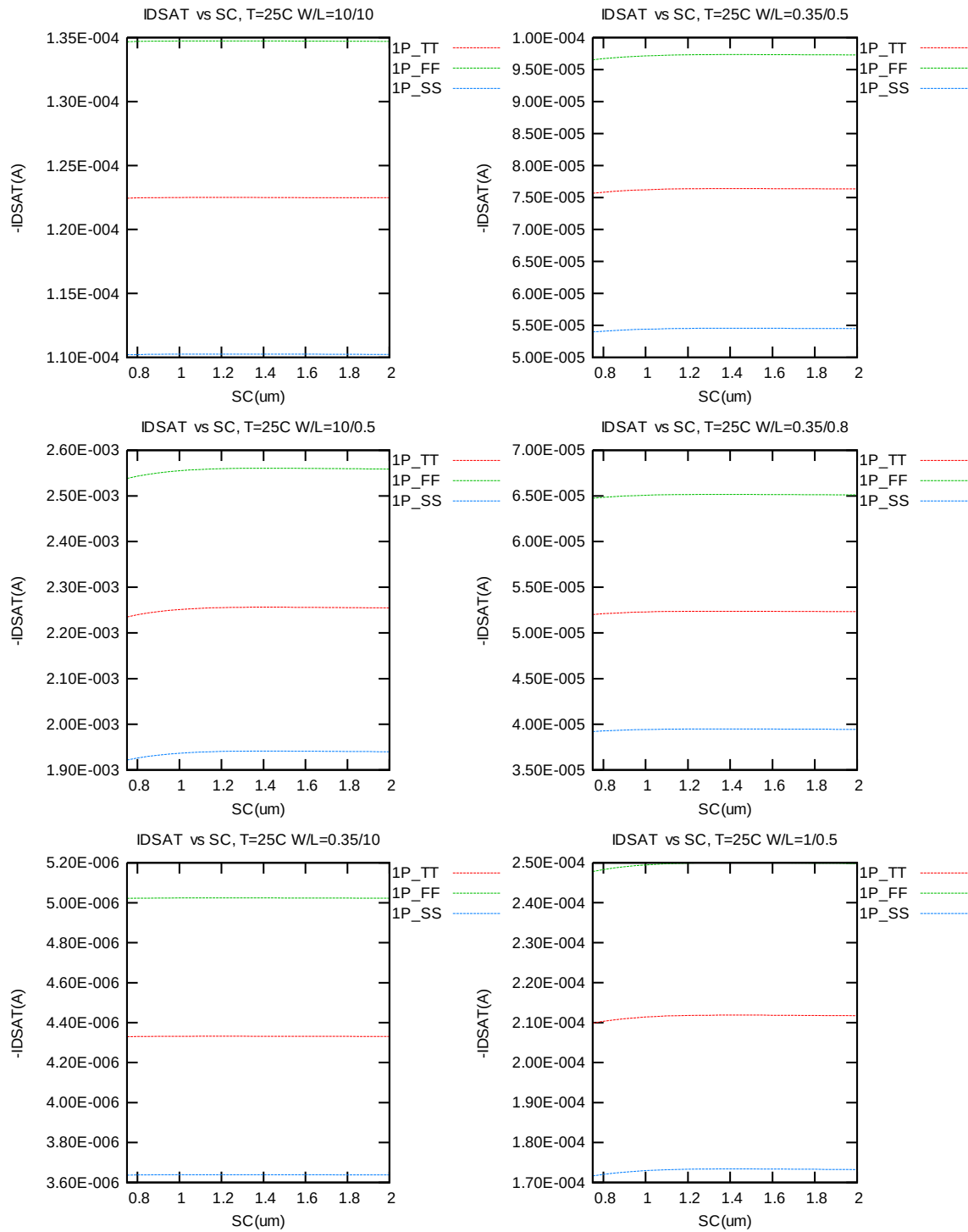


Figure 6-7 Saturation current I_{ON} vs. SC for PMOS

6.3 V_T and I_{ON} Variation v. SC for both N/PMOS

Delta V_T is defined as $V_T(SC) - V_T(2\mu m)$ for NMOS and $-[V_T(SC) - V_T(2\mu m)]$ for PMOS. Delta I_{ON} is defined as $[I_{ON}(SC) - I_{ON}(2\mu m)]/I_{ON}(2\mu m) * 100\%$.

Delta V_T and I_{ON} vs. SC

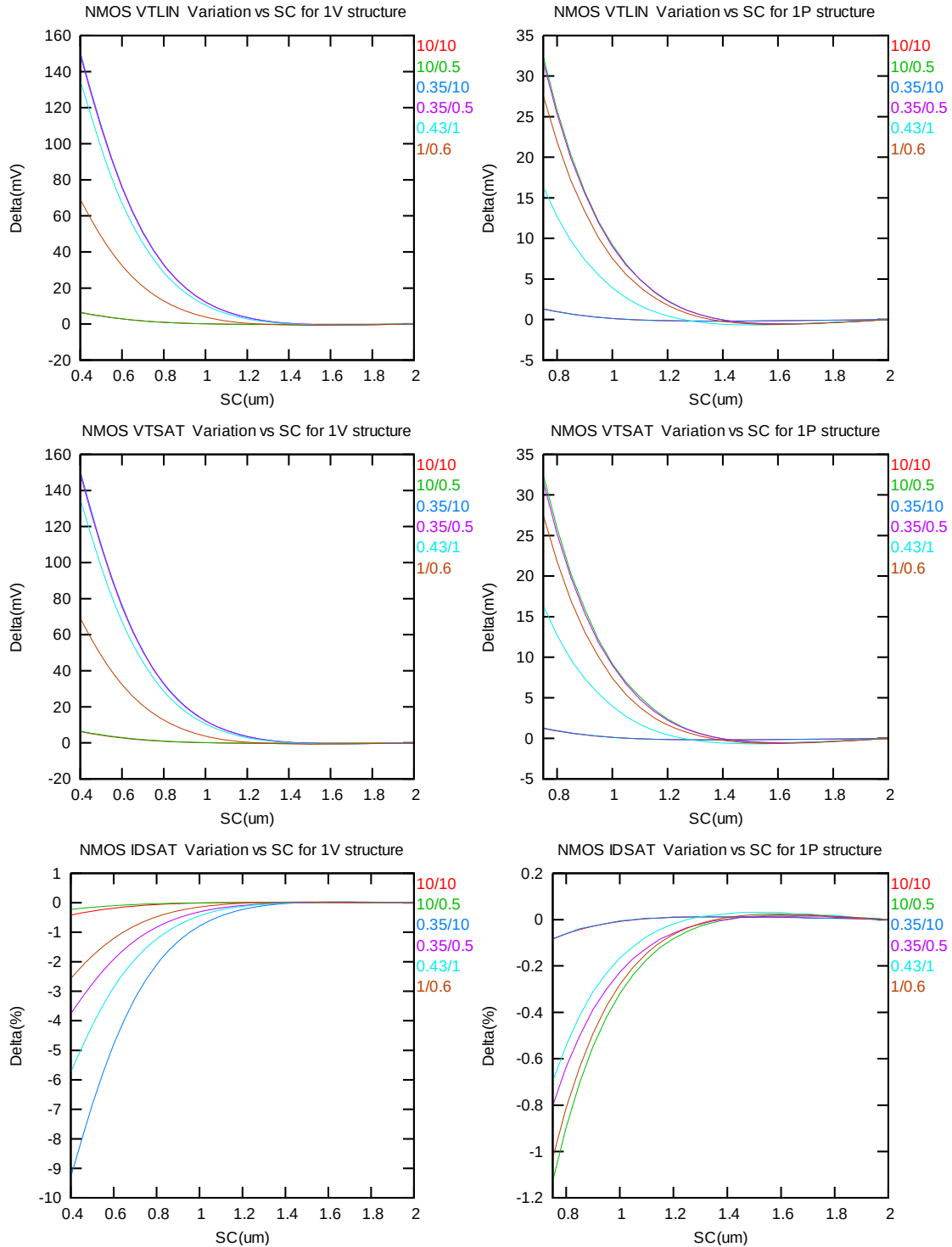


Figure 6-8 Delta V_T and I_{ON} vs. SC for NMOS

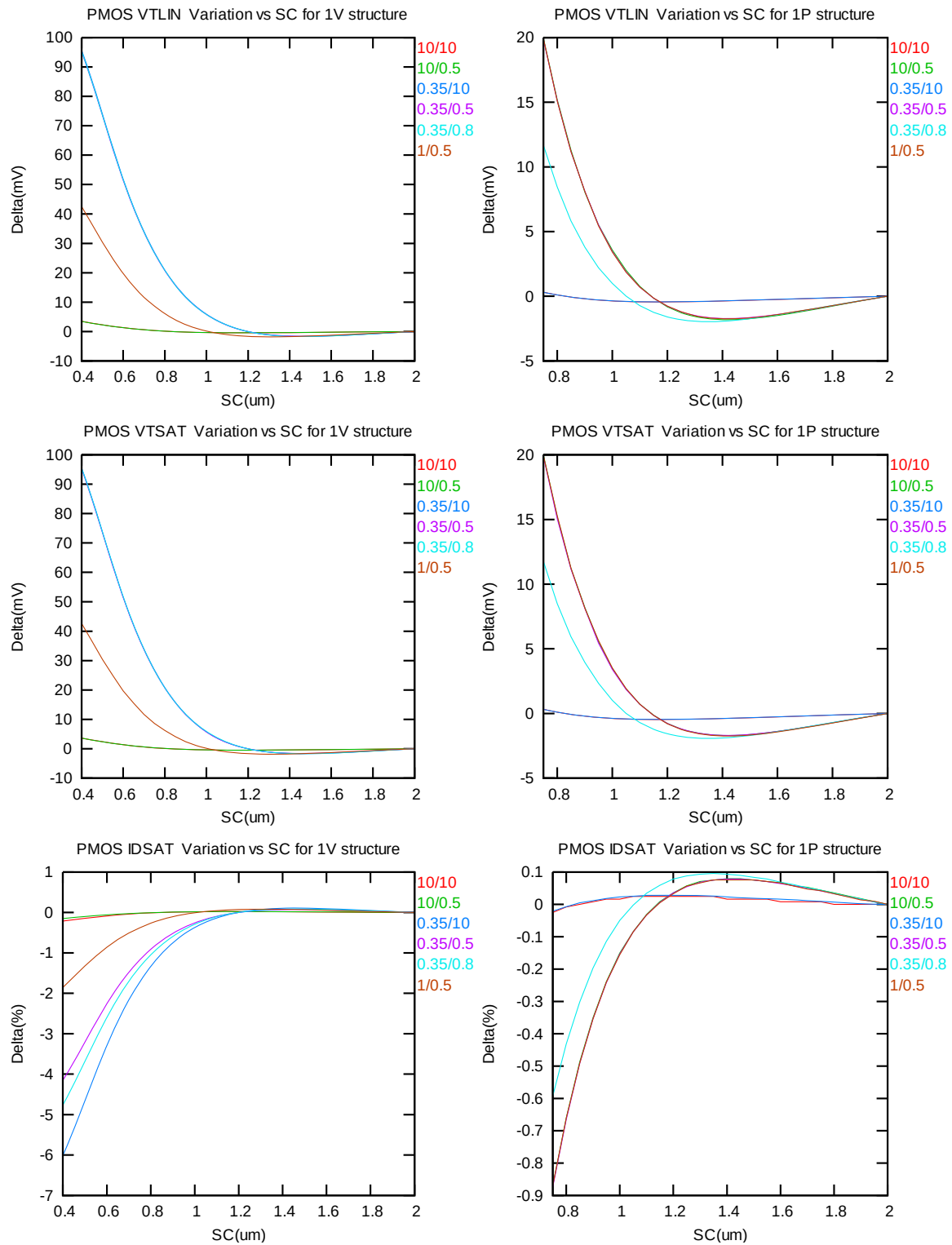


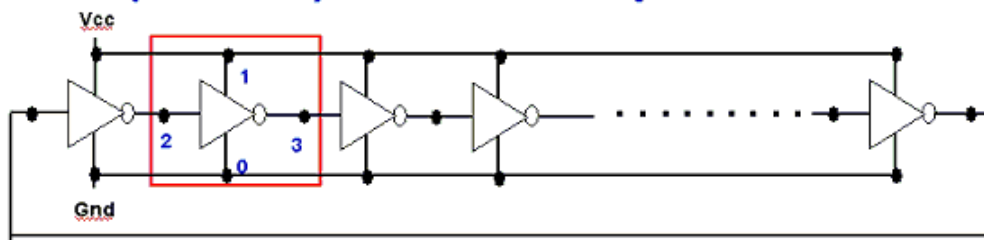
Figure 6-9 Delta V_T and I_{ON} vs. SC for PMOS

7 Dynamic Performance

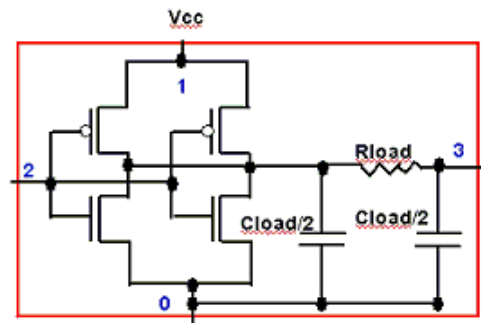
7.1 Test Circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configurations is adopted.

R.O. (Inverter) circuit description



Ring Oscillator circuit



single stage R.O. circuit

Figure 7-1 Ring oscillator netlist description

7.2 Dynamic Performance

Table 7-1 Ring oscillator (inverter) structure description

R.O. (Inverter) Structure description		
Contact size	0.144 um X 0.144 um	
Polysilicon gate to contact spacing	0.14 um	
Contact to diffusion edge spacing	0.06 um	
Without RC loading	R = 0 ohm	C = 0 fF
Inverter		
Logic gate type	Inverter	
Channel width for n-ch MOSFET	2.5 um X 2	
Channel length for n-ch MOSFET	0.5 um	
Channel width for p-ch MOSFET	3.75 um X 2	
Channel Length for p-ch MOSFET	0.5 um	

Table 7-2 Electrical parameters for ring oscillator (ps/stage)

Parameter	Condition	Unit	Target	Centered Value
T_{pd}	$V_{cc} = 5V$	ps/stage	67.50	67.47

The simulation results at different temperatures and using different worst-case models are listed below.

Table 7-3 Ring oscillator simulation results (ps/stage)

Inverter, R=0, C=0f, Fanout=1, Vcc=5V				Inverter, R=0, C=0f, Fanout=1, T=25C			
Temp(C)	TT	FF	SS	Vcc(V)	TT	FF	SS
-40	50.190	43.635	59.348	3.5	79.278	66.273	97.205
0	55.587	48.089	65.991	4	69.949	59.239	84.694
25	58.887	50.807	70.061	4.5	63.543	54.374	76.200
85	66.379	56.968	79.289	5	58.887	50.807	70.061
125	70.888	60.672	84.861	5.5	55.322	48.053	65.389

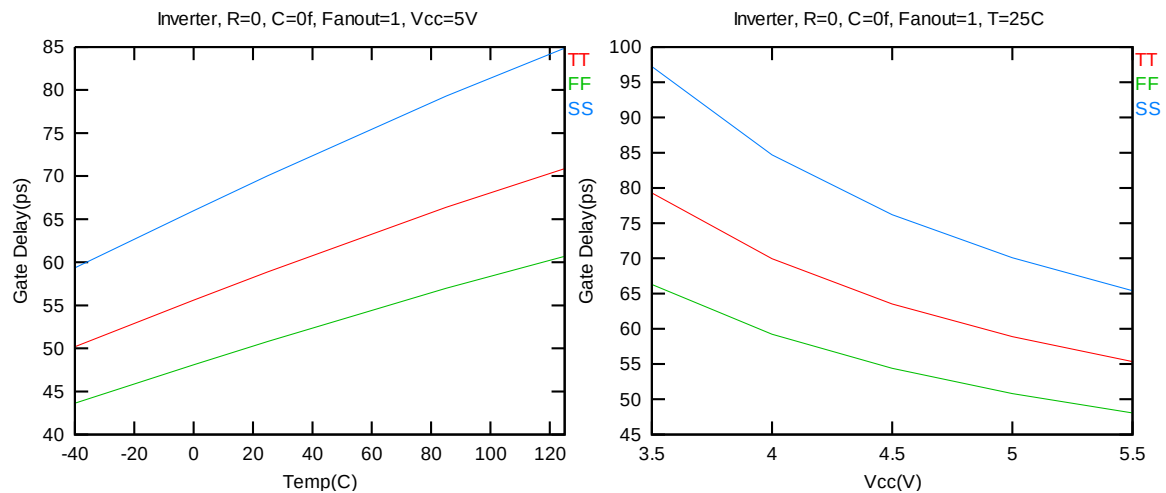


Figure 7-2 Ring oscillator (inverter) simulation

The (Inverter) ring oscillator gate delay simulation vs. width

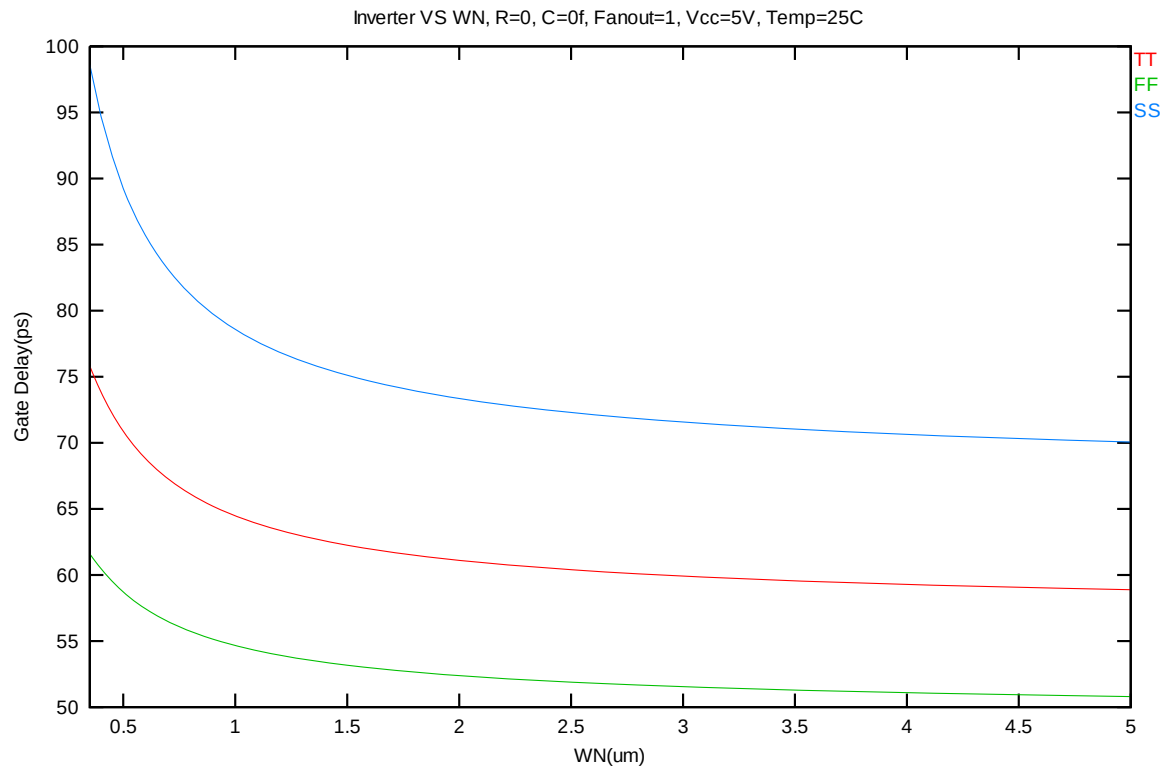


Figure 7-3 Ring oscillator (inverter) gate delay simulation vs. width

8 Noise Characteristics

8.1 Verification Plots

NMOS, $W=10\mu\text{m}$ $L=10\mu\text{m}$

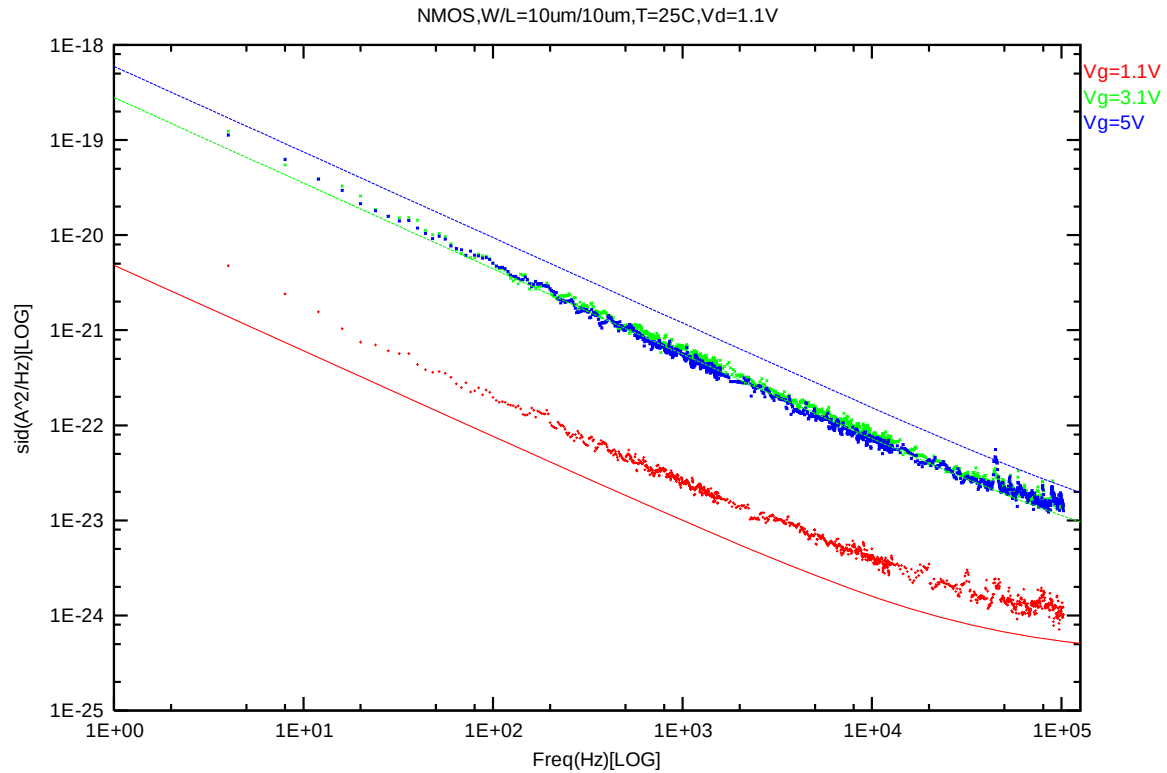


Figure 8-1 NMOS W/L=10um/10um @ $V_d=1.1\text{V}$

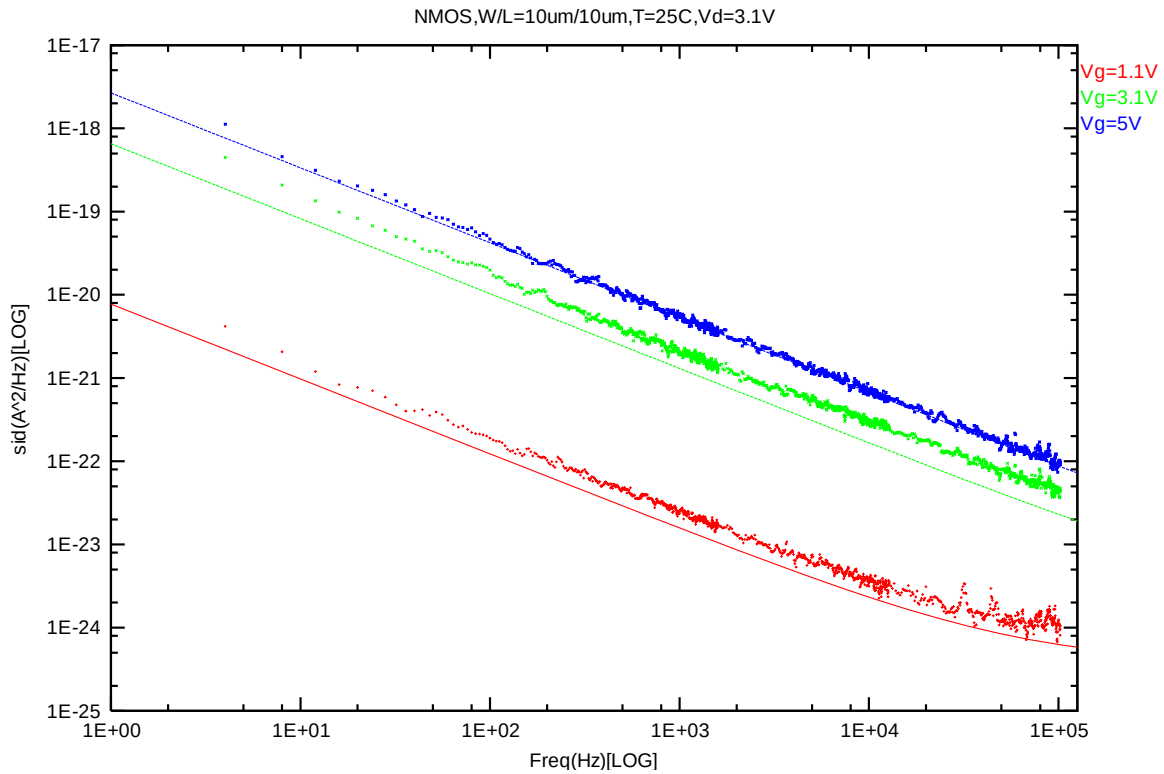


Figure 8-2 NMOS W/L=10um/10um @ Vd=3.1 V

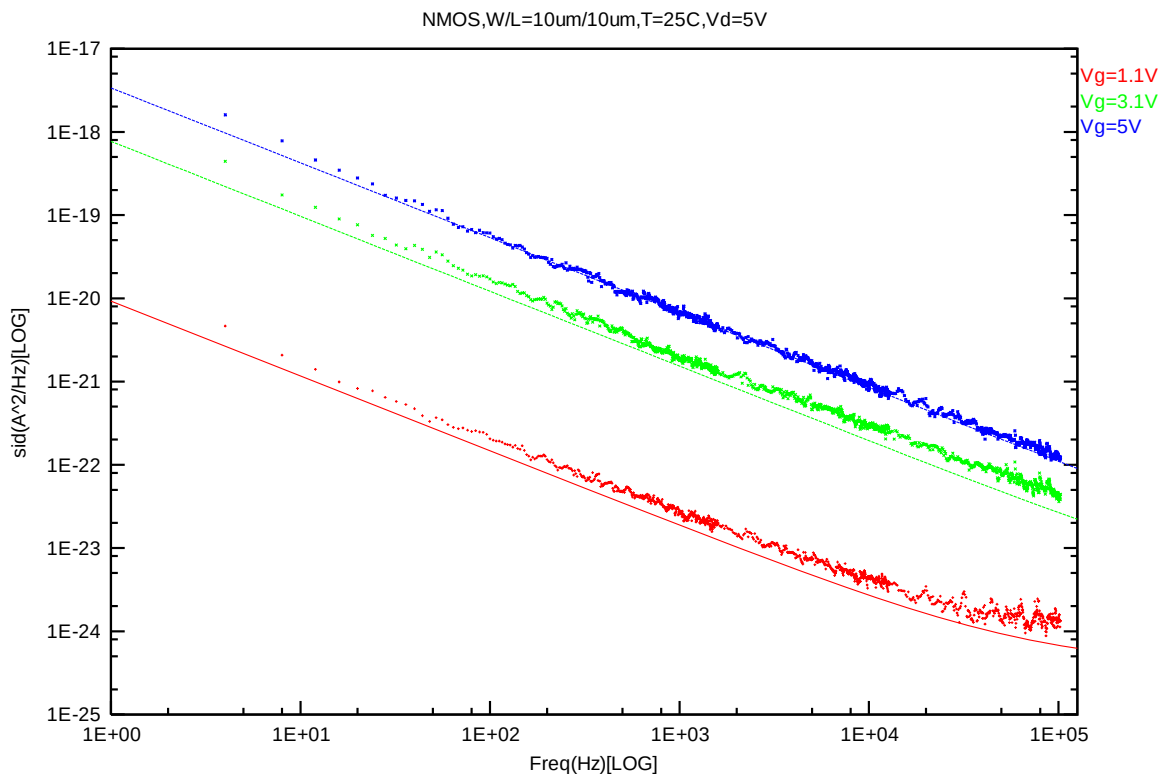


Figure 8-3 NMOS W/L=10um/10um @ Vd=5 V

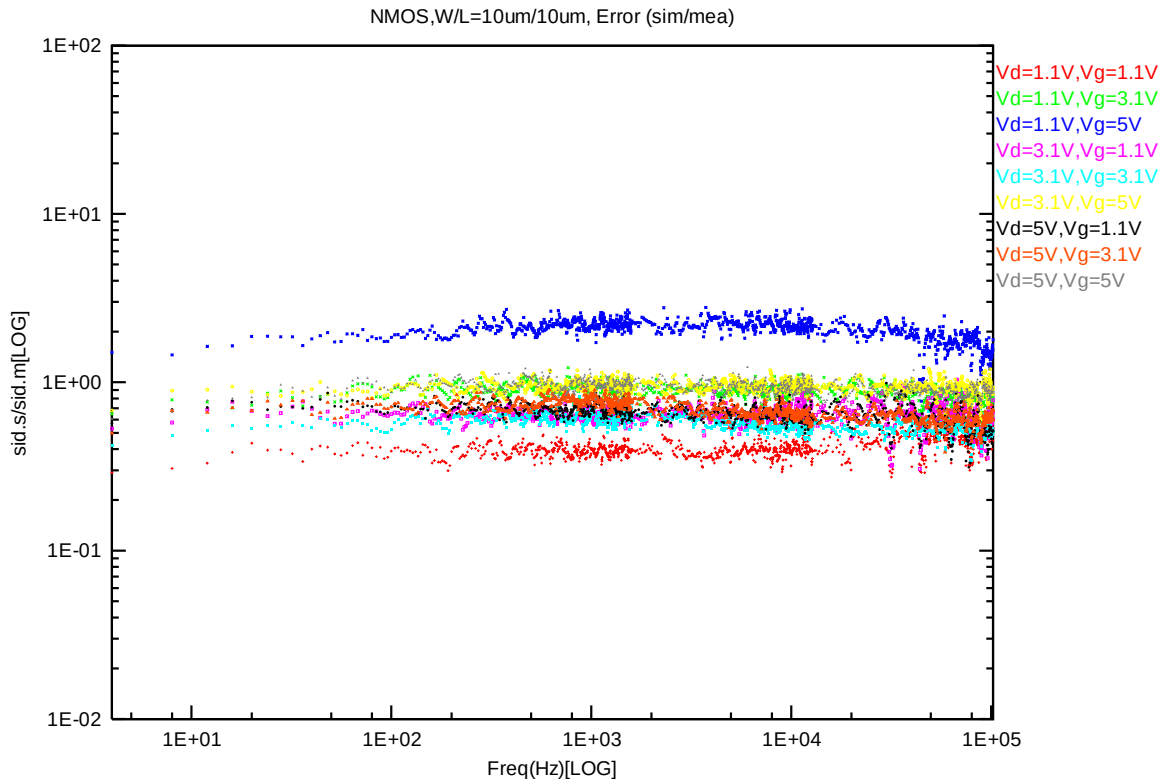


Figure 8-4 NMOS W/L=10um/10um error distribution plot

NMOS, W=10um L=1um

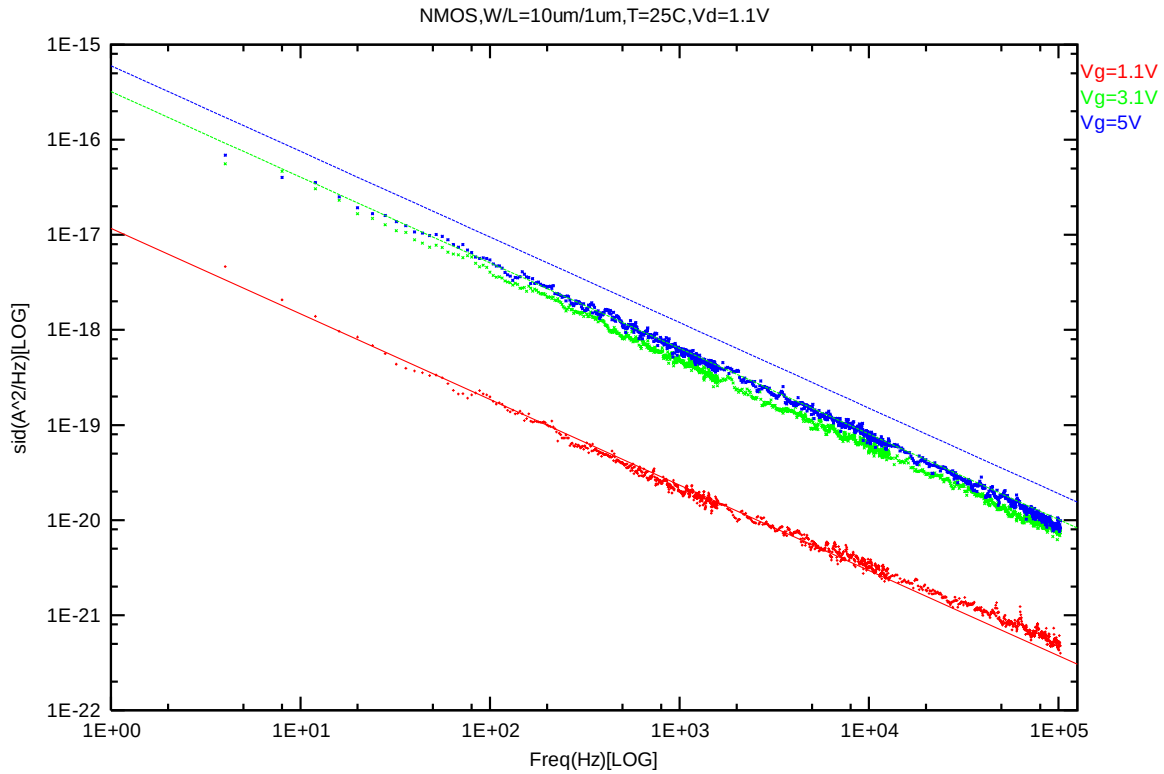


Figure 8-5 NMOS W/L=10um/1um @ Vd=1.1 V

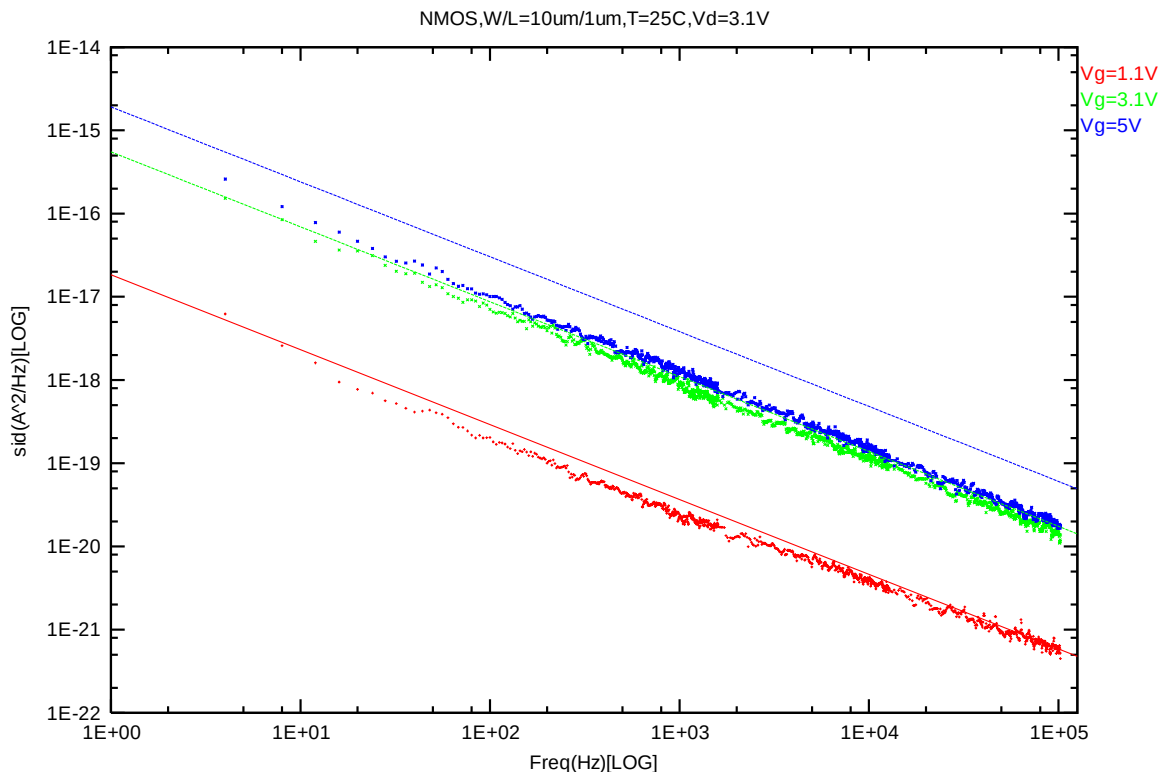


Figure 8-6 NMOS W/L=10um/1um @ Vd=3.1 V

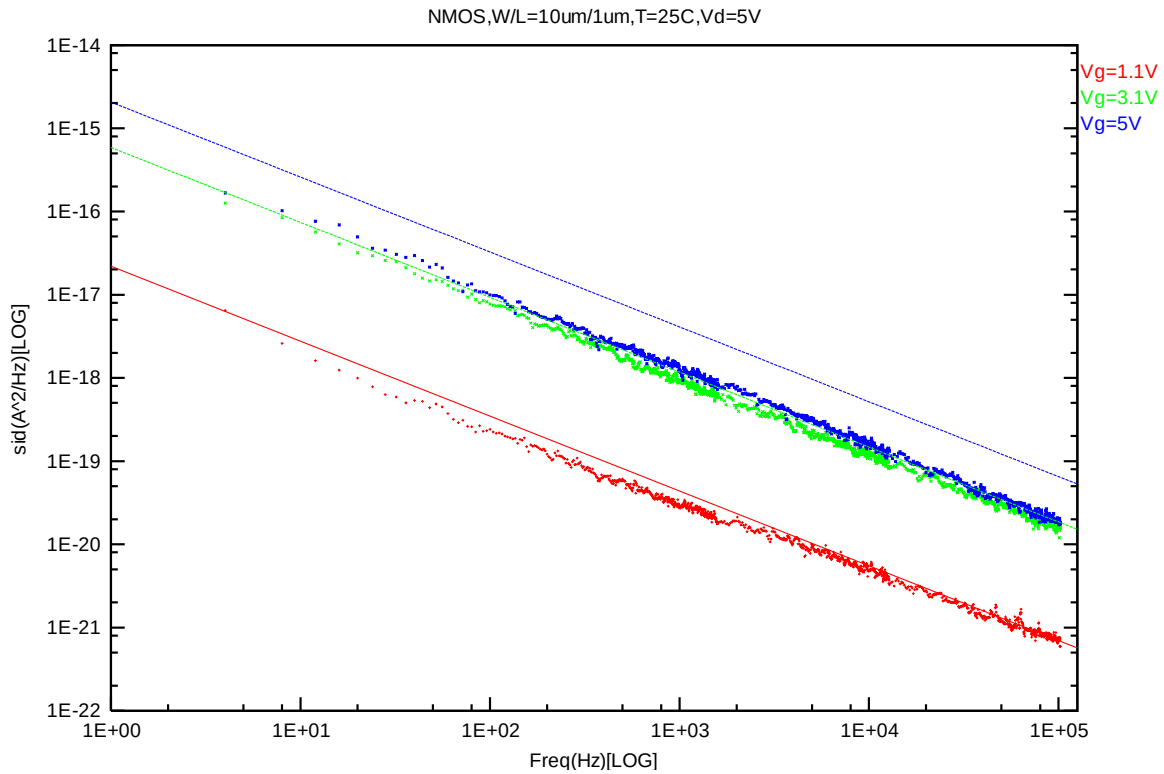


Figure 8-7 NMOS W/L=10um/1um @ Vd=5 V

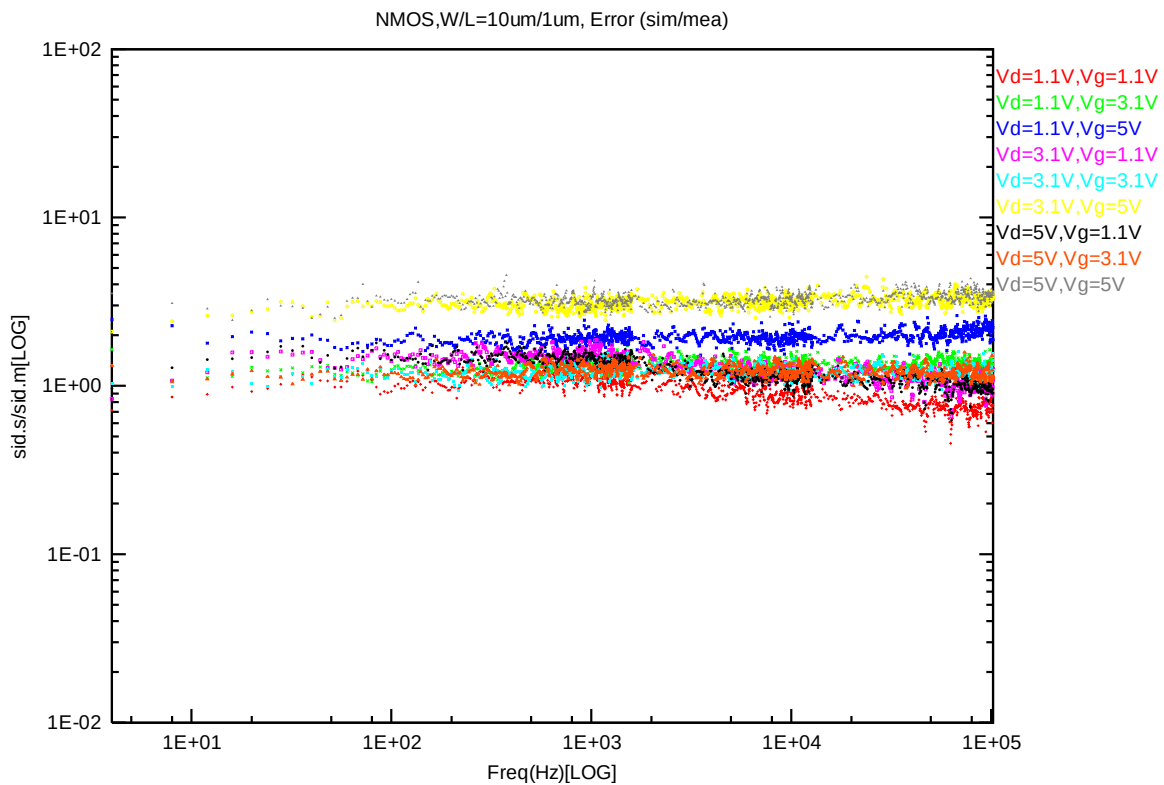


Figure 8-8 NMOS W/L=10um/1um error distribution plot

NMOS, W=10um L=0.5um

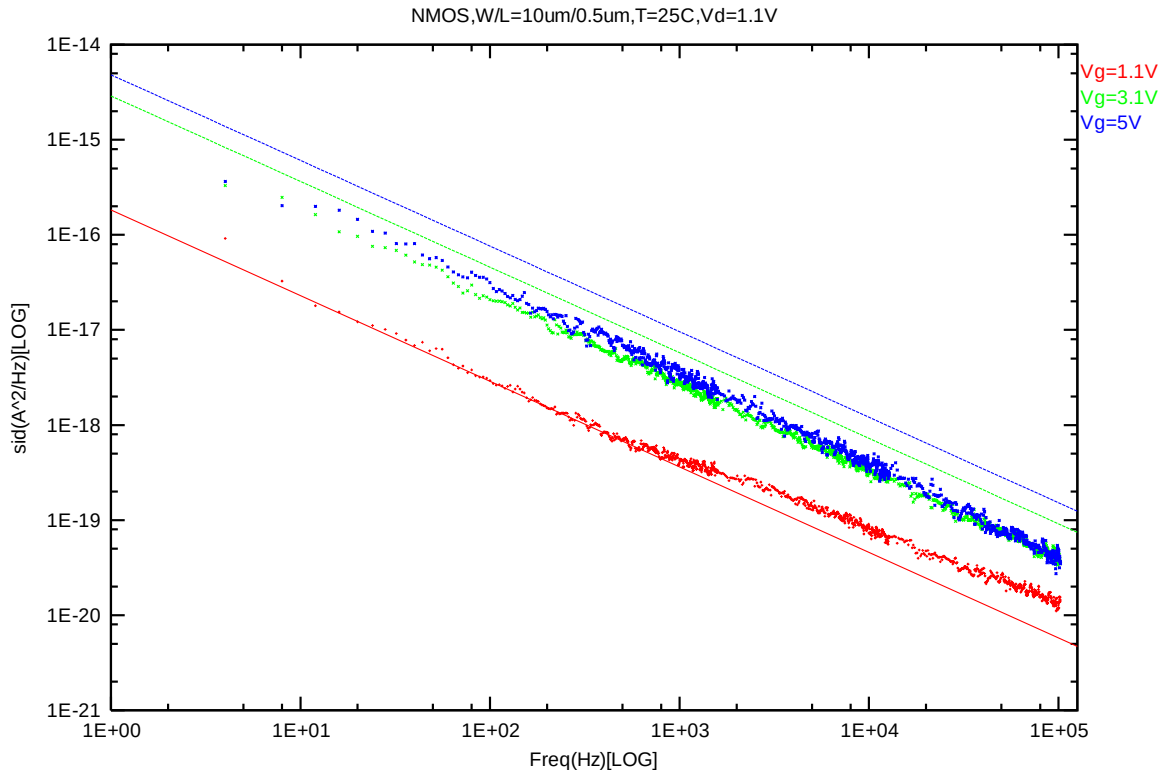


Figure 8-9 NMOS W/L=10um/0.5um @ Vd=1.1 V

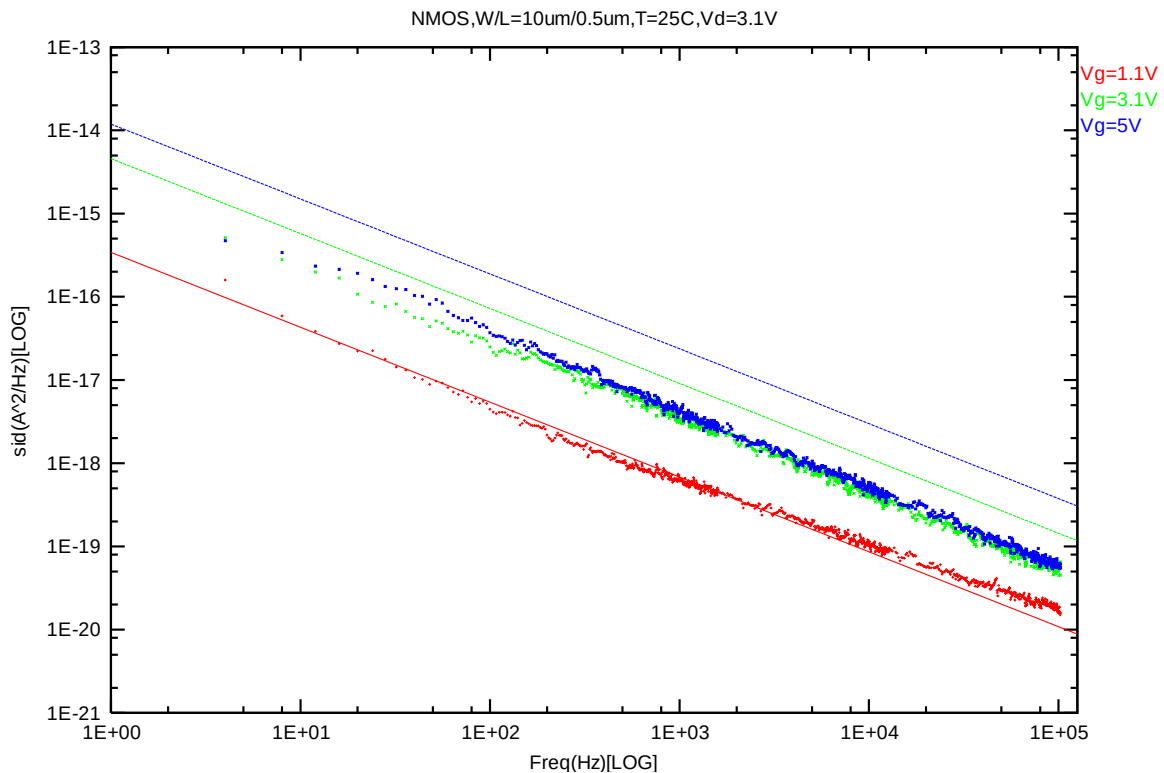


Figure 8-10 NMOS W/L=10um/0.5um @ Vd=3.1 V

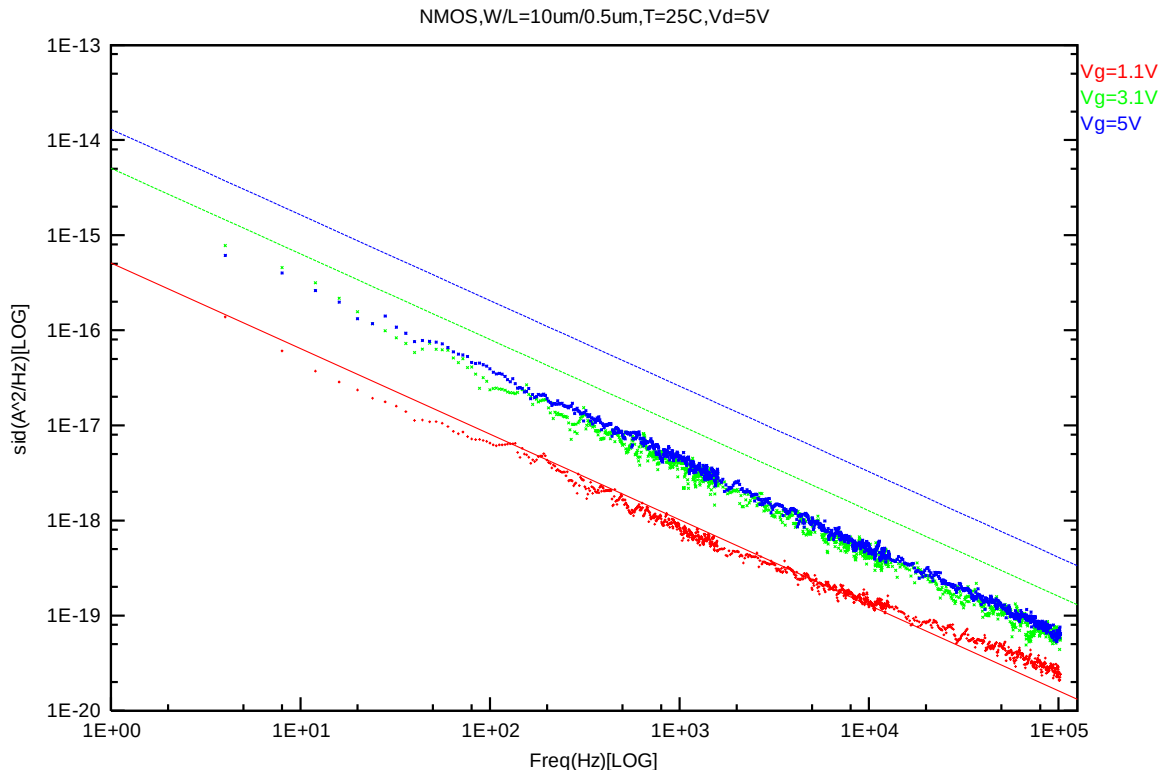


Figure 8-11 NMOS W/L=10um/0.5um @ Vd=5 V

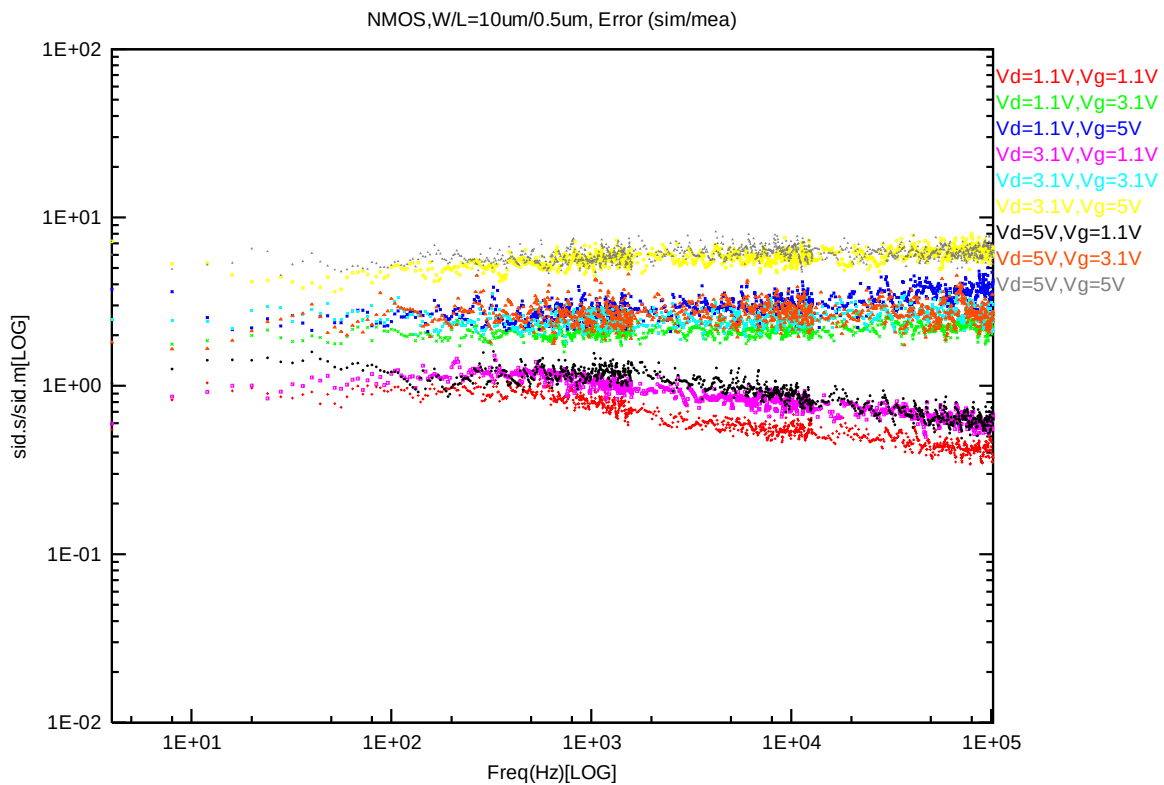


Figure 8-12 NMOS W/L=10um/0.5um error distribution plot

PMOS, W=10um L=10um

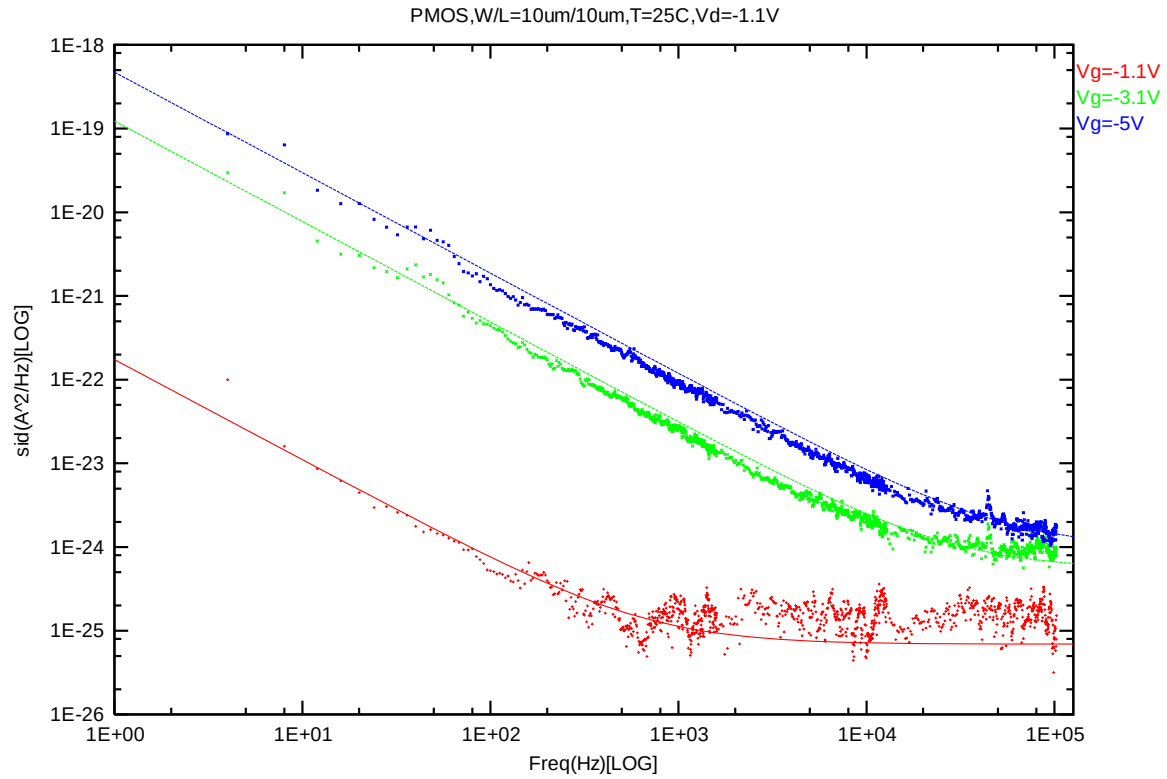


Figure 8-13 PMOS W/L=10um/10um @ Vd=-1.1 V

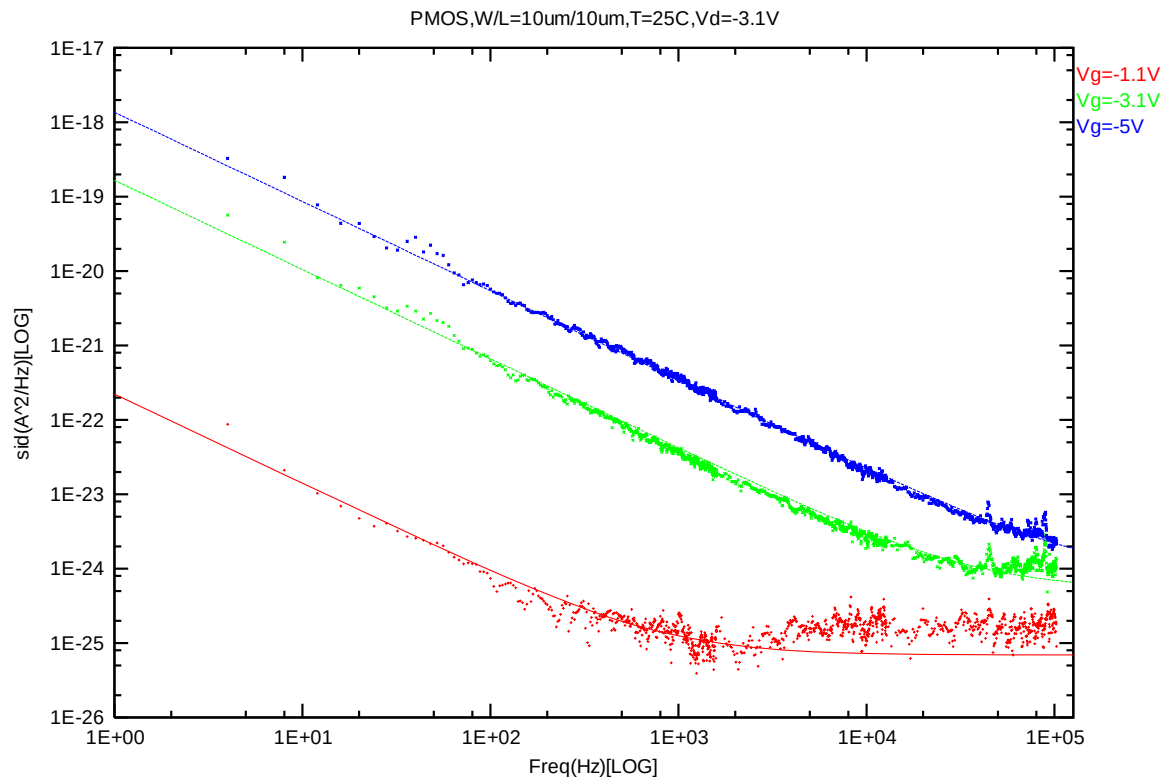
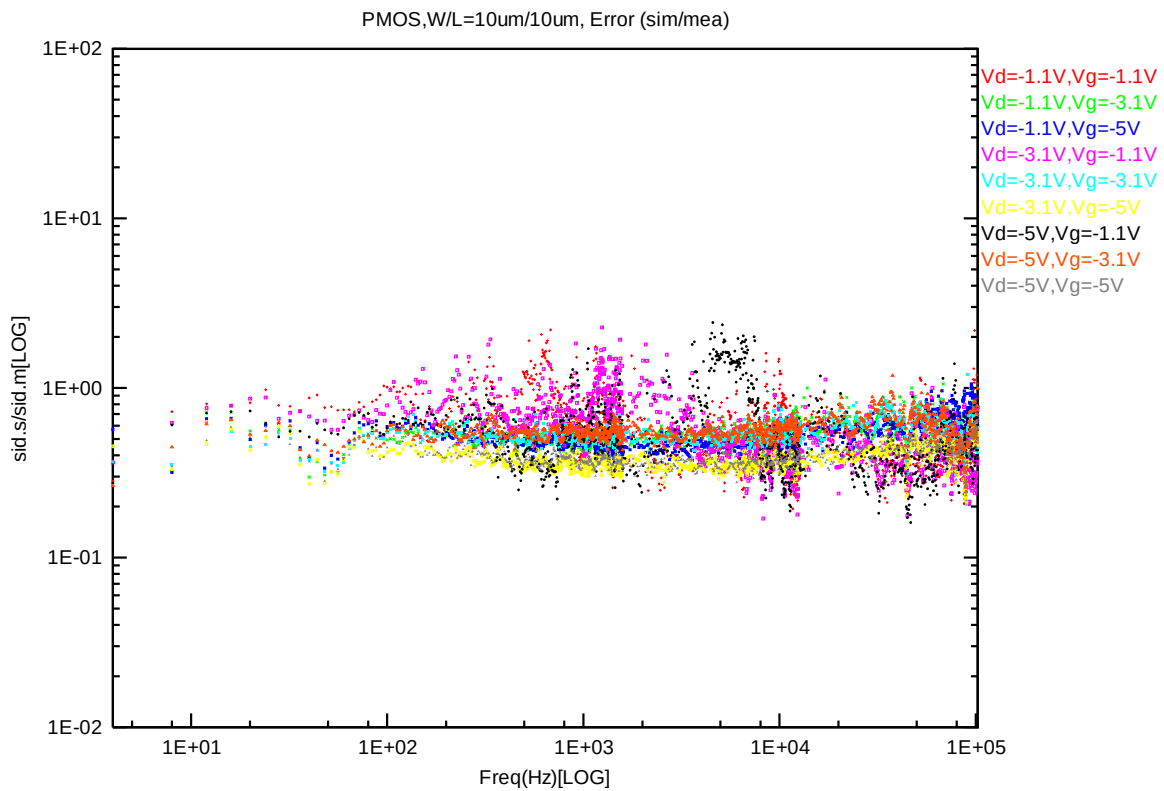
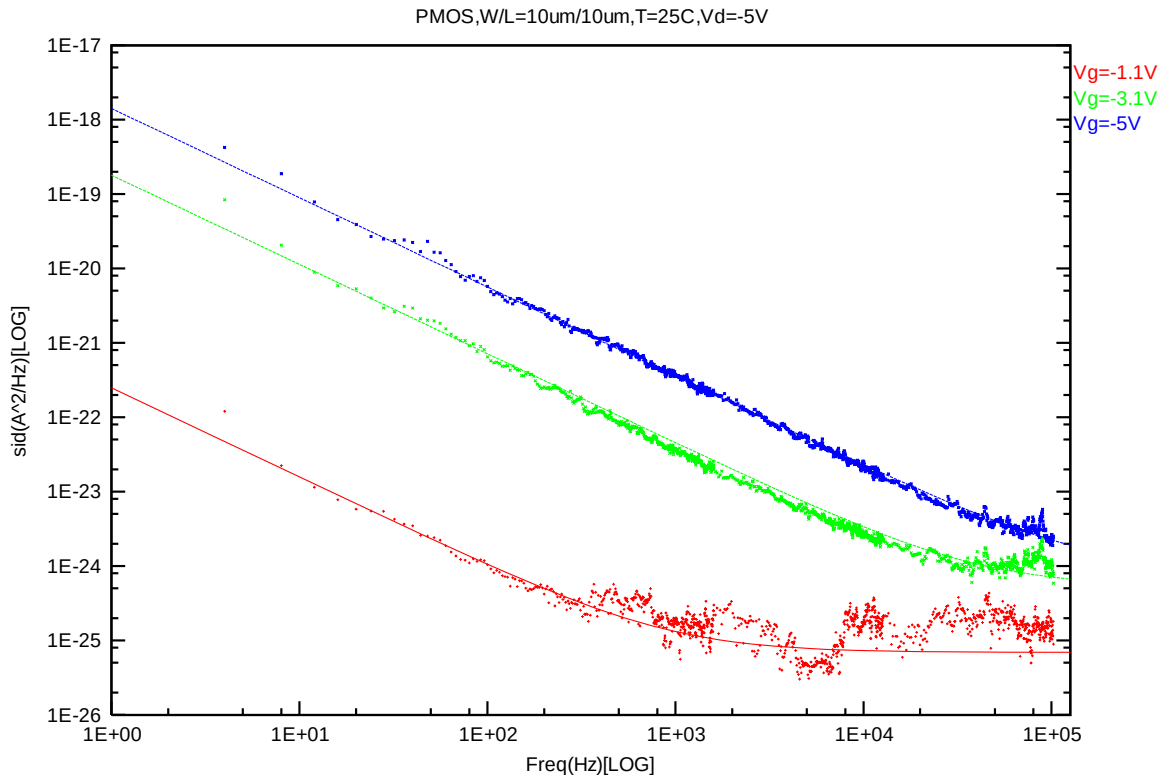


Figure 8-14 PMOS W/L=10um/10um @ Vd=-3.1 V



PMOS, W=10um L=1um

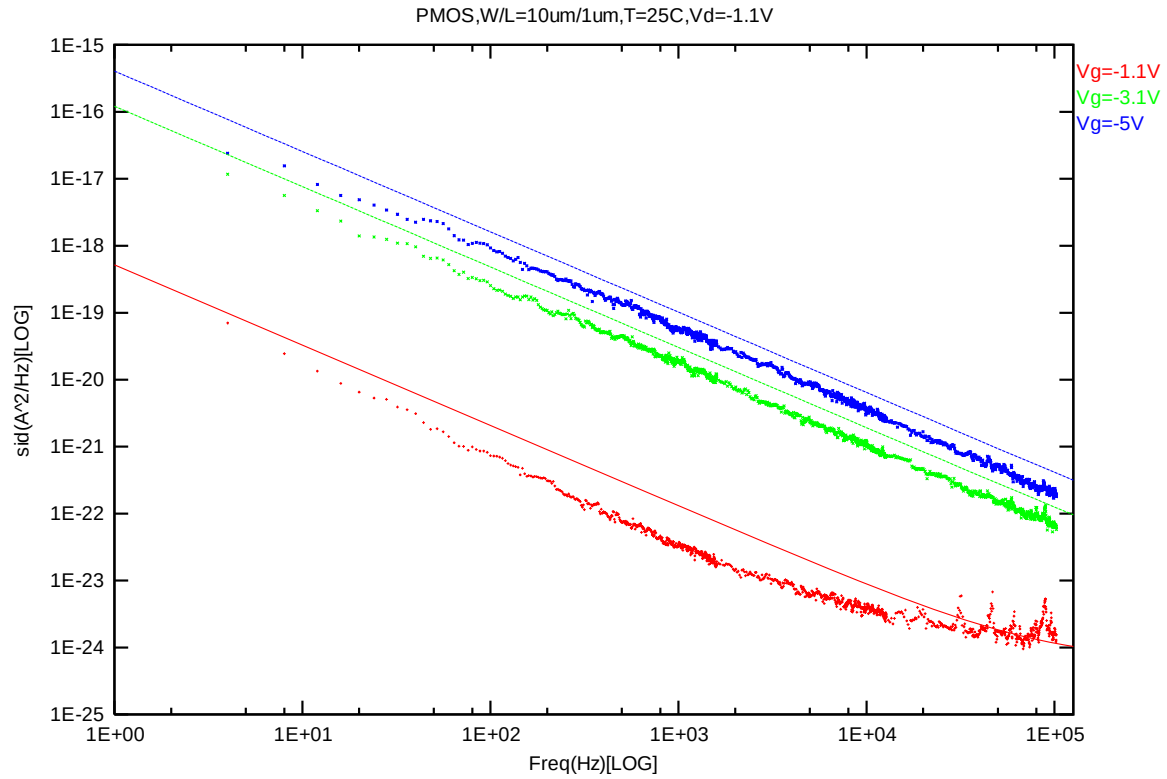


Figure 8-17 PMOS W/L=10um/1um @ Vd=-1.1 V

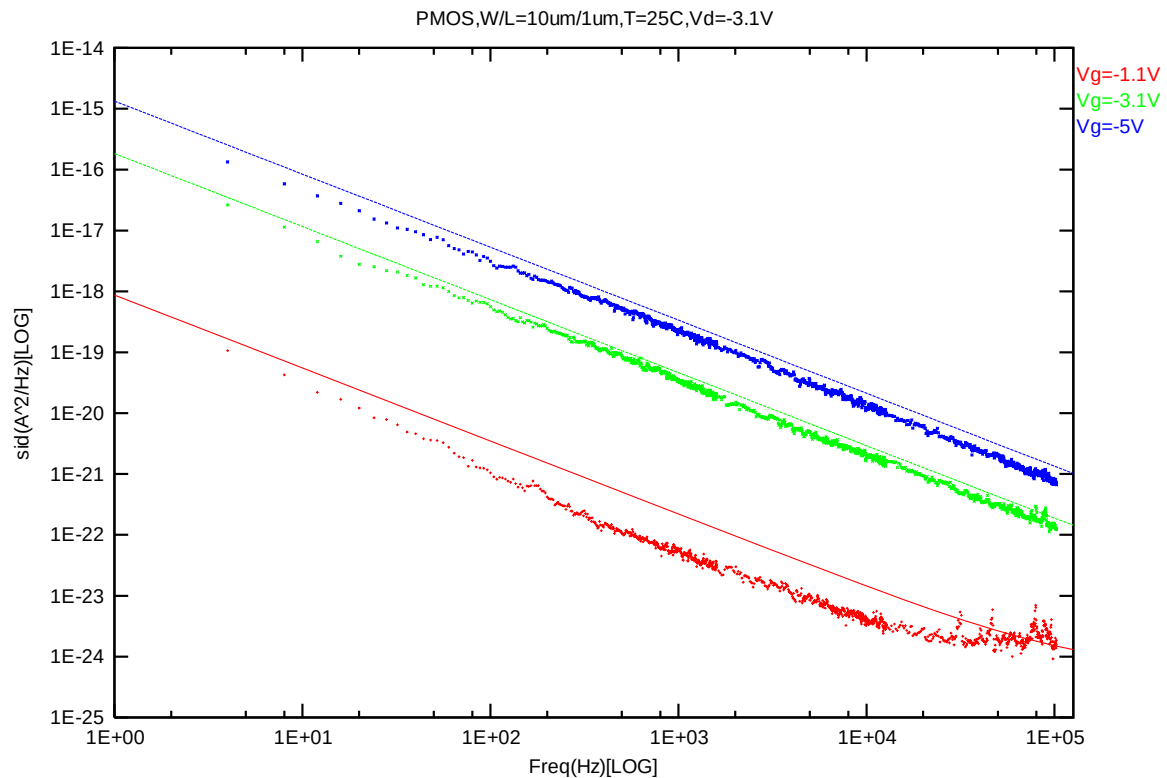


Figure 8-18 PMOS W/L=10um/1um @ Vd=-3.1 V

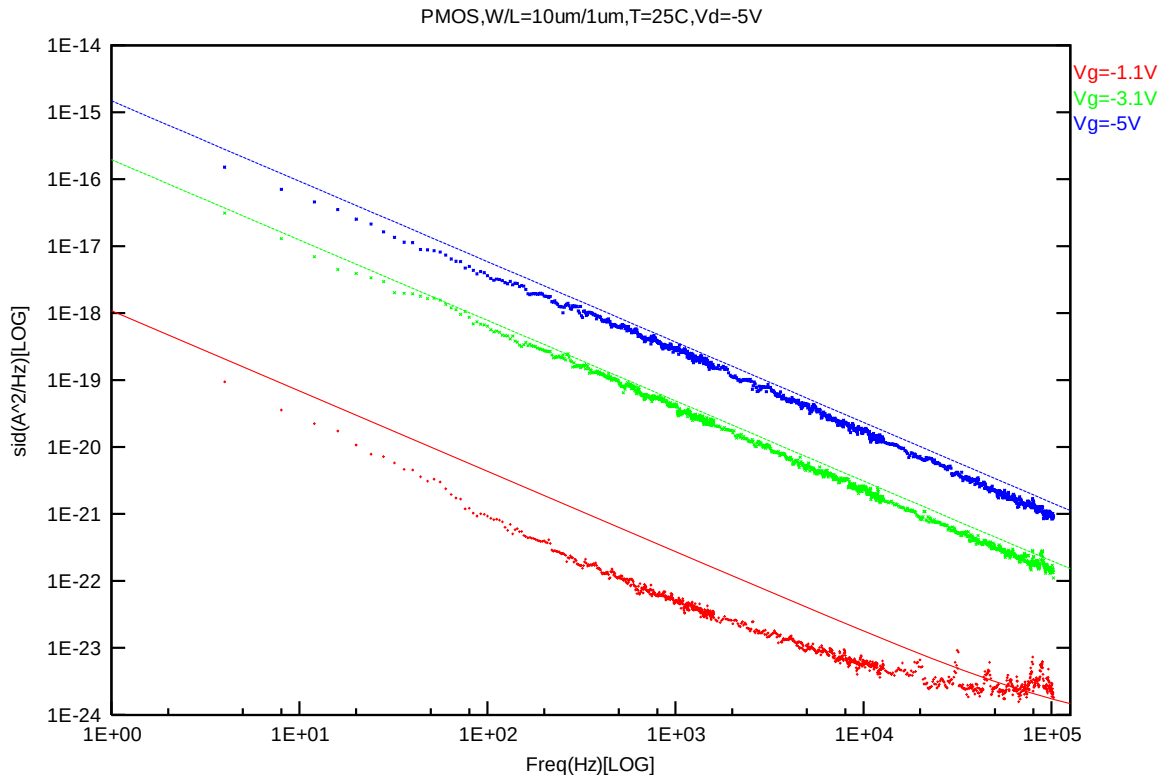


Figure 8-19 PMOS W/L=10um/1um @ Vd=-5 V

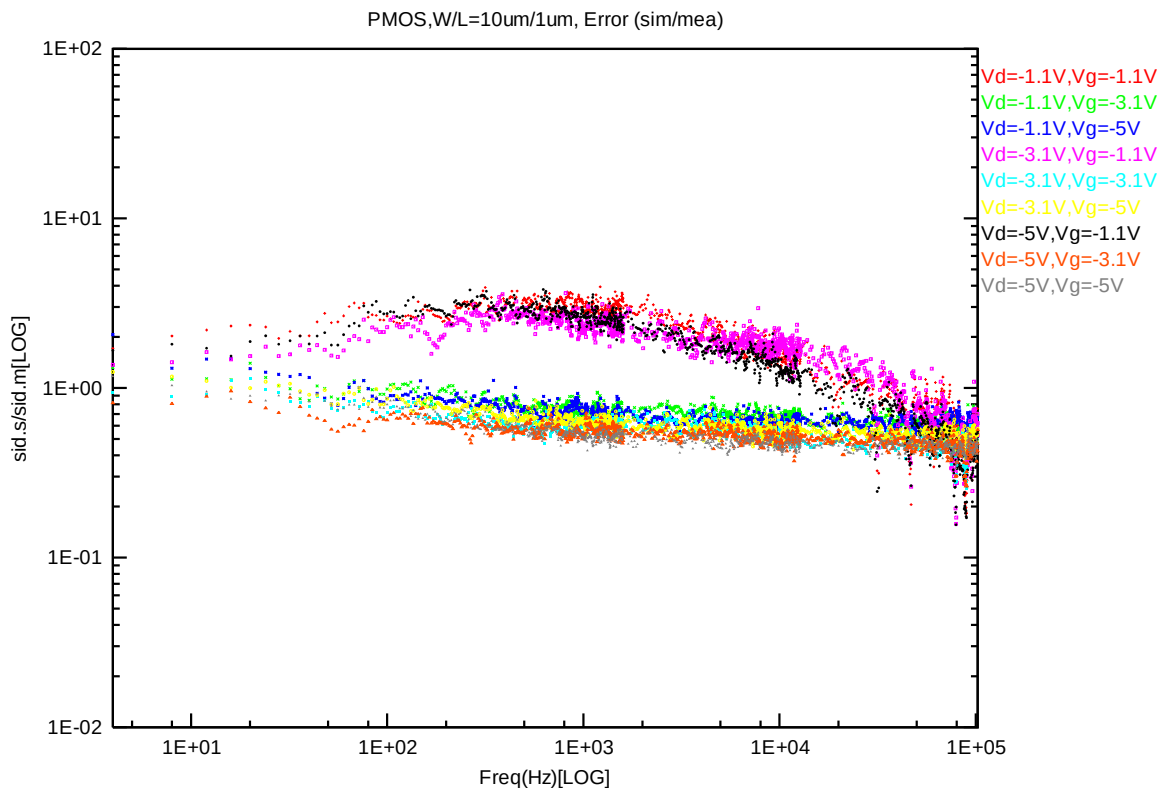


Figure 8-20 PMOS W/L=10um/1um error distribution plot

PMOS, W=10um L=0.5um

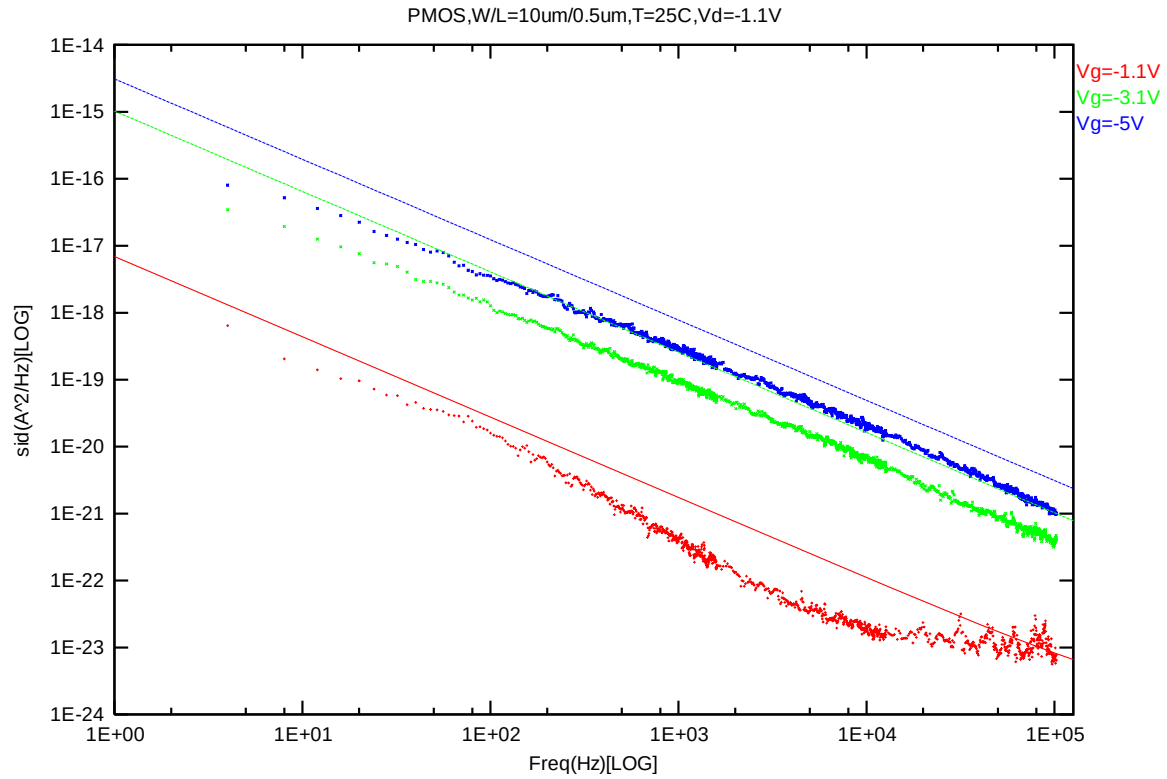


Figure 8-21 PMOS W/L=10um/0.5um @ Vd=-1.1 V

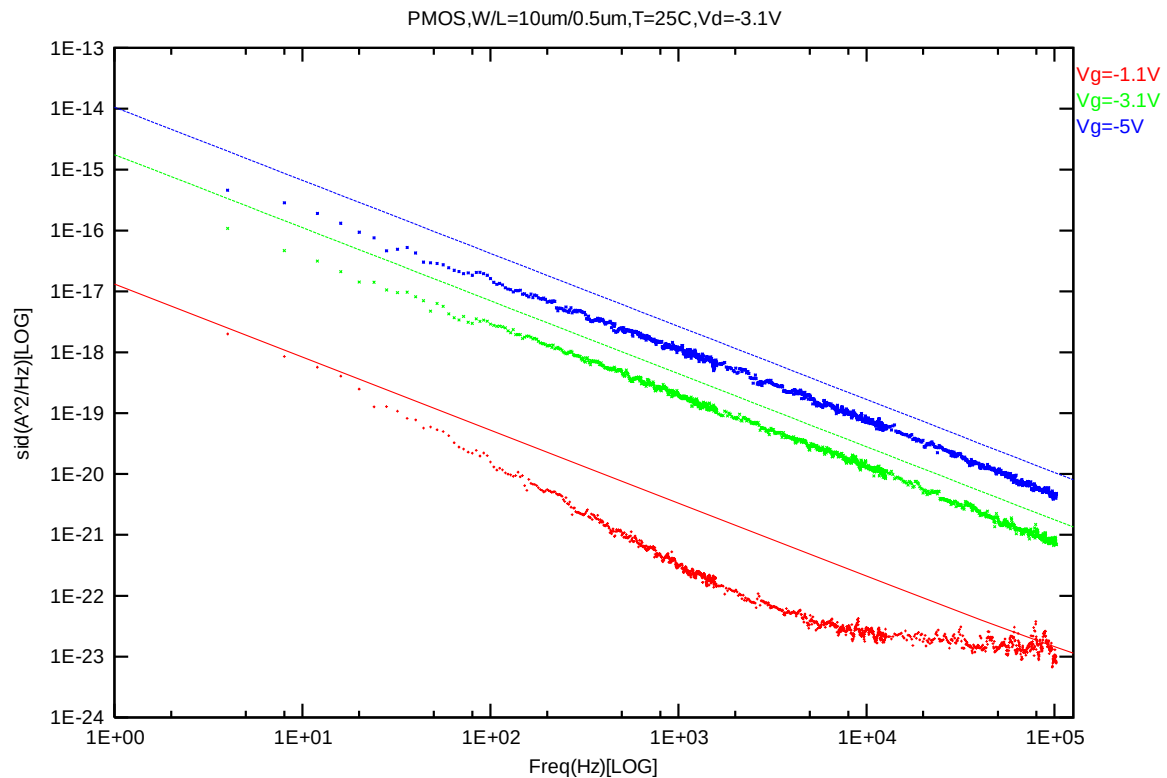


Figure 8-22 PMOS W/L=10um/0.5um @ Vd=-3.1 V

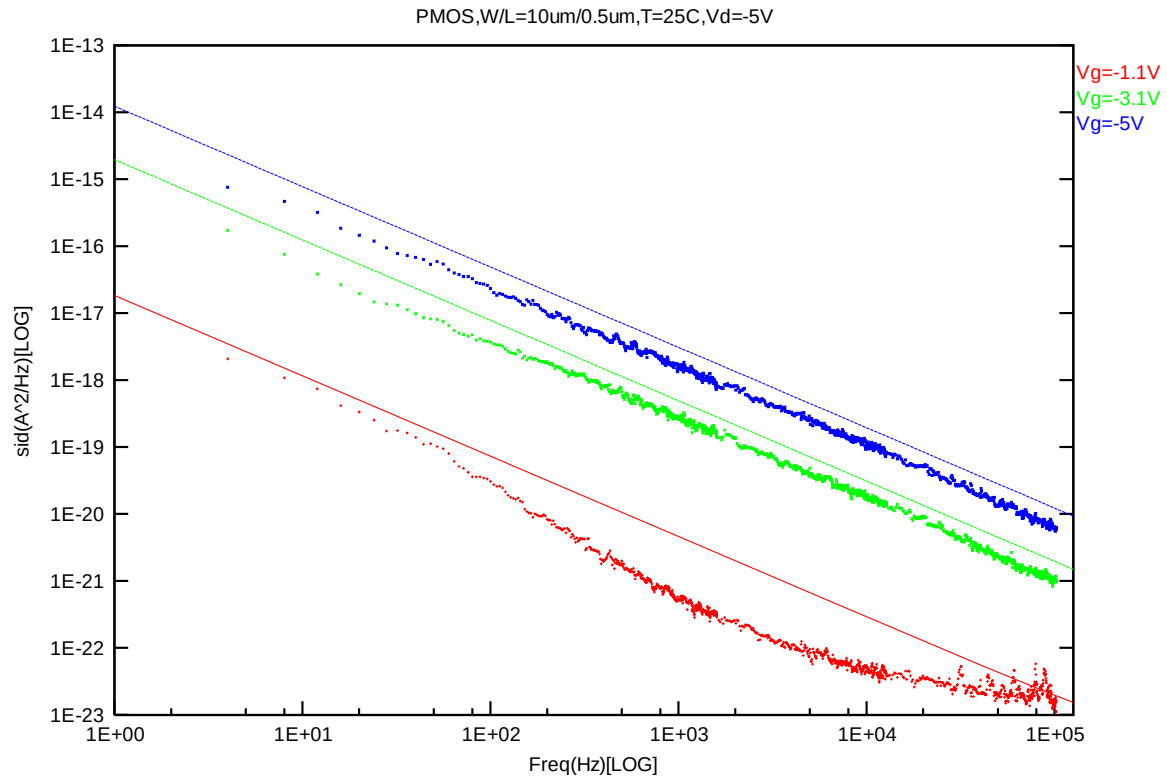


Figure 8-23 PMOS W/L=10um/0.5um @ Vd=-5 V

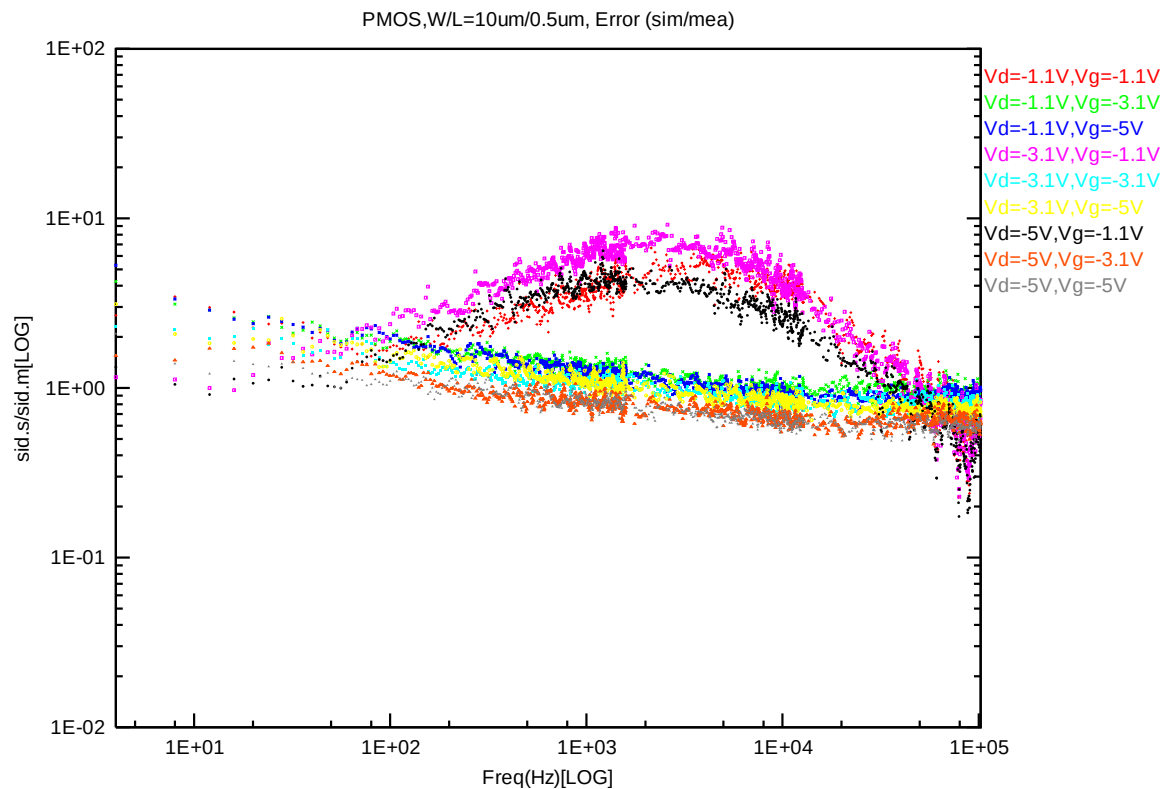


Figure 8-24 PMOS W/L=10um/0.5um error distribution plot

8.2 Trend Plots at Frequency=100Hz

A. NMOS

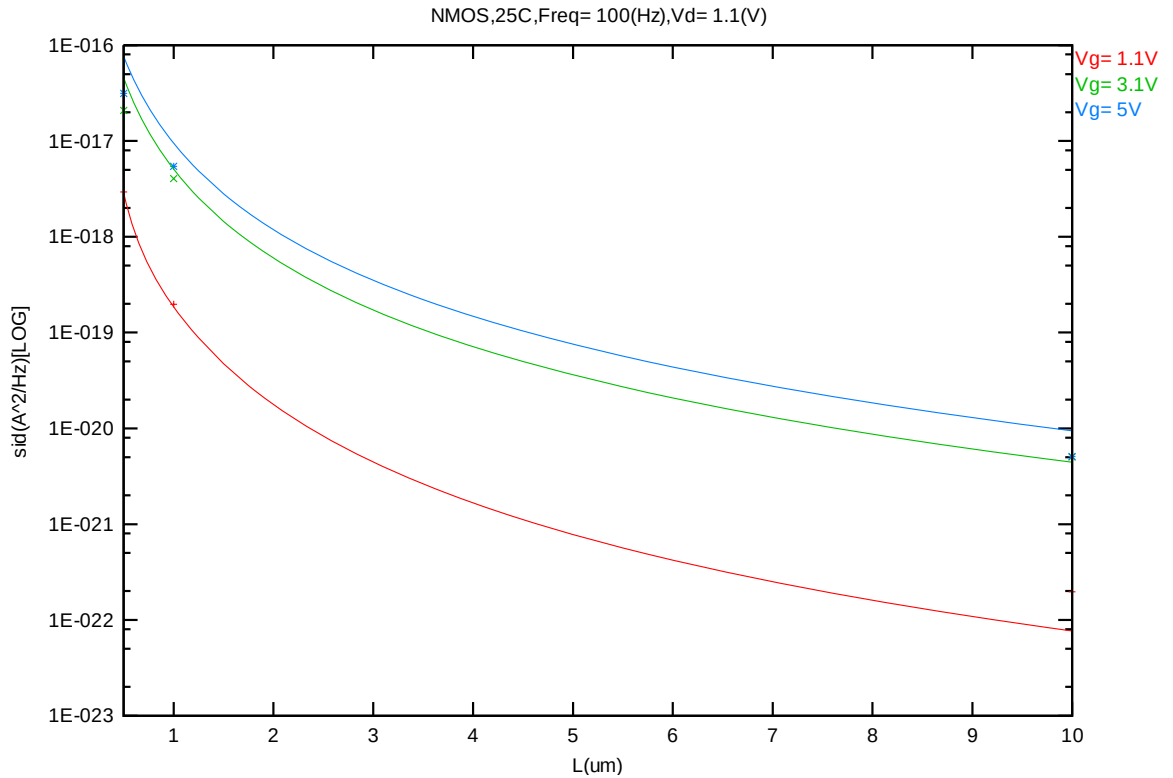


Figure 8-25 Noise vs. Length for NMOS @ Vd=1.1 V

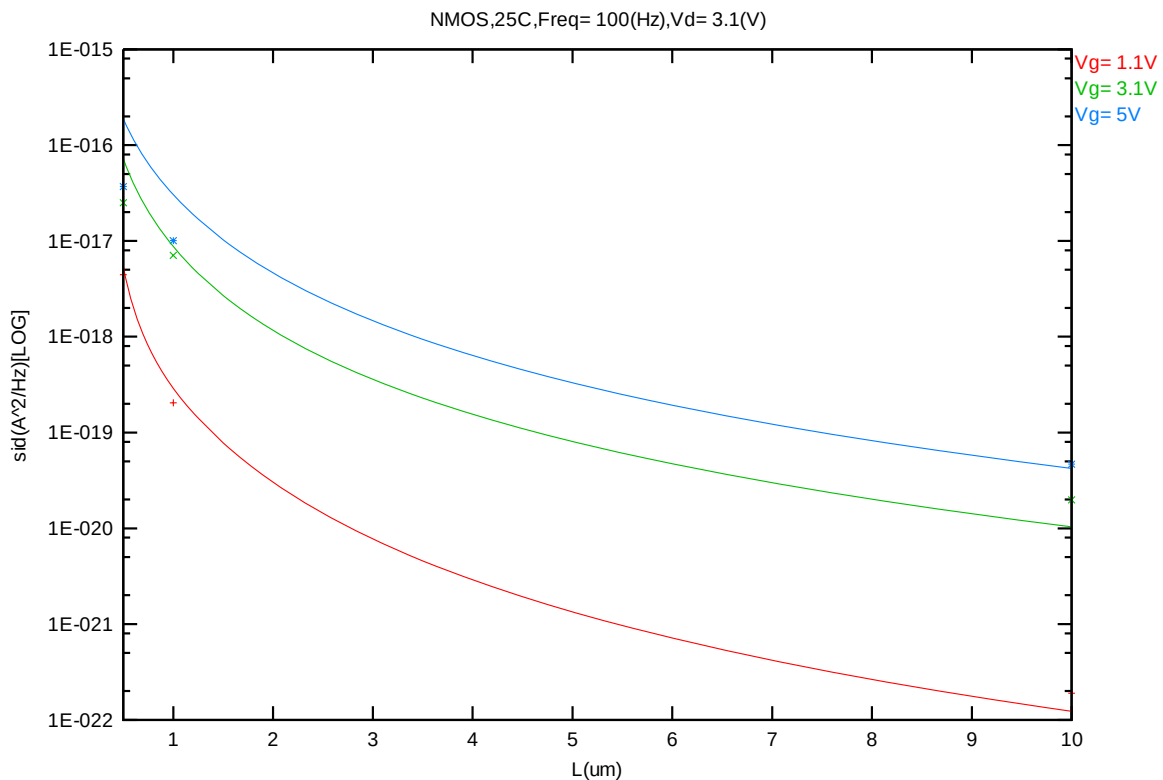


Figure 8-26 Noise vs. Length for NMOS @ Vd=3.1 V

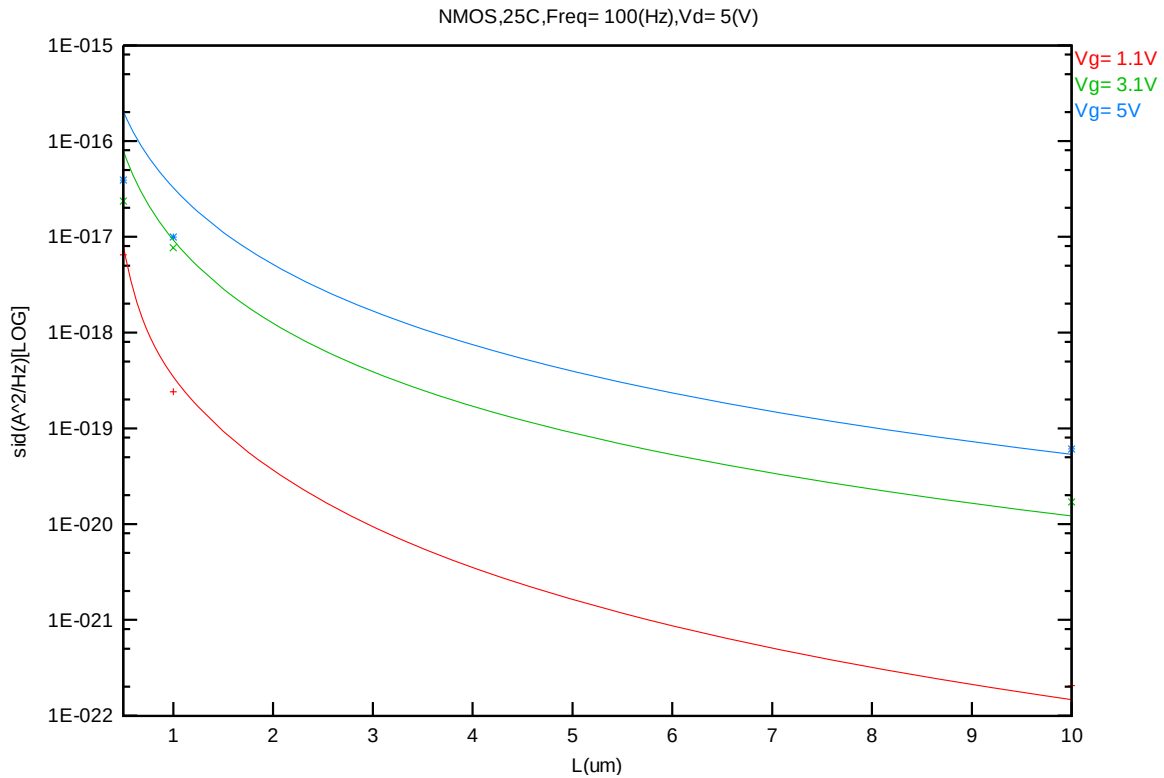


Figure 8-27 Noise vs. Length for NMOS @ Vd=5 V

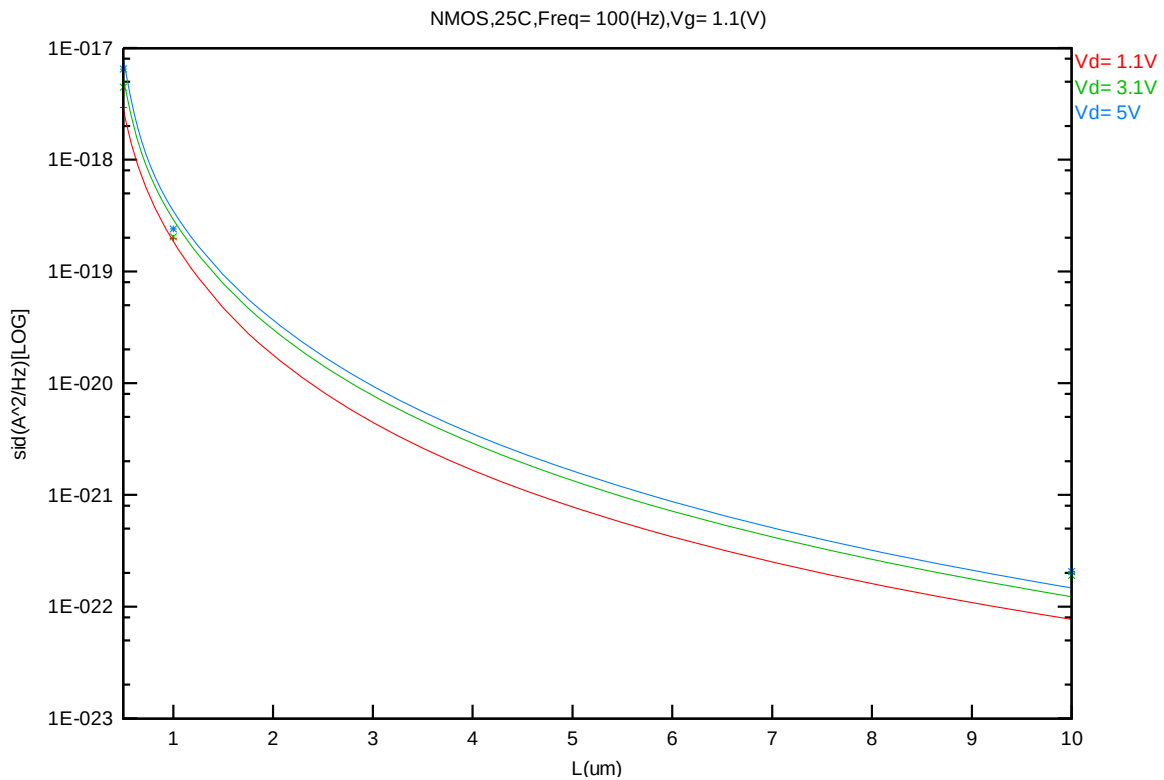


Figure 8-28 Noise vs. Length for NMOS @ Vg=1.1 V

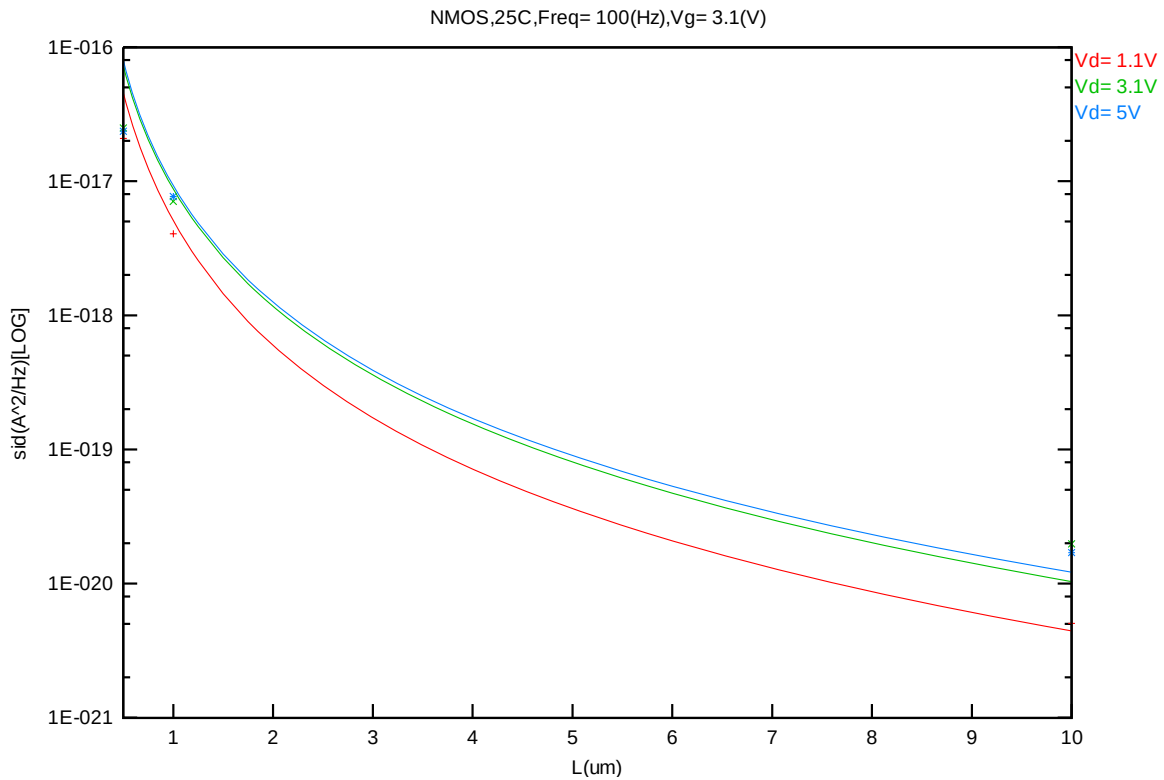


Figure 8-29 Noise vs. Length for NMOS @ Vg=3.1 V

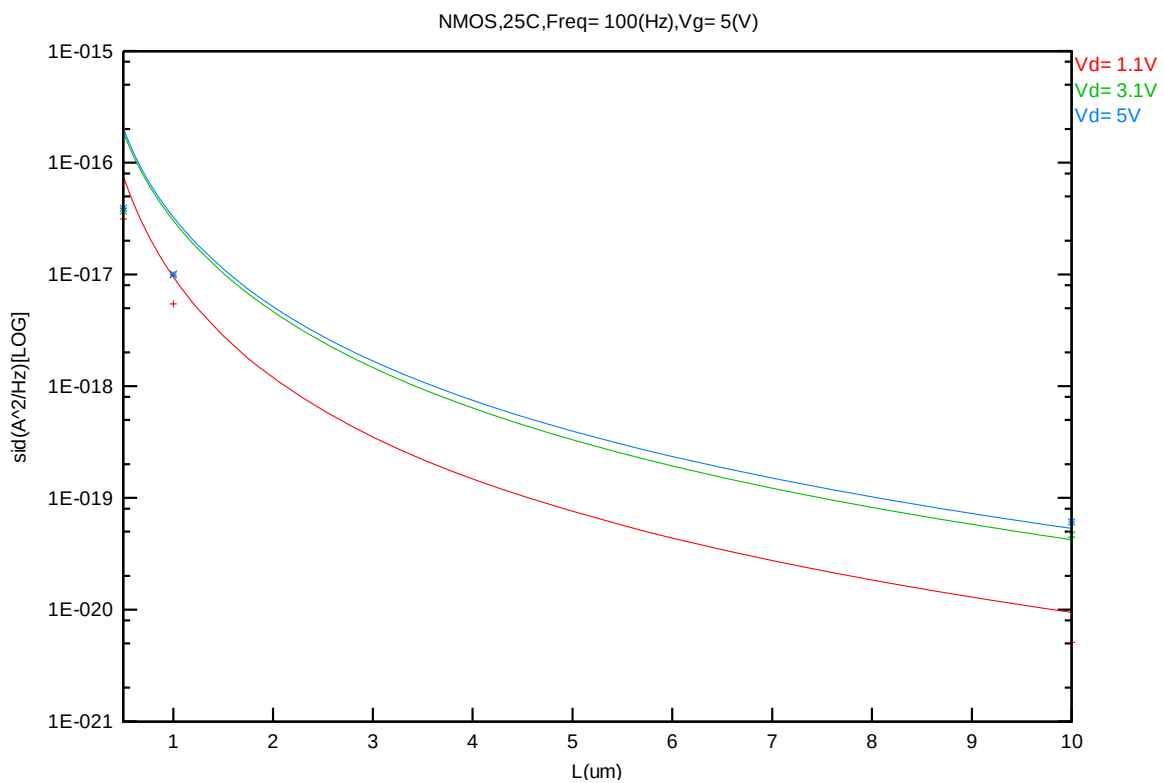


Figure 8-30 Noise vs. Length for NMOS @ Vg=5 V

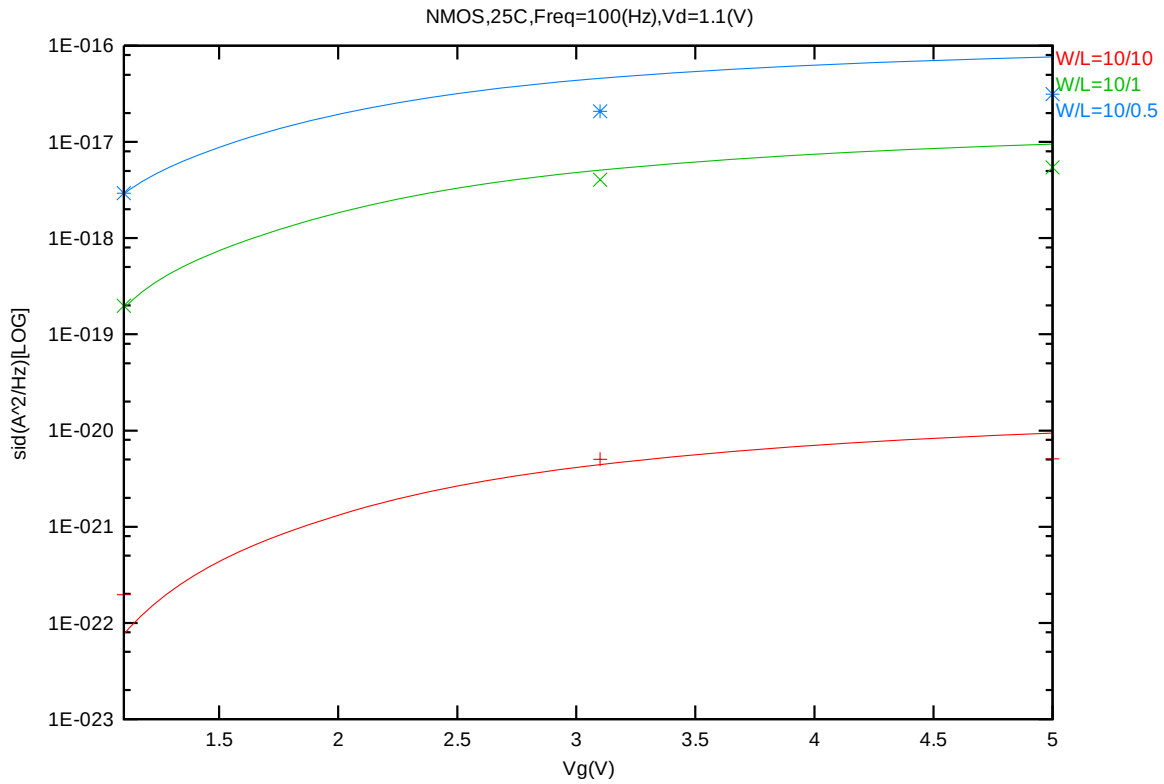


Figure 8-31 Noise vs. Vg for NMOS @ Vd=1.1 V

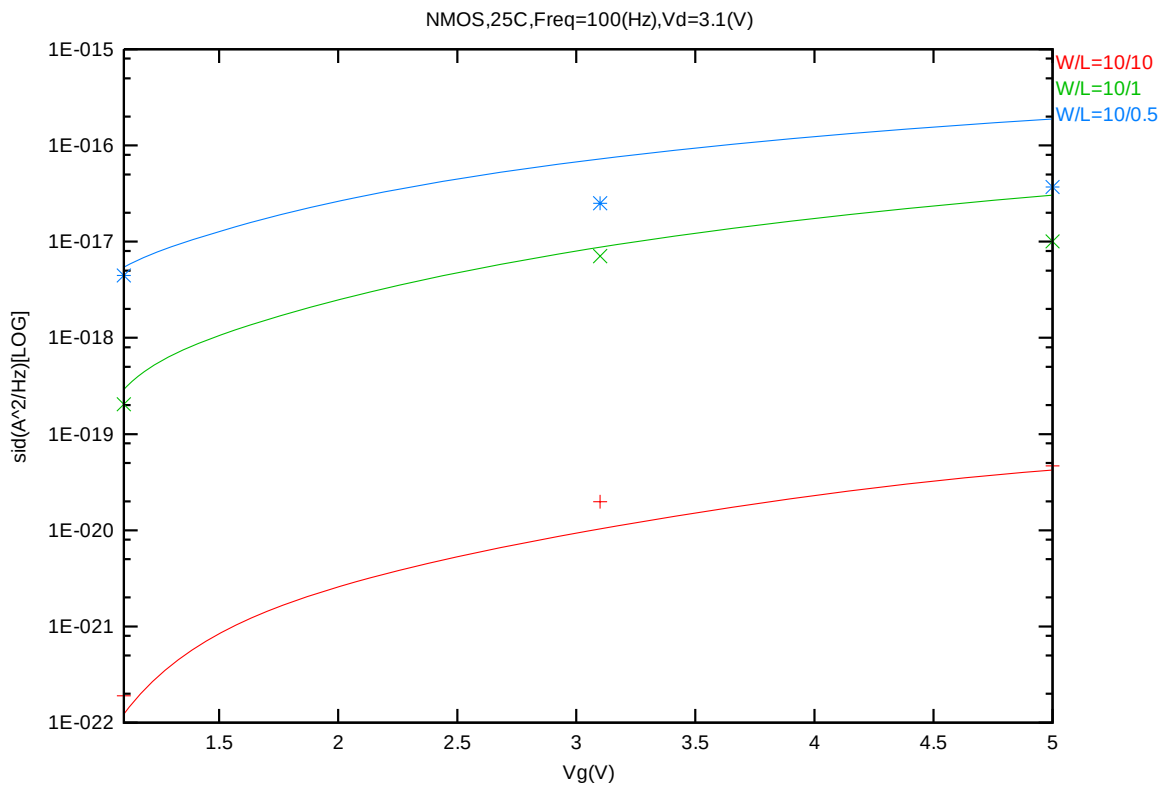


Figure 8-32 Noise vs. Vg for NMOS @ Vd=3.1 V

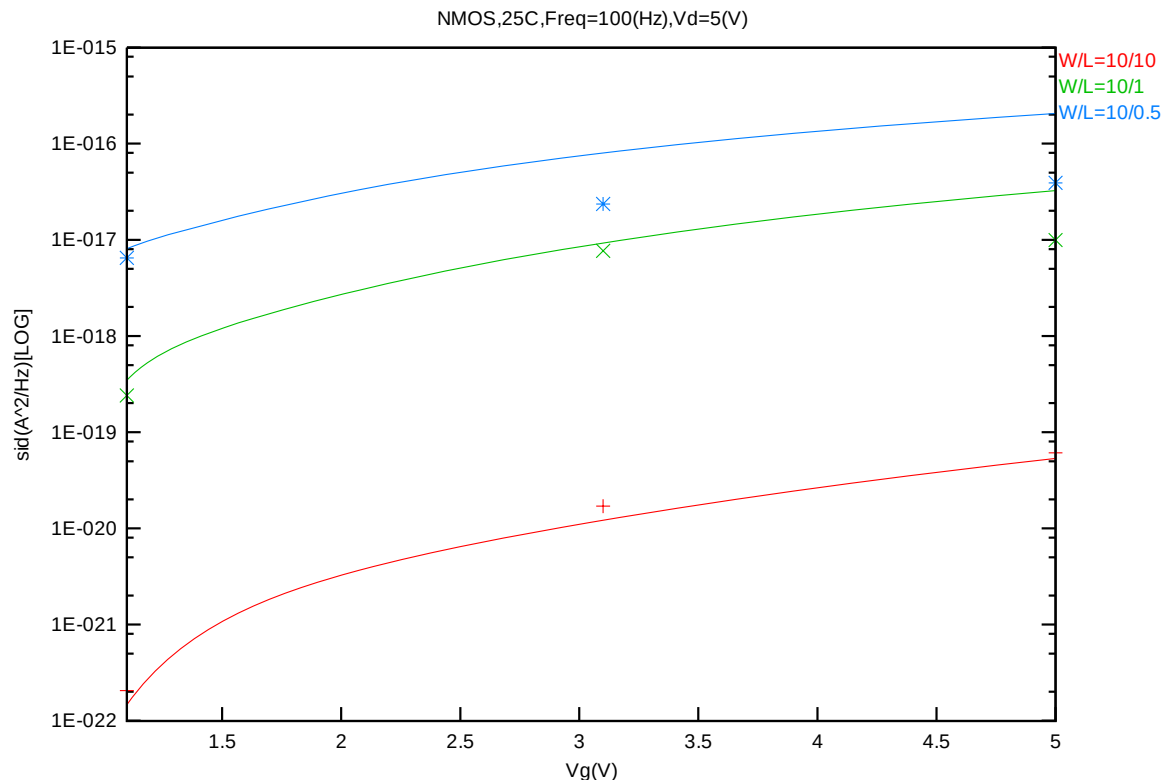


Figure 8-33 Noise vs. V_g for NMOS @ $V_d=5$ V

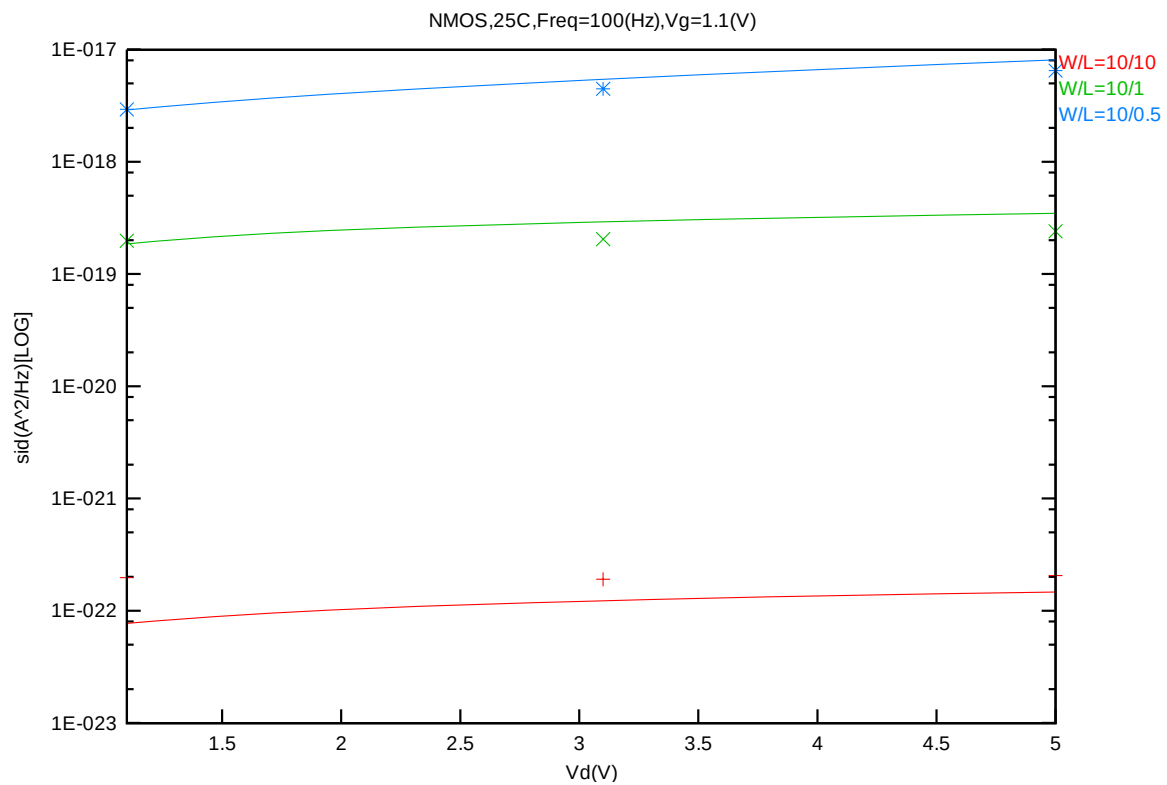


Figure 8-34 Noise vs. V_d for NMOS @ $V_g=1.1$ V

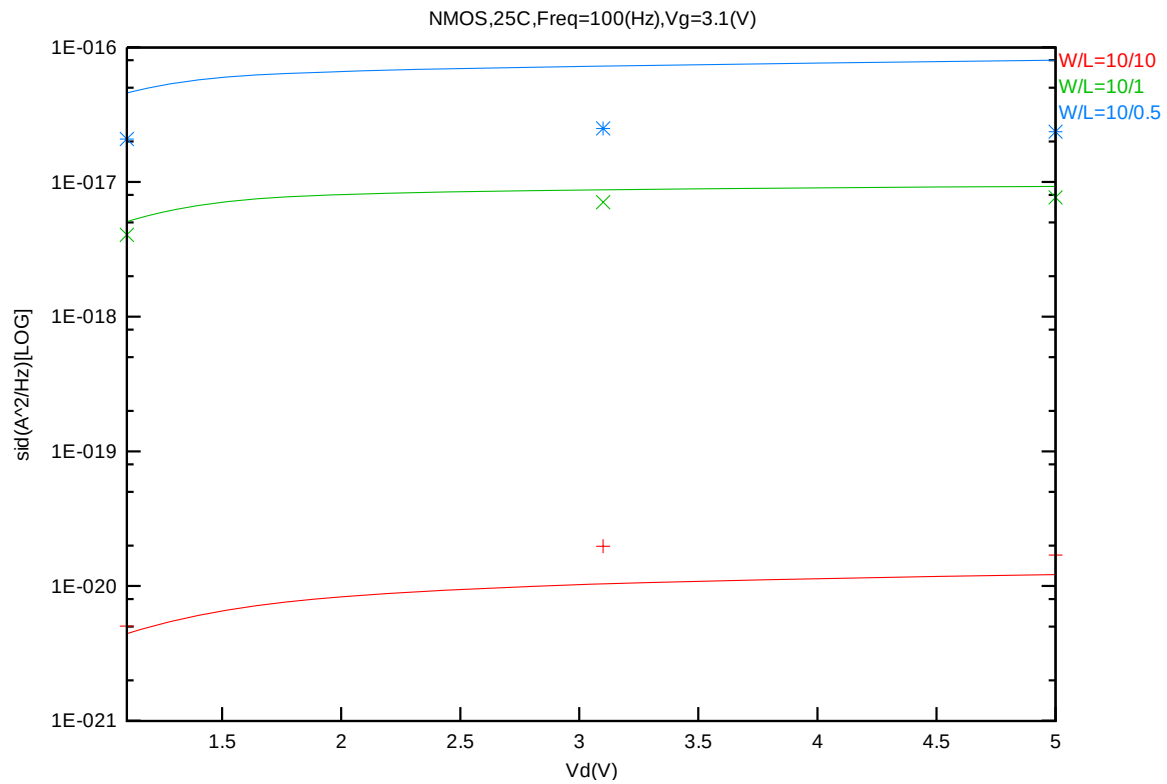


Figure 8-35 Noise vs. Vd for NMOS @ Vg=3.1 V

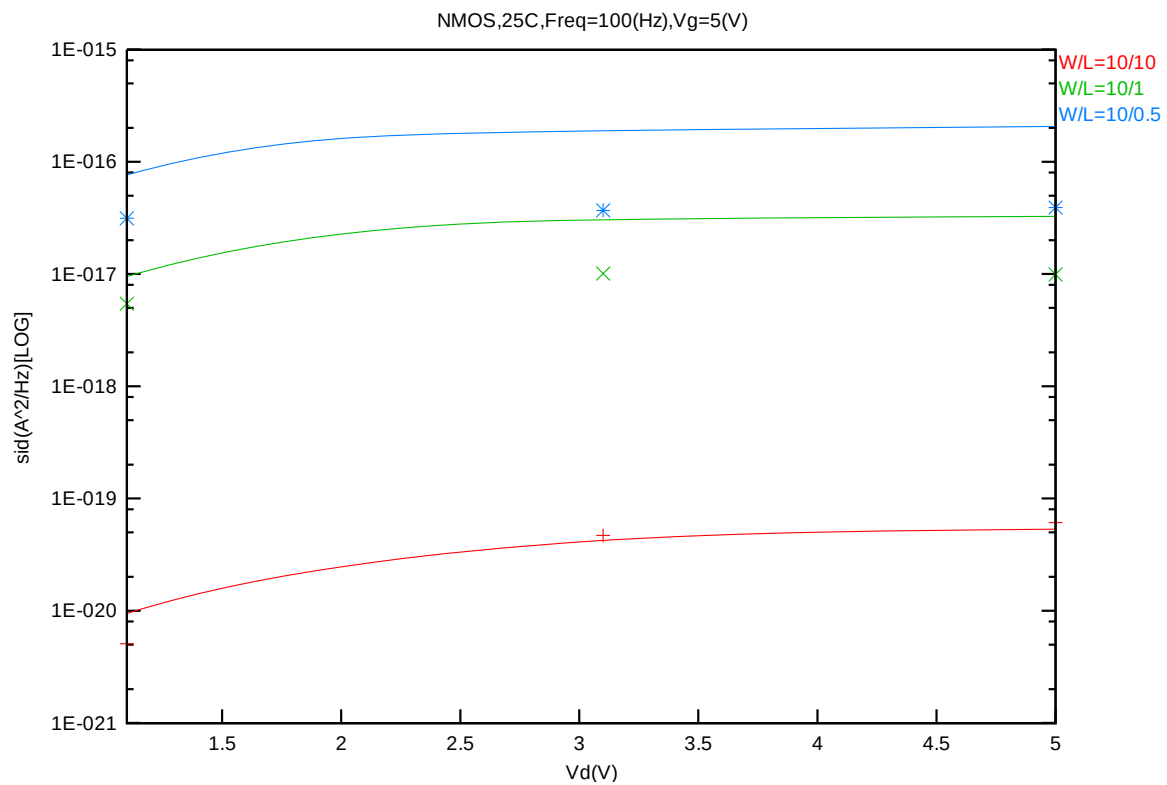


Figure 8-36 Noise vs. Vd for NMOS @ Vg=5 V

B. PMOS

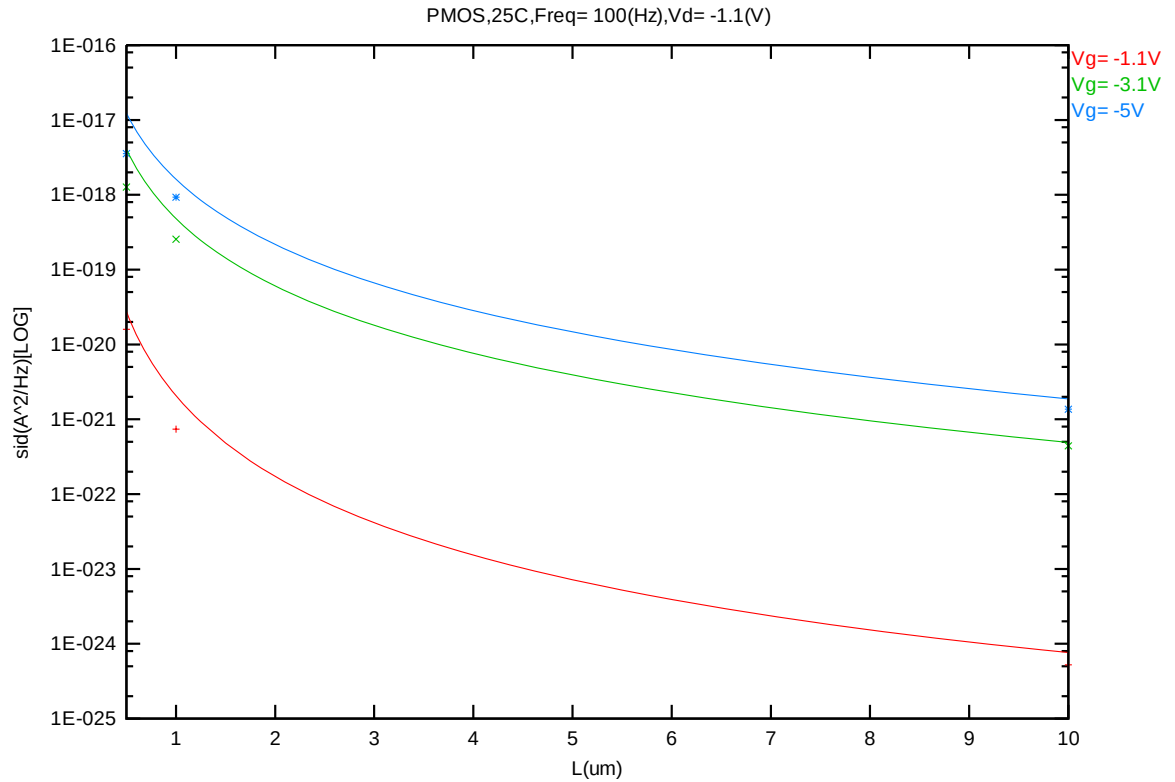


Figure 8-37 Noise vs. Length for PMOS @ Vd=-1.1 V

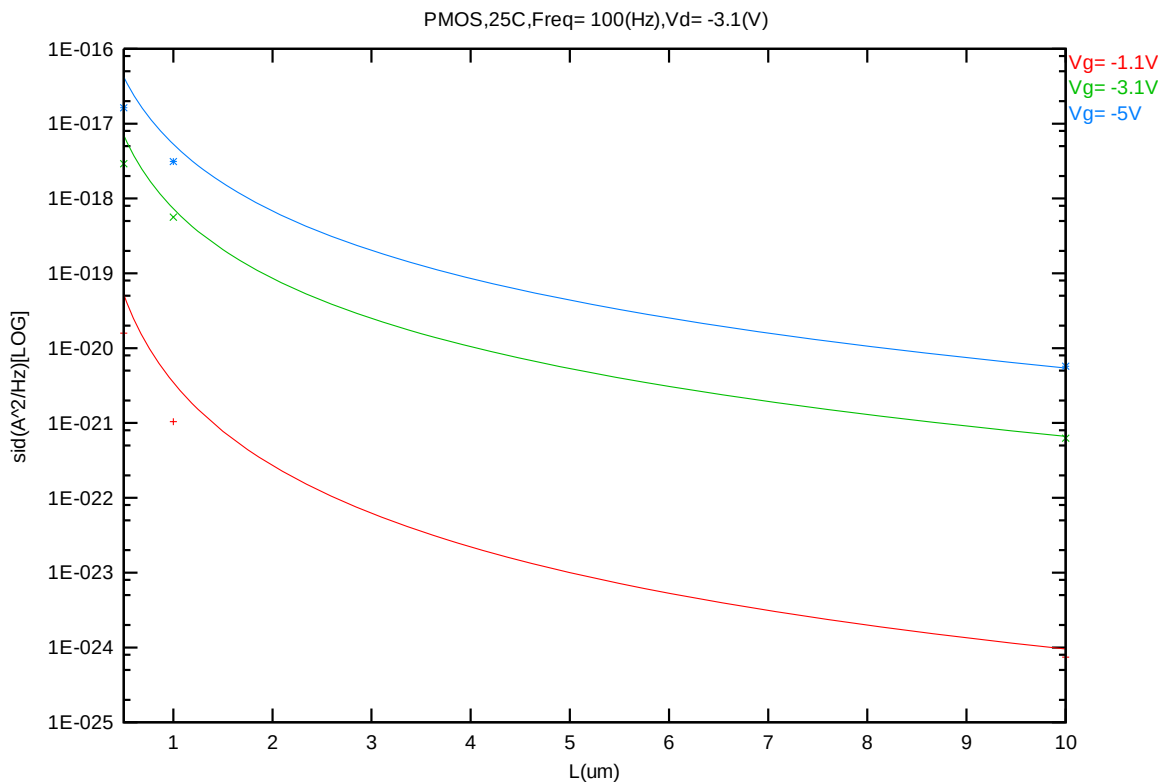


Figure 8-38 Noise vs. Length for PMOS @ Vd=-3.1 V

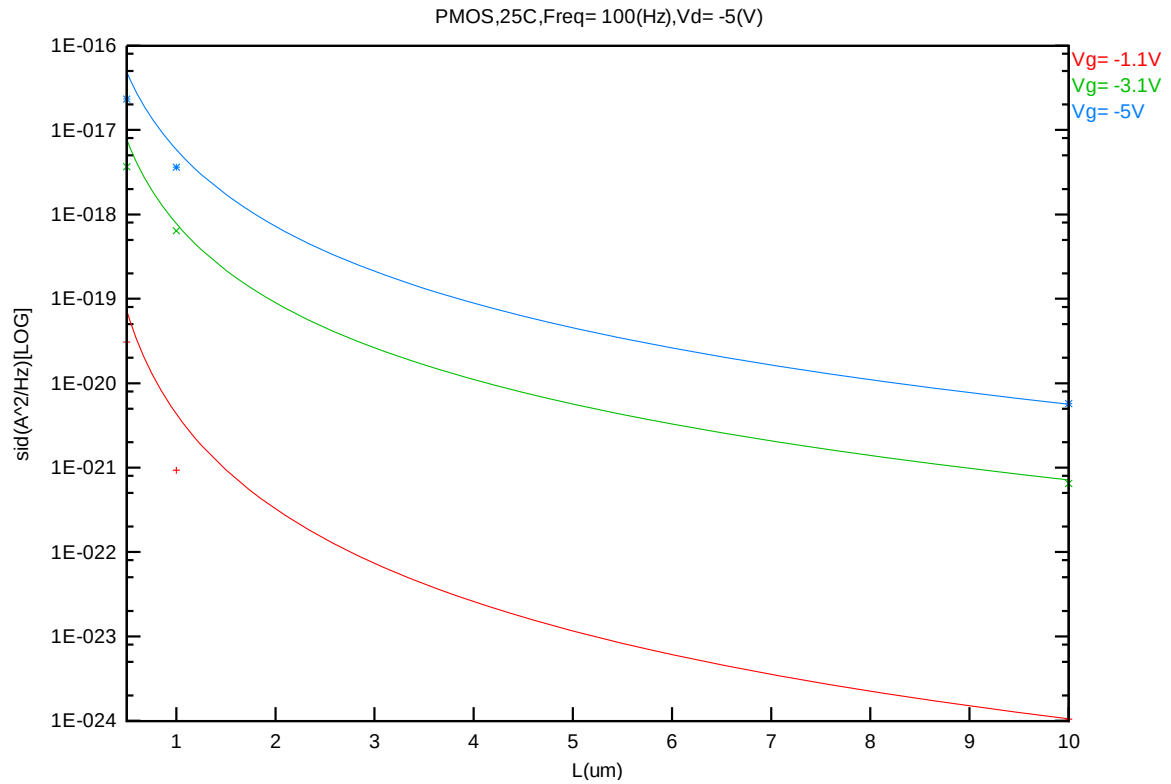


Figure 8-39 Noise vs. Length for PMOS @ Vd=-5 V

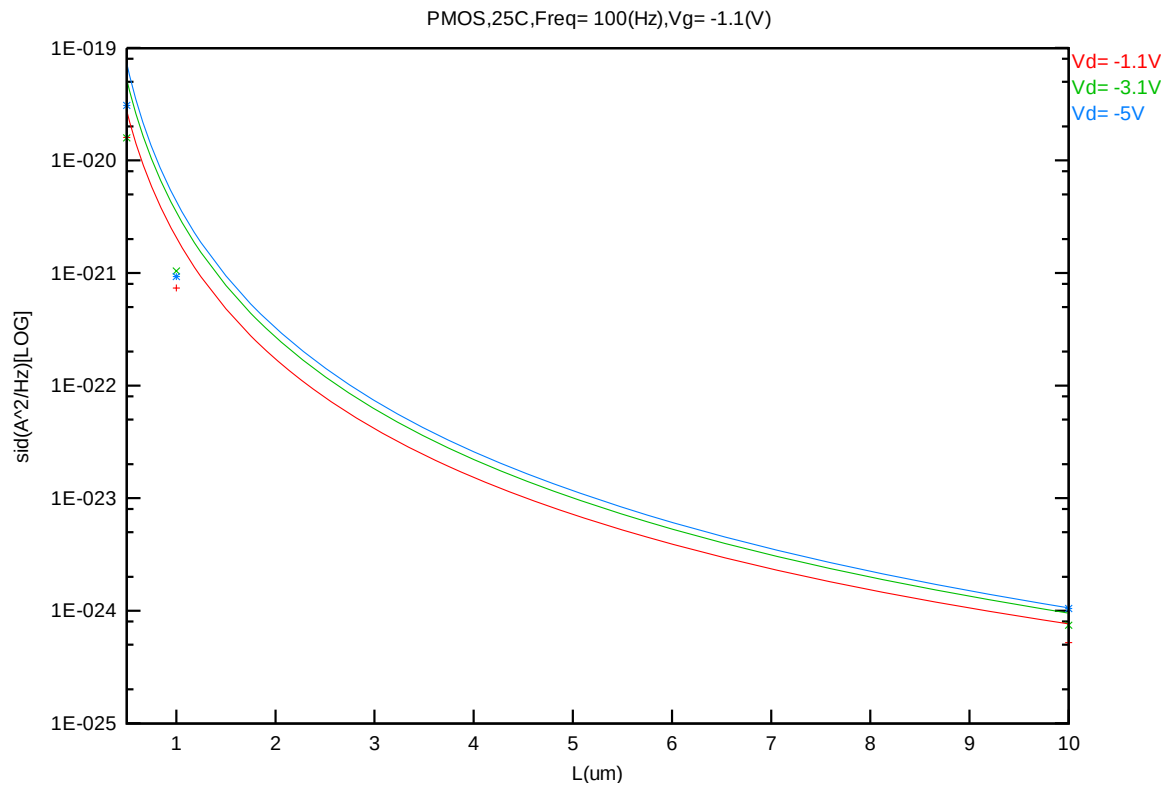


Figure 8-40 Noise vs. Length for PMOS @ Vg=-1.1 V

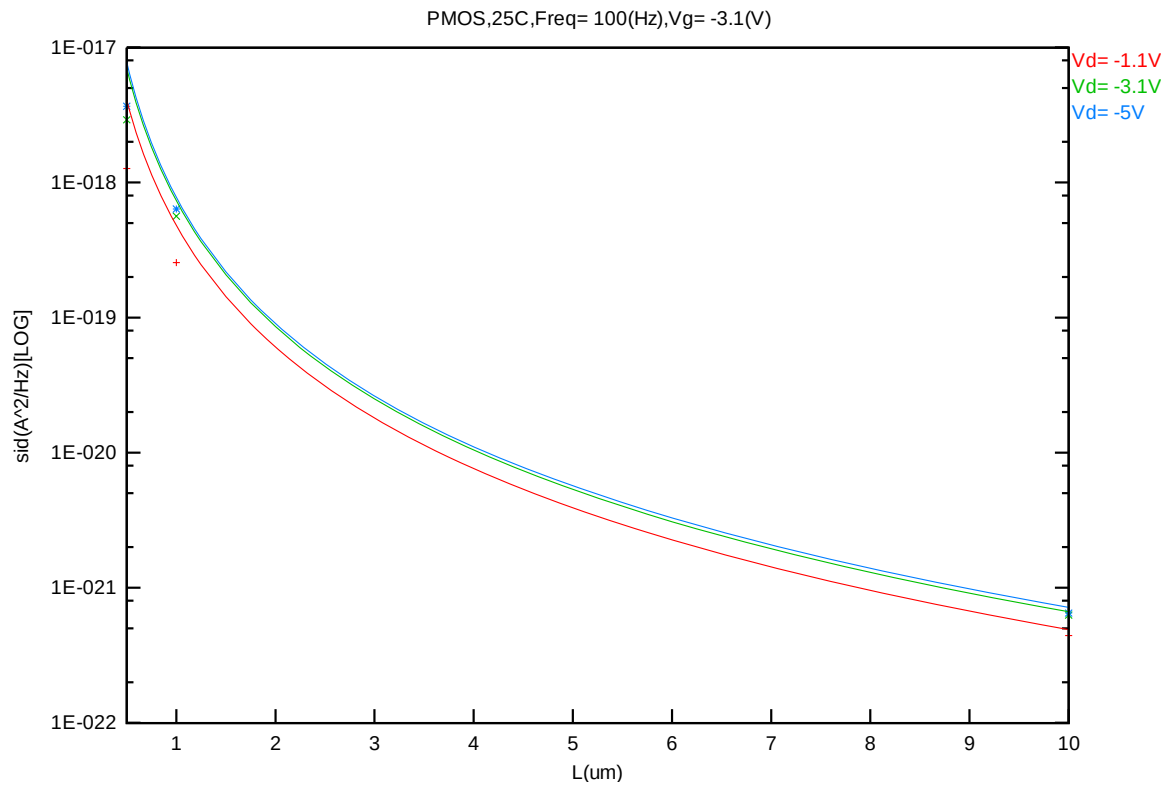


Figure 8-41 Noise vs. Length for PMOS @ Vg=-3.1 V

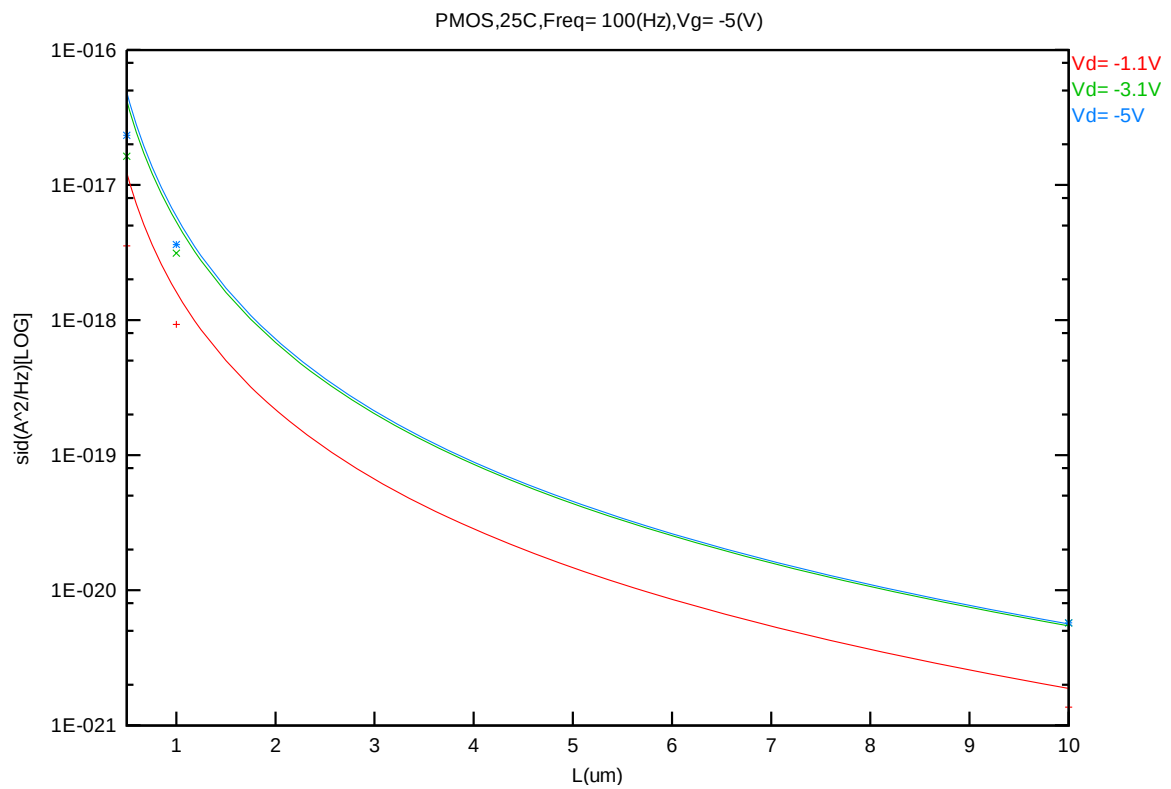


Figure 8-42 Noise vs. Length for PMOS @ Vg=-5 V

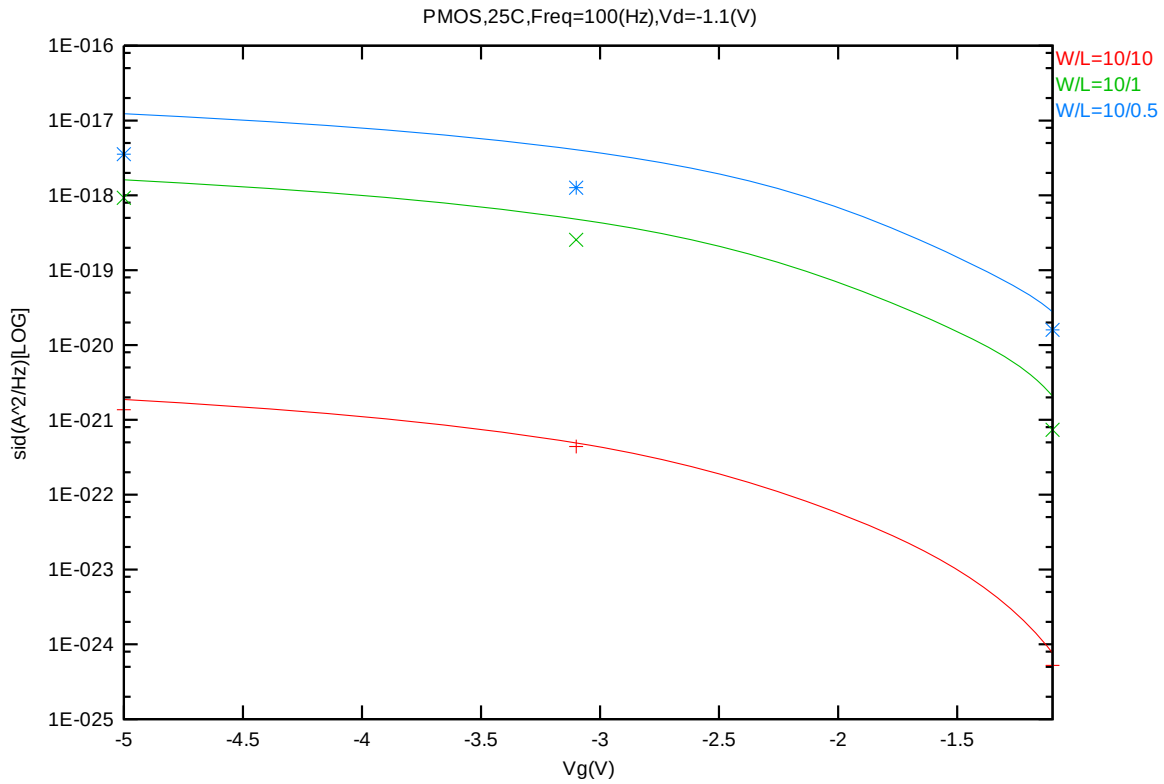


Figure 8-43 Noise vs. V_g for PMOS @ $V_d=-1.1$ V

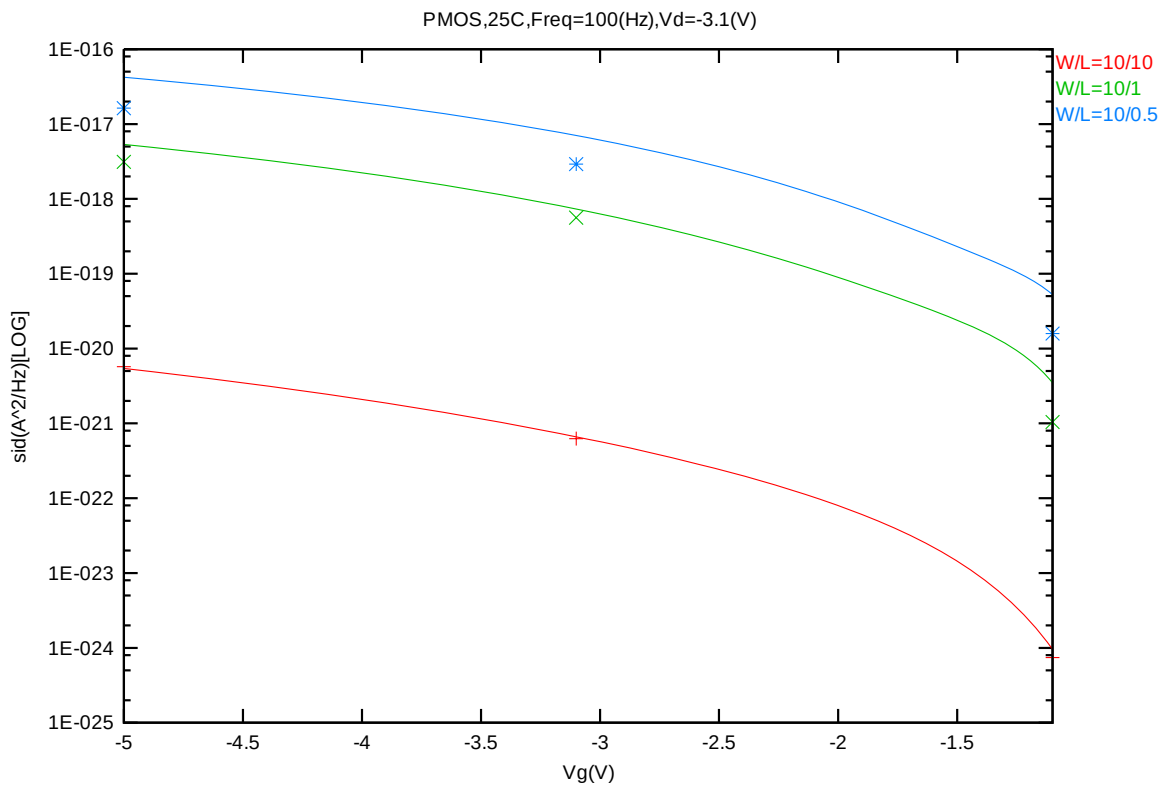


Figure 8-44 Noise vs. V_g for PMOS @ $V_d=-3.1$ V

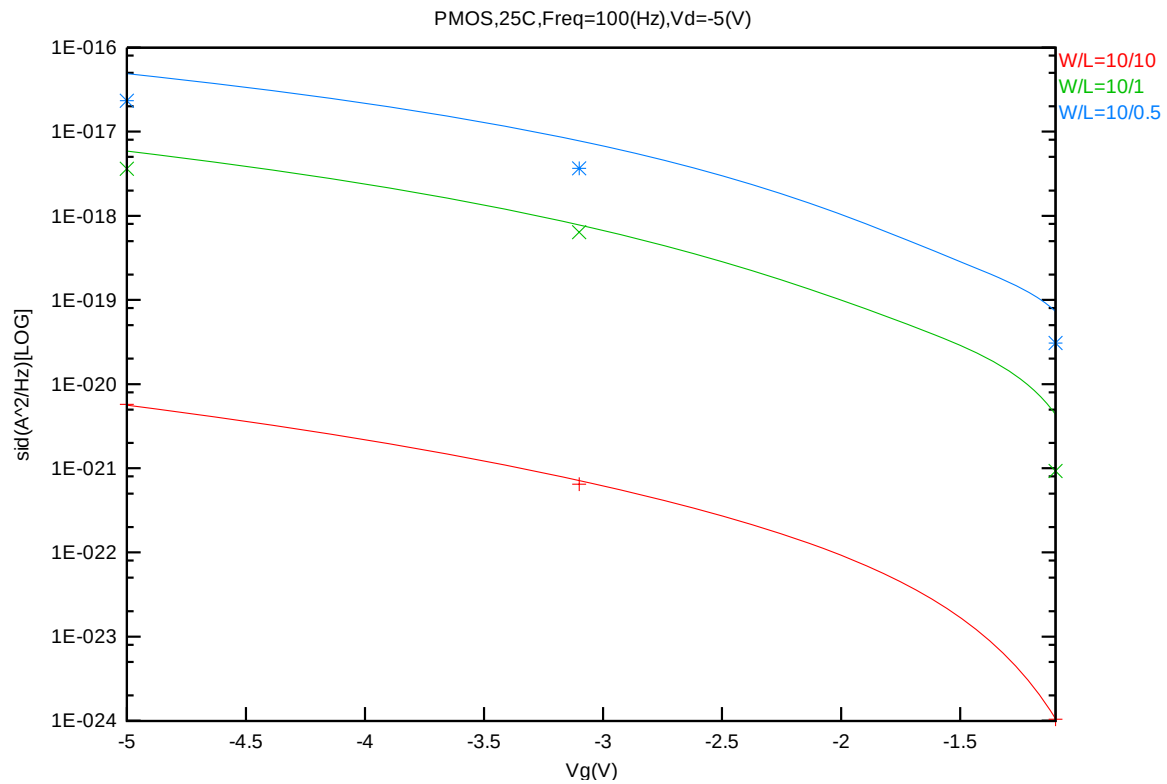


Figure 8-45 Noise vs. Vg for PMOS @ Vd=-5 V

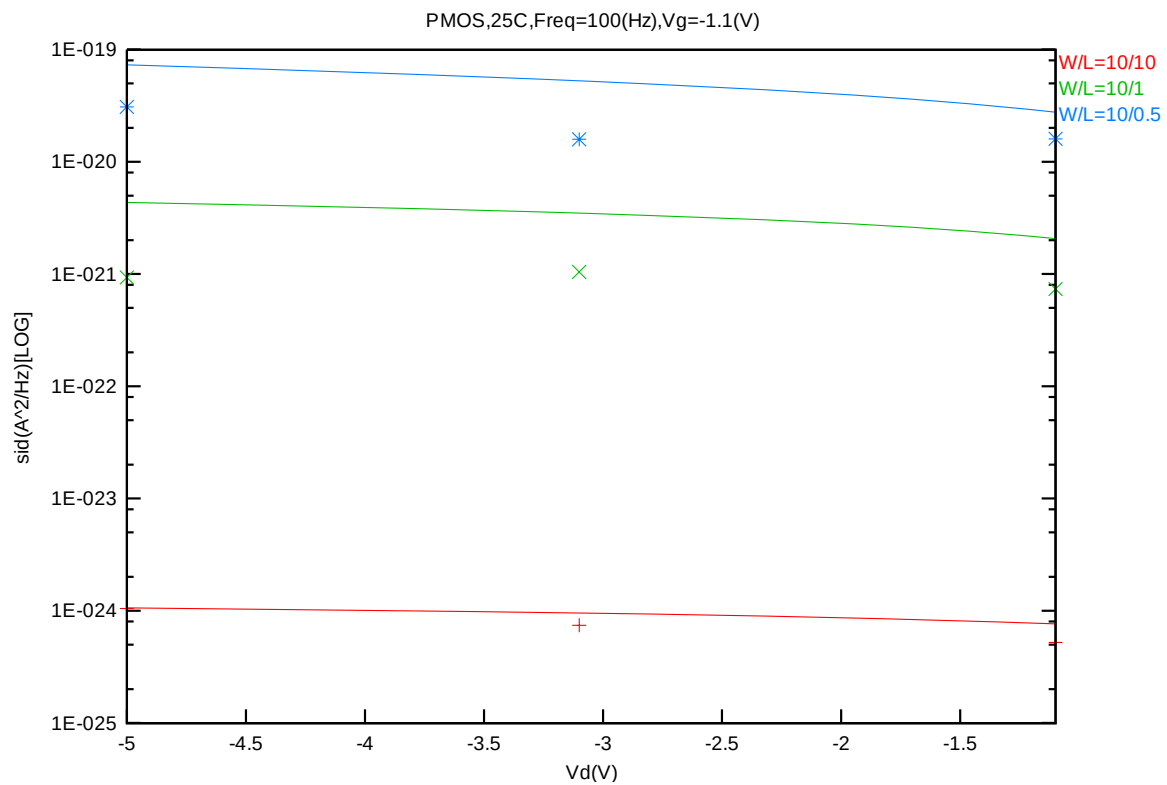


Figure 8-46 Noise vs. Vd for PMOS @ Vg=-1.1 V

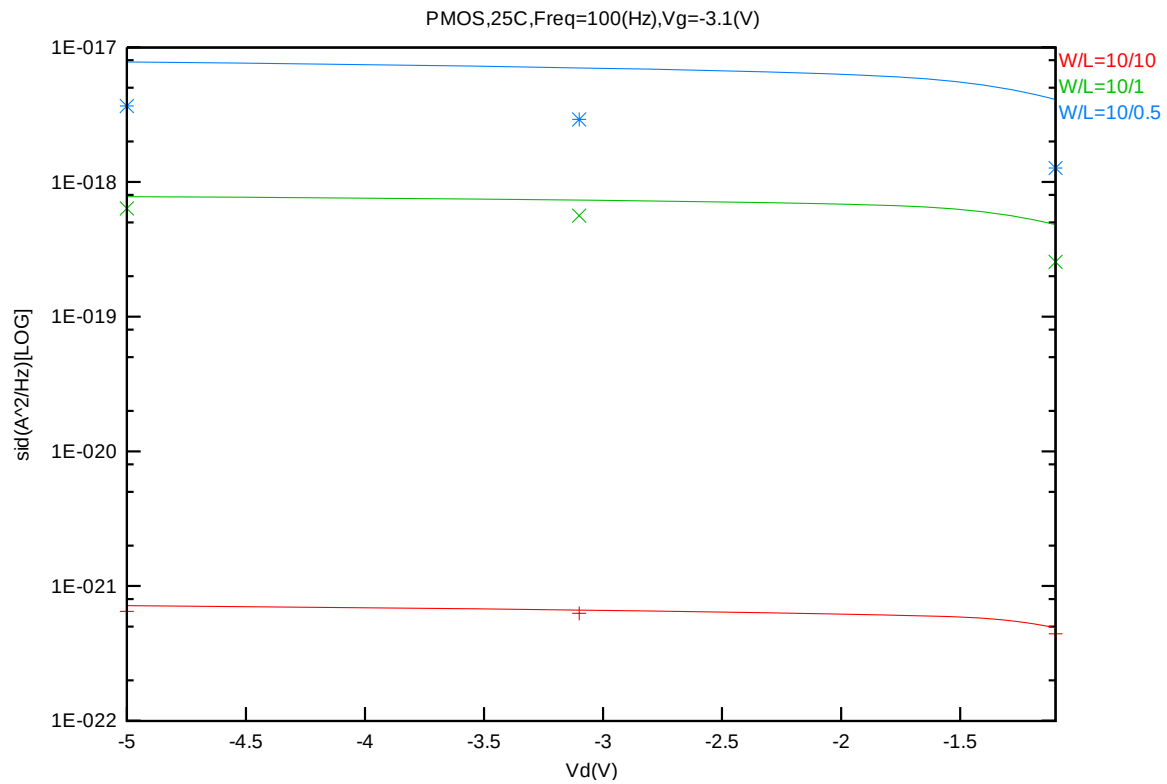


Figure 8-47 Noise vs. Vd for PMOS @ Vg=-3.1 V

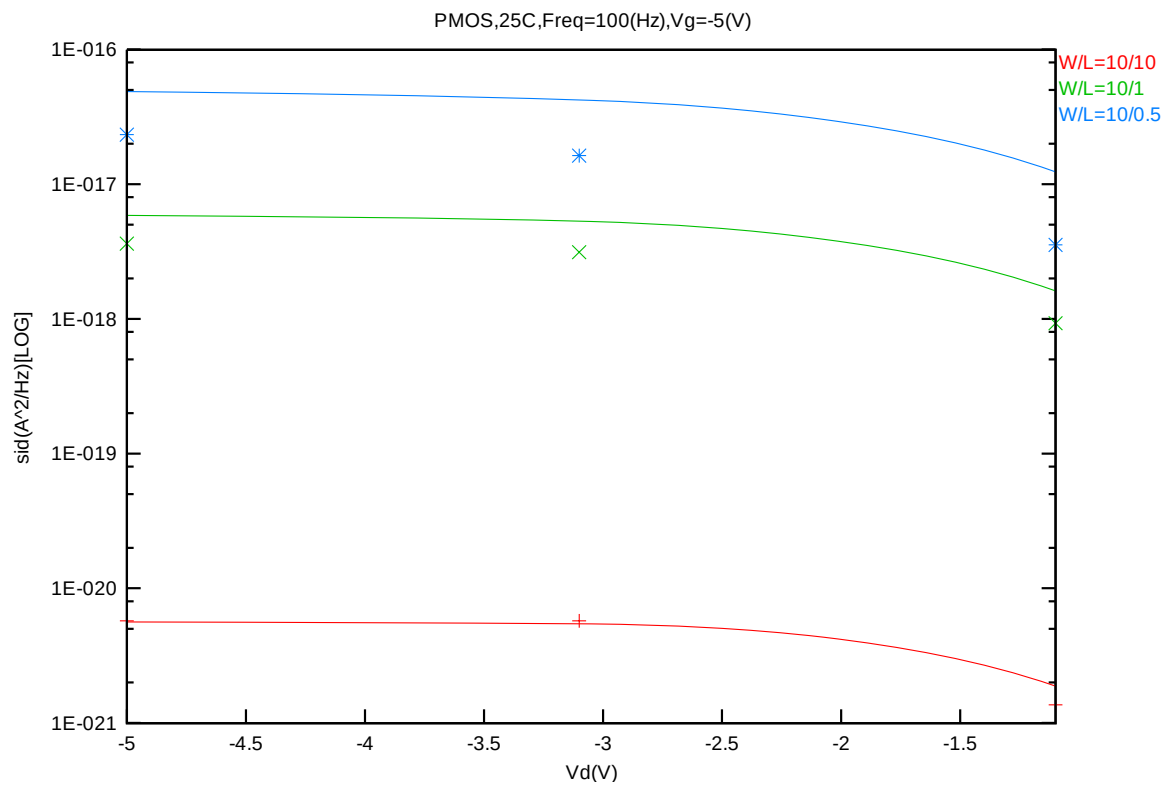


Figure 8-48 Noise vs. Vd for PMOS @ Vg=-5 V

9 Appendix

9.1 HSPICE Warning Message

Table 9-1 hspice warning message

NO.	Warning Message	Explanation
1.	-	-

9.2 ELDO Warning Message

Table 9-2 eldo warning message

NO.	Warning Message	Explanation
1.	-	-

9.3 spectre Warning Message

Table 9-3 hspice warning message

NO.	Warning Message	Explanation
1.	-	-